Software manual

TruTops CAD

TruTops from Version 8

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Edition **2015-12-04**

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TRUMPF

Chapter 1

Basics

1. Starting and exiting TruTops

1.1 Starting TruTops

- Select >*Start* >*Programs* >*TRUMPF.NET* >*TruTops*.

1.2 Exiting TruTops

- Select Exit TruTops .

2. Online help, software version, tips and tricks from the Internet

2.1 To call up the online help

Displaying the software manual

1. Select >*Help* >*Help* or press the <F1> key.

   The software manual of the relevant application (CAD, Nest, Laser, Punch...) opens (.pdf-file with bookmarks).

Displaying the Read me file

2. Select >*Help* >*Readme*.

2.2 Obtaining tips and tricks directly from the Internet

TruTops offers the option of accessing the Internet and thus the TRUMPF homepage directly. The homepage among other things has tips and tricks for programming and useful information about the product.

➢ Select >*Help* >*TruTops on the Internet*.

2.3 Showing software version

Displaying all software versions

1. Select the *?* symbol on the blue bar.

   The "Version information" mask appears.

2. Select *OK* to close.

Displaying the software version of individual applications

3. Select >*Help* >*Info....*

   The "Version" mask is displayed.

4. Select "Version" tab.

3. Applications in TruTops

After the installation of TruTops, every application is separately available as a separate "tab".

Every application can be opened several times, closed and reopened individually, closed permanently ("delete"), renamed or moved individually.

All individual modifications (including modifications to the interface, such as background color, adapted toolbars ...) are saved user-specifically and are loaded automatically when TruTops is opened.

Applications currently available in TruTops:

- TruTops CAD: drawing parts or loading them from other CAD systems, setting basic data (e.g. material).
- TruTops Tube CAD: drawing parts for tube processing or loading parts from other CAD systems, setting basic data (e. g. material).
- Nest: nesting parts, setting basic data (e.g. material).
- TruTops Laser: defining laser processing, setting basic data for the laser processing (e.g. slat layout, Cateye ...).
- TruTops Punch: defining punching or multi-purpose processing, setting basis data for the punching or multi-purpose processing (e.g. clamp data, repositioning cylinders ...).
- TruTops Tube: defining tube processing, setting basic data for the tube processing (e.g. clamping technique, measuring programs ...).
- TruTops Bend: defining bend processing, setting basic data for the bending program.

3.1 Applications, closing and reopening individually

Note

If an application has been closed, the individual user settings of the application are retained.

Closing an application

1. Right click on the tab of the application that is to be closed.

   The context menu is displayed.
2. Select >*Close*.

Reopen application?

3. Right click on the tab of any open application.

   The context menu is displayed.
4. Click >*Open*.
5. Select the application to be opened.

3.2 Applications, opening several times ("copy")

1. Right click on the tab of any application.

   The context menu is displayed.
2. Select >*New*.

   All available applications are displayed in another submenu.
3. Mark the application that is to be opened (i.e. copied) once more.

   The copied application is opened. The name is extended by an additive.

3.3 Application, closing permanently ("deleting")

Notes

- If an application is closed permanently, it is no longer opened automatically when TruTops is started. The individual user settings are lost.
- The application can be reopened using the context menu (right click on the tab of any open application, select >*New*).

1. Right click on the tab of the application that is to be closed permanently.

   The context menu is displayed.
2. Select >*Delete*.

3.4 Application, renaming

1. Right click on the tab of the desired application.

   The context menu is displayed.
2. Select >*Rename*.
3. Enter the new name of the application.
4. Press *OK*.

3.5 Switching language for all applications

1. Select *languages*.

   The "Language selection" mask is displayed.
2. Select the desired language.
3. Press *OK*.

The new language becomes effective the next time TruTops is started.

3.6 Closing open files

Files currently open in the applications (e.g. drawings, nesting jobs, processed tubes) can be closed without having to close and reopen the application itself.

➢ Select *>File >Close*.

The working range of the application is empty.

The application status is the same as directly after starting it up.

4. User-defined settings

TruTops can be set individually depending on the user (e.g. type and number of applications that are opened, individually created toolbars, size of the symbols...).

User-defined settings **within** an application are retained during an update as well as during a new installation of TruTops.

The type and number of applications that are opened when starting TruTops remain unchanged after updating TruTops. Each application is started once after being installed or whenever a new user logs on.

4.1 Copying and transferring user-defined settings

User-defined settings (e.g. type and number of applications that are opened, personal interface settings in the individual applications) can be copied and transferred as follows:

- Copy settings of a user and provide them to another user.
- Transfer the settings of a user from one work station (computer) to another work station (computer).
- Copy interface settings of an application (e.g. of "CAD 1") and transfer them to the "CAD 2" application (on the same computer).

Conditions

- Administrator rights and copyrights are required in all user folders in order to copy the user-defined settings.
- User-defined settings of TruTops can be copied and transferred to another computer only for the same user.

1. Select >*Start* >*Programs* >*TRUMPF.NET* >*TruTops Administration* >*TruTops Config*.

   The Microsoft Windows Explorer is started. The folder in which the user settings are saved is displayed automatically.

   Example: English operating system with "millerha" as user → path 'C:\Documents and Settings\millerha\Application Data \TRUMPF'" is opened.

   The 'TRUMPF' folder has a subfolder for TruTops and further subfolders for every application. In addition, the data is stored as per the product versions.

2. Either

   ➢ To transfer the settings to another computer: copy the entire 'TRUMPF' folder and paste it in the appropriate folder on another computer.

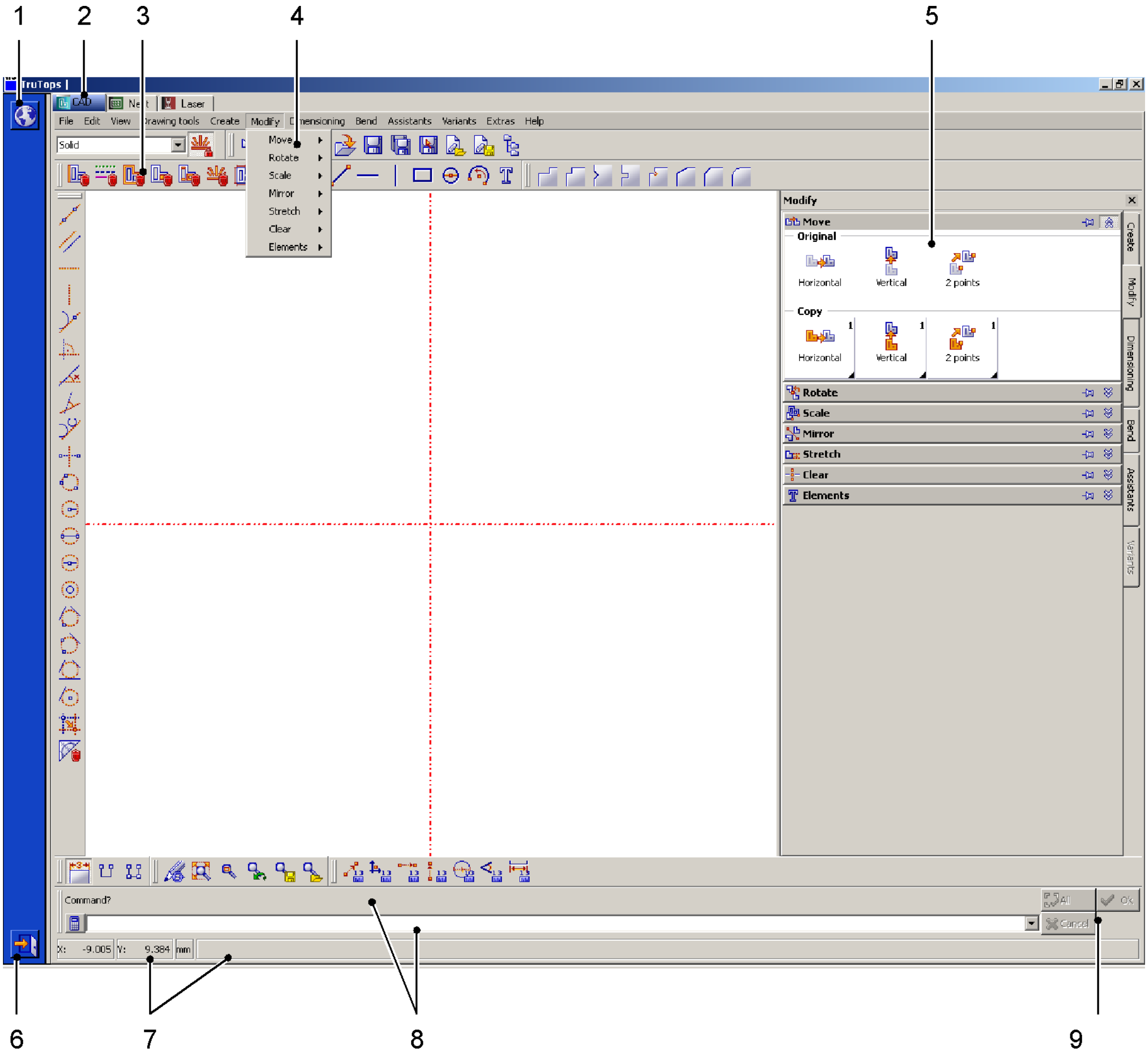
or

- To provide the settings to another user on the same computer: copy the entire 'TRUMPF' folder and store it in the intermediate storage at a suitable location.
- Log in another user on to the computer.
- Paste the entire 'TRUMPF' folder in the appropriate folder of another user.

or

- To copy (e.g.) the settings of the "CAD 1" application and transfer them to "CAD 2" application: copy the 'x.usr' file in the folder of the "CAD 1" application and paste it in the folder of the "CAD 2" application.

5. Menus, function browsers and toolbars



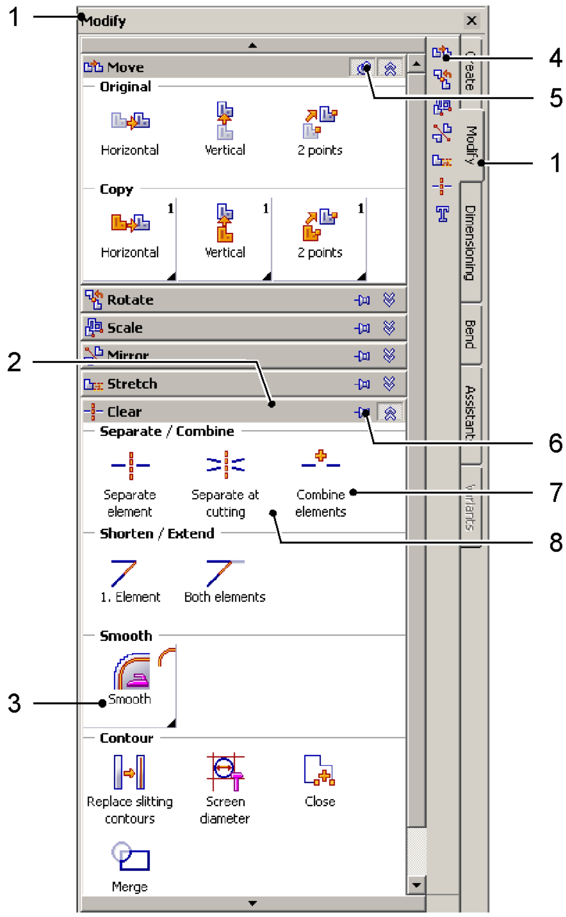
1 Language switchover
2 Application
3 Toolbars (can be customized, can be moved to the desired position using the sizing tool)
4 Main menu bar with main menu and submenus
5 Function browser
6 Exiting TruTops
7 Status bar and message line
8 Command and input line (can be moved to the desired position using the sizing tool)
9 Command block

TruTops user interface (here: CAD application is open) Fig. 50209

Main menus and submenus

The main menus and submenus contain **all** functions available in the relevant application (CAD, Nest, Laser, Punch ...).

Function browser



1 Function category (corresponding to an entry in the main menu)
2 Function group (corresponding to a submenu)
3 Multi-function button
4 Quick navigation bar (can be displayed and hidden)
5 Pin, pressed (function group is permanently open)
6 Pin, open (function group is closed as soon as another one is opened)
7 Button text (can be displayed and hidden)
8 Scroll with the mouse wheel

Function navigator Fig. 50208

The function browser contains the most important (TruTops-specific) functions. The function browser is subdivided into separate function categories (e.g. "Modify"). The function categories are further divided into function groups (e.g. "Move").

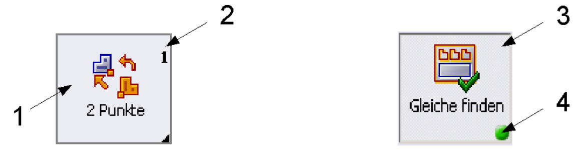
The function browser can be dragged to the desired width. Frequently used function groups within a function category can be "pinned", i.e. can be opened permanently.

Multi-function buttons

Some of the buttons in the function browser function as multi-function buttons or options buttons.

Multi-function buttons are marked with a triangle in the bottom right. They display their current function in the function browser. The current function can be changed with a right click (submenu).

Options buttons switch functions on or off.



1	Multi-function button	3	Options button
2	Displays the current function (here: create 1 copy and move by 2 points)	4	Green: Function is activated. Gray: Function is deactivated

Multi-function buttons, options buttons   Fig. 50212

Toolbars

Toolbars allow quick access to the most important functions.

Depending on the user, they can be created individually, can be added to commands, can be moved and downsized. The toolbars can thus be adapted to typical programming tasks at any time.

Status bar and message line

- The current X and Y position of the mouse pointer is displayed in the status bar (including the measuring unit).
- The message line displays the ongoing action. Examples:
  - `Reading of file.... finished`
  - `Drawing deleted!`

Command line and input line

- The command line gives instructions about what is to be done. Examples:
  - `Enter first point`
  - `Identify object to be rearranged`
  - `Mark element or click on "OK"`
- The required data can be entered in the input line. Each entry must be confirmed with the enter key (↲) on the keyboard. Examples of entries:
  - X and Y coordinates of points.
  - Angle.

5.1 Displaying and hiding the function browser

Note

In the Bend application, the function browser cannot be hidden.

➢ Activate or deactivate >*View* >*Toolbars* >*Function browser*.

5.2 Displaying and hiding button texts in the function browser

The button texts in the function browser can be hidden and displayed again.

If the button texts are hidden, one can only work with icons and tool tips.

1. Either
   - Select >*Tools* >*Options...*.

   or

   - Select >*View* >*Toolbars* >*Adapt...*.
   - Select the "Options" tab.
2. Activate or deactivate "Show icon and text".
3. Select *Close*.

5.3 Displaying and hiding quick navigation bar

Function groups can be opened very quickly with the quick navigation bar (4) (see "Fig. 50208", pg. 1-15).

1. Either
   - Select >*Tools* >*Options...*.

   or

   - Select >*View* >*Toolbars* >*Adapt...*.
   - Select the "Options" tab.
2. Activate or deactivate "Show Quick navigation bar".
3. Select *Close*.

5.4 Displaying and hiding toolbars

1 Sizing tool 2 Button

"File" icon bar Fig. 50210

1. Select >*View* >*Toolbars*.
2. Activate or deactivate the desired toolbars.

5.5 Moving or downsizing toolbars

Moving toolbars

1. Using the mouse, move the toolbar to the desired position with the sizing tool (= vertical lines on the left side).

Downsizing toolbars

2. To downsize the toolbar on the left: use the mouse to move the toolbar with the sizing tool (= vertical lines on the left side) to the left until the toolbar on the left is downsized.

   or

   ➢ Using the mouse, move the toolbar with the sizing tool (= vertical lines on the left side) to the right until the toolbar is downsized.

   Two arrows are displayed on the right side of the downsized toolbar (>>).

1 Arrows to select the hidden functions

Minimized "File" icon bar Fig. 50211

3. To select the functions that are (now) hidden: click >> and select the desired function.

5.6 Modifying existing toolbars, creating own toolbars

In addition to the already existing toolbars, the user can create his own toolbars and equip them with the required buttons.

Alternatively, unnecessary buttons can be deleted and missing buttons can be added to the existing toolbars.

Creating a toolbar

1. Select >*View* >*Toolbars* >*Adapt...*.
2. Select *New* in the "Toolbars" tab.
3. Enter the name of the user-defined toolbar and select *OK*.

   The toolbar that is (still) empty is displayed automatically under the main menu bar.

Adding a button to the toolbar

4. Select the "Commands" tab.
5. In the "Categories" field, click on the desired category.
6. Drag the desired command from the "Commands" field into the new toolbar.
7. After creating all the desired toolbars, select *Close*.

Removing a button again?

8. If the "Adapt" mask is no longer open: select >*View >Toolbars >Adapt...* again.
9. Drag the buttons to be removed out of the toolbar using the mouse and drop them.

5.7 Renaming a toolbar

1. Select >*View >Toolbars >Adapt...*.
2. Mark the icon bar that is to be renamed in the "Toolbars" tab.
3. Select *Rename*.
4. Enter the new name of the toolbar and select *OK*.
5. Select *Close*.

5.8 Deleting a toolbar

1. Select >*View >Toolbars >Adapt...*.
2. Mark the icon bar that is to be deleted in the "Toolbars" tab.
3. Select *Delete*.
4. Select *Close*.

5.9 Resetting toolbars to their original status

All individual modifications (e.g. user-defined toolbars, moved toolbars) are cancelled when resetting the toolbars. The originally installed status is restored. (Exception: size of the symbols.)

1. Select >*View >Toolbars >Adapt...*.
2. In the "Toolbars" tab, select *Reset...*.

5.10 Modifying the size of the symbols

The size of the symbols in the toolbars and the function browser can be modified. (However, the size of the symbols in the menus and the quick navigation bar cannot be modified.)

1. Either
   - Select >*Tools >Options...*.

or

- ➢ Select >*View* >*Toolbars* >*Adapt...*.
- ➢ Select "Options" tab.

2. Activate or deactivate "Large symbols".
3. Select *Close*.

5.11 Displaying overlapping contours

➢ Select >*View*, >*Highlight overlapping contours*.

The overlapping contours are displayed in cyan and all intersecting points as yellow squares. The contours are separated at the intersecting points.

5.12 Reactivating disabled optional message boxes

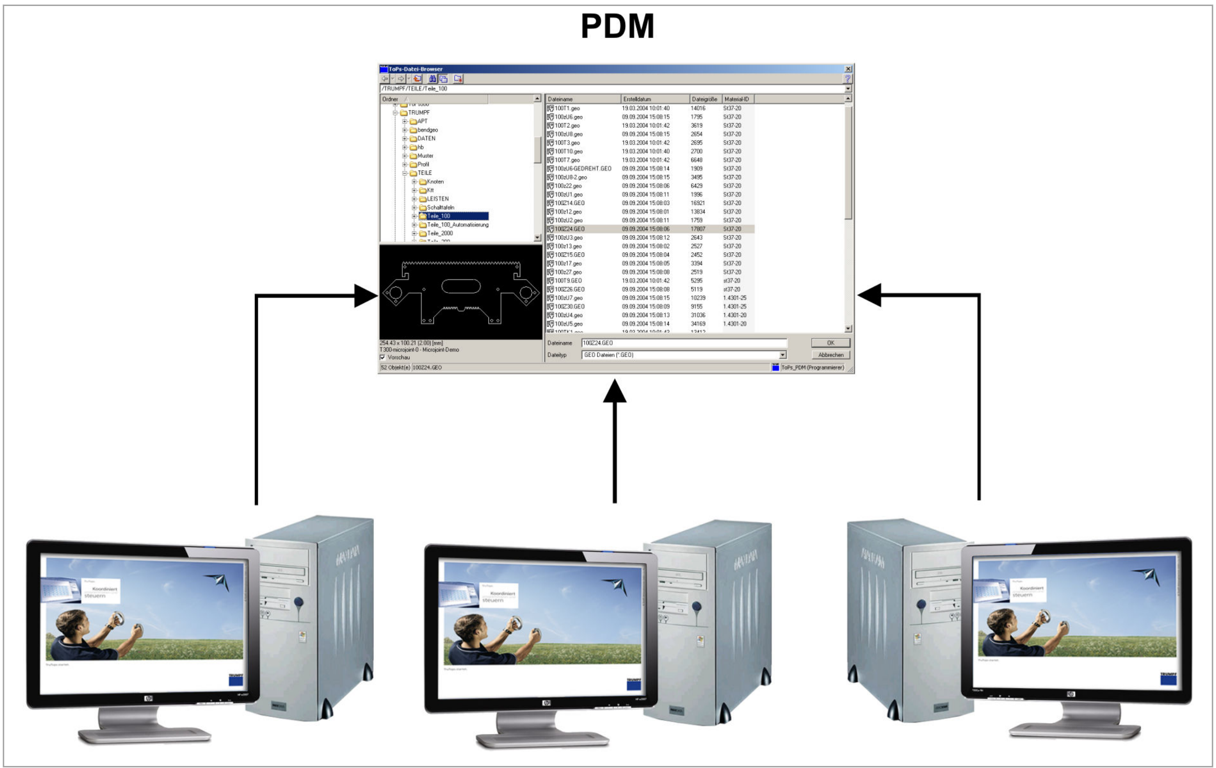
1. Select >*Extras* >*Modify data...*.
2. Select >*User* >*Configuration*.

   The "Configuration" mask is displayed.
3. Select "Show optional message boxes again".
4. Press *Modify*.

6. TruTops with PDM (Product Data Management)

6.1 What is PDM?

With the PDM Data Management, files can be saved and managed centrally for all products.



Data Management with PDM Fig. 56726

The advantages of PDM

- PDM is a central storage location for any number of TruTops work stations and various TruTops products.
- All TruTops products work using PDM.
- The dependencies between files are displayed for all products.
- You can delete and export files, recognize dependencies and view file properties with a mouse click.

- You can use the new TruTops file browser to search for parts quickly and specifically using the advanced search criteria in PDM.
- There are three options available as interfaces to older TruTops versions and other software products which are not PDM-compatible:
  - Importing and exporting files and directories via the TruTops file browser.
  - Importing and exporting files and directories with the Windows prompt or automated by means of a script.
  - Direct access via an enabled Windows folder or network drive.

6.2 TruTops with PDM, TruTops without PDM – a comparison

Characteristic	TruTops with PDM	TruTops with file system
The system checks whether the files are up to date.	Yes	-
The dependencies between files are monitored for all products.	Yes	-
Searching for and displaying files by means of TruTops-specific characteristics (e.g. material) is supported.	Yes, fast	Yes, slow
Heterogeneous data management (metric and inch) is supported.	Yes	Yes
File attributes ("Properties") created with different character sets (European, Japanese, ASCII…) can be saved (display in TruTops still not correct).	Yes	-

Comparison of TruTops with PDM and TruTops with file system Tab. 1-1

When working without the PDM, all functions of the TruTops file browser are available.

7. PDM functions, calling up via prompt/ scripts (pdmCLI.exe)

The following PDM functions can be called up via the Windows prompt or can be executed in an automated manner via script files (*.bat).

Function	Command	Remark
Exporting files/directories from a directory	Get	(see "Files and folders, importing/exporting (PDM)", pg. 1-41)
Importing files into a folder	Put	(see "Files and folders, importing/exporting (PDM)", pg. 1-41)
Deleting files from a folder	Delete	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Listing files in a folder	List	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Generating lists of attributes	ListAttributes	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Creating a new (sub) folder	Mkdir	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Deleting an empty (sub) folder	Rmdir	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Copying a file or folder into a target folder	Copy	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Moving a file or folder to a target folder	Move	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Redetermining file properties and dependencies	Rescan	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)
Executing commands contained in a script file	Batch	(see "Examples of PDM functions which can be performed with pdmCLI.exe", pg. 1-24)

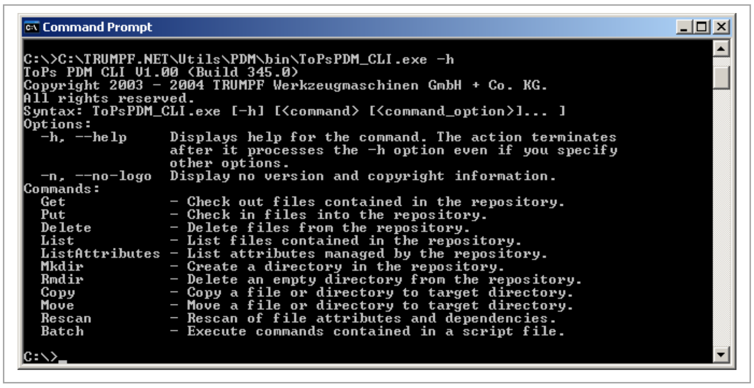
Tab. 1-2

7.1 Calling up help for pdmCLI.exe

1. Start Windows prompt.
2. Go to the '...:\TRUMPF.NET\Utils\PDM2\bin' folder.
3. Enter the following command:

```
pdmCLI.exe -h
```

Help for pdmCLI.exe is displayed:

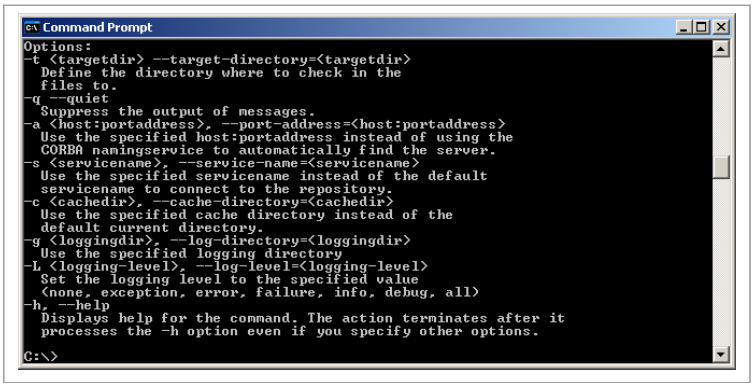


Help for pdmCLI.exe Fig. 43561

4. To call up help for a specific command (e.g. "Put"), enter the following command:

   `pdmCLI.exe -h Put`

   Help for the specific command is displayed:



Help on the "Put" command Fig. 43562

7.2 Examples of PDM functions which can be performed with pdmCLI.exe

Notes

- Write **everything in a single line** when entering commands in the command prompt or in script files.
- Be sure that slashes, backslashes, and spaces are correct.
- Replace `server name` with the actual name of the server.
- Only those parts highlighted in **bold** can vary from the commands shown in the examples. Everything before and after it remains the same.

Opening command prompts, creating script files

1. Either
   - Start Windows command prompt.
   - Go to the folder '…:\TRUMPF.NET\Utils\PDM2\bin''.

   or

   - Write `cd C:\TRUMPF.NET\Utils\PDM2\bin` in a self-written BAT shell file (*.bat).
   - Extend the following (example) command lines.

Deleting file

2. To delete the 'box.mi' file in the '/TRUMPF.NET/ TEST_DIR_1/' folder, for example, enter the following command:

   pdmCLI.exe **Delete /TRUMPF.NET/ TEST_DIR_1/box.mi** --service-name=TruTopsPDM2 --port-address=servername:9999

3. Press <Enter> or <Return> (↲).

Listing files in a folder

4. To list all files in the '/TRUMPF.NET/ TEST_DIR_1' folder, for example, enter the following command:

   pdmCLI.exe **List /TRUMPF.NET/TEST_DIR_1** --service-name=TruTopsPDM2 --port-address=servername:9999

5. Press <Enter> or <Return> (↲).

Generating lists of attributes

6. To create a list of attributes, enter the following command:

   pdmCLI.exe **ListAttributes** --service-name=TruTopsPDM2 --port-address=servername:9999

7. Press <Enter> or <Return> (↲).

Creating a new (sub) folder

8. To create a folder '/TRUMPF.NET/TEST_DIR_1', for example, enter the following command:

   pdmCLI.exe **Mkdir /TRUMPF.NET/TEST_DIR_1** --service-name=TruTopsPDM2 --port-address=servername:9999

9. Press <Enter> or <Return> (↲).

Deleting a (sub) folder

10. To delete the folder '/TRUMPF.NET/TEST_DIR_1', for example, enter the following command:

    pdmCLI.exe **Rmdir /TRUMPF.NET/TEST_DIR_1** --service-name=TruTopsPDM2 --port-address=servername:9999

11. Press <Enter> or <Return> (↲).

Copying a single file into a different folder

12. To copy the 'box.mi' file in the '/TRUMPF.NET/ TEST_DIR_1' folder to the '/TRUMPF.NET/ TEST_DIR_2' folder, for example, enter the following command:

    pdmCLI.exe **Copy /TRUMPF.NET/TEST_DIR_1/ box.mi --target-directory=/TRUMPF.NET/ TEST_DIR_2** --service-name=TruTopsPDM2 -- port-address=servername:9999

13. Press <Enter> or <Return> (↲).

Copying several files into a different folder

14. To copy all files from the '/TRUMPF.NET/ TEST_DIR_1' folder into the '/TRUMPF.NET/ TEST_DIR_2' folder, for example, enter the following command:

pdmCLI.exe **Copy /TRUMPF.NET/TEST_DIR_1 --target-directory=/TRUMPF.NET/TEST_DIR_2** --service-name=ToPs_PDM --port-address=servername:9999

15. Press <Enter> or <Return> (↲).

Moving a single file into a different folder

16. To move the 'box.mi' file from the '/TRUMPF.NET/TEST_DIR_1/' to the '/TRUMPF.NET/TEST_DIR_2' folder, for example, enter the following command:

    pdmCLI.exe **Move /TRUMPF.NET/ TEST_DIR_1/ box.mi --target-directory=/TRUMPF.NET/ TEST_DIR_2** --service-name=TruTopsPDM2 -- port-address=servername:9999

17. Press <Enter> or <Return> (↲).

Moving a folder

18. To move the '/TRUMPF.NET/TEST_DIR_1' folder into the '/TRUMPF.NET/TEST_DIR_2' folder, for example, enter the following command:

    pdmCLI.exe **Move /TRUMPF.NET/ TEST_DIR_1 --target-directory=/TRUMPF.NET/TEST_DIR_2** --service-name=TruTopsPDM2 --port-address=servername:9999

19. Press <Enter> or <Return> (↲).

Redetermining file properties and dependencies

20. Enter the following command:

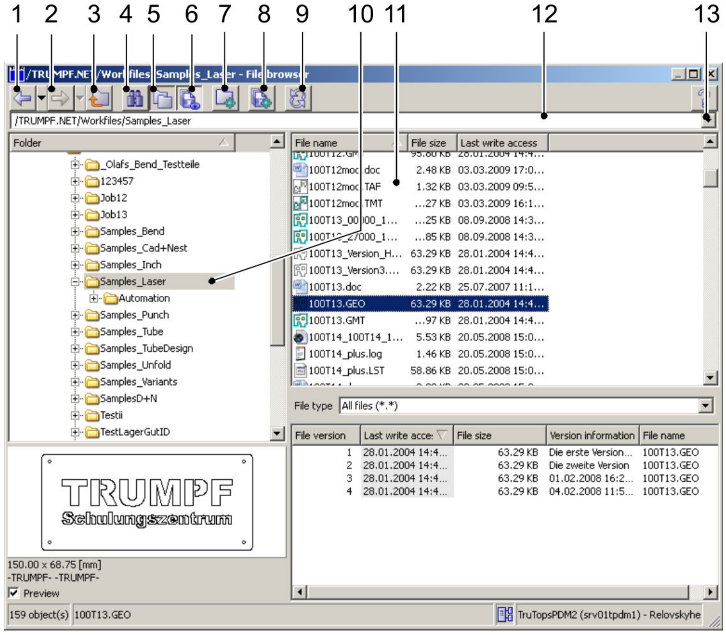
    pdmCLI.exe **Rescan** --service-name=TruTopsPDM2 --port-address=servername:9999

21. Press <Enter> or <Return> (↲).

8. TruTops file browser

The TruTops file browser features the following functions:

- Convenient and simple operation with a user-friendly interface (Windows standard).
- Advanced search functions: various additional configurable search criteria assist you in systematic part management.
- All files in the TruTops file browser are displayed in a folder structure.
- The dependencies between files are displayed (PDM).
- Simple import and export of files (PDM).
- Creating and restoring file versions (PDM).



1	Navigate back	8	Create version
2	Navigate forwards	9	Update view
3	Go to next higher directory	10	Navigator
4	Filter view (search)	11	Directory contents (file list)
5	Folder view	12	Current path
6	Show and hide version window	13	Show directories from which a file has already been opened
7	Create new folder		

TruTops file browser (here via File, Open, with PDM) Fig. 53347

Quick access to folders

Quick access to a folder in the TruTops file browser is possible as follows:

- Open file browser (e.g. via >*File* >*Open*).
- Keep the <Shift> key pressed and press the <Tab> key twice **or** click directly in the file list using the mouse.
- Enter the first letters of the desired folder until the required folder is marked.

Last selected folder

TruTops shows the current path under the toolbar (9).

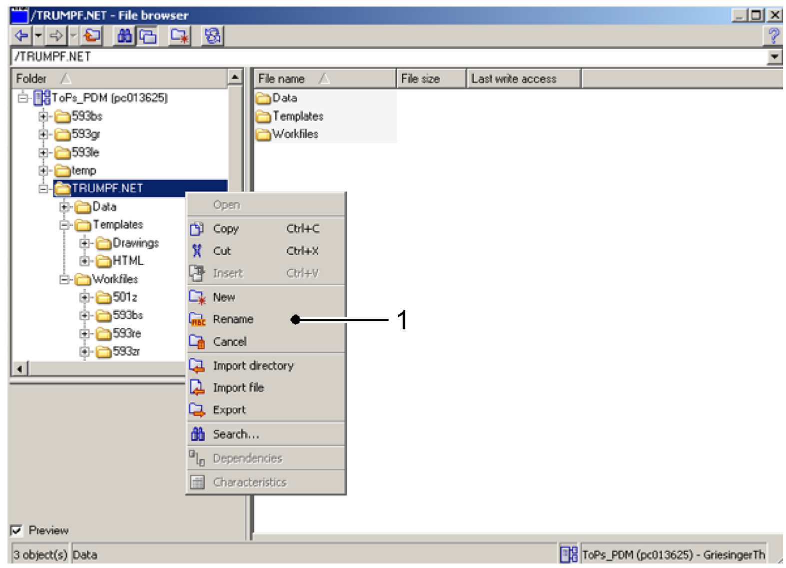
To show folders from which a file has already been loaded: open the current path with ▼ (11).

File sorting options

The displayed files can be sorted according to the required detail by clicking on the detail (e.g. "File size").

The detail used for sorting is marked in gray. Clicking repeatedly on the detail modifies the sorting direction (upward/downward).

Context menu of the TruTops file browser



1 Shortcut Menu

Context menu of TruTops file browser (here via Open file, with PDM) Fig. 41364

Note

Menu items that cannot be selected at the relevant point are grayed out.

8.1 Opening TruTops file browser

The TruTops file browser is automatically opened when, for example, a file is opened via *>File >Open…* or a file is saved via *>File >Save*.

Additionally, the TruTops file browser can be opened independent from the loading or saving of a file:

➢ In an arbitrary TruTops application: select *>File >File browser...*.

or

➢ Select *>Start >Programs >TRUMPF.NET >PDM Browser*.

8.2 Creating a new folder

1. Open file browser (e.g.via >*File* >*File browser*).
2. Either
   - ➢ Select  (Create *new folder*).

   or

   - ➢ Open the context menu with the right mouse button.
   - ➢ Select >*New*.
3. Enter the name of the new folder.
4. Press <Enter> or <Return> (↵).
5. If required, refresh view with  (*Refresh*) or <F5>.

8.3 Deleting folders

1. Open file browser (e.g.via >*File* >*File browser*).
2. Select the folder to be deleted.
3. Either
   - ➢ Press <Entf> or <Del>.

   or

   - ➢ Open the context menu with the right mouse button.
   - ➢ Select >*Delete*.
4. Confirm the query with *OK*.

8.4 Renaming files or folders

1. Open file browser (e.g.via >*File* >*File browser*).
2. Either
   - ➢ Go to the folder to be renamed.
   - ➢ Select the folder.

   or

   - ➢ Go to the folder in which files are to be renamed.
   - ➢ Select the file.
3. Either
   - ➢ Click on the folder or the file name once.

   or

   - ➢ Open the context menu with the right mouse button.
   - ➢ Select >*Rename*.
4. Enter a new name.

5. Confirm the new name with <Enter> or <Return> (↲) or click outside the folder or file name.

8.5 Deleting and restoring files, emptying the recycle bin (PDM)

When working with PDM, deleted files are moved to the recycle bin of the file browser. These files can be restored from the recycle bin if required (including all attributes such as material, material thickness etc.).

The folder name of the recycle bin is 'RecycleBin' and can be modified (at the PDM server) by the PDM administrator.

When working in the file system, a simple query appears asking whether the file should be deleted permanently.

Deleting files

1. Open file browser (e.g.via >*File* >*File browser*).
2. Go to the folder from which files are to be deleted.
3. Select the file(s) to be deleted. To mark several files: press <Strg> or <Ctrl> pressed.
4. Either
   - ➢ Press <Entf> or <Del>.

   or

   - ➢ Open the context menu with the right mouse button.
   - ➢ Select *Delete*.
5. Confirm the query.

   When working with PDM, the files (including the path specifications and the extension '.trash.zip') are moved to the recycle bin of the file browser.

   Tip

   To delete files **permanently** (then they are **not** moved to the recycle bin and can**not** be restored): press the Shift key when deleting files.

Restoring deleted files

6. Go to the recycle bin of the TruTops file browser ('RecycleBin').
7. Select the file to be restored (i.e. the complete file specification with the file name and the extension '.trash.zip'). To mark several files: press <Strg> or <Ctrl> pressed.
8. Open the context menu with the right mouse button.
9. Select >*Restore*.

Emptying recycle bin

10. Go to the recycle bin of the TruTops file browser ('RecycleBin').
11. To delete single files in the recycle bin permanently: select the files. To mark several files: press <Strg> or <Ctrl> pressed.

12. To delete all files in the recycle bin permanently: select files with <Strg>+<A> or <Ctrl>+<A>.
13. To delete the complete recycle bin permanently: select the recycle bin ('RecycleBin').

Note

If the entire recycle bin is deleted (folder including its content), it is automatically recreated the next time a file is deleted.

14. Either
    - Press <Entf> or <Del>.

    or

    - Open the context menu with the right mouse button.
    - Select >*Delete*.
15. Confirm the query.

Renaming recycle bin

Note

The recycle bin can be renamed only on the PDM server.

16. Select >*Start* >*Programs* >*TRUMPF.NET* >*PDM Administration* >*PDM Admin*.
17. Double click in the `RECYCLE_BIN_NAME` line in the table cell of the "ParamValue" column (default content of the cell: `RecycleBin`).
18. Rename `RecycleBin` as desired.
19. Select *Exit*.
20. Select >*Start* >*Settings* >*Control panel* >*Administration* >*Services*.
21. Select *TruTopsPDM2*.
22. Select *Restart service*.

8.6 Searching for files

1. Open file browser (e.g.via >*File* >*File browser*).
2. Either
    - Select .

    or

    - Open the context menu with the right mouse button.
    - Select *Search...*.
3. Set filters (the filters can be combined) and enter the searched names (case insensitive). If required, use place holders (see "Searching with the help of place holders (wildcards)", pg. 1-67).
4. Open the list field "Search in:" with ▼.

5. Select the folder where the search should be started.
6. If required, select "Search sub-folder".
7. To start the search: select *Search*.
8. To switch on the folder view again: select .

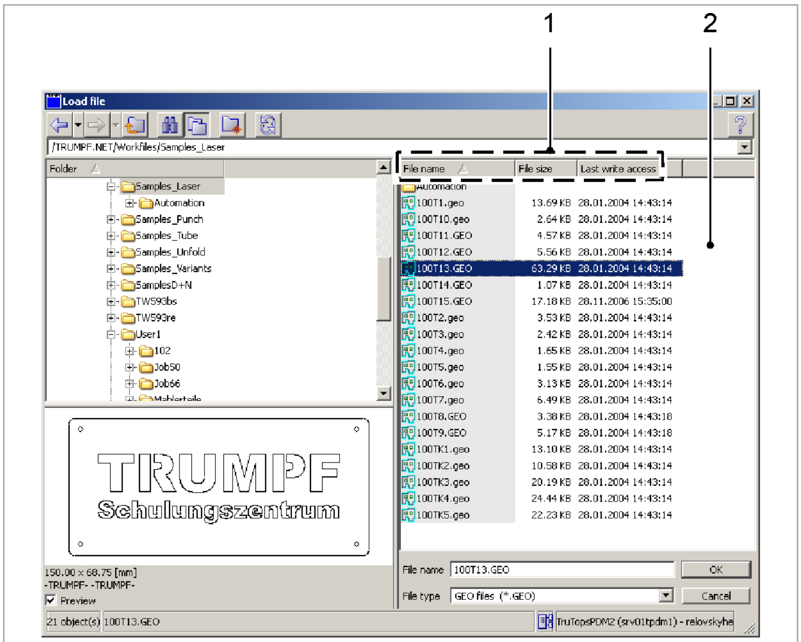
8.7 Adding other details to the list field of the files

By default, the TruTops file browser displays the "File name", the "File size" and the "Last write access" (sorted in ascending order of file names) in the file view.

The displayed details can be expanded (e.g. with "Machine name" or with "Sheet thickness").

Note

The more details the TruTops file browser displays, the more time it requires to create the view.



1 Standard information in the list header
2 Files list

TruTops File Browser Fig. 41788

Adding details

1. To add details: right click in the list header (1).
2. Either
   ➢ Select the desired detail.

or

- ➢ Select >*More…*.
- ➢ Select the desired details.
- ➢ Press *OK*.

Moving details in the list field to the desired position

3. Keep the mouse button pressed and drag the details one after another to the desired position.
4. To sort files by a specific detail: click on the detail.

The detail used for sorting is marked in gray. Clicking repeatedly on the detail modifies the sorting direction (upward/downward).

8.8 Opening files with the Geo Viewer

File formats from TruTops (e.g. *.GEO, *.TMT, *.ROT …) can be opened with the Geo Viewer in PDM. The appropriate shortcut required to open them with the Geo Viewer is created automatically when installing TruTops.

Opening a file with the Geo Viewer

1. Open file browser (e.g.via >*File* >*File browser*).
2. Search for the file to be opened with the Geo Viewer.
3. Select the file.
4. If the file browser was opened via >*File* >*Open* or >*File* >*Save*: right click to open the context menu, select >*Open*.
5. If the independent file browser is opened: double click on the file.

Closing Geo Viewer

6. Close Geo Viewer with *OK*.

8.9 Sending files through e-mail

1. Open file browser (e.g.via >*File* >*File browser*).
2. Search for the file(s) to be sent.
3. Select the file(s).
4. Open the context menu with the right mouse button.
5. Select >*Send to...*.
6. Select >*Mail recipient*.

An e-mail mask containing the selected file as an attachment is displayed. The e-mail can be processed further and sent.

8.10 Zipping files

1. Open file browser (e.g.via >*File* >*File browser*).

2. Search for the file(s) to be zipped.
3. Select the file(s).
4. Open the context menu with the right mouse button.
5. Select >*Send to...*.
6. Select >*Zip-File*.

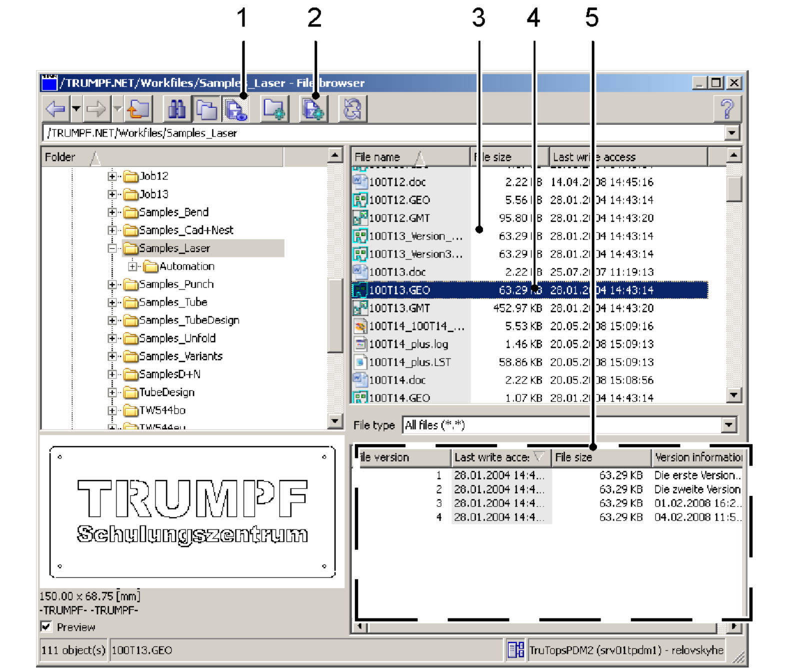
A zip file containing all the selected files is created in PDM. The name of the zip file comprises the name of the file selected first (without the file extension) and the new file extension '.tpdm.zip'. In one more step, the zip file can, for example, be exported into the file system or be sent to a mail recipient (see relevant sections).

8.11 Sending files from PDM to "My Documents" in the file system

1. Open file browser (e.g.via >*File* >*File browser*).
2. Search for the file(s) to be zipped.
3. Select the file(s).
4. Open the context menu with the right mouse button.
5. Select >*Send to...*.
6. Select >*My Documents*.

9. Creating and restoring file versions (PDM)

9.1 Displaying and hiding the version window



1 Hide and show version window
2 Button for creating a file version
3 Directory contents (file list)
4 File, of which a version is to be created and archived
5 Version window (archive)

File versions in version window Fig. 50219

File versions are displayed in a separate version window. In the TruTops file browser (with PDM), the version window appears by default.

➢ To hide and then view the version window again: select (*Version window*).

9.2 Creating a file version

1. Open file browser (e.g.via *>File >File browser*).
2. Select the file from which a version is to be created
3. Either
   ➢ Select (*Create version*).

or

- ➢ Open the context menu with the right mouse button.
- ➢ Select >*Create version*.

The "Properties... - Version x" mask, "Version information" tab is displayed.

4. Enter any information you wish about the new version.
5. Press *OK*. (*Cancel* cancels the entering of information, however not the creation of the file version.)

   The file version is displayed in a separate version window (see "Fig. 50219", pg. 1-35).

9.3 Restoring a file version

Note

If a file version is to be modified (e.g. contours are deleted or added), the file version must be copied from the version window into the normal directory. File versions cannot be opened directly from the version window.

1. Open file browser (e.g.via >*File* >*File browser*).
2. Select the file from which the versions have been created

   All file versions of the selected file are displayed in the version window (see "Fig. 50219", pg. 1-35).
3. Select the file version to be recovered.
4. Open the context menu with the right mouse button.
5. Either
   - ➢ In order to copy the file version from the version window into the normal directory **under its original file name** (so that the file version can be opened with the appropriate application): select >*Restore*.
   - ➢ Replying to a query:
   - − *Yes*: A version is created from the file from the normal directory and it is stored in the version window.
   - − *No*: The file in the normal directory is overwritten with the file version.

   or

   - ➢ In order to copy the file version into the normal directory **under a different file name** (so that the file version can be opened with the appropriate application): select >*Restore under...*.

     The "New file name" mask is displayed.
   - ➢ Change the suggested file name if required.

6. Press *OK*.

The restored file is displayed in the normal directory and can be opened with the appropriate application (e.g. TruTops CAD).

9.4 Deleting a file version

1. Open file browser (e.g.via >*File* >*File browser*).
2. Select the file from which the versions have been created

   All file versions that have been created from the selected file are displayed in the version window (see "Fig. 50219", pg. 1-35).
3. Select the file version to be deleted.
4. Open the context menu with the right mouse button.
5. Select >*Delete*.

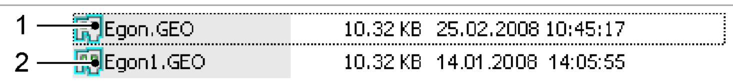
9.5 Tips & Tricks

Archiving the copies of a file

Earlier, variants of a file had to be saved as **copies** under different names (see "Fig. 53348", pg. 1-37).

But now, variants of a file can be archived together as **file versions** under **one** name. Only the current status of a file is displayed in the file list of the TruTops file browser and the file versions are in the version window (see "Fig. 50219", pg. 1-35).

The following gives an example of how an (old) copy of a file ('egon1.geo') can be created as a file version of the current status ('egon.geo').



1 Latest status of the file 2 Copy of a file (old status)

Variants of a file in the file list Fig. 53348

1. Open file browser (e.g.via >*File* >*File browser*).
2. Create a file version of 'egon1.geo' (right click to open the context menu, select >*Create version*).

   A file version is stored in the version window of 'egon1.geo' which is identical to 'egon1.geo'.
3. Open 'egon.geo' and save it as 'egon1.geo'.

When overwriting, 'egon1.geo' in the file list is overwritten. The file version of 'egon1.geo' in the version window is retained.

'Egon1.geo' has the current status. The old status of the file version is located in the version window.

4. Delete 'egon.geo'.
5. Rename 'egon1.geo' as 'egon.geo.'
6. If several files are to be archived together:
   - Create a version of the oldest file and overwrite it with the second oldest file. (This ensures that the file versions in the version window are stored in the right sequence).
   - Repeat the process till all file versions in the version window have been archived together.
   - Then rename the file in the file list whose versions are currently present.

10. File Properties

10.1 Viewing file properties

Note

The properties of files which have been written on an operating system with a different language and which contain special characters may be displayed incorrectly.

- Either
  - (With PDM) show properties of a file when checking the dependencies (see "Dependencies Between Files", pg. 1-45).

  or

  - Open file browser (e.g.via >*File* >*File browser*).
  - Search for the file whose properties are to be displayed.
  - Mark the file.
  - Open the context menu with the right mouse button.
  - Select >*Properties*.

    The "Properties..." screen, "General" tab is displayed.
  - To view all the information for a file: select the "File info" tab.
  - Select "Mask attributes that are not occupied" as required.

10.2 Selecting the unprocessed material ID and raw material from the drawing data

In addition to the unprocessed material, the raw material can now also be selected. This unprocessed material is then preset in TruTops Punch. TruTops Bend does not have a sheet layout, i.e. the unprocessed material and raw material are not requested in TruTops Bend until a bend is created. This selection should now be made in TruTops CAD, provided the unprocessed material is available in the database. The advantage is that this selected raw material is supported by TruTops Work and TruTops Unfold.

Selecting an unprocessed material ID

1. Select >*File*, >*Properties...*.

   The "File information" mask is opened.
2. Select the drop-down arrow for the unprocessed material ID.

   The "Select material" mask is opened.

3. Enter %Al%, for example, between the two percent signs (below the list field, first list field: Material ID, second list field: Stock ID).

   The list field displays all entries that begin with these letters. The material selection is thereby filtered and restricted.

   Note

   The selection can be limited even further using the "Z" filter field. If, for example, `<=3` is entered here, all unprocessed materials with a material thickness of 3 mm or less are displayed. `>=3` displays all unprocessed materials with a thickness of 3 mm or more.

4. Select Unprocessed material.

   The unprocessed material is applied.

Selecting the raw material

5. Select the raw material drop-down arrow.

   The "Selection" mask is opened.

   Note

   If the raw material does not match the unprocessed material or vice versa, the corresponding raw material or unprocessed material is automatically used. If there is no corresponding unprocessed/raw material, the relevant field remains empty and must be reselected.

6. Select raw material.

   The raw material is adopted.

11. Files and folders, importing/exporting (PDM)

11.1 Files, importing/exporting via Windows Explorer (PDM)

The PDM file storage can be used like a normally enabled Windows folder or an enabled network drive. This enables for example, saving drawings from a CAD system directly in PDM.

Connecting the PDM network drive

1. Either
   - ➢ Start Windows Explorer.
   - ➢ Select >*Extras* >*Connect network drive …*.
   - ➢ Select Drive.
   - ➢ Enter `\\Servername\trumpf_pdm2` under "Folder" or using *Search* search for '\\Servername\trumpf_pdm2'. (Server name = name of the server on which PDM has been installed.)
   - ➢ Select *Finish*.

   or

   - ➢ Start the Windows command prompt via >*Start* >*Programs* >*Accessories* >*Command prompt*.
   - ➢ Enter `NET USE P: \\Servername\trumpf_pdm2`. (Server name = name of the server on which PDM has been installed. Any free drive letter can be selected as the drive letter.)

Importing/exporting files

2. Use the selected drive to store files and directories in PDM or to read them out from PDM.

11.2 Importing files and single folders using the file browser (PDM)

Files and folders are imported using the TruTops file browser as follows:

- PDM adopts the folder structure from the file system and ensures the correct sequence. All sub-folders are also imported and their structures are adopted. The target folder in the PDM can be located anywhere.
- TruTops determines the existing dependencies between the files and saves them.

Notes

- If a *.GEO file is moved to a different folder after being imported, TruTops deletes the dependency to all sheets to which the *.GEO was nested.

- Dependencies can only be established to files that are also in the PDM. No dependencies are created if, for example, a nested sheet (*.TAF) is imported **before** the parts (*.GEO) are imported.
- Depending on the file format, not all dependencies between the files can be determined when importing. For example, dependencies to .pdf files are set in TruTops when they are created. No dependencies can be read from .PDF files when exporting from PDM or re-importing into PDM.

Selecting an import folder

1. Open file browser (e.g.via >*File* >*File browser*).
2. Go to the folder **where files are to be imported**.
3. Open the context menu with the right mouse button.

Import folder

4. Select >*Import directory*.

   The "Import directory" mask is displayed.
5. Go to the folder to be imported.
6. Mark the folder.

Importing files

Note

So that dependencies can be established between files, always import files **chronologically in the sequence of their creation** or import whole folders at a time.

7. Select >*Import file*.

   The "Import file(s)" mask is displayed.

   The mask has the same functions as the file browser (e. g. adding further details to the list field of the files, searching for files in the filter view...; see section "TruTops file browser").
8. To restrict the file selection in the folder view ( ): open the "File type" list field with ▼.
9. Select file type.
10. Mark the searched files. To mark several files: press <Strg> or <Ctrl> pressed.
11. Select *OK*.

11.3 Folders, importing several folders in a single job step (PDM)

Several folders (that are independent of each other) can be imported into the TruTops PDM in a single job step via the "PDM importer".

Additionally, a different target folder can be selected in TruTops PDM for each folder. TruTops determines the existing dependencies between the files and saves them.

Select folder

1. Select >*Start* >*Programs* >*TRUMPF.NET* >*PDM Importer*.
2. In the "PDM directory import" mask, mark the folder to be imported.
3. Transfer the folder to the right list field with .
4. To import other folders: repeat steps 2 and 3.

Defining target paths

5. Right click on the folder icon under "Target path".

   The "Selection of PDM target path" is displayed.
6. Select the target folder.
7. Select *OK*.

Delete folder from the import list

8. Mark the folder to be deleted.
9. Select .

Updating file dependencies after import

Note

When importing folders, the dependencies between files are automatically checked and entered in the PDM. A .TAF'. for example, knows which '.GEO' it uses. A '.GEO', however, does not know the '.TAF' in which it is being used.

If all '.TAF' files are imported from one folder first and then the corresponding 'GEO' files belonging to them are imported from another folder, the dependencies cannot be entered during the import.

10. To set the dependencies correctly after the import: select "Update file dependencies after import".

Starting import

11. If all folders to be imported have been transferred into the right-hand list field and the target folders have been defined: select *Start directory import* .
12. After the import: select *Exit*.

11.4 Files, exporting via the TruTops file browser (PDM)

1. Open file browser (e.g.via >*File* >*File browser*).
2. Either
   - Go to the folder to be exported.

   or

   - Change to the folder from which files are to be exported.
   - Select the file(s) which are to be exported. To mark several files: press <Strg> or <Ctrl> pressed.
3. Open the context menu with the right mouse button.
4. Select >*Export*.

   The "Target path for export..." mask is displayed.
5. Go to the folder to which the files are to be exported.
6. Mark the folder.

7. Select *OK*.

11.5 Importing/exporting files using the prompt/script (PDM)

Files and folders can also be imported into the PDM or exported from the PDM via the Windows command prompt or automated via shell scripts (*.bat) (see "PDM functions, calling up via prompt/scripts (pdmCLI.exe)", pg. 1-23).

Exporting/importing using the prompt

1. Start Windows command prompt.
2. Go to the folder '…:\TRUMPF.NET\Utils\PDM2\bin'.

Exporting a file

3. To export files, enter the following command (enter everything **in one line**, and be sure that slashes or backslashes and spaces are correct). Replace the **highlighted** example names with actual names:

   pdmCLI.exe Get --service-name=ToPs_PDM --port-address=**servername**:9999 --target-directory=**C:\target directory /source directory/example file.geo**

4. Press <Enter> or <Return> (↲).

Exporting a directory

5. To export directories, enter the following command (enter everything **in one line**, and be sure that slashes or backslashes and spaces are correct). Replace the **highlighted** example names with actual names:

   pdmCLI.exe Get --service-name=TruTopsPDM2 --port-address=**servername**:9999 --recursive --use-entity-directory --target-directory=**C:\target directory°/source directory/example directory**

6. Press <Enter> or <Return> (↲).

Importing files

7. To import files, enter the following command (enter everything **in one line**, and be sure that slashes or backslashes and spaces are correct). Replace the **highlighted** example names with actual names:

   pdmCLI.exe Put --service-name=TruTopsPDM2 --port-address=**servername**:9999 --target-directory=/**target directory°C:\source directory\°example file.geo**

8. Press <Enter> or <Return> (↲).

Automated export/import using script files

9. Write cd **C**:\TRUMPF.NET\Utils\PDM2\bin in a self-written BAT shell file (*.bat).
10. Add the above (exemplary) command lines.

12. Dependencies Between Files

12.1 Checking dependencies between files (PDM)

Files can, on the one hand, form the basis for other files and, on the other hand, be based on other files. Checking these dependencies is important whenever, for example, a file needs to be modified or deleted.

Example: The file is the basis for other files

- On which sheets (*.TAF) has a particular part (*.GEO) already been nested?
- In which NC programs (*.LST) does a particular GEO file (*.GEO) occur?

Example: the file uses the other following files:

- Which parts (*.GEO) have been nested on a particular sheet (*.TAF)?

Note

If a *.GEO file has been moved to a different folder after being imported, TruTops deletes the dependency to all sheets on which the *.GEO was nested.

1. Open file browser (e.g.via >*File* >*File browser*).
2. Search for the file whose dependencies are to be checked.
3. Mark the file.
4. Open the context menu with the right mouse button.
5. Select >*Dependencies* (if >*Dependencies* has been grayed out, the file has no dependencies.)
6. To check which other files the file uses: select the "Used..." tab.
7. To check which other files are based on the file: select the "Basis for..." tab.

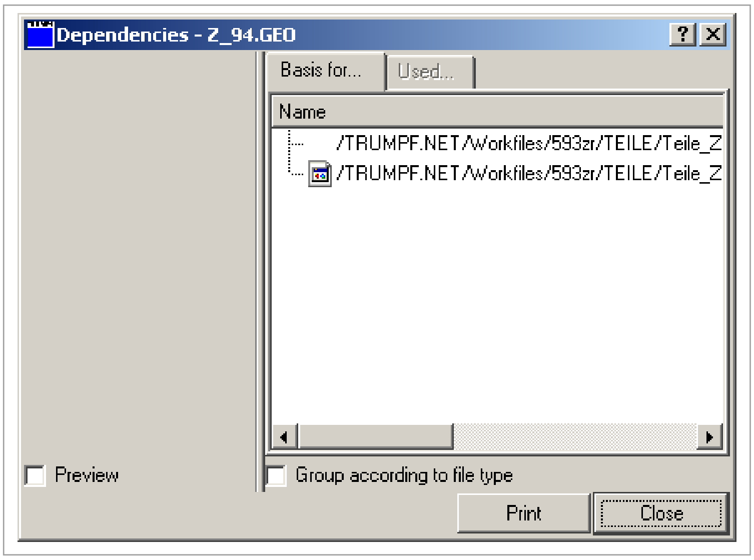


Fig. 41785

8. To group the display according to file types: select "Group according to file type".
9. To view a preview of dependent files: select "Preview" (self-holding) and select the desired file.

12.2 Deleting, exporting dependent files, displaying properties

Viewing the "Dependencies" mask

1. Open file browser (e.g.via >*File* >*File browser*).
2. Search for the file whose dependencies are to be checked.
3. Mark the file.
4. Open the context menu with the right mouse button.
5. Select >*Dependencies* (if >*Dependencies* has been grayed out, the file has no dependencies.)

Delete dependent files

6. In the "Dependencies" mask, select the file to be deleted.
7. Open the context menu with the right mouse button.
8. Select >*Delete*.

Export dependent files

9. Select the file(s) which are to be exported. To mark several files: press <Strg> or <Ctrl> pressed.
10. Open the context menu with the right mouse button.
11. Select >*Export*.

    The "Target path for export..." mask is displayed.

12. Switch to the directory to which you want to export.
13. Select the directory.
14. Press *OK*.

Displaying properties of dependent files

15. Mark the file(s) whose properties are to be displayed. To mark several files: press <Strg> or <Ctrl> pressed.
16. Open the context menu with the right mouse button.

17. Select >*Properties*.

The properties of the file are displayed.

13. Updated status of the files

13.1 Checking the updated status of the files automatically when loading and saving in TruTops (PDM)

➢ To automatically check the updated status of the files when loading them in TruTops: set the value of the "PDMWarning-Mode" variables in the public system rules to the desired value (see TruTops Laser or TruTops Punch software manual, chapter "Setting basic data"):
   - 0 = Do not check updated status.
   - 1 = Check updated status when loading.
   - 2 = Check updated status when saving.
   - 3 = Check updated status when loading and when saving.

14. Characteristics: a property of elements and contours

A characteristic is a property which one or more elements or contours have.

Characteristics are created in TruTops CAD, saved in the geo file and can be evaluated in other TruTops applications.

Advantages:

- Rows of holes and hole grids can be machined more easily:

  Rows of holes that have been created in TruTops CAD through >*Create* >*Macros* >*Row of holes* can also be evaluated in other TruTops applications. The conversion into rows of holes is no longer required. As a result, it is also possible to create rows of holes which are not axially parallel. The NC program is shortened.
- In TruTops CAD, Geometries with the characteristic Property are interpreted as **a unit**. All actions are applied to **the entire unit**, reducing the drawing effort considerably.

14.1 Switching "Characteristic fixed" on or off - TruTops behavior

"Characteristic fixed" is active:

- Certain elements are combined to form a unit with the characteristic property. If the "Characteristic fixed" is active, then a certain action is applied either to the complete unit (i.e. to **all** elements) or the action is rejected.

  For example, a **single element** of the "Single square hole" unit cannot be deleted. The **entire single hole** is deleted.
- All **impermissible actions** result in the contour or the element being displayed in pink. The colored highlight is kept up to the next redrawing.

  Examples:

  A single stroke cannot be stretched. The single stroke is displayed in pink.

  It is not possible to shorten elements for single strokes. The elements are displayed in pink.
- Clicking on an element results in the selection of **the entire unit** (all elements). When boxing in an element using the mouse, an action is executed only if **the entire unit** has been boxed in.

"Characteristic fixed" is **not** active:

- If "Characteristic fixed" is **not** active, the elements of contours can be arbitrarily modified with the Property characteristic. A single element of the single square hole can be deleted, for example. During modification, the Property characteristic is lost and the display color "orange" is reset to the normal line color.

Example	"Characteristic fixed"		Delete		Effect
	active	not active	Element	Contour	
Single square hole	X		X		The **entire** single hole is deleted.
	X			X	The **entire** single hole is deleted.
		X	X		The **single element** of the single hole is deleted. The display color changes from orange to the normal line color. The characteristic is dissolved.
		X		X	The **entire** single hole is deleted.
Row of square holes	X		X		The **entire** row of holes is deleted.
	X			X	The row of holes remains unchanged. The clicked contour of the row of holes is displayed in pink. If the drawing is updated ( ), the original state is restored.
		X	X		The **clicked element of the contour** is deleted. The remaining holes are assembled into smaller rows of holes. (Example: row of holes with 10 holes, an element of hole 3 is deleted: one of the new rows of holes consists of holes 1 and 2, the other consists of holes 4 to 10.)
		X		X	The **clicked contour** is deleted. The remaining holes are assembled into smaller rows of holes. (Example: row of holes with 10 holes, an element of hole 3 is deleted: one of the new rows of holes consists of holes 1 and 2, the other consists of holes 4 to 10.)

Tab. 1-3

Switching "Characteristic fixed" on and off

1. Select *>Edit >Characteristic fixed*.

   or

   ➢ Select *Fix characteristics*. (If "Characteristic fixed" is activated, the button is pressed).

Highlighting characteristics

2. Select *Highlight characteristics*.

   The characteristics are displayed in orange.

Removing characteristics

3. Select *>Edit >Delete >Characteristics*.

   or

   ➢ Select *Delete characteristics*.

14.2 Notches: Advanced options through the "Characteristic" property

The "Characteristic" property automatically increases the number of options for shaping and processing notches:

- Notches can be enhanced by roundings, bevels or sub-notches.
- Roundings, bevels, separating cuts or sub-notches can be removed from a notch. TruTops CAD joins adjacent notch elements automatically.
- If a notch is deleted entirely, the adjacent elements are joined again.
- Notches can be copied, moved, rotated, mirrored and stretched.
- The length of end elements of notches can be changed.

15. Screen display

15.1 Adapting window size

The window size can be adapted in a stepless manner. This is advantageous, for example, when working with online documentation simultaneously, or when the start bar lies over the TruTops command bar.

➢ Drag the window frame to the desired size using the mouse.

15.2 Increasing or reducing the work surface

Increasing the work surface

1. Click the vertical separating line between the blue bar and the work surface.

   The blue bar is hidden.

Reducing the work surface

2. Click the blue line at the left edge of the screen.

   The blue bar appears.

15.3 Modifying color scheme

The following color schemes are possible:

- Bright (white) background.
- Dark (black) background.
- Blue background.
- Gray background.

Each color scheme is effective only for the current application (CAD, Nest, Laser, Punch ...).

1. Either
   - ➢ Select >*Tools* >*Options...*.

   or

   - ➢ Select >*View* >*Toolbars* >*Adapt...*.
   - ➢ Select the "Options" tab.
2. Open the color scheme with ▼ and mark the desired color scheme.

15.4 Moving masks and messages on the interface

In TruTops, masks and messages can be freely moved on the interface. This is advantageous, for example, when a message and the objects under the message need to be visible.

1. Click on the title bar of the mask or the message.
2. Hold down the mouse button and move to the desired position.

16. Display of drawings

16.1 Restructuring display of drawings

- Either
  - Select >*View* >*Zoom* >*Refresh*.

  or

  - Press <F5> button.

  or

  - Select *Refresh view*.

16.2 Displaying drawing completely

- Either
  - Select >*View* >*Zoom* >*All*.

  or

  - Press <Ctrl>+<T>.

  or

  - Select *Total view*.

16.3 Displaying sheet completely

This view can be displayed only in Laser and Punch applications.

- Either
  - Select >*View* >*Zoom* >*Sheet*.

  or

  - Select *Total sheet view*.

16.4 Displaying detail view

Boxing in the desired detail

1. Either
   - Select >*View* >*Zoom* >*Detail*.

   or

   - Select *Detail view*.

2. Click on the first point in the box with the mouse pointer (cursor).
3. Click on the opposite point in the box.

Zooming in (detailing)

4. Right click on the desired detail in the graphics area.

   The graphics is zoomed in on. The pick point in the graphics area becomes the new center.

Zooming out

5. Use the mouse wheel or the middle mouse button to click on the desired detail in the graphics area.

   The graphic is zoomed out from. The pick point in the graphics area becomes the new center.

16.5 Moving screen section

- Either
  - Select >*View* >*Zoom*.
  - Select >*Up*, >*Down*, >*Left* or >*Right*.

  or

  - Position the mouse pointer (cursor) as sizing tool in the working range.
  - Keep the mouse wheel (or its middle button) pressed.
  - Move the mouse pointer.

Tip

Depending on the usual method, the <Ctrl> button can be pressed. This however is not necessary.

16.6 Zooming drawing

- Either
  - For stepless zooming: position the mouse pointer (cursor) in the working range.
  - Keep the right mouse button pressed.
  - Move the mouse downwards (zoom out) or upwards (zoom in).

  or

  - To zoom stepwise: position the mouse pointer (cursor) in the working range.
  - Keep the <Ctrl> key pressed.
  - Zoom the view in or out using the mouse wheel.

16.7 Getting last view

TruTops always memorizes the last view that was set using a button or the mouse.

- Either
  - Select >*View* >*Zoom* >*Last zoom*.

  or

  - Select *Show last view*.

16.8 Memorizing and recalling view

Memorize view

1. Use the mouse or the *Total view* or *Detail view* buttons to set the view that is to be saved.
2. Either
   - Select >*View* >*Zoom* >*Memorize zoom*.

   or

   - Select *Memorize view*.

Getting saved view

3. Either
   - Select >*View* >*Zoom* >*Get zoom*.

   or

   - Select *Get view*.

16.9 Displaying and hiding outline points

- Either
  - Select >*View* >*All points* or >*View* >*Open points*.

  or

  - Select *Show all points* or *Show open points*.

Contour point	Description
Green	Closed contour point
Red	Open contour point
Yellow	More than two contour elements attached to a point

Tab. 1-4

17. Selection preview, selecting components

17.1 Working with selection preview

With selection preview, actions which are performed with the mouse are visible immediately on the cursor or on the screen.

Function	Screen/cursor display	Example
Measuring	The current measurement result is shown on the cursor	Length, point, circle measurements
Selection of components (elements, bendings, contours etc.)	Components are highlighted in color	Deleting contours, creating bending lines, stretching
Selection of components	The number of selected components is shown on the cursor	Deleting elements, moving, smoothing
Corner selection	The corner and both corner elements are highlighted in color	Creating roundings, bevels, notches
Position selection	Special element points (see the following table) are caught and are displayed on the cursor	Creating points, line 2 points, curve center radius

Selection preview options Tab. 1-5

Special element points:

Point	Affected elements	Symbol
End Point	Line, bevel, curve, rounding	Square
Make Midpoint	Circle, construction circle, curve, rounding	+ (color: cyan)
Intersection	All Entities	Text: "x" and large circle
Element center	Line, bevel, curve, rounding	Text: "1/2"
0 degree position	Circle, construction circle, curve, rounding	Text: "0"
90 degree position	Circle, construction circle, curve, rounding	Text: "90"
180 degree position	Circle, construction circle, curve, rounding	Text: "180"
270 degree position	Circle, construction circle, curve, rounding	Text: "270"
Other element point	All Entities	Small circle

Point	Affected elements	Symbol
Virtual corner (original position of a corner that was beveled, rounded or notched)	Bevel, rounding, notch	Square

Element points Tab. 1-6

Activating selection preview

➢ Select >*View* >*Selection preview*.

Selection preview is active (a check mark appears in front of the menu item).

17.2 Selecting components

Elements, contours, bendings or points can be selected and modified individually (individual selection). Multiple selection is a further option. For this, they are firstly selected one after the other and are then modified.

Multiple selection

Components are selected one after the other and are then modified for multiple selection. Selecting the component once again undoes the selection.

1. Function, e.g. delete, select elements.
2. Press the <Ctrl> key.

   The * symbol on the cursor shows that multiple selection mode is active.
3. Select the elements to be deleted.

   The selected elements are marked in color. If >*Selection preview* is active, the number of selected elements is displayed on the cursor.
4. Select *OK*.

The selected elements are deleted.

Fixed multiple selection

When this mode is selected, a component cannot be deselected once it has been selected. This is only possible by changing over to multiple selection mode.

1. Function, e.g. delete, select elements.
2. Press the <Shift> key.

   The + symbol on the cursor shows that fixed multiple selection mode is active.
3. Select the elements to be deleted.

   The selected elements are marked in color. If >*Selection preview* is active, the number of selected elements is displayed on the cursor.
4. Select *OK*.

The selected elements are deleted.

Switching between multiple selection and fixed multiple selection

1. Select components.
2. Press the <Ctrl> or <Shift> key to switch to the other mode.

The selected mode is indicated by the symbol on the cursor.

Switching back to individual selection

- Either
  - Select >*Edit* >*Unmark*.

  or

  - Release the <Ctrl> or <Shift> key.
  - Click on the selected components again.

  The components are displayed in black again.

Marking all components

All drawn components are marked using this function. Individual components can then be deselected.

➢ Select *>Edit >Mark all*.

All components are marked.

Tip

If *All* is used, the selected function, for example "Delete", will be immediately applied to all components.

18. Operating TruTops with the mouse

18.1 Automatically placing the mouse pointer in masks

1. Select *>Extras >Modify data...*.
2. Select *>User >Configuration*.

   The "Configuration" mask is displayed.
3. Select "Place mouse pointer at OK".

   or

   ➢ Select "Place mouse pointer at the next area of action".
4. Press *Modify*.

19. Operating TruTops using keyboard

19.1 Using Windows keyboard commands and shortcuts

TruTops can now be operated using the usual Windows keyboard commands or shortcuts:

- All **menu items** linked with a shortcut have been appropriately marked (e.g. >*File* >*Open...*: <Ctrl>+<O>).
- The keyboard commands for the **Command block** (9) (see "Fig. 50209", pg. 1-14) are displayed in the appropriate tool tip:
  - *Total view*: <Ctrl>+<Enter> (↲) or <Ctrl>+<Return> (↲).
  - *End*: <Esc>.
  - *Ok*: <Enter> or <Return> (↲).

Example 1

1. To save a file: press <Ctrl>+<S> (instead of selecting >*File* >*Save*).

Example 2

2. To undo an action: press <Ctrl>+<Z> (instead of selecting >*Edit* >*Undo*).

19.2 Entering numerical values and coordinates

Entering decimal values

1. When entering decimals, use a point as separator. (Example: Enter `30.5`.)
2. Press <Enter> or <Return> (↲).

Entering coordinates

3. Enter X and Y coordinates separated by a comma. (Example: X = "10 mm", Y = "20,5 mm": enter `10,20.5`.)
4. Press <Enter> or <Return> (↲).

19.3 Entering texts in input fields

1. Click into the input box with the mouse pointer (cursor).
2. Enter characters one after the other.
3. To confirm the input: Press <Enter> or <Return> (↲).

19.4 Using key functions of TruTops

During initial installation and update of TruTops, the Windows key functions are automatically set in the individual applications. However, one can also work with the TruTops key functions on request.

Note

The key functions must be separately set in each existing application.

1. Select the application in which the TruTops key functions are to be set.
2. Select >*Extras* >*Modify data...*.
3. Select >*User* >*Configuration*.

   The "Configuration" mask is displayed.
4. Select "Set TruTops key functions".
5. Select *Modify*.

Key	**Configuration**	
	TruTops key functions	**Windows key function**
↲	Jump to next input field.	Confirm mask; is the same as pressing *Ok*.
<↑>, <↓>	-	Delete characters in input fields.
<SHIFT>+<Backspace>	Delete characters in input fields.	Delete characters in input fields.
<Tab>	Jump to next input field. All the field contents are automatically marked.	
<Tab>+<→> or <Tab>+<←>	Jump to next input field. Automatic marking of all the field contents is removed.	
Keep <SHIFT> pressed, mark the area using <→> or <←>.	Mark selected field contents from the cursor position onwards.	
Keep <SHIFT> pressed + <Pos 1> (German keyboard) or <Home> (English keyboard).	Mark from the cursor position to the start of the input field.	
Keep <SHIFT> pressed + <Ende> (German keyboard) or <End> (English keyboard).	Mark from the cursor position to the end of the field.	
(Field contents are marked) <Entf> (German keyboard) or <Del> (English keyboard)	Delete marked contents of field.	
(Field contents are marked) Press desired character.	Marked field contents are replaced with the character pressed.	
<SHIFT>+<Tab>	-	Jump to previous input field.
<Esc>	Delete characters in input fields.	Close mask; is the same as pressing: *Cancel*.

TruTops or Windows key functions Tab. 1-7

19.5 Using "Copy" – "Paste"

The "copy and paste" function is available in all TruTops input fields.

Depending on the position of the cursor in the input field, the contents are copied either completely or only partly into the intermediate memory.

When pasting, the copied contents appear at the point where the cursor is located.

Contents of the intermediate memory can be pasted from other programs and into the TruTops input fields and vice versa.

➢ Copy and paste as described in the following table.

Copy		**Pasting into empty input field**	
Cursor position I	**Key combination**	**Key combination**	**Result**
Example 123 I	<CTRL> + <C>	<CTRL> + <V>	Example123
I Example123	or	or	Example123
Exam p le123	<Strg> + <C>	<Strg> + <V>	ple123

Copying and pasting Tab. 1-8

19.6 Using keyboard buffer of the command line (repeating entries)

Repeated entries and functions do not need to be reentered.

The keyboard buffer saves entries or functions which have been entered into the command line or which have run in the background. The last commands can be opened and scrolled through using ▼:

1. Open the list field of the command line with ▼.
2. Mark the desired command or the desired entry.
3. Press <Enter> or <Return> (↵).

The command or the entry is repeated.

20. Calculator

You can carry out the following actions using the calculator:

- Entering numerical values and coordinates.
- Executing arithmetic operations and entering the result of the arithmetic operations simultaneously.

The calculator is automatically activated during a new installation and a TruTops update. It can be operated with the mouse as well as (to some extent) with the keyboard in all modules. It is displayed automatically on double-clicking in an input field.

Function	Using with the mouse	Using with the keyboard
Opening the calculator	Double click in an input field for numerical values.	(Not possible)
Deleting an entry	*C / Cl*	(Not possible)
Exiting the calculator without adopting entries	Cancel	<Esc> (possible only if the Windows key functions have been set)
Closing calculator, adopting entry	*OK*	<Strg/Ctrl>+<Enter/Return> (↲)
Squaring	x^2	(Not possible)
Pi	*Pi*	(Not possible)
Multiply	*	<*>
Divide	/	</>
Adding	+	<+>
Subtracting	-	<->
Result	=	↲

Calculator functions Tab. 1-9

20.1 Deactivating calculator permanently

The calculator is automatically activated during a new installation and a TruTops update.

➢ To deactivate the calculator permanently: set the value of the "ST_NumInputWithMouse" variable in the public rules to `0` (under >*Tools* >*Modify data...* >*Rules* >*System*).

20.2 Entering numerical values in input fields

1. Position the mouse pointer (cursor) in the input field for numerical value.
2. Double click.

   The "easy" calculator is displayed:

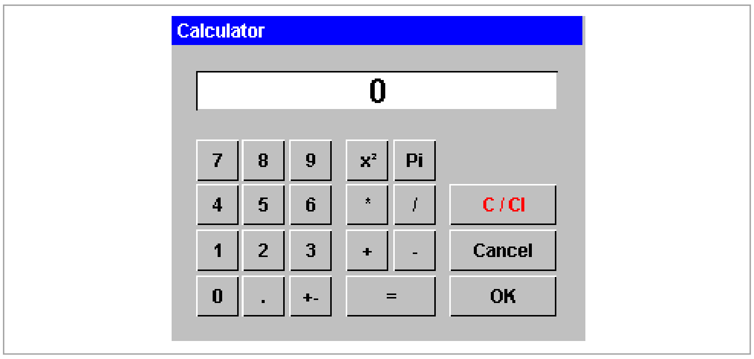


Fig. 20085

3. Click on the desired figures and arithmetic operations or enter them using the keyboard.
4. Press *OK*.

The calculator is closed.

20.3 Entering X and Y coordinates of points

1. Select *Calculator* (on the left near the command line).

   The calculator extended by "Coordinate" is displayed.
2. Enter the numerical value of the X coordinate in the upper input field.
3. Select *→X*.

   The value is adopted in the lower input field.
4. Enter the numerical value of the Y coordinate in the upper input field.
5. Select *→Y*.

   The value is adopted in the lower input field.
6. Press *OK*.

The calculator is closed.

21. Searching with the help of place holders (wildcards)

Place holders simplify the search, when a designation (e.g. a file name) or a number (e.g. a technology table) is not exactly known.

The following place holders can be used in the TruTops file browser:

- `?` = **1** arbitrary character.
- `*` = arbitrary character string.

In case of database accesses (under >*Tools* >*Modify data...*) and when selecting the technology table, the percent sign ("%") is a place holder for arbitrary characters:

- `1%` = `1`, subsequently none, one or several arbitrary characters.
- `%1` = none, one or several arbitrary characters, subsequently `1`.

22. Message and confirmation masks

22.1 Reactivating hidden message and confirmation masks

1. Select >*Extras* >*Modify data...*.
2. Select >*User* >*Configuration*.

   The "Configuration" mask is displayed.
3. Select "Show optional message boxes again".
4. Press *Modify*.

23. Print

23.1 Printing screen content

The present screen can be printed with this function.

1. Load the file whose screen content is to be printed.
2. Select *>File >Print >Screen...*.

   The standard printer mask is displayed.
3. If required, modify the settings.
4. Press *OK*.

The screen content is printed.

23.2 Configuring line thickness when printing screen contents

If the screen contents are printed on a printer with high resolution, the lines may be hard to see.

1. Open the file *topsprn.ini* in the Windows folder using a text editor (such as Notepad).

   Note

   The entry *LineWidthResolution* specifies what proportion of the printer's resolution is used for a line width of 1 pixel. With printer resolution of 600dpi, the default value of 150 results in a line thickness of 4 pixels.

   Higher values produce finer lines, lower values produce thicker lines.
2. Adapt the *LineWidthResolution* entry.

23.3 Printing GEO files as *.HPGL files

TruTops CAD can print GEO files in the '*.HPGL' format (= graphical output of plotters).

1. Load the desired file in the '*.GEO' or '*.DXF' format.
2. Select *>File >Print >Plotting...*.

   The "Plotting" mask is displayed.
3. Make the desired settings.

4. Press *OK*.

The 'GEO' is printed in the '*.HPGL' format.

24. Using auxiliary tools

24.1 Drawing auxiliary lines

Drawing auxiliary line through 2 points

1. Select *>Aux. tools*,*>Construction line >Two points*.
2. Follow the instructions in the command line.

Drawing an auxiliary line parallel to the existing auxiliary line

Condition

- The drawing contains at least one auxiliary line.

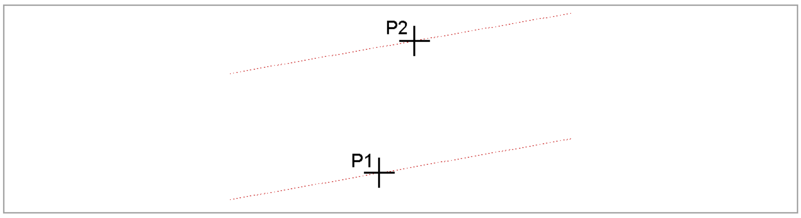


Fig. 4726

1. Select *>Auxiliary tools >Auxiliary line >Parallel*.
2. Click on the auxiliary line (P1) for which a parallel line is to be drawn.
3. For one parallel auxiliary line: click on the point (P2) on the parallel auxiliary line.

   or

   ➢ For several parallel auxiliary lines: enter all distances at which parallel auxiliary lines are to be drawn (separate numerical values through blanks. Do not use any commas. Example: 10 20 30).
4. ↲Press .
5. Click on the side on which the parallel line(s) should appear.

Drawing horizontal or vertical auxiliary lines

1. Select *>Aux. tools >Construction line >Horizontal*.

or

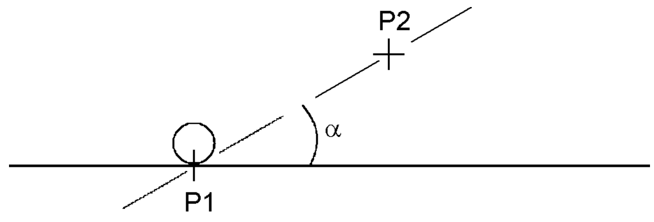
➢ Select >*Aux. tools* >*Construction line* >*Vertical*.

2. Follow the instructions in the command line.

Drawing an auxiliary line as a tangent at the arc element and point

Condition

- The drawing contains at least one arc element or circle.

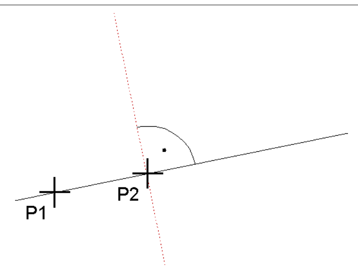


P1 Approximate tangent point P2 End point of the tangent

Fig. 27381

1. Select >*Auxiliary tools* >*Auxiliary line* >*Tangent at point*.
2. Follow the instructions in the command line.

Drawing an auxiliary line at right angles to an element

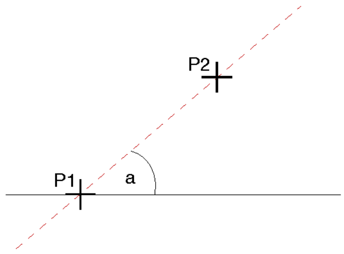


P1 Point on element P2 Start point of the auxiliary line at right angles

Fig. 4729

1. Select >*Aux. tools* >*Construction line* >*Perpendicular*.
2. Follow the instructions in the command line.

Drawing an auxiliary line through point and angle of inclination

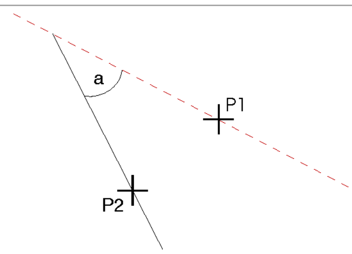


P1 Start point of the auxiliary line A Angle
P2 Point on the auxiliary line

Fig. 4730

1. Select >*Aux. tools* >*Construction line* >*Angle X*.
2. Follow the instructions in the command line.

Drawing an auxiliary line through point and angle to the base line

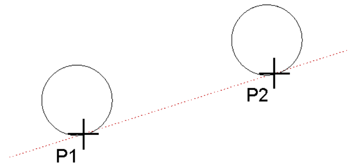


P1 Point on auxiliary line A Angle between auxiliary line and element
P2 Point on base line

Fig. 4732

1. Select >*Auxiliary tools* >*Auxiliary line* >*Angle to line*.
2. Follow the instructions in the command line.

Drawing an auxiliary line through tangent at 2 arc elements



P1 First tangent point P2 Second tangent point

Fig. 4731

1. Select >*Auxiliary tools* >*Auxiliary line* >*Tangent at 2 arcs*.
2. Follow the instructions in the command line.

Dividing an element with an auxiliary line

Application: dividing an element with a vertical construction line between two random points (P1) and (P2)

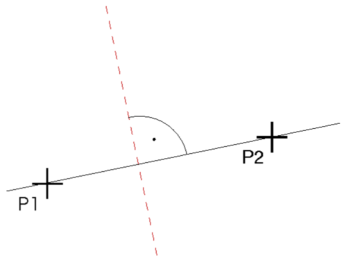


Fig. 4733

1. Select >*Aux. tools* >*Construction line* >*Division*.
2. Either
   - ➢ Click on the first point (P1).

   or

   - ➢ Enter the division factor (between 0 and 1).

     Division factor 0.5: The auxiliary line is placed at the center of the base element.

     Division factor 0.2 means that the auxiliary line is placed after one fifth of the base element.

   - ➢ Click on the first point.
3. Click on the second point (P2).

24.2 Drawing auxiliary circle

Drawing an auxiliary circle through three points

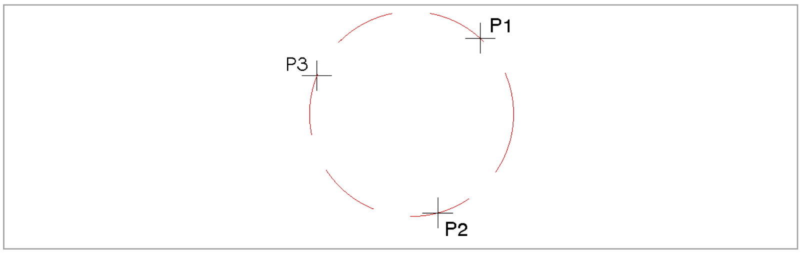


Fig. 4734

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Three points*.
2. Follow the instructions in the command line.

Drawing auxiliary circle through center and radius

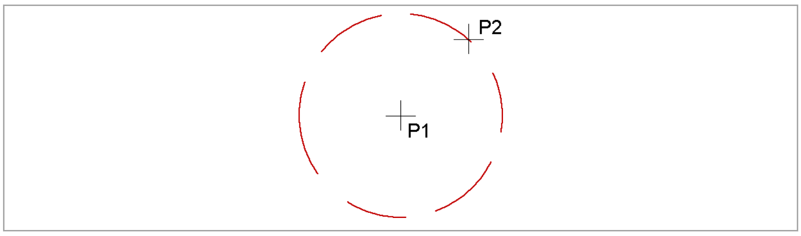
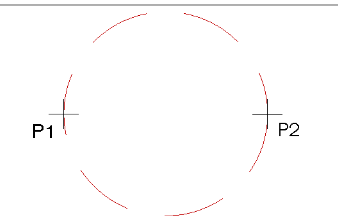


Fig. 4735

1. Select >*Auxiliary circle* >*Auxiliary circle* >*Center radius*.
2. Follow the instructions in the command line.

Drawing auxiliary circle through diameter



P1 Start point of the diameter P2 End point of the diameter

Fig. 4736

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Diameter*.
2. Follow the instructions in the command line.

Drawing auxiliary circle through center and diameter

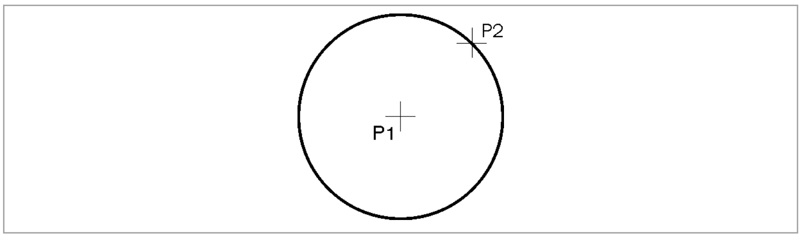


Fig. 4709

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Center - diameter*.
2. Follow the instructions in the command line.

Drawing concentric auxiliary circles

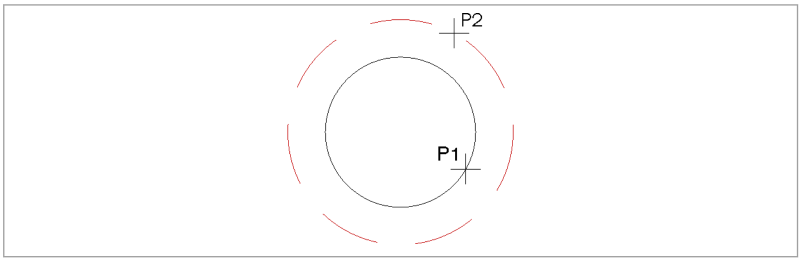


Fig. 4737

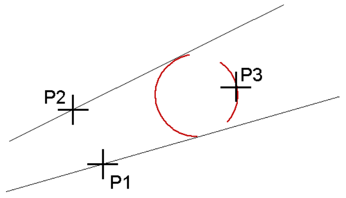
1. Select >*Auxiliary tools* >*Auxiliary circle* >*Concentric*.
2. Click on the base circle (P1).
3. Click on a point (P2) on the arc.

   or

   ➢ Enter the distance.

   Positive value: concentric circle outside the basic circle. Negative value: concentric circle inside the basic circle).

Drawing auxiliary circle tangential to two elements



P1 Point on first tangent

P2 Point on second tangent

P3 Point on circle line

Fig. 4738

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Two tangents - one point*.
2. Follow the instructions in the command line.

Drawing auxiliary circle tangential to an element and two points

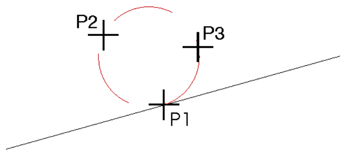
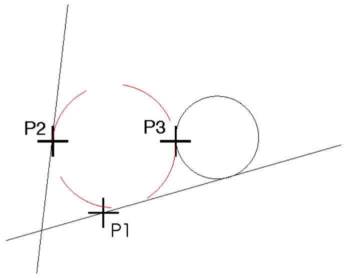


Fig. 4739

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Tangent - two points*.
2. Click on approximate tangent point (P1).
3. Click on a point (P2) on the arc.
4. Click on a point (P3) on the arc.

Drawing auxiliary circle tangential to three elements

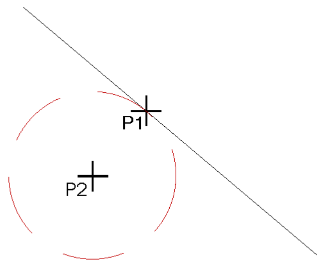


P1 Tangent point on first element
P2 Tangent point on second element
P3 Tangent point on third element

Fig. 4740

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Three tangents*.
2. Follow the instructions in the command line.

Drawing auxiliary circle tangential to an element



P1 Tangent point
P2 Center of the auxiliary circle

Fig. 4741

1. Select >*Auxiliary tools* >*Auxiliary circle* >*Tangent - center*.
2. Follow the instructions in the command line.

24.3 Auxiliary geometry

Note

Auxiliary geometries can be deleted partially or completely.

Auxiliary geometry, deleting

1. Select >*Auxiliary tools* >*Delete auxiliary geometry*.
2. Select *All* if the auxiliary geometry is to be completely deleted.

   or

   ➢ Click on individual elements if particular auxiliary geometries are to be deleted.
3. Select *Total view* to rebuild the interface after the auxiliary geometries are deleted.

   or

   ➢ Change the window setting.

24.4 Measuring

Measuring the distance between 2 points

1. Select >*Auxiliary tools* >*Measure* >*2 points*.
2. Follow the instructions in the command line.

Measuring coordinates of a point

1. Select >*Auxiliary tools* >*Measure* >*Point*.
2. Follow the instructions in the command line.

Measuring horizontal or vertical distance between 2 points

1. Select >*Auxiliary tools* >*Measure* >*Horizontal* or >*Vertical*.
2. Follow the instructions in the command line.

Measuring coordinates, radius and diameter of a circle

1. Select >*Auxiliary tools* >*Measure* >*Circle*.
2. Follow the instructions in the command line.

Measuring angle

1. Select >*Auxiliary tools* >*Measure* >*Angle*.
2. Follow the instructions in the command line.

Measuring length

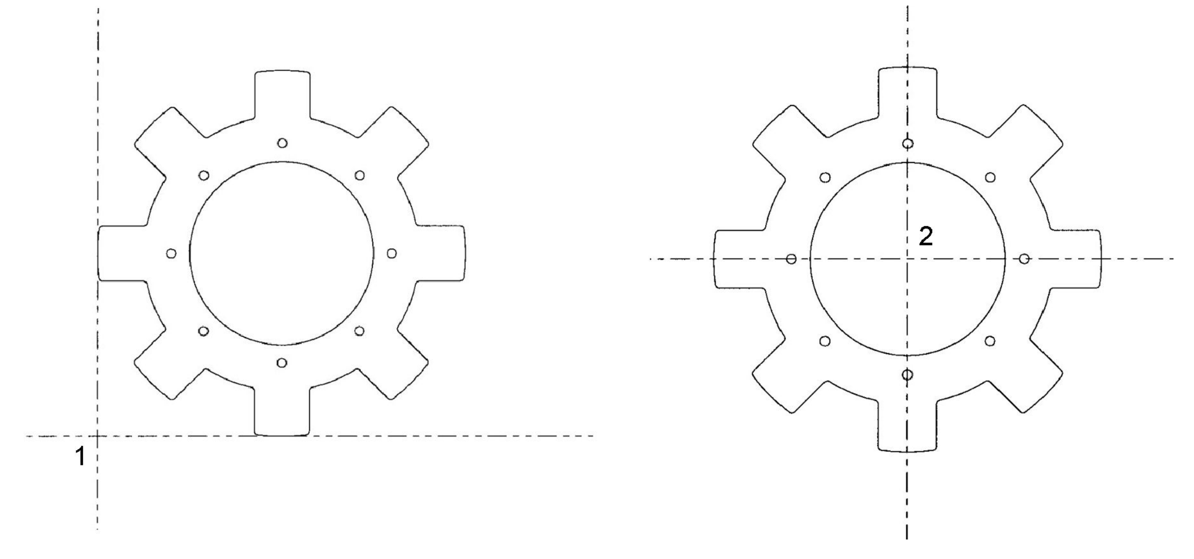
1. Select >*Auxiliary tools* >*Measure* >*Length*.
2. Follow the instructions in the command line.

24.5 Zero point of the coordinate system

Displacing zero point

Note

If the zero point of the coordinate system has been displaced and the *.GEO is then saved and reloaded, the zero point is located again at the original place: in the lower left corner of the circumscribing rectangle of the loaded '*.GEO' (1).



1 Zero point of the coordinate system in the lower left corner of the circumscribing rectangle
2 Offset zero point

Fig. 30651

1. Select >*Auxiliary tools* >*Displace zero point*.
2. Click on the zero point.

25. Help in the case of problems

25.1 Diagnostics for program crash

A so-called "Crash-Dump" of the process is written in a file every time TruTops crashes. This helps the TruTops development analyze the cause of the crash.

1. Following a crash, change to the following directory:e.g. in TruTops Laser '...:\TRUMPF.NET\Applications\ToPs100\bin.'

   In this directory you will find files with the following names:

   'CRASH_20YYMMDD_hhmmss.DMP'.

   'ERRORLOG_20YYMMDD_hhmmss.TXT'.

   The time stamp of both files is the time of the crash: 20YYMMDD = year, month, day and hhmmss = hour, minute, second.

2. Send both files and a description of what action cause the crash to technical customer service.

Chapter 2

Setting basic data

1. The database as a knowledge-base

Technological knowledge that TRUMPF has developed for TRUMPF machines over the years, and which has been continuously optimized, has been integrated into the TruTops database.

It contains all data that TruTops accesses in the different programming phases.

- machine data.
- Material data.
- Technology tables.
- Rules.
- NC output data (NC cycles, formats, times ...).
-

Current application parameters

The technology tables and the rules contain the current application parameters that have been released by the TRUMPF Process Development Department. The updates of the data collections are released at regular intervals.

Knowledge base

The database is an upgradeable knowledge base. Newly acquired knowledge in the field of processing technology can be entered here. TruTops accesses this data the next time an NC program is generated.

Working with the database

The database offers the following options:

- Creating data records (creating complete new ones or copying the existing ones).
- Changing data records (modifying entries).
- Deleting available data records.

Maintaining data

- Adding data about your material.
-
- Checking the values stored in the database.

The database must be configured for the individual machine in the Data Management. Only then are the required functions available for processing in TruTops.

1.1 Backup of data

Note

The data will be saved using the "Backup & restore" tool.

➢ (See the TruTops installation manual)

1.2 To import data into the database

Some data (e.g. a new set of rules, new technology tables, ...) cannot be imported directly into the TruTops database without an update.

Condition

- The file extension corresponds to the current database version.

1. Load the storage medium.
2. Select *>Extras >Modify data...*.
3. Select *>User >Database... >Modify...*.

   The file browser opens.
4. Change the directory and mark the database file.
5. Press *OK*.

The new data will be read into the database.

1.3 To modify the database

To modify the database, any database script (*.exe file) can be launched, to fix faults in the database or to meet customer requirements (e.g. to change data).

Database scripts can be obtained from technical customer service.

1. Select *>Extras >Modify data...*.
2. Select *>User >Database... >Modify...*.

   The file browser opens.
3. Select the database script.

All changes are shown in a log after the modification.

1.4 To send a database via email

Notes

- With this procedure, only application-specific data can be sent. To send the complete database, use the "Backup & restore" tool.
- When working with a network database (MSDE), sending of the TruTops database through email can only be started by the database server.

- The email program for sending the database must be MAPI-compatible.

Modifying file size

Note

By default, when sending the database by email, it will be broken up into 4 MB files. This file size can be changed.

1. Select >*Extras* >*Modify data...*.
2. Select >*Rules* >*System*.
3. Find the variable "DbComprSplitSizeInKb" and enter the required file size under "Value".

To generate emails

4. Select >*User* >*Database...* >*Send...*.

   The "Send database via email" mask is displayed.
5. Fill in the input fields.
6. Press *OK*.

   The database will be compressed and broken up into parts. The required number of emails will be created in the email program.
7. To send emails.

2. TRUMPF function

Certain functions are password-protected. Only open and modify functions under the instructions of technical customer service.

Note

The daily password must be requested from technical customer service. It is only valid for the day on which it is issued.

3. Data management

All default values for machines and components have already been set. The options available depend on the machine selected.

3.1 To open the data management

1. Start TruTops.

Note

The Data Management is usually opened for the currently active machine. The machine cannot be changed within the open Data Management.

3. If required, change the machine (via >*Tools* >*Change machine*).
4. To open the data management, select >*Extras* >*Modify data...*.

4. Machine data

The machine data is the basic requirement for the generation of NC programs. TruTops can be configured for different machines with an NC interface.

The technical data for the machine, the laser and the control system is stored in the database. TruTops accesses this data when generating an NC program.

NOTICE

Incorrect values in the machine data can lead to inoperable NC programs.

- Check that all entries and modifications are correct.
- If required: contact Technical Service.

All the data necessary for your machine can be found in the current operator's manual and the current programming manual of your machine.

4.1 To copy or delete a machine

Copying machine

In order to be able to create various configurations from a machine, it can be copied (several times).

1. Start Windows Explorer.
2. Go to the following directory:

   <LW>\TRUMPF.NET\Utils\DbTools\MachineCopy'.

   (LW = drive letter)
3. Double-click on the 'machineCopy.bat' shell script.

   The "Copy machine" mask is displayed.
4. Mark the desired language and select *OK*.

   Note

   The correct ODBC entry can be called up via >*Help* >*Info*"Files". The current database will be shown in the first line, enclosed in round brackets.

   The "Copy/Delete machine" mask is displayed.
6. Mark the machine to be copied.
7. Select *Copy*.

The machine is copied. A number is attached in front of the type no. ("Type") and increases consecutively (number for the first copy: "1").

8. Select *Exit*.
9. Select *>Extras >Modify data...*.
10. Configure the machine.

Deleting machine

1. Start Windows Explorer.
2. Go to the following directory:

   <LW>\TRUMPF.NET\Utils\DbTools\MachineCopy'.

   (LW = drive letter)
3. Start the machineCopy.bat script file (double click).

   The "Copy machine" mask is displayed.
4. Mark the desired language and select *OK*.

   The "Copy/Delete machine" mask is displayed.

6. Mark the machine to be deleted.
7. Select *Delete*.
8. Select *Exit*.

5. Material data

With the exception of sheet margins, the material data is machine-independent. Additions or changes will not be overwritten when TruTops is updated.

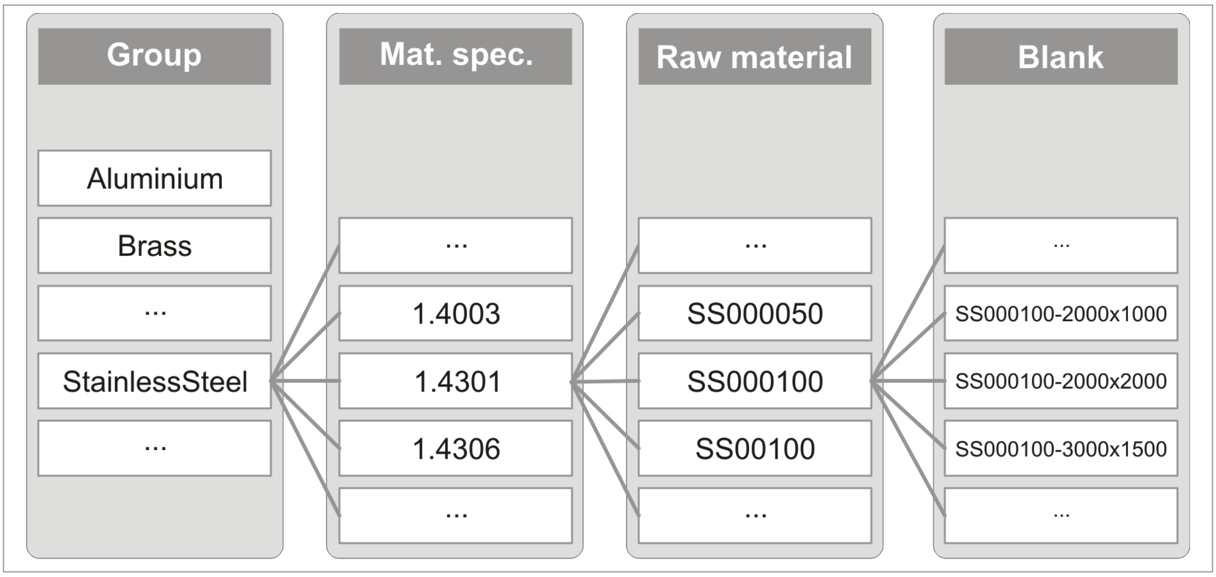
When a new raw material is created, all values which are important for machining in TruTops must also be created. This is why every material is assigned an alias, whose values should be used instead.

TruTops uses the following layers of material data:

Term	Important properties	Description
Group	e.g. aluminum, steel, stainless steel,...	Materials are assigned to various materials groups.
Material	e.g.: ▪ Tensile strength ▪ E-modulus ▪ Admitted processing	The material is the basis for the raw material. If no bend allowances have been entered for the associated unprocessed material, then the bend allowances from the bend allowance alias will be used.
Raw material	▪ Material ▪ Material thickness ▪ Film ▪ Tempering	A raw material is a material of a defined thickness.
Blank	▪ X dimension ▪ Y dimension	A blank is a raw material with defined dimensions (=°sheet). By specifying a stock ID, the warehouse management can precisely determine the bin location of the blank and make it available for processing. The stock ID is output in the NC program and in other TRUMPF-specific file formats (e.g. *.taf, *.pdf, *.job, *.tmt ...).

Material data levels Tab. 2-1

All levels of the material data have a clear relationship to one another.



Material data levels

Fig. 60476

Unprocessed material ID

The standard raw material ID consists of the name of the standard material group and the material thickness in 1/100°mm as well as information regarding the material surface.

An 11-character raw material ID is automatically created for new raw materials.

Position	Variable	Explanation
1-4		Abbreviation if the material group.
5-8		Material thickness 4-digit in 1/100 mm (0.1 in = 0100).
9	Film type	
	-	None
	f	One-sided, ready-to-laser
	D	Two-sided, ready-to-laser
	p	One-sided, points
	I	Two-sided, points
	c	One-sided, vaporize
	K	Two-sided, vaporize
10	Surface coating	
	-	None
	I	Electrolytically galvanized
	H	Hot-dip galvanized
	Z	Galvanized
	B	Primed
	O	Oxidized

Position	Variable	Explanation
11	High surface quality required?	
	-	Standard stripper level
	K	Stripper level +1 mm (only TruTops Punch) Note: The letter "K" for the variable is adapted to other languages.

Meaning of the raw material ID Tab. 2-2

Example: **SS000100f-K**

- SS00: Material group StainlessSteel 00 (1.4301).
- 0100: Material thickness 1.0 mm.
- f: foil type, one-sided, ready-to-laser.
- -: no surface coating.
- K: high surface quality is required (stripper level +1 mm).

Stock ID The standard stock IDs correspond to the standard raw material IDs with a sheet size of 2000 x 1000 mm.

A 21-character stock ID is automatically created for new raw blanks.

Position	Explanation
1-11	Corresponds to the raw material ID.
12	Dummy.
13-21	Sheet dimension (X dimension x Y dimension) in mm or inch (e.g. 96°in = 0960)

Meaning of the stock ID Tab. 2-3

Example: **SS000100f-K-2000x1000**

- SS000100f-K: corresponds to the raw material ID.
- 2000x1000: sheet size with 2000 mm in X direction and 1000 mm in Y direction.

Example in inch: **SS000100f-K-0800x0400**

- SS000100f-K: corresponds to the raw material ID.
- 0800x0400: Sheet dimension with 80°inch in X direction and 40° inch in Y direction.

5.1 To modify material data

1. Select *>Extras >Modify data...*.
2. Select *material*.
3. Select the data record.

   The data is entered in the mask head.

4. Modify data.
5. Press *Modify*.

5.2 To delete material data

Note

TRUMPF materials cannot be deleted.

NOTICE

This is because faulty assignments will result between materials and technology tables in TruTops if original material data from TRUMPF is deleted!

➢ Do not delete original material files from TRUMPF .

1. Select *>Extras >Modify data...*.
2. Select *material*.
3. Select the data record.

   The data is entered in the mask head.
4. Select *Delete*.
5. Confirm message with *Yes*.

   The material, the unprocessed material or the blank will be deleted in the selection field and in the database.

6. System rules

Internal and public sets of rules

Apart from the internal TRUMPF, password-protected rules, there is also a public set of system rules available. The variables in the system rules concern all machines and all of TruTops.

6.1 To change the value of a variable

Notes

- In the internal, password-protected rules, variables can only be changed with the help of technical customer service.
- Changes to the public rules will be retained after a new installation or an updates of TruTops.

NOTICE

Deleted or incorrect variables can lead to inoperable NC programs!

- ➢ Do not delete TRUMPF variables.
- ➢ Check all entries and changes.
- ➢ If required: contact Technical Service.

Open rules

1. Select >*Extras* >*Modify data...*.

   Note

   The variable shown in the machine rules depends on the active machine.

2. Either

   ➢ To open the machine rules, select>*Rules* >*Machine*.

   or

   ➢ To open the system rules, select >*Rules* >*System*.

   A screen with all of the available variables will be displayed.

   Tip

   By entering filter criteria, the number of variables displayed can be limited.

Display the value and comment of the variable

3. Mark variables.

   The value of the variables will be displayed under "value". In the comment text underneath, information about the marked variable will be displayed.

Modify the value

4. Modify the value.
5. Press *Modify*.

If the value entered is invalid, an error message will be displayed.

If it is necessary to restart the active TruTops application, for the changes to take effect, a message will be displayed.

6.2 To create a variable

Not all variables have already been created when TruTops is delivered. As required, some can be added.

Condition

- The machines for which the copied variables should apply must be active.

Open rules

1. Select >*Extras >Modify data...*.

   Note

   The variable shown in the machine rules depends on the active machine.

2. To open the machine rules, select>*Rules >Machine*.

   A screen with all of the available variables will be displayed.

   Tip

   By entering filter criteria, the number of variables displayed can be limited.

Creating a variable

3. Press *Create*.

   The "Selection" mask is displayed.

4. Mark variables.

   The "Selection" screen will be closed.

Enter value

5. Enter the value.
6. Press *Create*.

   The value will be entered into the list.

7. Select *Return*.

6.3 To copy a variable

In the machine rules, variables for one machine can be copied and used for another.

Condition

- The machines for which the copied variables should apply must be active.

Open rules

1. Select >*Extras* >*Modify data...*.

   Note

   The variable shown in the machine rules depends on the active machine.

2. To open the machine rules, select>*Rules* >*Machine*.

   A screen with all of the available variables will be displayed.

   Tip

   By entering filter criteria, the number of variables displayed can be limited.

Variable, copying

3. Press *Create*.

   The "Selection" mask will be displayed. It contains variables, which are already in the database for other machines, but which have not yet been defined for the active machine.

4. Mark variables.

   The "Selection" screen will be closed.

Enter value

5. Enter the value.
6. Press *Create*.

   The value will be entered into the list.

7. Select *Return*.

6.4 To display the details for a variable

Other information is stored for the variables apart from the descriptive text, such as the range of values, the test expression or whether they will be overwritten by a TruTops update.

Open rules

1. Select >*Extras* >*Modify data...*.

   Note

   The variable shown in the machine rules depends on the active machine.

2. Either
   - To open the machine rules, select>*Rules* >*Machine*.

or

- To open the system rules, select >*Rules* >*System*.

A screen with all of the available variables will be displayed.

Tip

By entering filter criteria, the number of variables displayed can be limited.

Display the value and comment of the variable

3. Mark variables.

   The value of the variables will be displayed under "value". In the comment text underneath, information about the marked variable will be displayed.

Displaying details

4. Select *Info*.

Details about the variables are displayed.

6.5 To delete a variable

NOTICE

Deleted or incorrect variables can lead to inoperable NC programs!

- Do not delete TRUMPF variables.
- Check all entries and changes.
- If required: contact Technical Service.

Open rules

1. Select >*Extras* >*Modify data...*.

Note

The variable shown in the machine rules depends on the active machine.

2. Either
   - To open the machine rules, select>*Rules* >*Machine*.

   or

   - To open the system rules, select >*Rules* >*System*.

   A screen with all of the available variables will be displayed.

Tip

By entering filter criteria, the number of variables displayed can be limited.

Display the value and comment of the variable

3. Mark variables.

   The value of the variables will be displayed under "value". In the comment text underneath, information about the marked variable will be displayed.

Deleting a variable 4. Press *Delete*.

6.6 To assign the values to colors and line types

If colors or line types can be defined for variables in the rules, always the same digits are used for the respective color and line type.

Value	Color	Line type
-1	Transparent	-
0	Black, if background is black. White, if background is white.	Solid
1	White, if background is black. Black if background is white.	Dashed
2	Red	Dotted
3	Yellow	Dot and dash
4	Green	-
5	Cyan	-
6	Blue	-
7	Magenta	-

Tab. 2-4

6.7 Description of variables in the system rules

Variable	Description
BarcodeCommon	▪ 0: TruTops does not output a bar code for the program name in the setup plan (*.html, *.pdf). (standard) ▪ 1: TruTops outputs a bar code for the program name in the setup plan (*.html, *.pdf).
BarcodeEncoding	The coding of the barcode.
BarcodeFont	The font of the barcode.
BarcodeSinglePart	▪ 0: TruTops does not output bar codes for single parts in the setup plan (*.html, *.pdf). (standard) ▪ 1: TruTops outputs the part ID of each single part as bar code in the setup plan (*.html, *.pdf). ▪ 2: TruTops outputs the drawing notes of each single part as bar-code in the setup plan (*.html, *.pdf).
BarcodeSize	The font size of the barcode.
CadCheckContourIntersectionOnSave	1: When saving in TruTops CAD, a check for overlapping contours is done.

Variable	Description
CheckForGmtWhenLoadGeo_FsMode [1] CheckForGmtWhenLoadGeo_PdmMode [2]	
CheckForGmtWhenLoadGeo_Option	Relates to variables "CheckForGmtWhenLoadGeo_FsMode" and "CheckForGmtWhenLoadGeo_PdmMode". If *.gmt files are also to be suggested which were produced for other machines, remove `machineName=ST_MaschinenName` from the value.
ClickZoomFactor	Factor for zoom when clicking with center or right mouse click.
CompanyLogo	The path where the company logo is stored, which will be shown in the setup plan. Note: The file name of the company logo must begin with "System". Example: 'C:\trumpf\kundendaten\system_mylogo.gif'
DimensioningColor	Color of the dimensioning ▪ 3: Yellow, (standard) Other colors: (see "To assign the values to colors and line types", pg. 2-18)
DxfIgnoreInvisibleElements	Loading of DXF files with/without invisible elements ▪ 0: Dxf files are loaded with invisible elements. ▪ 1: Dxf files are loaded without invisible elements.
EnableRedrawAcceleration	Accelerated redraw. ▪ 0: Deactivated ▪ 1: Activated
FactorElectircPowerPiercing	Part of cutting performance for the piercing power output (negative: standard).
LabelPrinterLayoutFile	Path to the layout file to print a label.
LabelPrinterModeCommon LabelPrinterModePart	Producing labels. The "LabelPrinterModeCommon" variable relates to the NC program, the "LabelPrinterModePart" variable to the part. ▪ 0: No labels are produced. ▪ 1: Labels are produced.
LaserAutoRuleSelect	If only one matching technology table or a matching set of rules is found, this will automatically be activated. ▪ 0: Yes ▪ 1: No
LaserFormatString	This defines which information is shown for the active laser. ▪ $N: Text "Laser". ▪ $L: LTT. ▪ $G: Gas type. ▪ $M: Material. ▪ $R: Laser rules. ▪ $I: Process rule remark. Example $N $L $R ($I): "Laser T2D-5532 T2D-5532-1 (round off)" will be displayed.

1 When working with TruTops in the file system.

2 When working with TruTops in the PDM system.

Variable	Description
LaserShowAllRulesDefault	Only show suitable rules for the technology table ▪ 0: No ▪ 1: Yes
LicenceServerPeekTimeSpanInSeconds	Time span in seconds in which the TruTops network license server is to be queried. ▪ -: If no value is entered, then the server will not be queried.
LOG_PATH	The path where log files of the backup are placed.
MaterialRawInchMetricSelection	▪ 0: Only display materials in the active system of dimensions. ▪ 1: Show all material.
MaxDoubleClickTimeInMs	Time in ms during which a mouse click is interpreted as a double-click.
NestResultLogText	Write .erg files. ▪ 0: No ▪ 1: Yes
NestResultShowText	Show .erg files as result of nesting. ▪ 0: No ▪ 1: Yes
NestShowParametersAfterCreatingJob	Display parameter dialog **after** assembling the job. ▪ 0: No ▪ 1: For free geometry nester ▪ 2: For rectangle nester.
NestShowParametersBeforeCreatingJob	Display parameter dialog **before** assembling the job. ▪ 0: No ▪ 1: For free geometry nester ▪ 2: For rectangle nester.
NestTextInFeedback	Show texts in the feedback. ▪ 0: No ▪ 1: Yes
NestWriteTafFileCopies	▪ 0: Do not write copies of sheets. ▪ 1: Write each individual sheet in taf files when saving.
NrOfFilesInCadRingBuf	Size of the ring buffer in the CAD part
PDM_minimalDatabaseSpace	Minimum required database memory (MB)
PDM_minimalDiscSpace	Minimum required space on hard disk for file storage (MB).
PDMWarningMode	Specification when PDM warnings are to be issued. ▪ 0: No warnings if files are used which are no longer up to date, or if files are overwritten which have lots of dependencies.
	▪ 1: Warn if files that are based on files that have been modified in the meanwhile are to be loaded. The name of the relevant file is specified. Example: – A 'sample.dxf' file is used to generate a 'sample.geo' file. – The 'sample.geo' file is nested with other GEO files on a sheet. – The 'sample.dxf' file will be changed. – If the sheet is re-loaded, a message saying that the 'sample.dxf' file has been modified will be displayed.

Variable	Description
	■ 2: Warn if, when saving a file, another file is affected by the modifications. The name of the relevant file is specified. Example: − A 'sample.geo' file will be generated and saved. − The 'sample.geo' file will be nested on a 'sample.taf' sheet. − The 'sample.geo' file will be modified and saved again. A message saying that the 'sample.taf' file is also affected by the modifications will be displayed.
	■ 5: Warn if files that are based on modified files that have been intermediately modified are to be loaded. The names of all the relevant files are listed. Example: − A 'sample.dxf' file is used to generate a 'sample.geo' file. − The 'sample.geo' file is nested with other GEO files on a sheet. − The 'sample.dxf' file will be changed. − If the sheet is re-loaded, a message saying that the 'sample.dxf' file has been modified and the 'sample.geo' file is no longer up to date will be displayed.
	■ 6: Warn if, when saving a file, other files are affected by the modifications. The names of all the relevant files are listed. Example: − A 'sample.geo' file will be generated and saved. − The 'sample.geo' will be saved as 'sample.mtl' mini nest and nested on a 'sample.taf' sheet. − The 'sample.geo' file will be modified and saved again. A message saying that the 'sample.mtl' and 'sample.taf' files are also affected by the modifications will be displayed.
	■ 7: Warn if files that are based on files that have been modified in the meanwhile are to be loaded and warn if when saving a file, other files are also affected by the modifications. The names of all the relevant files are listed. For examples, see value = 5 and value = 6.
PrinterDialog	Printing for setup plans. ■ 1: The screen to select the printer will be displayed. ■ 3: The setup plan will be printed on the default printer with no further interaction.
Sc_KennzNotBearbGeo	Evaluation of the settings for contours which cannot be processed. The settings can be modified under *General 2* in the parameters for the laser rules. Modify settings: . ■ 0: The settings are evaluated. ■ 1: The settings are evaluated. (standard)
Sc_Krit_Geo	Evaluation of the color information during laser cutting. Cyan colored points will be point marked. Yellow lines will be marked. ■ 0: Color information will be evaluated. ■ 1: Color information will not be evaluated. (standard)

<table>
<tr><th>Variable</th><th>Description</th></tr>
<tr><td>ScrollcustomerNameDefaultOf
ScrollmaterialDefaultOf
ScrollmaterialIDDefaultOf
ScrollstockIDDefaultOf
ScrolltruMachineNameDefaultOf
ScrolluserDefaultOf</td><td>"Scroll" variables are used to configure content of selection list boxes (scrolls).

Example: Scrolluser ("Operator" selection list box in the "Create NC program" screen).
▪ ScrolluserDefaultOf: default value 100. TruTops searches the database and determines the 100 last used user names.
▪ ScrolluserMaxNumber: default value 19. Of the 100 last used user names, the 19 most frequent are determined and shown in the selection list box.
▪ ScrolluserSortOrder: default value TIME. The 19 most frequently used user names are sorted in descending order by date.
− AZ: alphabetical sorting from A in ascending order.
− ZA: alphabetical sorting from Z in descending order.

Other selection list boxes (scrolls) are configured analogously:
▪ ScrollcustomerNameDefaultOf: "Customer/Job number" selection list box in the "Create NC program" screen. Customer names are displayed.
▪ ScrollmaterialDefaultOf: material
▪ ScrollmaterialIDDefaultOf: unprocessed material identity numbers
▪ ScrollstockIDDefaultOf: stock identity numbers
▪ ScrolltruMachineNameDefaultOf: machine names</td></tr>
<tr><td>ScrollDrawingNoteMaxNumber</td><td>Maximum number of entries in the selection list box for drawing notes</td></tr>
<tr><td>ScrollDrawingNoteSortOrder</td><td>The sort order for the selection list box with drawing comments.
▪ AZ: Alphabetical sort, from A upwards.
▪ ZA: Alphabetical sort, from Z downwards.
▪ TIME: Sort from the last entry used, downwards.</td></tr>
<tr><td>ScrollOrderNumberMaxNumber</td><td>The maximum number of entries in the selection list box for job numbers</td></tr>
<tr><td>ScrollOrderNumberSortOrder</td><td>The sort order for the selection list box with order numbers.
▪ AZ: Alphabetical sort, from A upwards.
▪ ZA: Alphabetical sort, from Z downwards.</td></tr>
<tr><td>ScrollPartIdNumberMaxNumber</td><td>Maximum number of entries in the selection list box for part identity numbers</td></tr>
<tr><td>ScrollPartIdNumberSortOrder</td><td>The sort order for the selection list box with part identity numbers.
▪ AZ: Alphabetical sort, from A upwards.
▪ ZA: Alphabetical sort, from Z downwards.
▪ TIME: Sort from the last entry used, downwards.</td></tr>
<tr><td>SetupPlanTimeformatGeneralData</td><td>Format for total time in the setup plan.
▪ %#H : %M : %S [h:min:s]: Example, (standard)</td></tr>
<tr><td>SetupPlanTimeformatSinglePart</td><td>Format for the single part time in the setup plan.
▪ %q2 min: Example, (standard)</td></tr>
<tr><td>ST_BendTypeAutoCorrect</td><td>1: Bending method and technology (pre/final bend) are automatically corrected.</td></tr>
<tr><td>ST_BmtPath</td><td>For *.bmt files:

Path to the folder which is displayed by default when opening or saving the file type.</td></tr>
</table>

Variable	Description
ST_BmtPathModal	For *.bmt files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_DwgfPathModal	For *.dwgf files: Path to the folder which is displayed by default when opening or saving the file type.
ST_DwgPath	For *.dwgf files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_DxfPath	For *.dxf files: Path to the folder which is displayed by default when opening or saving the file type.
ST_DxfPathModal	For *.dxf files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_FsLoaderMask	To activate the command line file selector: set the value to `M_LoaderKommandoZeile`.
ST_GraPath	For *.gra files (only in the Nest application): Path to the folder which is displayed by default when opening or saving the file type.
ST_GraPathModal	For *.gra files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_IgsPath	For *.igs files (only in the Nest application): Path to the folder which is displayed by default when opening or saving the file type.
ST_IgsPathModal	For *.igs files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_IgsSaveExt	For *.igs files: File ending when saving. ▪ IGS: (standard)
ST_IgsSelect	For *.igs files: *Reg. expr. for the file display*
ST_MiPath	For *.mi files (only in the Nest application): Path to the folder which is displayed by default when opening or saving the file type.

Variable	Description
ST_MiPathModal	For *.mi files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_MiSaveExt	For *.mi files: File ending when saving. ▪ IGS: (standard)
ST_MiSelect	For *.mi files: *Reg. expr. for the file display*
ST_NumInputWithMouse	A calculator will be displayed for input fields where a numerical value can be entered. ▪ 0: Deactivated ▪ 1: Activated
ST_ReplaceArcWithLineError	Replace arcs with a single line when loading a drawing if the maximum chordal error is smaller than the given tolerance value. ▪ 0.001: (standard)
ST_ReplaceArcWithPolygonDistance	Indicates the maximum permissible error for the replacement of a banana with a large radius (see ST_ReplaceArcWithPolygonRadius). If this value is reduced, then more lines are generated. ▪ 0.001: (standard)
ST_ReplaceArcWithPolygonMinLength	This value indicates the minimum length of a line when replacing an arc that is too large with a line (see ST_ReplaceArcWithPolygonRadius). ▪ 4.0: (standard)
ST_ReplaceArcWithPolygonRadius	Replace arcs with a line during loading of a drawing if the radius of the arc is larger than the given radius. The fewest possible lines are always generated. If the replacement fails, the arc remains unaffected. ▪ 10000: (standard)
ST_SelRingBufAnz	Ring buffer size
ST_TtfPath	For *.ttf files: Path to the folder which is displayed by default when opening or saving the file type.
ST_TtfPathModal	For *.ttf files: The last selected folder will be displayed again when opening or closing. ▪ 0: No ▪ 1: Yes
ST_VlgSelRingBufAnz	Ring buffer size for templates
ST_WindowPanFaktor	Factor and direction of the movement arrows for the graphics window
TcPartBitmapHeight	The size of the single part image in the html setup plan in the Y direction, in pixels.
TcPartBitmapMonochrome	The mode of the single part images in the *.html setup plan. ▪ 0: Colored ▪ 1: Black & white
TcPartBitmapWidth	The size of the single part image in the html setup plan in the X direction, in pixels.

Variable	Description
TcSheetBitmapHeight	The height of the sheet image in the *.html setup plan in pixels. ▪ 440: (standard)
TcSheetBitmapMonochrome	The mode of the sheet image file in the *.html setup plan. ▪ 0: Colored ▪ 1: Black & white, (standard)
TcSheetBitmapWidth	The width of the sheet image in the *.html setup plan in pixels. ▪ 580: (standard)
'TOPS_PWD'	Working directory for ToPs. Overwrites the INI environment entry TOPS_PWD. (If empty, then the last working directory will be used)
TxSrTafC_WRITE_CRG_EXT	To generate an event file for TC-CELL. ▪ -: If no value is entered, no event file will be produced. ▪ CEL: For example, an event file will be generated for TC-CELL with the file ending '*.cel'.
UseAdvancedCalculator	To active the extended calculator. ▪ 0: No ▪ 1: Yes

Description of variables in the system rules Tab. 2-5

7. NC post-processing programs, external programs

7.1 To enter programs

NC post-processing programs or external programs can be entered and used in TruTops. The possible settings are explained in the following example of an NC post-processing program.

Examples:

- *.html setup plans can be edited, each time they are generated, by an NC program using Microsoft Word.
- *.lst files can be edited, each time they are generated, by an NC program using Word.

Opening Management

1. Select *>Extras >Modify data...*.
2. Select *>User >Programs*.

   The "External Program Settings" mask is displayed.

Setting

3. Under "User", enter `trumpf`.

   Note

   Any designation can be entered here which is not already in use.

4. Enter "Designation".

   Note

   As soon as an external program (in the example Microsoft Word) is opened in self-holding (modal) mode from within TruTops, or if a TruTops file is opened by an external program, it will no longer be possible to interact with TruTops. The external program must first be closed.

5. Either
   - ➢ If the program is to remain open or if the modified file is to be re-imported into the PDM file management system then, under "Modal", select *YES*.

   or

   - ➢ If the program is not to remain open then, under "Modal", select *NO*.
6. If the program (e.g. to do some editing) is to be opened in its own window then, under "Window", select*YES*.
7. If a logfile is to be created:

- Under "Show log", select *YES*.
- Under "Parameters", enter the following:

  `-l ` (in words: minus lower-case L, space), followed by the absolute path to the logfile.

  Example: `-l C:\TRUMPF.NET\workfiles\user1\logfile.log`

If the 'C:\TRUMPF.NET\workfiles\user1 \logfile.log' log file is created by the external program (must be programmed in this way), TruTops displays the log file automatically after the external program has been ended.

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8. If the external program operates with UNC paths (Windows default): under "Export" and "Re-import", select *NO*.

   It is not necessary to export and then re-import.

   Note

   If files exported from the PDM system are to be re-imported, the file names must not be changed.

9. If the external program does not operate with UNC paths then, under "PDM settings", make the following settings:
   - Set "export" to *Yes*.

     The files to be accessed in PDM will be exported to a temporary directory in the file system.
   - If the exported files are to be modified and are then to be available again in the PDM, then set "re-import" to *Yes*. (Only possible if "Modal" is active.)

     The modified files will be re-imported into PDM as soon as the program is closed.
   - If the program needs the files which were used to create the NC program (*.geo, *.mtl, *.gmt, ...),then set "export help files" to *Yes*.
10. To notify Microsoft Word of the location of the *lst file, under "Parameter" enter the variable `$(U)`.

    If for example, the NC program '/TRUMPF.NET/WORKFILES/USER1/SAMPLE.LST' is created on the PC4711 computer and TruTops with PDM, the contents of the variables reads "\\PC4711\TRUMPF.PDM2\TRUMPF.NET\WORKFILES\USER1\SAMPLE.LST".

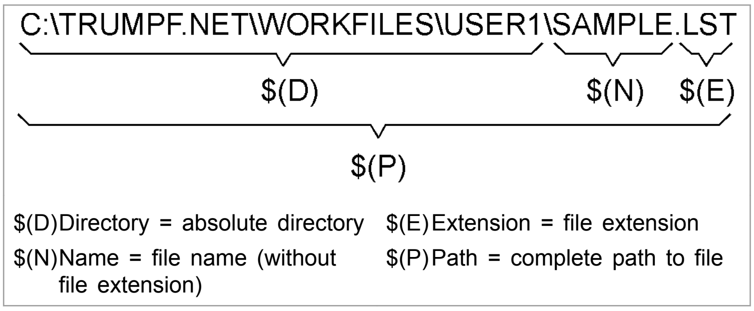


Fig. 42508

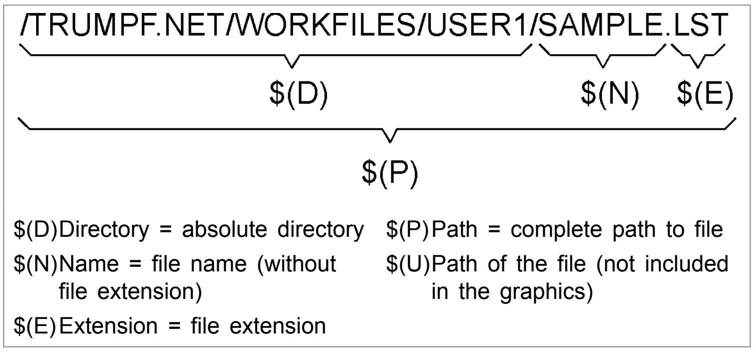


Fig. 43230

The variable contains "\\PC4711\TRUMPF.PDM2\TRUMPF.NET/WORKFILES/USER1/SAMPLE.LST".

11. Under "Program", enter the path and the name of the program.

    Tip

    In the example with Microsoft Word, the path and the program name would be `C:\Programme\Microsoft Office\OFFICE11\Winword.exe`.

12. Under "Work directory" enter the directory in which the program should be run

    Tip

    As a rule, `$(D)` is correct.

13. Press *Create*.

The program will be added to the selection field.

7.2 To execute programs

➢ .

7.3 Creating a collection call up for several NC post-processing programs

Several single NC post-processing programs can be consecutively called up in a defined sequence via the 'PostprocessorCollection.bat' NC post-processing program.

Any number of 'PostprocessorCollection.bat' files can be created. However, only one per machine can be activated as a standard.

Copying template

1. Start Windows Explorer.
2. Go to the '…\TRUMPF.NET\UTILS\Macros' directory.

Notes

- The copies must also be in the '…\TRUMPF.NET\UTILS\Macros' directory.
- The name of the file must not contain more than 14 characters.

3. Copy and rename 'PostprocessorCollectionTemplate.bat' in the required quantity (e.g. 'PostprocessorCollection_1.bat', 'PostprocessorCollection_2.bat').

Modifying template

4. Open the copy (e.g. 'PostprocessorCollection_1.bat') with an editor (e.g. Notepad).

Note

The line `notepad %A_LST_FILENAME_STRING%` in the following example of a *.bat file must be in **one** line.

5. Enter the NC post-processing programs to be activated in the desired order (see **highlighting**).

Example:

- Execute the NC post-processing program 'customer.pl'.
- Open NC program using an editor.
- Print the NC program using the default printer.

Corresponding entries in the *.bat file (e.g. 'PostprocessorCollection_1.bat'):

```
rem ----------------------------------------
set A_LST_FILENAME_STRING=%1
set A_TC_PERL_PATH=%2
set A_TC_TOOL_PATH=%3
rem ----------------------------------------


rem ----------------------------------------
rem Here comes the list of postprocessors
rem ----------------------------------------


%A_TC_PERL_PATH%\bin\perl
%A_TC_TOOL_PATH%\customer.pl
%A_LST_FILENAME_STRING%
notepad %A_LST_FILENAME_STRING%
notepad /p %A_LST_FILENAME_STRING%
```

```
rem ----------------------------------------
set A_LST_FILENAME_STRING=
set A_TC_PERL_PATH=
set A_TC_TOOL_PATH=
rem ----------------------------------------
```

Tip

The NC post-processing programs entered can be deactivate by commenting them out with "rem" at the beginning of the line.

6. Save changes.

Entering a collection call up as NC post-processing program

7. Open the Data Management of the machine for which the NC post-processing program is to be used.
8. Select *>User >Programs*.
9. Enter "User", `trumpf`.
10. Enter "Designation", `PostprocessorCollection_1` (example!).
11. Select "NC converter", *YES*.
12. Select "Modal", *YES*.
13. Select "Window", *YES*.
14. Select "Show log", *NO*.
15. Select "export", "re-import" and "export help files", *NO*.

Note

The following list must be extended or shortened according to the entered post-processing program.

16. Under "Parameters", enter

    `$(P) $(TOPS_PERL_PATH)$(TOPS_APP_UTILS_PATH)`

    eingeben.

    These transfer parameters are adapted to the 'PostprocessorCollectionTemplate.bat' template and mean the following in the above-mentioned example:

    - `$(P)`: Name of the generated NC program.
    - `$(TOPS_PERL_PATH)`: Path to PERL.
    - `$(TOPS_APP_UTILS_PATH)`: Path to the product-specific tools.
17. Under "Program": enter `$(TOPS_MACROS_PATH)\PostprocessorCollection_1.bat`.

Note

The NC post-processing program selected last is set as the standard for the active machine. The standard holds for all TruTops users.

18. Select *Create*.

    TruTops adds the NC post-processing program in the selection field.

    Tip

7.4 To modify programs

Opening Management

1. Select *>Extras >Modify data...*.
2. Select *>User >Programs*.

   The "External Program Settings" mask is displayed.

Modify parameter

3. Select the program.

   The program data is entered in the mask header.
4. Modify parameters (see "To enter programs", pg. 2-26).
5. Press *Modify*.

7.5 To delete programs

Note

Programms which are delivered with TruTops may not be deleted.

Opening Management

1. Select *>Extras >Modify data...*.
2. Select *>User >Programs*.

   The "External Program Settings" mask is displayed.

Delete

3. Select the program.

   The program data is entered in the mask header.
4. Press *Delete*.

The program will be deleted from the selection field.

8. User-defined programs

Note

User-defined programs are managed centrally for all applications.

8.1 To enter programs

Opening Management

1. Select >*Extras* >*Modify data...*.
2. Select >*User* >*TruTops functions*.

   The "User-defined TruTops functions" mask is displayed.

Create a program

3. Enter any name that you wish for the program under "Command name".

   Note

   The standard path for the user-defined programs is '…\TRUMPF.NET\Utils\UserFunctions'.

4. Enter the path and file name of the program which was supplied by TRUMPF under "Shell script".
5. Select the desired applications under "Modules with visible command".
6. Press *Create*.

8.2 To change programs

Opening Management

1. Select >*Extras* >*Modify data...*.
2. Select >*User* >*TruTops functions*.

   The "User-defined TruTops functions" mask is displayed.

Modifying data

3. Select the program.
4. Modify data.
5. Press *Modify*.

8.3 To run programs

1. Leave the data management via *Exit*.
2. Select >*File* >*Execute...*
3. Click on a user-defined program.

   The program is running.

Chapter 3

Opening and saving files

1. Saving TruTops CAD files

Geometry preparation when saving

When saving two-dimensional drawings, a geometry is automatically prepared to avoid inaccuracies. All geometry elements are analyzed and, if necessary, prepared:

- The inside and outside contours are determined.
- Lines that have been drawn twice are removed.
- Contour transitions which have not been drawn cleanly are cleaned.

Auxiliary lines are not saved

When saving a drawing as '*.GEO' or '*.DXF', the auxiliary lines and auxiliary circles are not saved.

Tip

To retain auxiliary lines and auxiliary circles as long as you are working with them, save the drawing as a workfile.

Zero point

When saving drawings as '*.GEO', the zero point of the drawing is automatically set to the bottom left corner. If a contour has no bottom left corner, the zero point is set to the bottom left corner of a circumscribing rectangle.

Tip

To retain the zero point of a drawing as long as you are working with it, save the drawing as a workfile .

1.1 Saving drawings as a workfile

Application: Quick intermediate save of the current status of the following files without geometry preparation:

- Drawings.
- Geometry templates for single holes, rows of holes, bolt hole circles and hole grids
- The auxiliary lines and auxiliary circles are retained. A workfile is always overwritten when another workfile is saved.

Directory of the workfile in TruTops CAD:

- The directory is set in the "TOPS_PWD" variable in the public system rules and can be changed there. (See the TruTops Laser or TruTops Punch software manuals, chapter "Setting basic data".)

Condition

- The drawing is open.

➢ Select *>File >Save workfile*.

The workfile is saved as 'Workfile_CAD.geo'.

The Windows user name and the computer name are added to the name of the workfile.

Example: '...\TRUMPF.NET\Workfiles\workfile_CAD_muellerhe_PC012345.geo'

(muellerhe = Windows user name, PC012345 = computer name.)

1.2 Saving drawing as GEO

Condition

- The drawing is open.

1. Enter the drawing data if required.
2. Either
   - ➢ Select *>File >Save*.
   - ➢ The drawing is saved with the unchanged name in the unchanged directory.

   or

   - ➢ Select *>File >Save as...*.
   - ➢ Change the directory in the TruTops file browser if required.
   - ➢ Select **.GEO* as "file type".
   - ➢ Enter the file name (without file extension).
   - ➢ Press *OK*.

     The drawing is saved under the new name in the (if required) new directory.

1.3 Entering drawing data

Drawing data means additional information that TruTops accesses, e.g. when machining a part.

The drawing data is a part of the file properties (see "File Properties", pg. 1-39).

Condition

- The drawing is open.

1. Select >*File*> >*Properties ...*.

   The "File information" mask is displayed.
2. Enter the desired information:
   - Open the list fields with ▼.
   - Mark the selection or enter the data manually.

   Tips

   - To display the "File information" mask automatically the next time a new geometry is saved: select "Request properties during saving",
   - With *Values last used*, the mask can be filled with the values used last at the press of a button. Only individual data must be modified then.

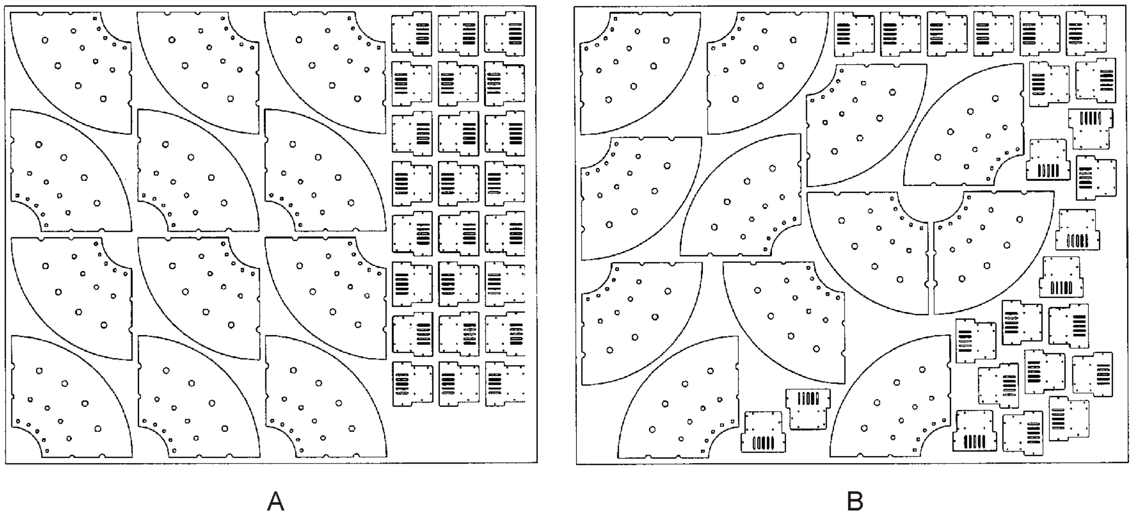
3. Select "General settings".

   Note

   Under "Quantity", a minimum and maximum quantity can be defined for a part. This data is then used accordingly in the nester.
4. If necessary, switch to "TwinLine/Options" and select the desired options:
   - If a part is suitable for common cuts (TwinLine): select "Suitable for TwinLine". Enter the number of parts in X or Y direction which are to form a TwinLine group.
   - To define a part as sample part: select "Sample part" and enter the number (for description of "Sample parts", see the software manuals of TruTops Laser or TruTops Punch, chapter "Nesting parts").
   - To divide sheets in grids in TruTops Laser/Punch and to place the part in the appropriate grid: select Displace "part in grid".

     With "Part in grid", the sheets are separated into grids (depending on the size and the shape of the part) and the parts are regularly transferred in it. In case of different parts, various grid divisions can appear within a sheet.

     "Part in grid" is suitable for L-shaped, rectangular and C-/V-shaped parts. Parts with circular outer contour are unsuitable". "Part in grid" can improve the material utilization and generate more homogeneous nesting patterns.



A Parts displaced in grid   B Parts not displaced in grid

Displacing parts in grid - example   Fig. 43220

5. Switch to "Technology" as required.

Note

Only enter the rules number if the part is to be machined according to very specific rules (e.g. if you have created rules specifically for this purpose).

6. To save the drawing data immediately: select *Save as…*.

Note

If the drawing is to be saved only later, the drawing data is also saved along with it.

7. Close the "File information" mask with *OK*.

1.4 Saving drawings as DXF

Condition

- The drawing is open.

1. Select *>File >Save as...*.
2. Change the directory in the TruTops file browser if required.
3. Select **.DXF* as "file type".
4. Enter the file name (without file extension).
5. Press *OK*.

1.5 Saving geometry templates as VLG for single holes, rows of holes and circles of holes

Condition

- A geometry template has been drawn using the drawing functions in TruTops

or

- A geometry template from a CAD system has been read in.

Entering the file name

1. Enter the drawing data if required.
2. Select >*File* >*Save*.

   or

   ➢ Select >*File* >*Save as...*.

   The file browser opens.
3. Select **.VLG* as "file type".
4. Enter the file name.
5. Press *OK*.

   A query is displayed, whether a new center (with respect to the zero point) should be defined.

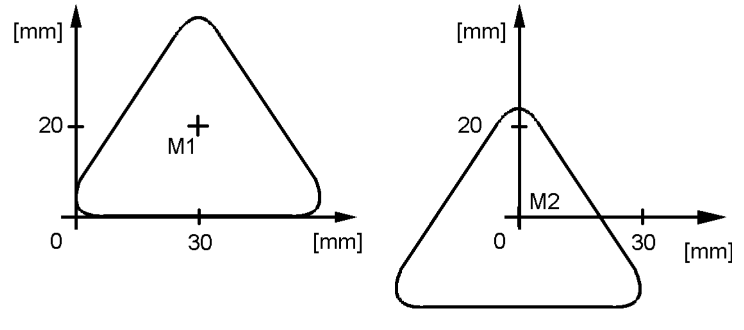
Define a new zero point?

6. If a new center is to be defined: confirm the query with *Yes*.
7. Enter the X and Y coordinates of the point which is to be the new zero point (example below: `30, 20`).

   or

   ➢ Click on the point which is to be the new zero point (example below: click on M).

   The zero point is redefined.



0 Old zero point
M1 Center of one's own geometry template
M2 Center and new zero point of one's own geometry template

Left: before moving; right: after moving Fig. 26986

8. To save the new zero point: select >*File* >*Save*.

9. Confirm the query as to whether the existing data should be overwritten.
10. Do **not** confirm the query as to whether a new center should be defined (select *No*.)

1.6 Saving punching or bending tools as WZG

Condition

- A punching tool has been drawn using the drawing functions in TruTops. (For importing punching tool data, see software manual of TruTops Punch.)

or

- A bending tool has been created using the bending tool assistant in TruTops(see "Drawing bending tool", pg. 6-7).

1. Enter the drawing data if required.
2. Select *>File >Save as...*.

   The file browser opens.
3. Select **.WZG* as "file type".

   Note

   Tool drawings are automatically saved in the '…\TRUMPF.NET\Data\PunchTools' (punching tools) or '…\TRUMPF.NET\Data\BendTools' (bending tools) directory. The directory cannot be changed.
4. Enter the file name (without file extension).
5. Press *OK*.

   The "Save tool" mask is displayed.
6. Select "Punching tool" and options for the punching tool.

   or

   ➢ Select "Bending tool" and options for the bending tool.
7. Press *OK*.

The drawings are saved as tools in the *.WZG format.

2. Opening TruTops CAD files

2.1 Opening workfile

Condition

- A workfile has been saved.

➢ Select >*File* >*Open workfile*.

2.2 Opening TruTops files

1. Select >*File* >*Open*.

   The file browser opens.

2. Select the file type:
   - 2D drawings of parts: **.GEO*.
   - Geometries with variants: <**.GMV*.
   - Geometry templates: **.VLG*.
   - Tools from TruTops Punch or TruTops Bend: **.WZG*.

   The file browser displays all files of the selected format which are located in the current directory.

3. Search for the file if necessary.
4. Mark the desired file.
5. Press *OK*.

2.3 Opening and vectorizing image files (optional)

Image files of types BMP, JPG, PNG, GIF, TIF or PDF can be opened, vectorized, and saved as *.GEO files. The purpose of vectorization is to convert a raster image into a vector image which can then be used for calculations (it contains closed contours and no contour lines which are not for production) or which is production-ready (can be used for calculations and contains suitable geometric elements).

The better the used drawing is, the better the vectorization process works. The following points should be observed:

Drawing	Remark
Black/white	Best results. When scanning drawings, make sure that the "Black/white" function is used.
Grayscale	Good results. When scanning drawings, make sure that the "Grayscale" function is used.
colored	Poor results.
Grid background	Poor results. The results may be able to be improved by adjusting the threshold value.

Tab. 3-1

Tips

Contour types	The GEO format recognizes only three types of segments: line, circle, and contour segment. If a circle is not recognized as such and a line string of contour segments is generated instead, the production of the part will be more complex as a result (e.g. multiple strokes and possibly multiple tools in the case of punching machines). It may therefore be advisable to replace these "line string circles" with geometric circles.
Drawings with reflective gloss	Using photographs as the drawing often results in unsightly reflective gloss which may be recognized as a contour in the vector image. In such cases, the functions "Generate black/white picture" (adjust threshold) and "Edit black/white picture" (strengthen or weaken contour) can help.
Technical drawings	Vectorizing dimension lines and units of measurement regularly causes problems. The dimension lines need to be separated from the workpiece and excluded from vectorization using the threshold. Or you can remove the dimensions and units from the drawing.
Any contours still open?	The steps described might not always result in success. In such cases, you need to adjust the drawing manually via "Start drawing program".
Overlaps and unnecessary lines	Superfluous lines and filled-in corners are often a problem for vectorization. It is advisable to remove these parts manually.
Work step-by-step	Whenever possible, one work step (e.g. generating the black/white image) should be completed before starting the next step. This ensures that the results of work steps are never lost.

Tab. 3-2

Vectorizing an image

1. Select >*File*, >*Open bit map...*

   The file browser opens.
2. Select "file type".

3. Select the desired file.
4. Press *OK*.

   The "Grid loading options" mask is opened

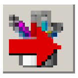
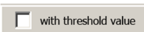
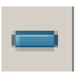
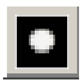
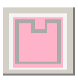
   Note

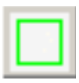
   The vectorization process must be initiated manually each time the parameters have been changed.

5. Process drawing.

   or

   ➢ Select the desired vectorization parameter.

Button	Options	Remark
Edit drawing		
	Start drawing program	The raster image can be opened and edited in an external drawing program. This is helpful in allowing you to close contours manually and to remove superfluous elements. This function must only be used if the automatic vectorization process is unsuccessful. The resulting drawing is added to the vectorization sequence again.
	Select drawing program	Use this button to select the drawing program to be used.
Generate black/white picture		
	Generate black/white picture with threshold value	This function defines the minimum level of brightness required for a line or surface to be included in the image. Everything below the threshold value will be suppressed and will have no effect during vectorization. This is useful with smudged hand drawings, for example, or when scanning has produced darkened areas. Checking the box for the "Contour "option allows the threshold to be adjusted for unevenly filled-in areas so that they form a single contour.
Edit black/white picture		
	Thickening	Lines are thickened.
	Thinner	Lines are drawn thinner.
	Invert	Black and white areas in the picture are exchanged. This function helps to remove shading in the raster image.
	Select region	Selection of a rectangular section of the raster image.
Set vectorization parameters		

Button	Options	Remark
	Connection distance	Open contours are connected if their end points are within the value set here. It is recommended to select a small value first and only increase it if necessary.
	Filter length	Superfluous detached contours that are less than the filter length are deleted during vectori-zation. It is recommended to select a small value first and only increase it if necessary.
	Accuracy	This function controls the maximum deviation of the generated contour from the raster image. High accuracy means that the generated contour will have more anchor points.
	Center line	Line drawing In a line drawing, the center line is used as a contour.
	Contour	Drawing with filled-in areas Here the edge of the filled-in areas is used as a contour
Set dimensions		
	Width/height	These define the dimensions of the bounding rectangle of the resultant geometry.
	Retain ratio	The ratio is maintained when alterations are made to the width or height.
Toolbar		
	Refresh view	This button is active only when vectorization parameters have been changed. When the but-ton is pressed, the drawing is revectorized using the set parameters.
	Show all points	Shows all points on the segment boundaries of the contours. In this way you can find out whether a circle was recognized as such or instead consists of many separate partial circles.
	Show raster image	Hide the display of the raster image so that the vector image can be seen more easily.
	Show vector image	Hide the display of the vector image so that the raster image can be seen more easily.

Tab. 3-3

6. Press *OK*.

   The drawing opens.

Show legend

Note

Depending on the function selected, either the imported drawing only will be displayed, or the resultant geometry, or both together. The legend provides a key to the meanings of the symbols and colors.

7. Click on the drawing surface.

8. Press Shift and F1.

   The legend is displayed.

Finishing vectorization

9. Press *OK*.

Vectorization is concluded and the geometry is accepted.

3. Opening foreign formats from CAD systems

Files can be loaded in the following formats:

External format	Supported by TruTops CAD 10.0
DWG (= file format from the AutoCAD CAD system).	DWG 2013
DXF (= a file format from CAD systems, e.g. AUTOCAD)	ASCII-DXF (all versions)
MI (= file format from the ME10 CAD system)	ASCII and binary format up to version 2.5
Optionally available: *.IGS (= file format of various CAD systems).	ASCII

External formats Tab. 3-4

All drawings loaded from external systems can be further processed with the drawing functions of TruTops CAD.

3.1 Loading 2D drawings in format DXF, DWG, IGS or MI

Conditions

- TruTops CAD can read in only 2D data with standardized Z coordinates. 3D drawings must be projected onto a plane and saved as 2D drawings.
- '*.IGS': The IGES standard stipulates a fixed format of 80 characters per line. Only drawings which comply with this standard can be read in.
- '*.MI': TruTops CAD can read only the uncompressed '*.MI' format as a 2D drawing or as a processing for a tube.
- '*.DXF': When loading DXF files, specific unknown colors can be changed via the rules entry *DxfReplaceColor*, e.g. to a fixed color or to an automatically selected color.
- The drawing may only contain machinable contours, in other words no drawing frames and headers and no auxiliary geometries.
- Bending lines are interpreted only in TruTops Bend. Otherwise, they are skipped.
- Only geometry elements recognized by TruTops CAD may be transferred.

1. Select >*File* >*Open*.

   The file browser opens.
2. Open the desired folder and select the file type.

Note

If the '*.MI' format is selected, the TruTops file browser displays all files that are located in the current directory and are not TruTops-specific (with and without file extension).

3. Mark the file and select *OK*.

The "... loading options" screen for the selected file format will be displayed.

3.2 Defining loading options for 2D drawings in DXF/DWG format

Notes

- Mutually exclusive loading options are grayed out accordingly.
- Invisible elements in the DXF files will not be loaded. If these have to be loaded, the entry *DxfIgnoreInvisibleElements* in the system rules has to be set to `0`.

Parameters	Description
"Transfer only specific layers"	By reading in particular layers it is possible to read in geometries of a layer specifically or leave them out specifically. As a result, time-consuming post-machining procedures (e. g. deleting dimensions) are no longer required.
"Adopt inactive layers"	Individual layers can be "deactivated" in AutoCad. They are then no longer displayed. The inactive layers can be loaded in TruTops.
"Contour preparation"	A tolerance can be entered for the drawing's contour preparation. Points which are not separated by the defined tolerance are merged into one point.
Automatic preparation	The automatic preparation attempts to recognize collision contours such as drawing frames, dimensionings, approach flags etc. The recognized part becomes white in color, and all other parts become blue. Advantage: The geometry offset can be easily removed via >*Edit*, >*Delete*, >*Selection*, "Blue".
"Smooth drawing"	During smoothing, TruTops replaces an existing contour (or contour section) with a new contour (or contour section). The aim is to reduce the number of single elements of a contour as far as possible without deviating too far from the original contour.
"Transfer block section"	All geometry elements which were defined as a block in '.DXF' or '.DWG' format while creating a drawing are adopted.
"Adopt only single block"	This option can be used to adopt a single block with geometry elements in TruTops and to ignore the remaining drawing. The block can be modified using the drawing functions in TruTops CAD and saved as a geometry template (*.). Geometry templates can be positioned in a drawing as often as desired through >*Create* >*Macros* (e.g. as hole grids).

Parameters	Description
Adopt only single layout	Single layouts (different views/cutouts of the model) can be stored in DXF and DWG files. These can be loaded individually in TruTops CAD.
"Transfer texts"	Defines whether the texts of the drawing should be transferred and which TruTops font should be used for this.
"B-splines"	A "B-spline" is a mathematical description of a curve. With this form of description, the curve is defined by a series of points and further parameters. If a file contains B-splines, they can be converted into the TruTops elements "lines" and "arcs".

Loading options for .DXF - description of parameters Tab. 3-5

Read in only particular layers?

1. Check "Transfer only specific layers".

   The "Existing layers" list field is displayed.
2. Mark the layer to be adopted.

   The layer is added to the "Layers for transfer" list field.
3. To remove an adopted layer again: execute the procedure in reverse order.

Load deactivated layers?

4. Check the box for "Display deactivated".

Adopt inactive layers?

5. Select "Adopt inactive layers".

Prepare contours?

6. Select "Contour preparation".
7. Enter the tolerance.

   Points which are not separated by the defined tolerance are merged into one point.

Automatic preparation?

Note

The preparation is now visible in the preview.

8. To prepare the DXF/DWG file automatically, select the "Prepare""automatic" checkbox.

   The collision contours are displayed in blue.

Smooth drawing?

Note

If contours have to be smoothed **individually** (with different parameters or several times if required), you can also smoothen them "manually" after loading .

9. Select "Smooth drawing".

   The "Smooth parameter" mask is displayed.
10. Either
    - ➢ Select the preset "coarse", "medium" or "fine-pitch" resolution (the respective data is displayed at the bottom).

    or

    - ➢ Select "user-defined".
    - ➢ Enter the desired data.

The coarser the resolution, the fewer elements are required. The new contour section deviates further away from the original contour.

The finer the resolution, the more elements needed. The new contour section is the closest to the original contour.

11. Press *OK*.

Read in block section?

12. Select "Transfer block section".

Adopt only single block?

13. Select "Adopt only single block".

    The block can be modified using the drawing functions in TruTops CAD and saved as a geometry template (*.). Geometry templates can be positioned in a drawing as often as desired through >*Create* >*Macros* (e.g. as hole grids).

Adopt only single layout?

14. Select "Adopt only single layout".

    A list of the available layouts is opened.

    If there are no layouts in the selected file, a message appears.

Transfer texts?

15. Select "Transfer texts".
16. Open the list field of text fonts with ▼.

    The file browser opens. It displays all available fonts.

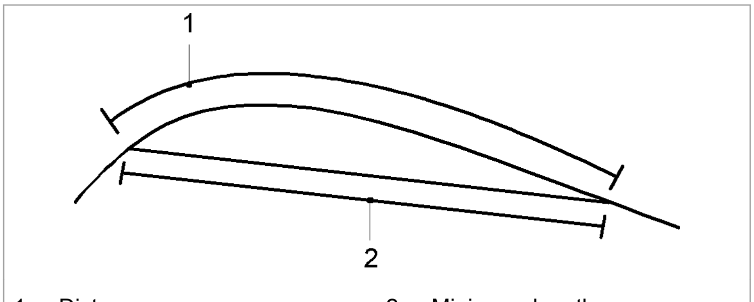
    By default, the fonts can be found in the '...\TRUMPF.NET \Data' directory. The directory cannot be changed.

17. Mark the font to be adopted.
18. Press *OK*.

Defining options for B-splines

19. Select "B-Splines".
20. Open the list with ▼.
21. To convert B-splines into a contour: Mark: *Interpolate*.
22. Enter the distance (1) and the minimum length (2).

    TruTops CAD converts the curve into a contour during the interpolation:



1 Distance 2 Minimum length

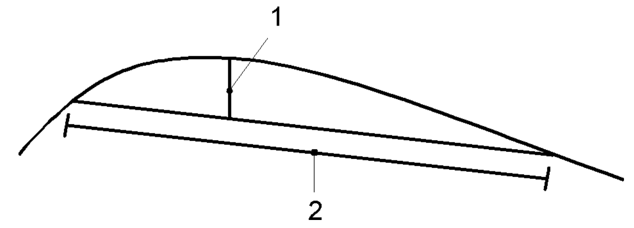
Interpolation of B-splines Fig. 25996

The "Distance" corresponds to the increment along the curve. The smaller this value is, the more exactly the interpolation will match the curve. A disadvantage is that the number of lines generated is increased.

"Minimum length": "Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

23. To convert B-splines into a contour consisting of lines and arcs, mark *Approximate*.
24. Enter the distance (1) and the minimum length (2).

TruTops CAD converts the curve into a contour of lines and arcs during the approximation:



1 Distance 2 Minimum length

Approximation of B-splines Fig. 25997

This parameter defines the maximum distance from the lines and arcs to the curve. The curve is divided until the distance between the approximated continuous line and the curve is less than the specified value.

"Minimum length":"Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

Display control polygon?

25. To display the control polygon of the B-splines, mark *Control polygon*.

The control polygon joins the control points of B-splines with a contour, thereby indicating the approximate course of the curve. The control polygon can be used to estimate the position of the B-splines.

Application example: A B-spline is too short and therefore cannot be converted. The control polygon can be used to estimate the point where the too-short B-spline lay.

Display file preview?

26. Select *Show selection*.

Once the loading options are defined

27. Press *OK*.

Report that errors have occurred?

- ➢ To load the file in spite of errors, press Press *OK*.
- ➢ . To display a list of the errors that have occurred, select *Check*. The error list specifies the line to which the error is assigned. There are errors that are assigned to only one line; other errors are assigned to a range in the file. If the name of the range is available, it is specified. Splines which have not been loaded are also included in the error list.
- ➢ If you do not wish to load the file, select *Cancel*.

If there are no errors, the 2D drawing is loaded in the '.DXF' or '.DWG' format.

3.3 Defining loading options for 2D drawings in IGS format

Parameters	Description
"Contour preparation"	A tolerance can be entered for the drawing's contour preparation. Points which are not separated by the defined tolerance are merged into one point.
"Smooth drawing"	During smoothing, TruTops replaces an existing contour (or contour section) with a new contour (or contour section). The aim is to reduce the number of single elements of a contour as far as possible without deviating too far from the original contour.
"B-splines"	A "B-spline" is a mathematical description of a curve. With this form of description, the curve is defined by a series of points and further parameters. If a file contains B-splines, they can be converted into the TruTops elements "lines" and "arcs".

Loading options for .IGS - description of parameters Tab. 3-6

Prepare contours?

Only activate this option if the DXF file to be loaded is (very) complex and contains "unnecessary things" such as drawing frames etc., which can easily be deleted if this option is activated. If the drawing is largely OK, the activation of this option can have negative side effects (e.g. yellow lines become white lines etc.).

1. Select "Contour preparation".
2. Enter the tolerance.

   Points which are not separated by the defined tolerance are merged into one point.

Smooth drawing?

Note

If contours have to be smoothed **individually** (with different parameters or several times if required), you can also smoothen them "manually" after loading .

3. Select "Smooth drawing".

   The "Smooth parameter" mask is displayed.

4. Either

   ➢ Select the preset "coarse", "medium" or "fine-pitch" resolution (the respective data is displayed at the bottom).

   or

   ➢ Select "user-defined".

➢ Enter the desired data.

The coarser the resolution, the fewer elements are required. The new contour section deviates further away from the original contour.

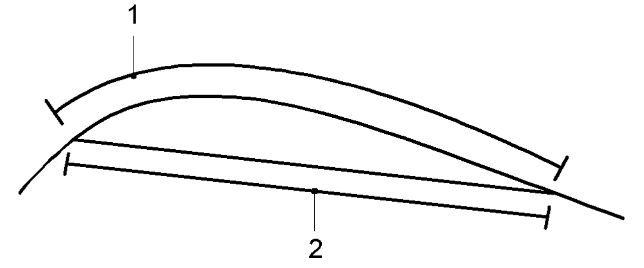
The finer the resolution, the more elements needed. The new contour section is the closest to the original contour.

5. Press *OK*.

Defining options for B-splines

6. Select "B-Splines".
7. Open the list with ▼.
8. To convert B-splines into a contour: Mark: *Interpolate*.
9. Enter the distance (1) and the minimum length (2).

   TruTops CAD converts the curve into a contour during the interpolation:



1 Distance 2 Minimum length

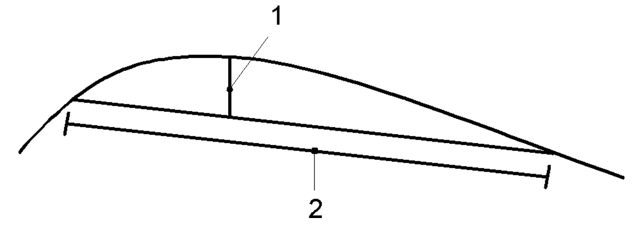
Interpolation of B-splines Fig. 25996

The "Distance" corresponds to the increment along the curve. The smaller this value is, the more exactly the interpolation will match the curve. A disadvantage is that the number of lines generated is increased.

"Minimum length": "Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

10. To convert B-splines into a contour consisting of lines and arcs, mark *Approximate*.
11. Enter the distance (1) and the minimum length (2).

    TruTops CAD converts the curve into a contour of lines and arcs during the approximation:



1 Distance 2 Minimum length

Approximation of B-splines Fig. 25997

This parameter defines the maximum distance from the lines and arcs to the curve. The curve is divided until the distance between the approximated continuous line and the curve is less than the specified value.

"Minimum length":"Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

Display file preview?

12. Select *Show selection*.
13. Press *OK*.

The 2D drawing is loaded in the '.IGS' format.

3.4 Defining loading options for 2D drawings in MI format

Parameters	Description
"Transfer only certain parts."	By reading in particular parts of a drawing it is possible to read in geometries specifically or leave them out specifically. As a result, time-consuming post-machining procedures (e. g. deleting dimensions) are no longer required.
"Contour preparation"	A tolerance can be entered for the drawing's contour preparation. Points which are not separated by the defined tolerance are merged into one point.
"Smooth drawing"	During smoothing, TruTops replaces an existing contour (or contour section) with a new contour (or contour section). The aim is to reduce the number of single elements of a contour as far as possible without deviating too far from the original contour.
"B-splines"	A "B-spline" is a mathematical description of a curve. With this form of description, the curve is defined by a series of points and further parameters. If a file contains B-splines, they can be converted into the TruTops elements "lines" and "arcs".
"Transfer detail drawings"	In the ME 10 program, it is possible to create detail drawings of a part. These detail drawings are given part names which begin with a point. Detail drawings of such parts are normally not loaded by TruTops CAD.

Parameters	Description
"Generate stroke characteristics"	Single holes with tool information ("stroke characteristics") can be defined and saved in the MI file in Solid Designer/Sheet Advisor. (Tapping tools are still not supported.) If the stroke characteristics are adopted, the single holes from Solid Designer have the property "Characteristic" (see "Characteristics: a property of elements and contours", pg. 1-49). TruTops Punch compares tool ID with tool ID and executes the single stroke with the desired tool (prerequisite: the names of the tools must be identical in Solid Designer and TruTops Punch). If TruTops Punch does not find a matching tool, the single hole is not processed. If the stroke characteristics are not adopted, TruTops compares the geometries and allocates appropriate tools (TruTops Punch) or machines the geometry with the laser (TruTops Laser, TruTops Tube). The tool ID is displayed at each single hole (as text; not machined). Tool information is stored differently in an MI file than in TruTops Punch: ▪ MI file: The dimensions e.g. of a rectangular tool are described by the ":HORIZ" value for the horizontal dimension and the ":VERT" value for the vertical dimension. The name of a rectangular tool is "RECT". ▪ TruTops Punch: Via *Macros*, *Parameters*, "Tools", tools are defined by tool type and the dimensions "Dimension 1", "Dimension 2", "Dimension 3 …". "Translation" rules must therefore be defined for each tool type of a MI file for automatic conversions. These tool-specific rules can be saved and applied to each MI file to be loaded. TruTops has a pre-defined set of rules for standard tools, which are adapted to the automatic transfer of TruTops tools in the Solid Designer.

Loading options for .MI - description of parameters — Tab. 3-7

Only load searched parts of the file?

1. Select "Only transfer specific parts".
2. Mark the desired part (to display a preview of the part: right click).
3. Use the arrow to add the marked part to "Parts for transfer".

Prepare contours?

4. Select "Contour preparation".
5. Enter the tolerance.

   Points which are not separated by the defined tolerance are merged into one point.

Smooth drawing?

Note

If contours have to be smoothed **individually** (with different parameters or several times if required), you can also smoothen them "manually" after loading .

6. Select "Smooth drawing".

   The "Smooth parameter" mask is displayed.

7. Either

   ➢ Select the preset "coarse", "medium" or "fine-pitch" resolution (the respective data is displayed at the bottom).

   or

   ➢ Select "user-defined".

➢ Enter the desired data.

The coarser the resolution, the fewer elements are required. The new contour section deviates further away from the original contour.

The finer the resolution, the more elements needed. The new contour section is the closest to the original contour.

8. Press *OK*.

Transfer texts?

9. Select "Transfer texts".
10. Open the list field of text fonts with ▼.

    The file browser opens. It displays all available fonts.

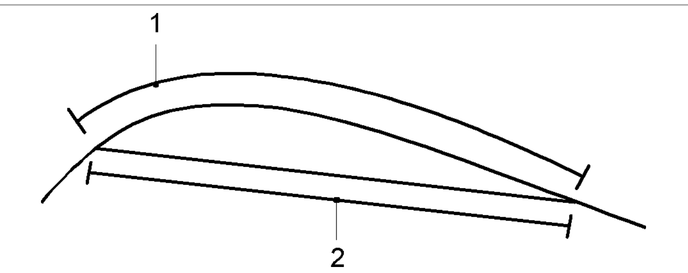
    By default, the fonts can be found in the '...\TRUMPF.NET \Data' directory. The directory cannot be changed.

11. Mark the font to be adopted.
12. Press *OK*.

Defining options for B-splines

13. Select "B-Splines".
14. Open the list with ▼.
15. To convert B-splines into a contour: Mark: *Interpolate*.
16. Enter the distance (1) and the minimum length (2).

    TruTops CAD converts the curve into a contour during the interpolation:



1 Distance 2 Minimum length

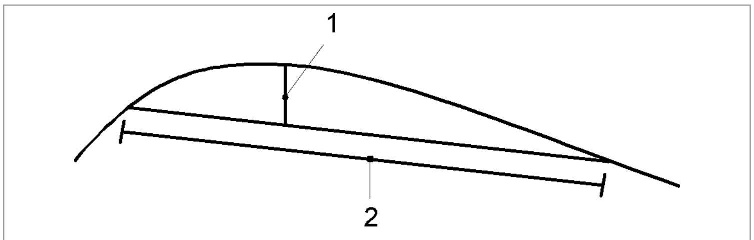
Interpolation of B-splines Fig. 25996

The "Distance" corresponds to the increment along the curve. The smaller this value is, the more exactly the interpolation will match the curve. A disadvantage is that the number of lines generated is increased.

"Minimum length": "Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

17. To convert B-splines into a contour consisting of lines and arcs, mark *Approximate*.
18. Enter the distance (1) and the minimum length (2).

    TruTops CAD converts the curve into a contour of lines and arcs during the approximation:



1 Distance 2 Minimum length

Approximation of B-splines Fig. 25997

This parameter defines the maximum distance from the lines and arcs to the curve. The curve is divided until the distance between the approximated continuous line and the curve is less than the specified value.

"Minimum length":"Each line must be longer than the value entered in "Min. length". The B-spline is not converted if the total length of the B-spline is shorter than the value entered in "Min. length".

Transfer detail drawings?

19. Select "Transfer detail drawings".

Adopting stroke characteristics from Solid Designer?

20. To transfer the stroke characteristics from Solid Designer, select "Generate stroke characteristics".
21. Select *Options*.

The "Rule overview" mask is displayed:

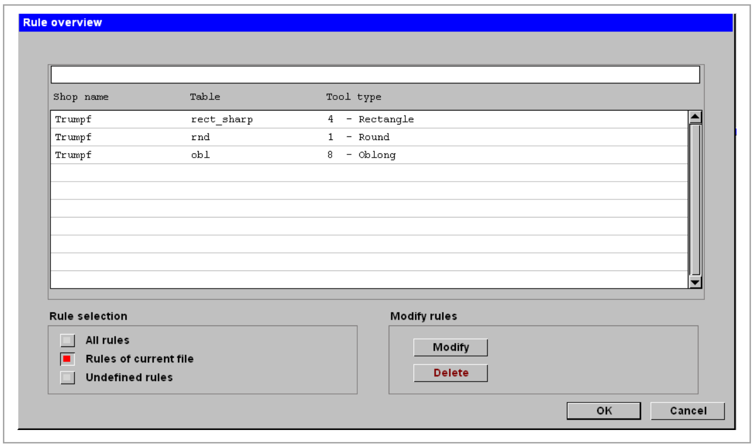


Fig. 43564en

"Shop name": Name of the database which is the source of the information. Normally (if the TruTops database is calibrated automatically) "Trumpf" is entered here.

"Table": Tool type from the Solid Designer. Corresponds more or less to the tool types in TruTops (e.g. "rnd" for a round tool). There is no direct allocation. The special tools from TruTops are thus distributed among various tables in the Solid Designer.

"Tool type": Tool type from TruTops.

22. Select tool-specific rules to be displayed in the mask:

- To display all rules ever defined (even if the respective tool does not exist in the current MI file), select "All rules".
- To display only those rules required for the current MI file, select "Rules of current file".
- To display all not yet defined rules generated automatically by TruTops when it did not find a matching rule for a tool type: select "Undefined rules".

23. To modify a tool-specific rule: mark the rule and select *Modify*.

    The "Tool parameters" mask is displayed.

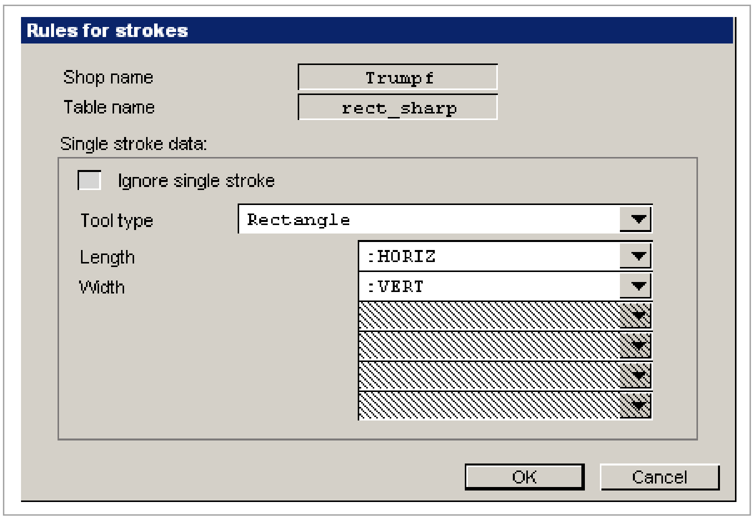


Fig. 43565en

24. Either
    - ➢ To hide the tool information from Solid Designer: select "Ignore single stroke" (all the following selection fields are grayed out).

    or

    - ➢ Open "Tool type" using ▼ and mark the tool type.
    - ➢ Open the list fields next to the tool dimensions used by TruTops (Dim. 1, Dim. 2…) with ▼ and assign them the dimension name from Solid Designer.
25. Confirm the changes with *OK*.
26. To delete a tool-specific rule, mark the rule and select *Delete*.

    If rules are changed, you are asked whether the changes are to be saved after loading the 2D drawing in '.MI' format.

    The changed rules apply to all following loading procedures concerning MI files if the question is answered in the affirmative.

    The changed rules apply to the current file and are subsequently discarded if the question is not answered in the affirmative.

All changes to the rules are discarded if the procedure is aborted.

The preceding zeros of the tool ID are not passed on by the Solid Designer/Sheet Advisor. To avoid error messages in the technology of TruTops Punch module, the preceding zeroes of the concerned tools in TruTops Punch must be deleted (technology module, >*Tools* menu, "Tools" mask).

Displaying the MI file preview

27. Select *Show selection*.
28. Press *OK*.

Once the loading options are defined

The 2D drawing is loaded in the '.MI' format.

4. Extracting parts and sheets from foreign formats (assistant for layout

4.1 Extracting parts from drawings in foreign format and saving as GEO

The assistant for the layout can be used to extract single parts from drawings in foreign format one after the other and to save them as *.GEO. The process can be repeated till all parts are extracted from a drawing.

If **all parts** from a drawing are to be saved as *.GEOs: select *>Assistants >Layout >Extract sheet >Sheet all* (see the next section).

If the sheet layout is not relevant, the option for saving the sheet file (*.TAF) can be deselected here.

Extracting selected elements/entire parts

Note

Single parts can be saved one after the other without having to reselect the function (*>Assistants >Layout >Extract parts >Part element*).

The process is ended with *Cancel* or by selecting a new function.

1. Load drawing in foreign format.
2. If required, select *>Assistants >Layout >Find equals*.

   (If a saved part is present in the drawing several times, the copies too are found in the same manner).
3. To mark **single elements** that should belong to one *.GEO: select *>Assistants >Layout >Extract part >Part element*.

   (*>Part element* is useful only if the parts in the drawing are not nested very close to one another.)

   or

   ➢ To save **full parts** as *.GEO (including all the inside contours, texts or points contained in it): select *>Assistants >Layout >Extract part >Part contour*.

   (*>Part contour* is useful only if the outer contour of the part is correctly drawn and has no gaps, for example.)
4. For *>Part element*: click all **elements** that should belong to *.GEO one after the other, and press the <Ctrl> key at the same time.

   or

   ➢ For *>Part contour*: click on the **outer contour** of the part that is to be saved as *.GEO.

   (Clicking several times while holding the <Ctrl> key makes it possible to deselect selected elements; see the section on multiple selections.)

Note

If the <Ctrl> key is **not** pressed, only **one** element can be selected. The assistant proceeds immediately with this selection.

5. After the selection is completed: Press <Enter> or <Return> (↲).

or

➢ Press *OK*.

The file browser opens.

6. Enter the file name for the new *.GEO.

The first time elements of a drawing are saved, the path from which the drawing was opened and the file type (*.geo) are present. In addition, the previous name with the extension `_1` is shown as file name. Example: 'J:\teile\test_1.geo' when 'J:\teile\test.dxf' was loaded. The index increases by one at each additional save (via >*Save* >*Selection* or >*Assistants* >*Layout*).

7. Press *OK*.

The *.GEO is saved.

The saved part obtains a characteristic that groups the elements of the part in the drawing into one unit (see "Characteristics: a property of elements and contours", pg. 1-49). In the "fixed" characteristic setting, certain actions are therefore either applied to the entire unit, (i.e. on **all** elements) or the actions are declined.

If *Highlight characteristics* is active, the saved parts with a characteristic are highlighted with a color in the loaded drawing.

If >*Find equals* is activated, copies of the part are searched for (independent of the position and turning position) and highlighted with a color as well.

Showing storage location and part names

8. Select >*Assistants* >*Layout* >*Show part names*.
9. Click on the part whose file name and storage location is to be shown.

The storage location and the file name are shown.

4.2 Extracting parts from drawings in foreign format and saving as sheet

The assistant for the layout can be used to extract selected parts from drawings in foreign format and to save them in a *.TAF (in other words, with sheet data included). The sheets saved in this way can be loaded and edited in "Nest", "Laser" und "Punch".

Showing layout assistant

1. Load drawing in foreign format.
2. If required, select >*Assistants* >*Layout* >*Find equals*.

   (If >*Find equals* is activated, all the copies of a part obtain the same name during automatic saving. If >*Find equals* is not activated, each copy obtains a different name.)

3. To save a **defined area** of the loaded drawing **with all the parts contained in it** as sheet geometry (*.TAF) and the parts as *.GEO: select >*Assistants* >*Layout* >*Extract sheet* >*Sheet box*. (The rectangle set up using the mouse leads to the sheet geometry.)

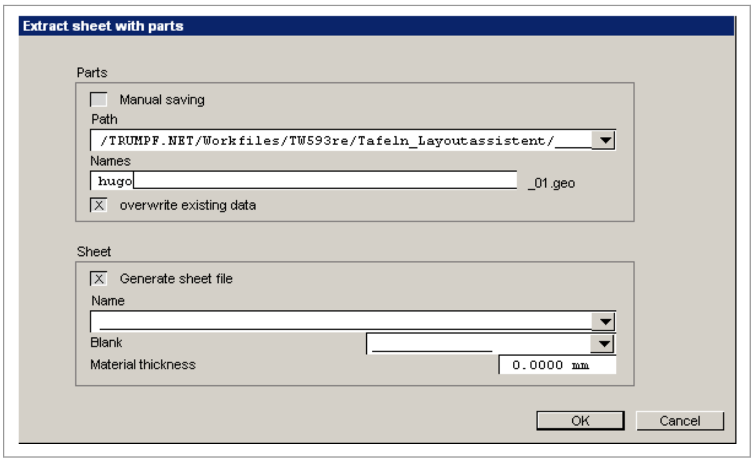
   or

   ➢ To save **all parts within a contour** on the loaded drawing as *.GEO and to adopt the contour (or its circumscribing rectangle) as sheet geometry: select >*Assistants* >*Layout* >*Extract sheet* >*Sheet contour*.

   or

   ➢ To save **all parts of the loaded drawing** as *.GEO and to adopt the circumscribing rectangle of the drawing as sheet geometry: select >*Assistants* >*Layout* >*Extract sheet* >*Sheet all*.

   The "Extract sheet with parts" mask is displayed.



Extracting sheet with parts Fig. 41322en

Defining part data

4. Either

   ➢ To manually save each single part after the definition of the sheet through >*File* >*Save as*: select "Parts""Save manually".

   or

   ➢ To give the individual parts a **single** defined name with a consecutive number (e.g. hugo_01.geo, hugo_02.geo...): open the "Path" selection field with ▼.

- In the TruTops file browser, select the folder in which the parts located on the sheet are to be saved.
- Press *OK*.
- Enter the "name" for the parts found (for example: hugo).
- If the *.GEOs are already present and are to be overwritten: select "Overwrite existing files".

Defining sheet data

5. To save the sheet as '*.TAF': select "Sheet""Generate sheet file".

   (If all parts from a drawing are to be saved as *.GEOs as quickly as possible and the sheet layout is not relevant: deselect "Generate sheet file".)

6. Open the "Name" selection field with ▼.

   The file browser opens.

7. Enter the name under which the extracted sheet (*.TAF) is to be saved.
8. Press *OK*.
9. Select material (the material thickness is automatically entered).
10. If the part and sheet data is specified: Press *OK*.

    The circumscribing rectangle of the drawing is saved as sheet geometry in case of the "Sheet all" area of application. Outer contours are presented in cyan.

    Is the sheet contour considered to be the outer contour of the part and all outer contours of the parts are considered to be inside contours of the "sheet part" by mistake?

    - Select *>Assistants >Layout >Reduce*.
    - TruTops changes the next inner contours to outer contours.
    - Repeat the reduce process if necessary.
    - When the selection of outer contours is correct: select *OK*.

Select area or mark contour?

11. Set up the desired area using the mouse on the loaded drawing.

    or

    - Click on the contour (= contour that should become the sheet geometry).

    Outer contours are presented in cyan.

    Note

    You can select or deselect parts by clicking several times or by boxing them in.

12. If the selection of the area/of the outer contours is complete: select *OK*.

Save parts manually?

13. If "Part""Save manually" has been activated and the marked part is to be saved manually: select *>File >Save as*.

    or

    - If the marked part is to be skipped (i.e.) not to be saved: select *>Assistants >Layout >Next part*.
14. Repeat manual saving/skipping until all parts have been either saved or skipped.

Enlarging the view

15. To enlarge the view on the selected sheet or (when saving parts manually) on the part to be saved: select *>Assistants >Layout >Zoom*.

    The area is displayed in enlarged form.

Showing storage location and part names

16. Select *>Assistants >Layout >Show part names*.
17. Mark the part whose file name and storage location is to be shown.

    The storage location and the file name are shown.

4.3 Filled-in selected parts

If several parts are selected via the layout assistant (extract sheet), these are filled in gray.

1. Select *>Assistants*, *>Layout*, *>Extract sheet*, *>Sheet box*.

   The "Extract sheet with parts" mask is opened.
2. Assign part name.
3. Press *OK*.
4. Box in parts that are to be extracted.

   The parts are filled in gray.
5. Press *OK*.

   The contours of the selected parts are displayed in red.

The selection can be saved.

Chapter 4

Geometries

1. Drawing geometries

Interfaces to CAD systems

In TruTops CAD, drawings can be created directly or imported from other systems. TRUMPF provides interfaces to foreign formats for this purpose (see "Opening foreign formats from CAD systems", pg. 3-14).

Modifying or adding CAD drawings

CAD drawings can be modified or added using the drawing functions of TruTops CAD.

1.1 Points

Drawing points

Notes

- Points that have to be dot marked have to be drawn in the color cyan (see "Modifying line attributes (color and line type)", pg. 4-45).
- Dot marked points can be defined as the base element for single holes, circles of holes and rows of holes .

1. Select >*Create* >*Special elements* >*Point*.
2. Follow the instructions in the command line.

1.2 Lines

Drawing a line via 2 points

1. Select >*Create* >*Lines* >*2 points*.
2. Follow the instructions in the command line.

Drawing parallel line to the available line

Condition

- The drawing contains at least one line.

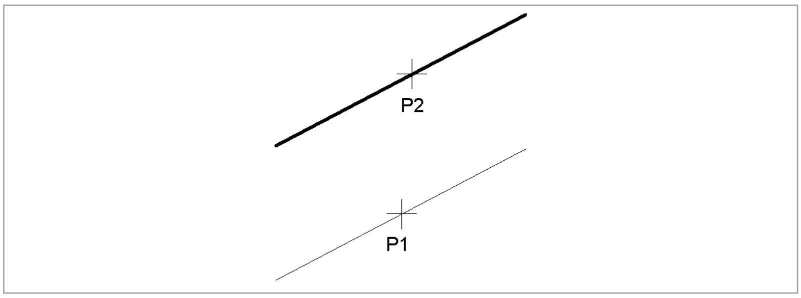


Fig. 4702

1. Select >*Create* >*Lines* >*Parallel*.
2. Click on the line (P1) for which a parallel line is to be drawn.
3. For **one** parallel line: click on the point (P2) on the parallel line.

   or

   ➢ For **several** parallel lines: enter all distances in which one parallel line is to be drawn. (Separate numerical values by blanks. Do not use any commas. Example: `10 20 30`.)
4. Press <Enter> or <Return> (↲).
5. Click on the side on which the parallel line(s) should appear.

Drawing horizontal or vertical lines

1. Select >*Create* >*Lines* >*Horizontal* or >*Vertical*.
2. Follow the instructions in the command line.

Drawing a line as a tangent at an arc element and point

Condition

- The drawing contains at least one arc element or circle.

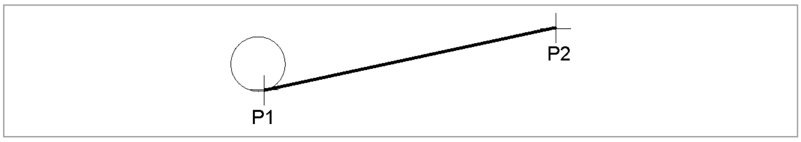


Fig. 4704

1. Select >*Create* >*Lines* >*Tangent through pnt*.
2. Follow the instructions in the command line.

Drawing a line at a right angle to another element

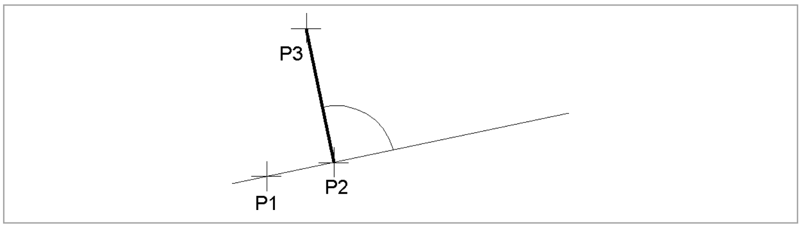


Fig. 4705

1. Select >*Create* >*Lines* >*At right angles*.
2. Follow the instructions in the command line.

Drawing a line with point, angle of inclination and length

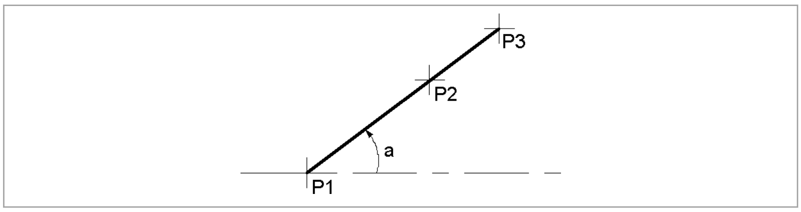


Fig. 4706

1. Select >*Create* >*Lines* >*Angle X*.
2. Follow the instructions in the command line.

Drawing a line via tangent at 2 arc elements

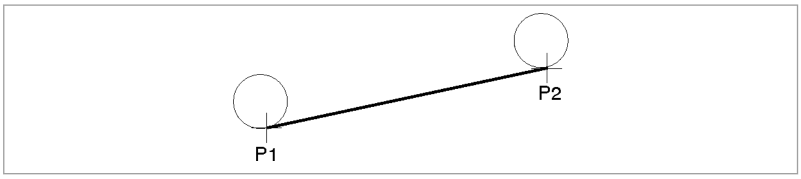


Fig. 4707

1. Select >*Create* >*Lines* >*Tangent at 2 arcs*.
2. Follow the instructions in the command line.

1.3 Circles

Drawing a circle via 3 points

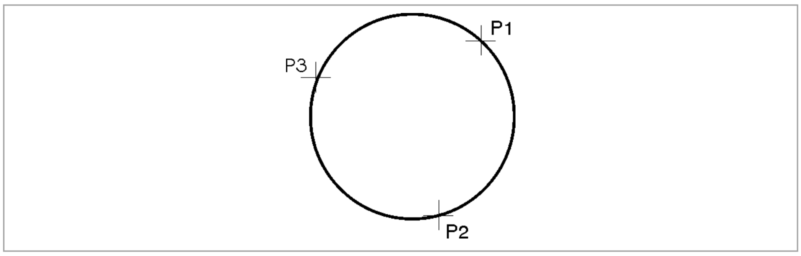


Fig. 4914

1. Select >*Create* >*Circles* >*3 points*.
2. Follow the instructions in the command line.

Drawing a circle via center and radius

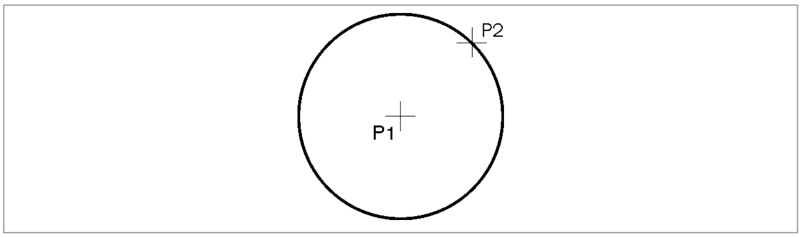


Fig. 4709

1. Select >*Create* >*Circles* >*Center radius*.
2. Follow the instructions in the command line.

Drawing a circle via diameter

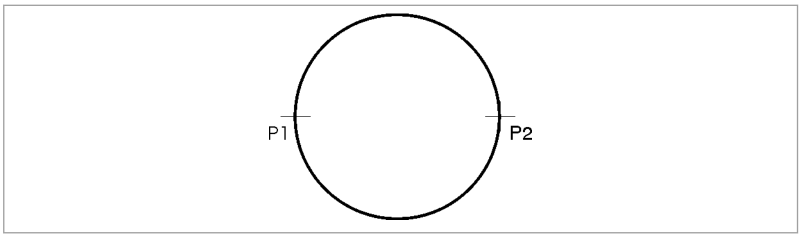


Fig. 4915

1. Select >*Create* >*Circles* >*Diameter*.
2. Follow the instructions in the command line.

Drawing a circle via center and radius

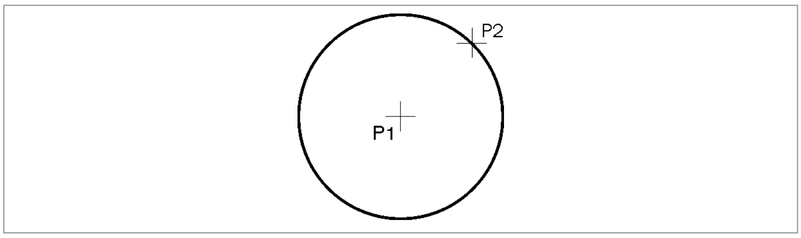


Fig. 4709

1. Select >*Create* >*Circles* >*Center diameter*.
2. Follow the instructions in the command line.

Drawing a circle tangential to 2 elements

Condition

- The drawing contains at least 2 elements.

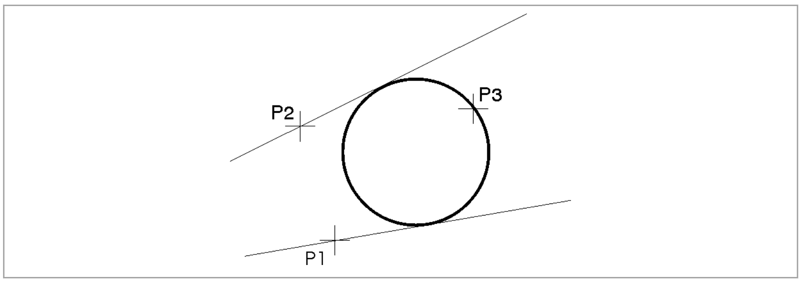


Fig. 4711

1. Select>*Create* >*Circles* >*2 tangents – 1 point*.

2. Follow the instructions in the command line.

Drawing concentric circles

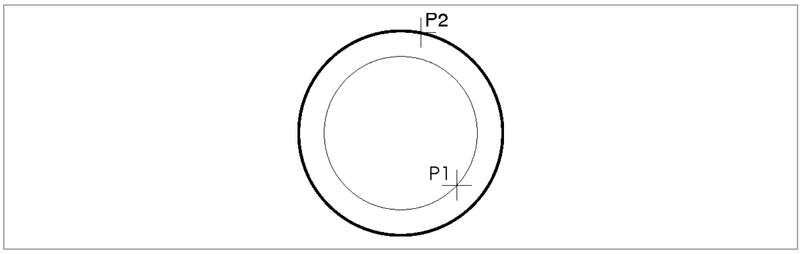


Fig. 4710

1. Select >*Create* >*Circles* >*Concentric*.
2. Click on the base circle (P1).
3. Click on a point (P2) on the arc.

   or

   ➢ Enter distance (positive value: concentric circle outside the base circle, negative value: concentric circle within the base circle)

1.4 Polygons

Polygons, drawing

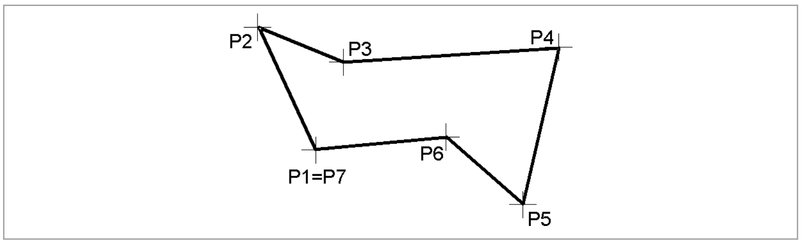


Fig. 4708

1. Select >*Create* >*Line* >*Polygon*.
2. Follow the instructions in the command line.

1.5 Rectangle

Drawing rectangles

1. Select >*Create* >*Lines* >*Rectangle*.
2. Follow the instructions in the command line.

1.6 Arcs

Notes

- Arcs are always drawn in counter-clockwise direction.
- If arcs are to be changed in one working step (e.g. new radius), they have to be previously converted into roundings (via >*Modify* >*Elements* >*Radius corner*).

Drawing an arc via 3 points

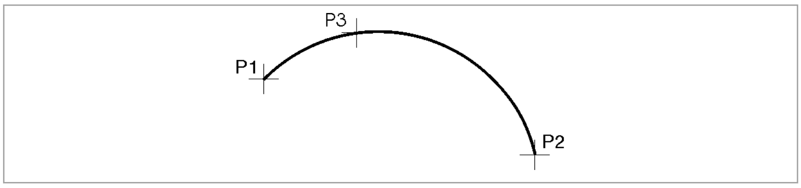


Fig. 4712

1. Select >*Create* >*Arcs* >*3 points*.
2. Follow the instructions in the command line.

Drawing an arc via center, start and end point

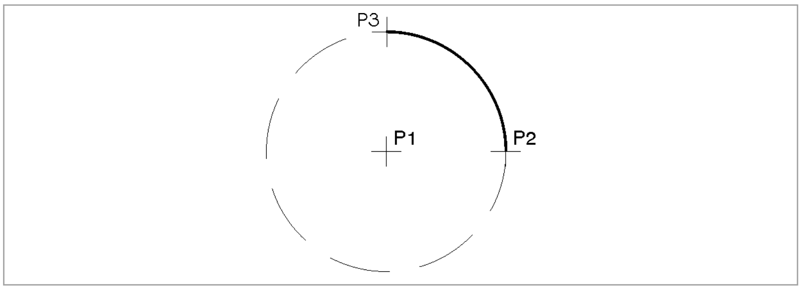


Fig. 4713

1. Select >*Create* >*Arcs* >*Center-Start-End*.
2. Follow the instructions in the command line.

Drawing an arc via diameter

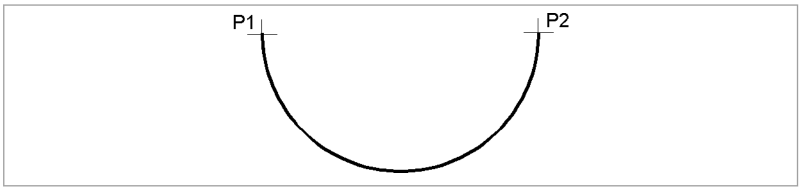


Fig. 4917

1. Select >*Create* >*Arcs* >*Diameter*.
2. Follow the instructions in the command line.

Drawing a concentric arc

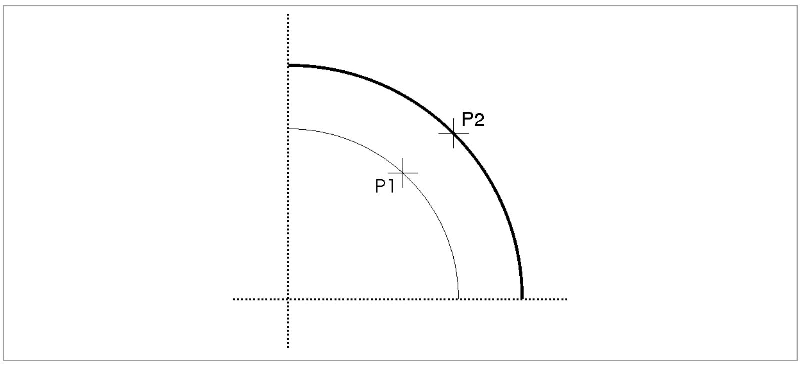


Fig. 4714

1. Select >*Create* >*Arcs* >*Concentric*.
2. Click on the point (P1) in the available arc element.
3. Click on the point (P2) in the new arc element.

   or

   ➢ Enter the distance between the concentric arcs:
   - To create a concentric circle outside the base circle: enter a positive value.
   - To create a concentric circle within the base circle: enter a negative value.

Drawing an arc via center, radius, start and final angle

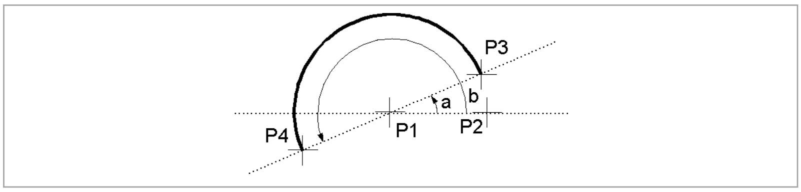


Fig. 4715

1. Select >*Create* >*Arcs* >*Center-Radius-Angle*.
2. Click on the center (P1).
3. Click on a point (P2) on the arc.

   or

   ➢ Enter the arc's radius.
4. Enter the start angle (a).

   or

   ➢ Click on a point (P3) on the arc.
5. Enter the final angle (b).

   or

   ➢ Click on a point (P4) on the arc.

Drawing a multicurve contour

A multicurve contour consists of several arcs that are connected together.

Notes

- Multicurve contours can only be attached to existing elements.
- If a construction circle is under the arc, no multicurve contour is created.

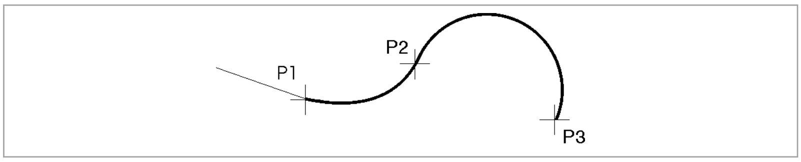


Fig. 4918

1. Select >*Create* >*Arcs* >*Multicurve contour*.

2. Click on the end point (P1) of an available element (arc or line).

   (P1) is also the start point of the arc.
3. Click on the end point (P2) of the arc.

   (P2) is then the start point of the following arc.
4. Click on the next arc point (P3), etc.

Converting arcs into roundings

1. Select >*Modify* >*Elements* >*Radius corner*.
2. Follow the instructions in the command line.

1.7 Notches

Notches can be defined as corner notches or as element notches. They can be created at line elements, arc elements and circle elements. Notches have the "attribute" property (see "Characteristics: a property of elements and contours", pg. 1-49).

Note

The contour at which the notch is to be defined must be available.

Corner notches

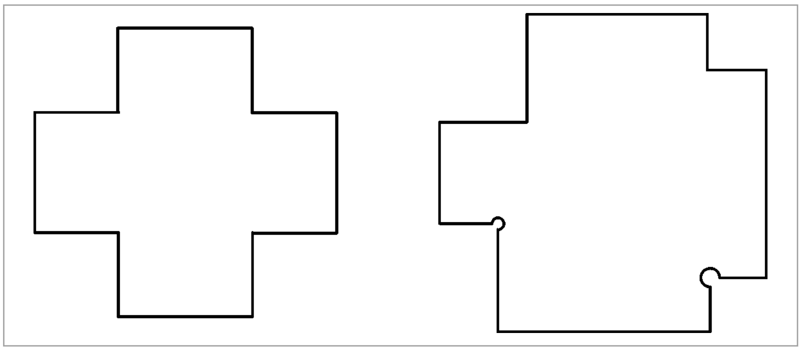


Fig. 19685

Element notches

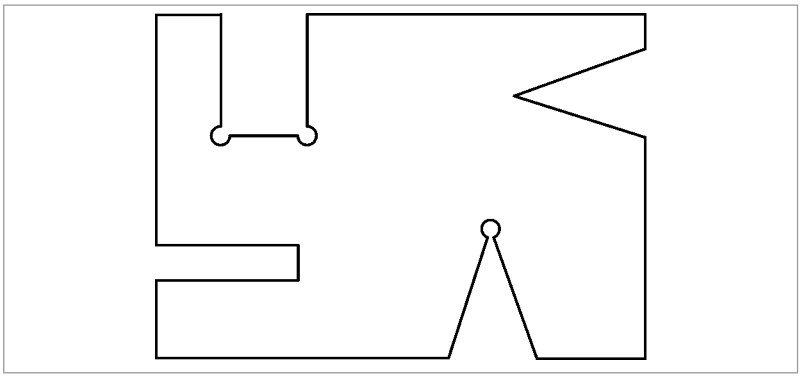
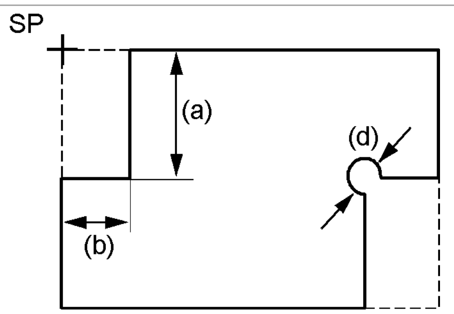


Fig. 19686

Creating corner notches with different distances



SP	Vertex	B	Notch depth of 2. element
A	Notch depth of 1. element	d	Bend relief diameter

Fig. 33472

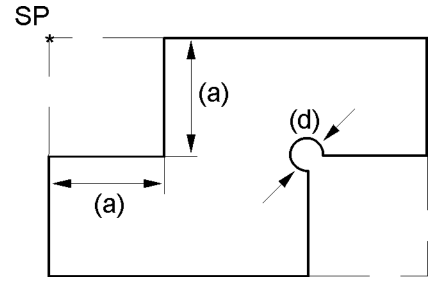
1. Select >*Create* >*Corners/elements* >*Rectangular corner notch*.
2. Follow the instructions in the command line. Before clicking on the elements, enter a bend relief diameter if necessary.

   For workpieces that will later be bent, the bend relief reduces the deformation of the corner.

   A bend relief diameter (d) >0 creates notches with pitch circle. A bend relief diameter (d) = 0 creates notches without pitch circle.

   If nothing is entered, the notch will also be created without pitch circle.

Creating corner notches with the same distances



SP	Vertex	d	Bend relief diameter
A	Notch depth of the first and the second element		

Fig. 19688

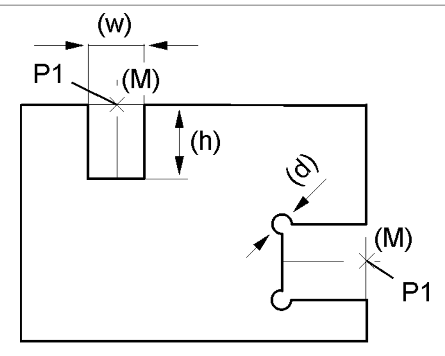
1. Select >*Create* >*Corners/elements* >*Square corner notch*.
2. Follow the instructions in the command line. Before clicking on the elements, enter a bend relief diameter if necessary.

   For workpieces that will later be bent, the bend relief reduces the deformation of the corner.

   A bend relief diameter (d) >0 creates notches with pitch circle. A bend relief diameter (d) = 0 creates notches without pitch circle.

   If nothing is entered, the notch will also be created without pitch circle.

Creating rectangular notches



P1	Element	h	Notch height
w	Notch width	d	Bend relief diameter
M	Notch center		

Fig. 19690

1. Select >*Create* >*Corners/elements* >*Rectangular notch*.

2. Follow the instructions in the command line. Before clicking on the elements, enter a bend relief diameter if necessary.

   For workpieces that will later be bent, the bend relief reduces the deformation of the corner.

   A bend relief diameter (d) >0 creates notches with pitch circle. A bend relief diameter (d) = 0 creates notches without pitch circle.

   If nothing is entered, the notch will also be created without pitch circle.

Creating triangular notches

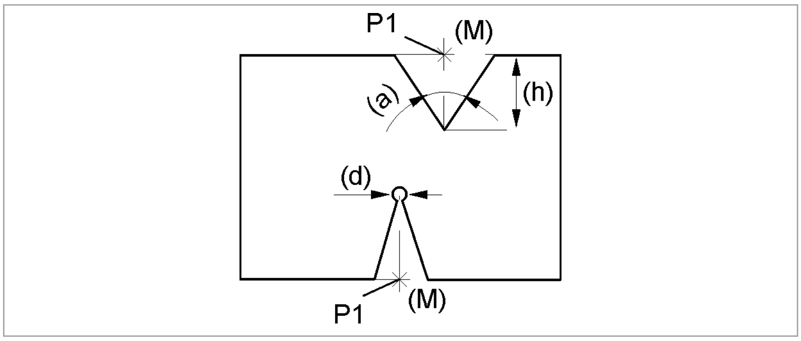


Fig. 19689

1. Select >*Create* >*Corners/elements* >*Triangular notch*.
2. Enter notch height (h).
3. Enter angle of triangle (α).
4. Enter the notch center (M).

   or

   ➢ Enter the bend relief diameter (d).

   For workpieces that will later be bent, the bend relief reduces the deformation of the corner.

   A bend relief diameter (d) >0 creates notches with pitch circle. A bend relief diameter (d) = 0 creates notches without pitch circle.

   If nothing is entered, the notch will also be created without pitch circle.

5. Click on the element (P1) at the position at which the notch is to be created.

Creating the bend relief later

For workpieces which are to be bent later, a bend relief reduces the deformation of the corner.

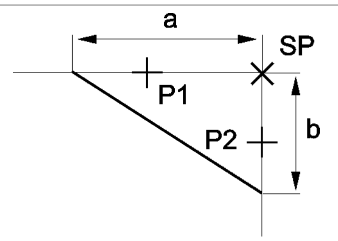
1. Select >*Create* >*Corner/element* >*Bend relief*
2. Follow the instructions in the command line. (Enter bend relief diameter >0.)

1.8 Bevels

Bevels have the "attribute" property (see "Characteristics: a property of elements and contours", pg. 1-49). When a real bevel (as opposed to a line) is deleted, the corner is closed.

When taking over external formats, bevels might be read as lines, since not all foreign formats differentiate between bevels and lines. Lines can be converted into real bevels (see "Converting lines", pg. 4-53)

Drawing a bevel with the distance to the vertex

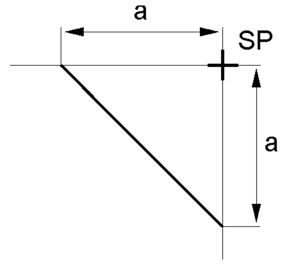


a	Distance between the bevel and the vertex (first element)	P2	Point on second element
b	Distance between the bevel and the vertex (second element)	SP	Vertex
P1	Point on first element		

Fig. 33468

1. Select >*Create* >*Corners/elements* >*X/Y bevel*.
2. Follow the instructions in the command line.

Drawing a bevel via a corner point



a Distance between the bevel and the vertex

SP Vertex

Fig. 33469

1. Select >*Create* >*Corners/elements* >*Square bevel*.
2. Follow the instructions in the command line.

1.9 Rounding corners

Corner round

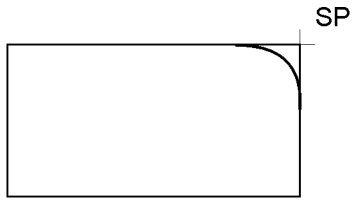


Fig. 33470

1. Select >*Create* >*Corner/element* >*Radius corner*
2. Follow the instructions in the command line.

Restoring corners

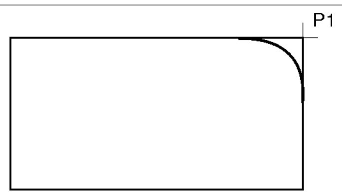


Fig. 4716

1. Select >*Edit* >*Delete* >*Element*.

2. Click on the rounding(s) to be deleted.

The corner(s) will be restored.

1.10 Drawing an equidistant to the contour

An equidistant is a second contour which runs parallel to the original contour. It is created with a defined distance to the original contour:

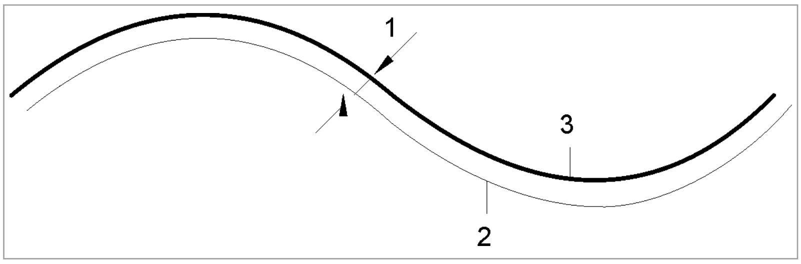


Fig. 33471

1. Select >*Create* >*Special elements* >*Equidistant*.
2. Follow the instructions in the command line.

1.11 Redrawing construction geometry

1. Trace the auxiliary geometries in such a manner that they lead to the desired geometry.

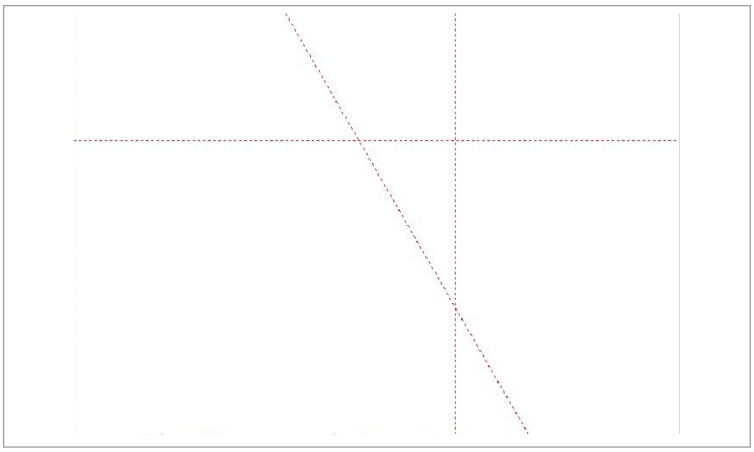
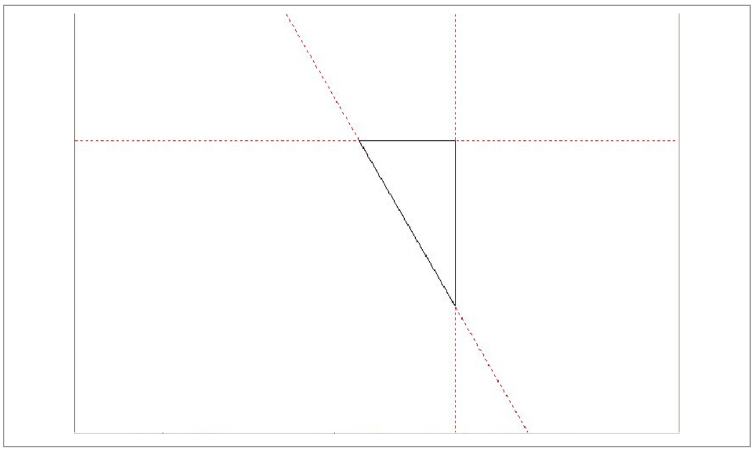


Fig. 51355

2. Select >*Create* >*Special elements* >*Trace auxiliary geometry*.
3. Click on the intersection of the two auxiliary geometries.

4. Click on the auxiliary geometry whose line is to be a part of the geometry.

   A line is drawn up to the next intersection.
5. Click on the next intersection and the next auxiliary geometry.
6. Continue till the geometry is traced.



Redrawn auxiliary lines Fig. 51356de

2. Generating single holes, circles of holes and rows of holes (macros)

You can use macros to create the following complex geometries in a simple way:

- Single holes
- Circles of holes
- Rows of holes
- Hole grids
- Any bend relief.

  (Any bend reliefs are single holes. They are calculated with contours that they cut.).

Circles of holes, rows of holes and hole grids consist of a base element which is repeated.

The following base elements are possible:

- Circles/ellipses/dot mark points
- Rectangles/squares
- Oblong holes
- Geometry templates of your own
- Drawings of forming tools
- Tool drawings

Overview of the procedure

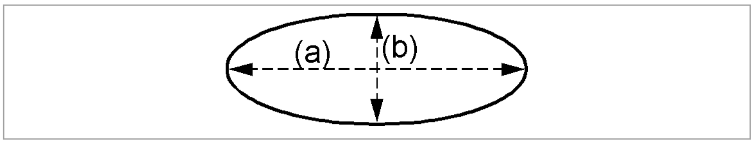
- Determine the base element and its parameters.
- Create a single hole, row of holes, circles of holes or hole grid.

2.1 Defining circle, ellipse or dot marked point as base element

Condition

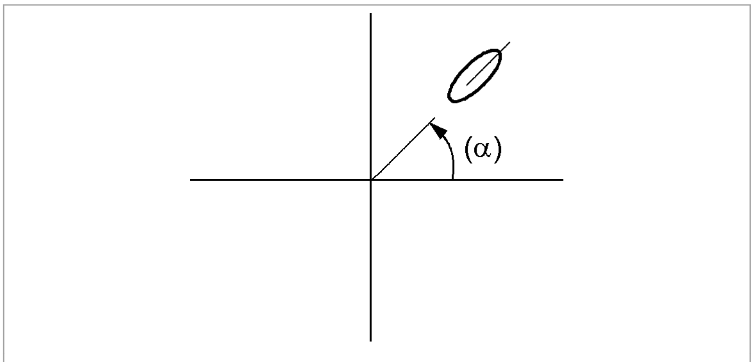
- The drawing is open.

1. Select >*Create* >*Macros* > *Parameter...*.
2. Select the "Circle" index card.
3. Select "Circle", "Ellipse" or "Dot mark pt.".
4. Enter the geometry data after selecting "Circle" or "Ellipse". The geometry data for an ellipse is defined as follows:
   - "Radius 1" = a/2
   - "Radius 2" = b/2



Geometry data for an ellipse Fig. 19721

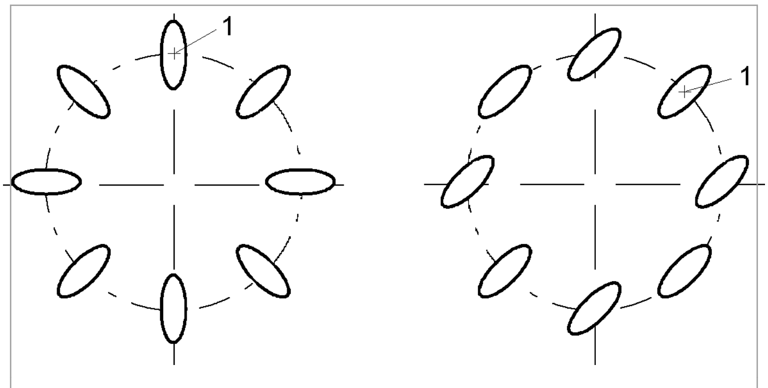
5. Enter the angle of inclination if an ellipse is to be rotated around itself.



α Angle of inclination

Example: angle of inclination 45° Fig. 19722

6. Check "Also rotate geometry" if an ellipse is to be rotated when creating rows of holes, circles of holes or hole grids:



1 Zero point of the ellipse

Left: ellipse is rotated, right: ellipse is not rotated Fig. 27160

7. Press *OK*.

   The base element is defined.

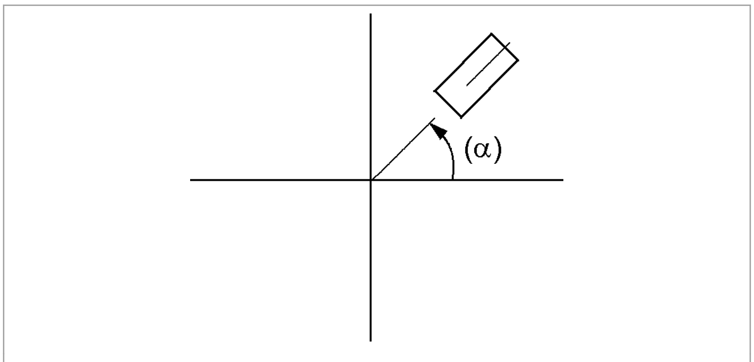
8. Create a single hole, circles of holes, row of holes or hole grid.

2.2 Defining rectangle or oblong hole as base element

Condition

- The drawing is open.

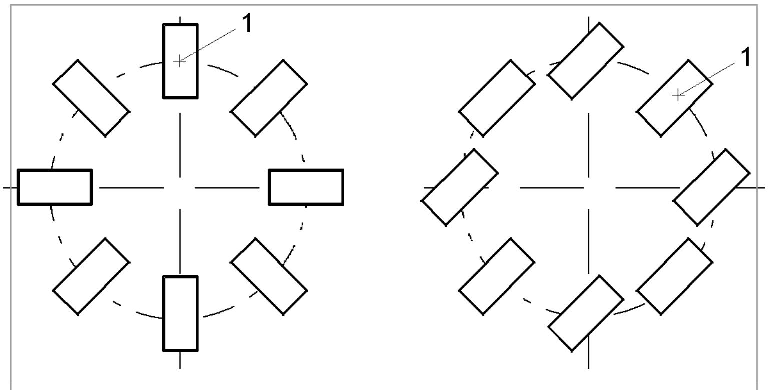
1. Select *>Create >Macros > Parameter...*.
2. Select the "Rectangle/Oblong hole" index card.
3. Select "Rectangle" or "Oblong".
4. Enter the length and width.
5. Enter the angle of inclination if the rectangle or oblong hole is to be rotated around itself:



α Angle of inclination

Example: angle of inclination 45° Fig. 19725

6. Enter the corner radius if the base element is a rectangle.
7. Check "Also rotate geometry" if the geometry is to be rotated when creating rows of holes, circles of holes or hole grids:



1 Zero point of the rectangle

Left: rectangle is rotated, right: rectangle is not rotated Fig. 27159

8. Press *OK*.

   The base element is defined.

9. Create a single hole, circles of holes, row of holes or hole grid.

2.3 Defining your own geometry templates as base element

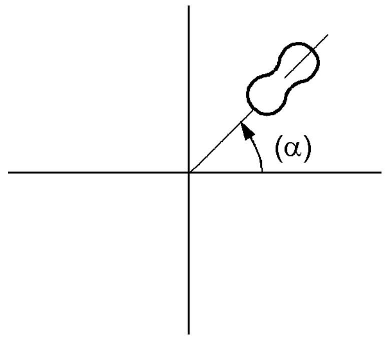
Condition

- The drawing is open.

1. Select >*Create* >*Macros* > *Parameter...*.
2. Select the "Template" index card.
3. To display the geometry template used last, open "Template" with ▼.

   The "Last templates" mask is displayed.
4. In order to select from all geometry templates, click on Selection.

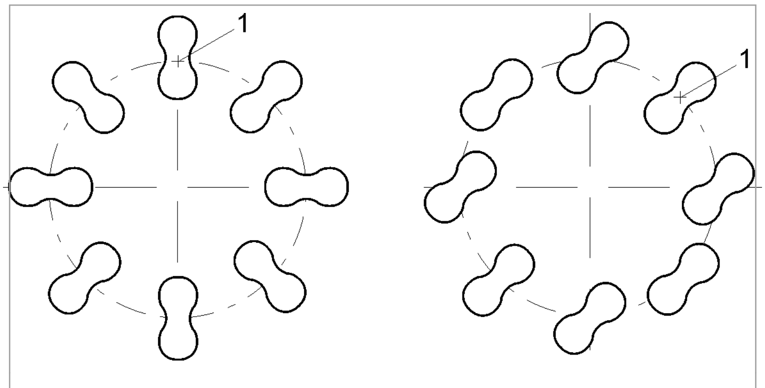
   The file browser opens.It displays all files in *.VLG format which are in the current directory.
5. Search for the file if necessary.
6. Mark the desired geometry template.
7. Press *OK*.
8. Enter an angle of inclination if the geometry template is to be rotated around itself:



α Angle of inclination

Example: angle of inclination 45° Fig. 27162

9. Check "Also rotate geometry" if the geometry template is also to be rotated when creating rows of holes, circles of holes or hole grids:



1 Zero point of the geometry template

Left: template is rotated, right: template is not rotated Fig. 27163

10. Press *OK*.

    The geometry template is defined as a base element.

11. Create a single hole, circles of holes, row of holes or hole grid.

2.4 Defining the drawing of a forming tool as base element

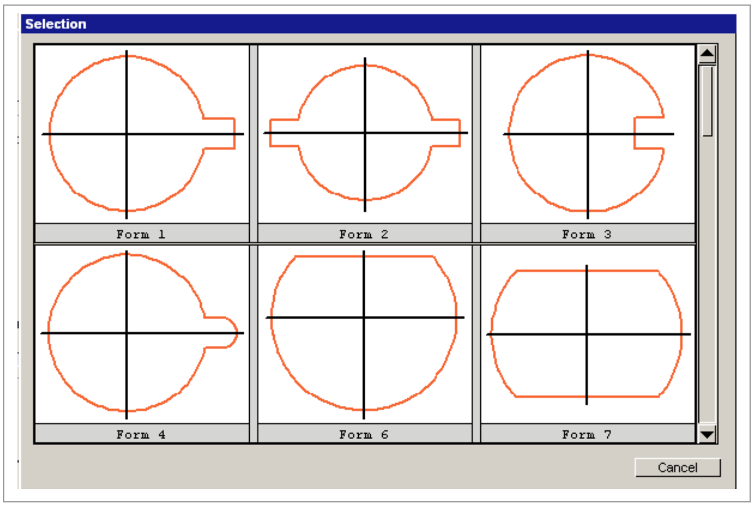
The geometries of forming tools from the TRUMPF punching tool catalog can be used for drawing.

Since they are drawings, it does not matter whether the created geometry is punched or machined with the laser.

Condition

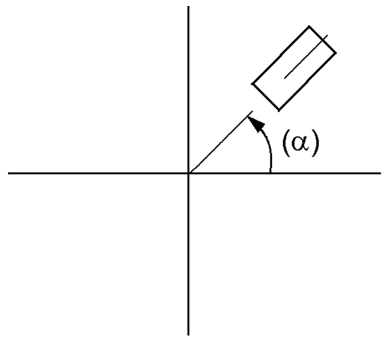
- The drawing is open.

1. Select >*Create* >*Macros* > *Parameter...*.
2. Select the "Form model" index card.
3. If the number of the forming tool is known:
   - Open "Form model type" with ▼.
   - Mark the number of the forming tool in the following mask.
4. If the number of the forming tool is not known:
   - Select *Selection*.
   - Mark the desired forming tool.



"Selection" mask Fig. 41304en

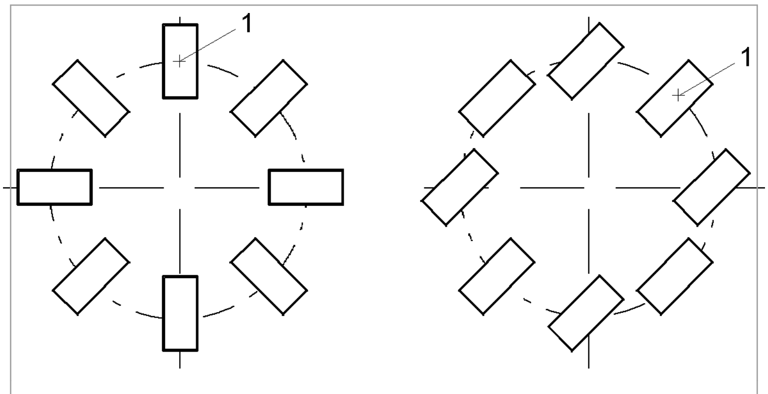
5. Enter the geometry data of the forming tool.
6. Enter the angle of inclination if the forming tool is to be rotated around itself:



α Angle of inclination

Example: angle of inclination 45° Fig. 19725

7. Check "Rotate form model" if the shaped tool is also to be rotated when creating rows of holes, circles of holes or hole grids:



1 Zero point of the shaped tool

Left: shaped tool is rotated, right: shaped tool is not rotated Fig. 27159

8. Press *OK*.

   The base element is defined.

9. Create a single hole, circles of holes, row of holes or hole grid.

2.5 Defining a tool drawing as base element

Condition

- The drawing is open.

Note

The function is only enabled in TruTops Punch.

1. Select >*Create* >*Macros* > *Parameter...*.
2. Select the "Tool" tab
3. Either
   - ➢ Select a tool.

   or

   - ➢ Select forming tool.
   - ➢ At "Display as", select the type of display; for example, graphic, point, identity number or comment.
   - ➢ Press *OK*.
   - ➢ Define the position of the stroke with the left mouse button.

   Depending on the selection, the stroke will be shown, for example, as identity number.

4. Press *OK*.

   The base element is defined.

5. Create a single hole, circles of holes, row of holes or hole grid.

2.6 Generating single hole via macro

Condition

- The base element is defined.
  - Circle, ellipse or dot mark point.
  - Rectangle or square.
  - Oblong hole.
  - A geometry template of your own.
  - Drawing of a shaped tool.
  - Tool drawing.

Creating single hole

1. Select >*Create* >*Macros* >*Single hole*.

   The base element is linked to the cursor.

   The base element can be rotated using the wheel mouse (increment of 10 °).

2. Either
   - ➢ Enter or click on the target point for the single hole.

     TruTops CAD sets the center of the base element on the target point.

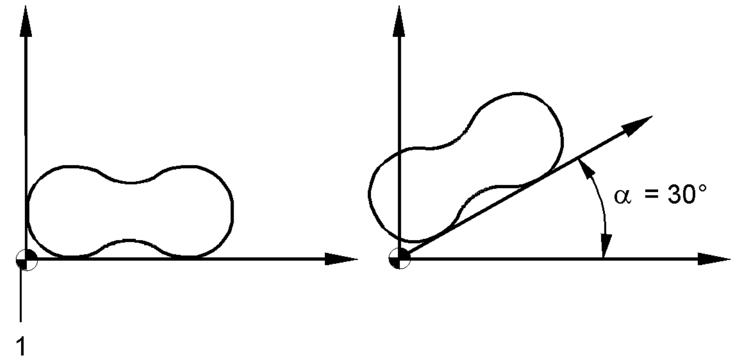
   or

   - ➢ Enter the angle of inclination of the base element. (TruTops CAD does not take into account the angle of inclination from the definition of the base element in this case. The base element appended to the cursor can be turned further with a mouse wheel).
   - ➢ Enter or click on the target point for the single hole.

   The center of the base element is set on the target point.

Generating further single holes

3. Enter or click on further target points.



1 Target point and zero point of the GEO file
α Angle of inclination for the single hole

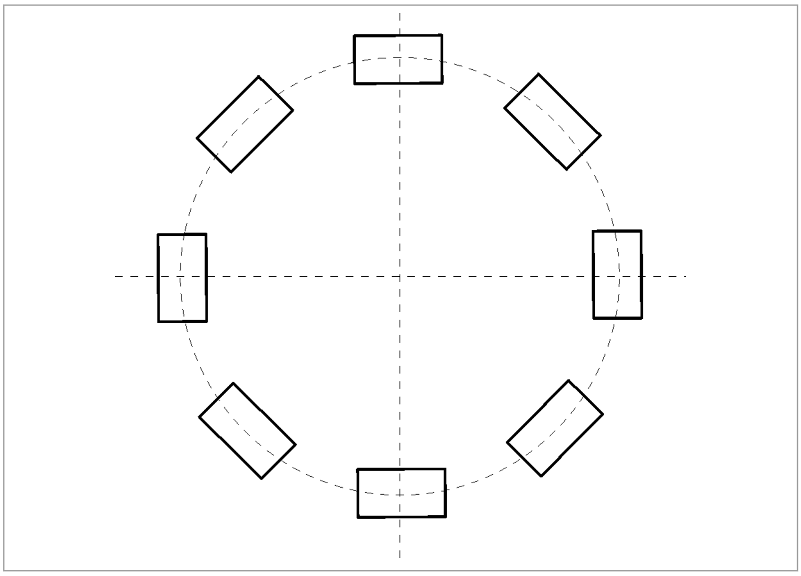
Fig. 26988

2.7 Generating a circle of holes via macro

A full circle or a partial circle can be created.

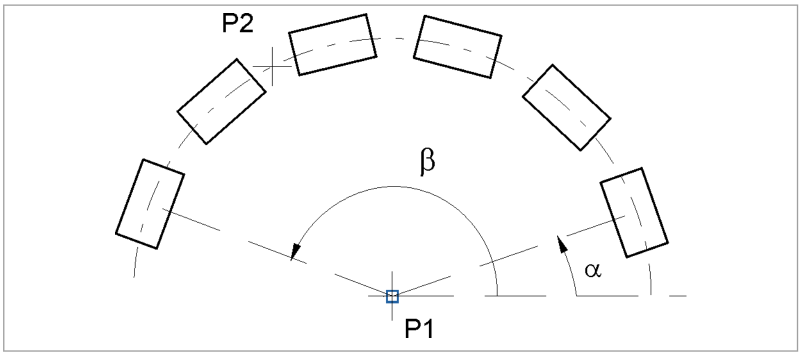
Condition

- The base element is defined.
  - Circle, ellipse or dot mark point.
  - Rectangle or square.
  - Oblong hole.
  - A geometry template of your own.
  - Drawing of a shaped tool.
  - Tool drawing.



Full circle

Fig. 4757



Part circle Fig. 4753

1. Select >*Create* >*Macros* >*Bolt hole circle*.
2. Click on the center (P1).
3. Enter the number of holes.
4. Enter the start angle (α).

   or

   ➢ Define the start angle (α) by means of two points.

   Note

   In case of a full circle, the end angle must be 360° larger than the start angle. (Example: start angle of 0°, end angle of 360°).
5. Enter the end angle (β).

   or

   ➢ Define the end angle (β) by means of two points.
6. Enter the radius.

   or

   ➢ Define the radius by means of two points.

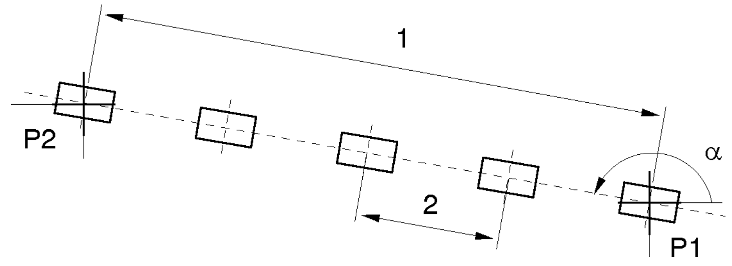
2.8 Generating a row of holes via macro

Application: Create several equally-spaced (2) holes of the same size on a straight line (1):

Condition

- The base element is defined.
  - Circle, ellipse or dot mark point.
  - Rectangle or square.
  - Oblong hole.
  - A geometry template of your own.

- Drawing of a shaped tool.
- Tool drawing.



1	Length of the row of holes	P2	End point
2	Distance between the holes	α	Angle of inclination
P1	Start point		

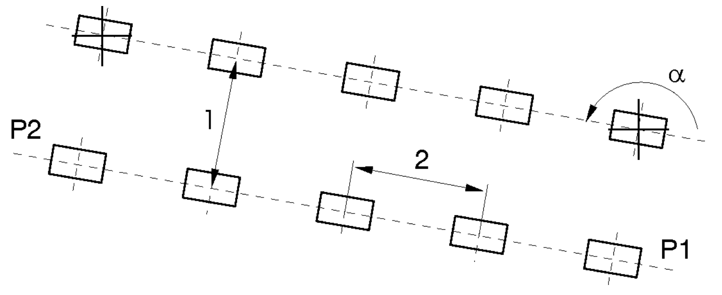
Row of holes — Fig. 4752

1. Select *>Create >Macros >Row of holes*.
2. Follow the instructions in the command line.

2.9 Generating hole grid via macro

Condition

- The base element is defined.
  - Circle, ellipse or dot mark point.
  - Rectangle or square.
  - Oblong hole.
  - A geometry template of your own.
  - Drawing of a shaped tool.
  - Tool drawing.



1	Distance between the rows of holes	P2	End point
2	Distance between the holes	α	Angle to X axis
P1	Start point		

Hole grid — Fig. 40779

1. Select >*Create* >*Macros* >*Hole grid*.
2. Follow the instructions in the command line.

2.10 Generating bend relief via macro

Any bend reliefs are single holes. They are calculated with contours that they cut.

Any bend reliefs can be used, for example, to create bend reliefs on notches later.

Any bend reliefs must not be circular. They can be square or rectangular and can be set at any positions of a closed contour (even without notches).

Conditions

- Any bend relief must consist of **precisely one** closed contour.
- Any bend relief must cut another closed contour.

Notes

- Any bend relief cannot be created at open contours.
- Free single holes that do not cut anything are not possible using this function.

1. Select >*Create* >*Macros* >*Any bend relief*.
2. Follow the instructions in the command line.

2.11 Delete duplicate single holes

"Duplicate single holes" are single holes which, for example, have two rows of holes in common. "Duplicate single holes" are equally sized, are positioned exactly over each other and are created by >*Create* >*Macros*. (Duplicate single holes which are **not** created by macros are automatically deleted during saving.)

Display duplicate single holes

1. Select >*Create* >*Macros* >*Display duplicate single holes*.

   or

   ➢ Select >*View* >*Display duplicate single holes*.

Deleting duplicate single holes

2. Select >*Create* >*Macros* >*Delete duplicate single holes*.

   or

   ➢ Select >*Edit* >*Delete* >*Delete duplicate single holes*.

In case needed, the rows of holes and circles of holes will be deleted. The duplicate single holes are removed.

Then the remaining holes are brought together into smaller rows of holes.

Example: for two rows of holes with one common hole, one row of holes is kept. The other is separated into two small rows of holes (respectively "left" and "right" of the other rows of holes).

3. Modifying drawn geometries

Geometry elements can be modified in two ways:

- Modifying the original element.
- Copying the original element. Modifying the copy.

3.1 Shifting geometries

Moving geometries horizontally or vertically

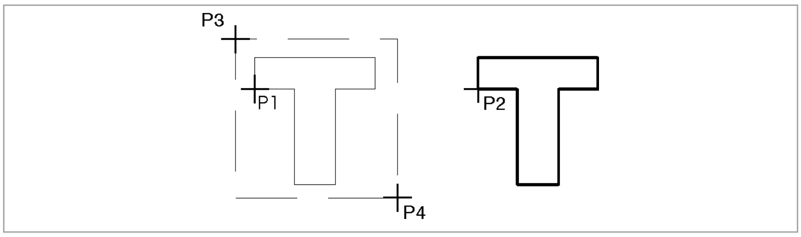


Fig. 27704

1. Either
   - ➢ To move the original geometry: select >*Modify* >*Move* >*Horizontal* or >*Vertical*.

   or

   - ➢ To create copies and to rotate and move them: select >*Modify* >*Move* >*Copy horizontally* or >*Copy vertically*.
   - ➢ Select the number of copies. (If the desired number is not offered: select >*Continue*.)
2. Either
   - ➢ Enter the distance for the movement.

   or

   - ➢ Click on the reference point (P1).
   - ➢ Click on the target position (P2).
3. Click on the element in the original geometry.

   or

   - ➢ Box in the entire original geometry using the mouse (P3, P4).

Moving geometries via 2 points

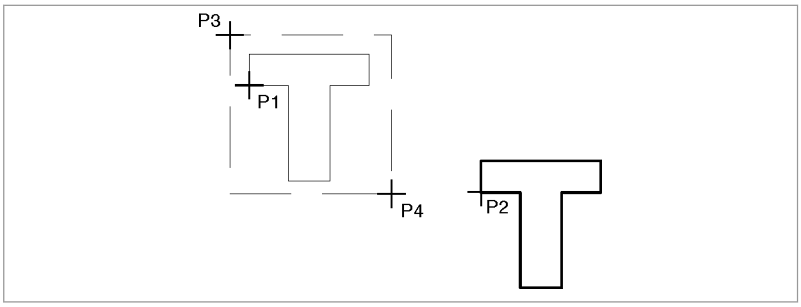


Fig. 27703

1. Either
   - ➢ To move the original geometry: select *>Modify >Move >2 points*.

   or

   - ➢ To create copies and to move them: select *>Modify >Move >Copy 2 points*.
   - ➢ Select the number of copies. (If the desired number is not offered: select *>Continue*.)
2. Click on the reference point (P1).
3. Click on the target position (P2).
4. Click on the element in the original geometry.

   or

   - ➢ Box in the entire original geometry using the mouse (P3, P4).

3.2 Rotating geometries

Rotating geometries around a center point

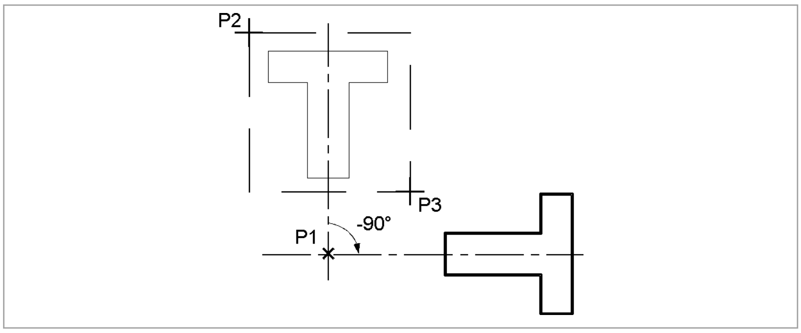


Fig. 28062

1. Either

- To rotate the original geometry: select >*Modify* >*Rotate* >*Center*.

or

- To create copies and to rotate them: select >*Modify* >*Rotate* >*Copy Center*.
- Select the number of copies. (If the desired number is not offered: select >*Continue*.)

2. Click on the center of rotation (P1).
3. Enter angle of rotation. E.g.: `-90`.
4. Click on the element in the original geometry.

or

- Box in the entire original geometry with the mouse (P2, P3).

Rotate geometries around 2 points

Rotating a geometry by two points corresponds to rotating and moving the geometry simultaneously.

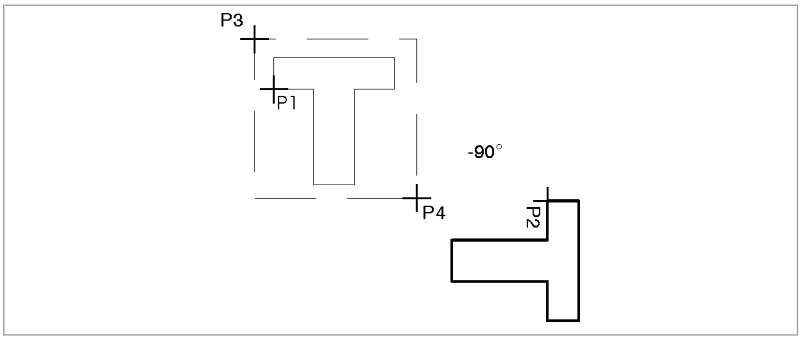


Fig. 28060

1. Either
   - To rotate the original geometry: select >*Modify* >*Rotate* >*2 points*.

   or

   - To create copies and to rotate them: select >*Modify* >*Rotate* >*Copy – 2 points*.
   - Select the number of copies. (If the desired number is not offered: select >*Continue*.)
2. Click on the reference point (P1).
3. Click on the target position (P2).
4. Enter angle of rotation. E.g.: `-90`.
5. Click on the element in the original geometry.

or

➢ Box in the entire original geometry using the mouse (P3, P4).

3.3 Modifying the scale of geometries

Modifying the scale of geometries around a center

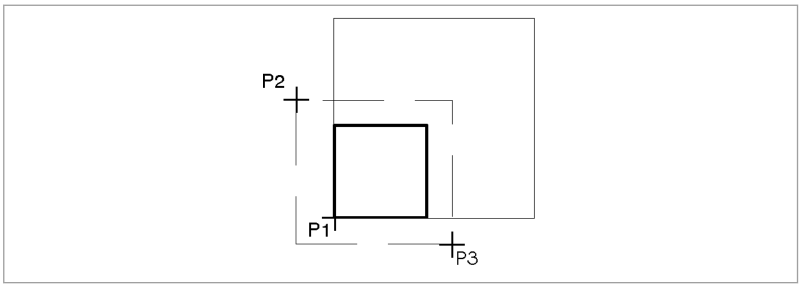


Fig. 28084

1. Either
   ➢ To modify the scale of the original geometry: select *>Modify >Scale >Center*.

   or

   ➢ To create copies and to modify their scales: select *>Modify >Scale >Copy Center*.
   ➢ Select the number of copies. (If the desired number is not offered: select >*Continue*.)
2. Click on the center (P1).
3. Enter the factor for enlargement or reduction.
4. Click on the element in the original geometry.

   or

   ➢ Box in the entire original geometry with the mouse (P2, P3).

Modifying the scale of geometries via 2 points

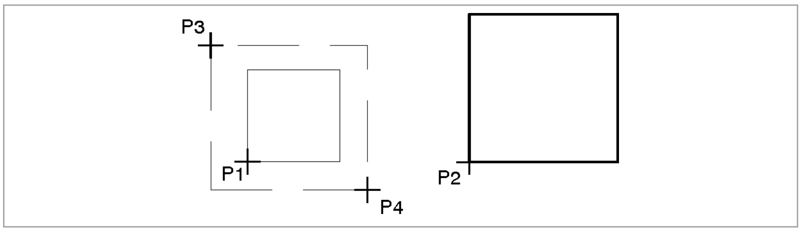


Fig. 28063

1. Either
   - ➢ To modify the scale of the original geometry: select *>Modify >Scale >2 points*.

   or

   - ➢ To create copies and to modify their scales: select *>Modify >Scale >Copy – 2 points*.
   - ➢ Select the number of copies. (If the desired number is not offered: select *>Continue*.)
2. Click on the reference point (P1).
3. Click on the target position (P2).
4. Enter the factor.
5. Click on the element in the original geometry.

   or

   - ➢ Box in the entire original geometry using the mouse (P3, P4).

3.4 Mirroring geometries

Mirroring geometries in a horizontal or vertical mirror line

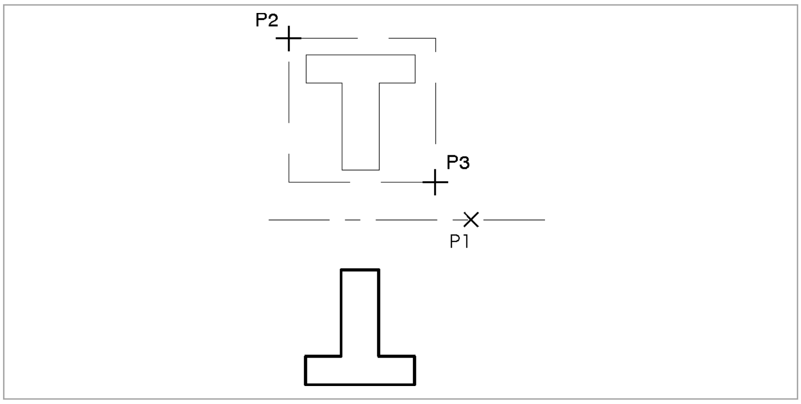


Fig. 28082

1. To mirror the original geometry: select *>Modify >Mirror >Horizontal* or *>Vertical*.

   or

   ➢ To create a copy and to mirror it: select *>Modify >Mirror >Copy horizontally* or *>Copy vertically*.
2. Click on a point (P1) on the mirror line.
3. Click on the element in the original geometry.

   or

   ➢ Box in the entire original geometry with the mouse (P2, P3).

Mirroring geometries around a symmetry point

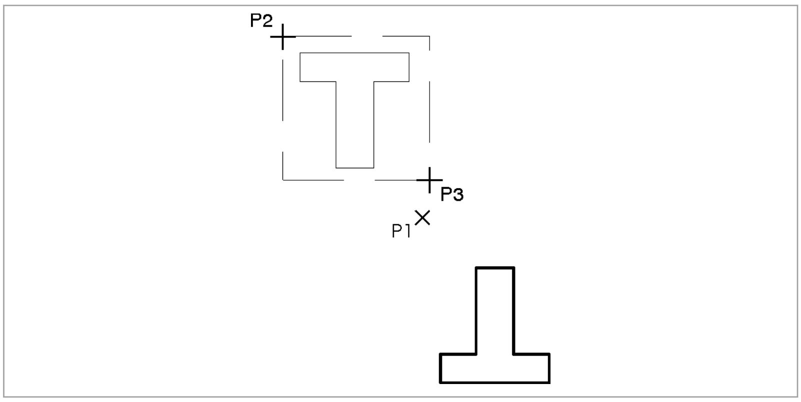


Fig. 28064

1. To mirror the original geometry: select *>Modify >Mirror >Center*.

   or

   ➢ To create a copy and to mirror it: select *>Modify >Mirror >Copy Center*.
2. Click on the symmetry point (P1).
3. Click on the element in the original geometry.

   or

   ➢ Box in the entire original geometry with the mouse (P2, P3).

Mirroring geometries around any axis

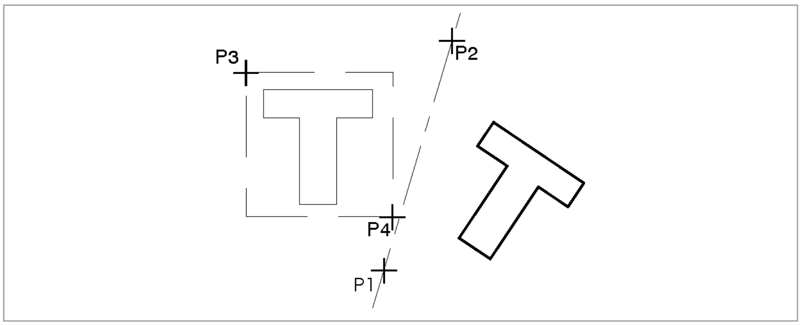


Fig. 28065

1. To mirror the original geometry: select *>Modify >Mirror >2 points*.

   or

   ➢ To create a copy and to mirror it: select *>Modify >Mirror >Copy – 2 points*.
2. Click on the first point (P1) on the mirror line.
3. Click on the second point (P2) on the mirror line.
4. Click on the element in the original geometry.

   or

   ➢ Box in the entire original geometry using the mouse (P3, P4).

3.5 Stretching geometries

Stretching geometry horizontal or vertical

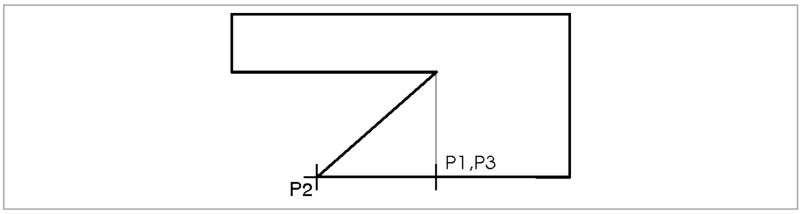


Fig. 4756

1. Select >*Modify* >*Stretch* >*Horizontal* or >*Vertical*.
2. Follow the instructions in the command line.

Note

Elements that are cut by the frame will be stretched. Elements that are entirely within the marking of the frame will be displaced.

3. To stretch several elements: use the mouse to box in all the elements which are to be stretched.

Stretching the geometry based on two points

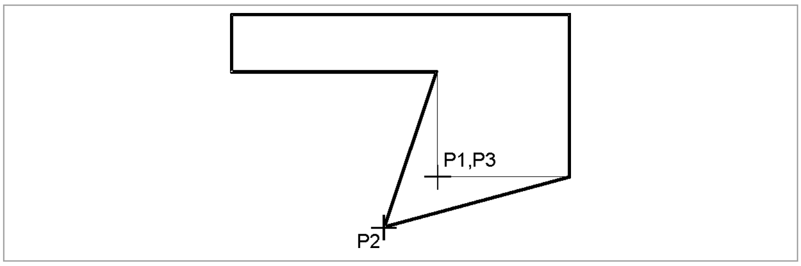


Fig. 4754

1. Select >*Modify* >*Stretch* >*2 points*.
2. Follow the instructions in the command line.

Note

Elements that are cut by the frame will be stretched. Elements that are entirely within the marking of the frame will be displaced.

3. To stretch several elements: use the mouse to box in all the elements which are to be stretched.

3.6 Modifying line attributes (color and line type)

The line attributes can be modified as follows:

- Preset line color or line type. All connecting drawn elements receive the new attributes.
- Modify the line color or line type of single (available) elements. The preset line attributes will not be changed by this.
- Globally modify lines of one color and/or line type (for example, convert all red lines to blue lines or all dashed lines into dotted lines). This function is especially useful if external drawings are prepared in which the line attributes have other meanings.

Preset line color and line type.

1. Open the "Current color and line type" with ▼.
2. In "Select line type", click on the desired line type and line color.
3. Press *OK*.

All connecting drawn elements receive the new attributes.

Modify the line color/line type of single (available) elements.

1. Select *>Modify >Elements >Color* or *>Line type*.
2. Click on the desired line color or line type.
3. One after the other, use the mouse to click on or box in all the elements whose line color or line type should be modified.

Globally modifying lines of one color and/or line type

1. Select *>Modify >Elements >Change color...* or *>Change line type...*
2. Click on the desired original color or original line type.
3. Click on the new color or the new line type.

4. Press *OK*.

For example, all red lines will be converted into blue lines or all dashed lines into dotted lines.

3.7 Modifying radiuses and diameters of roundings or circles

Note

Arcs whose radiuses are to be modified must be previously converted into roundings (via *>Modify >Elements >Radius corner*).

Modify rounding radius

1. Select *>Modify >Elements >Rounding radius*.
2. Enter the new rounding radius in [mm].
3. If only roundings with a particular radius should be changed: also enter the radius to be changed.
4. To modify individual roundings: click on the roundings one after the other.

   or

   ➢ Box in roundings with the mouse.
5. In order to modify all roundings: select *All*.

Modifying diameters of circles

1. Select *>Modify >Elements >Circle diameter*.
2. Enter the new circle diameter in mm.
3. If circles with a particular diameter should be changed: also enter the diameter to be changed.
4. To modify individual circles: click on the circles one after the other.

   or

   ➢ Box in circles with the mouse.
5. In order to modify all circles: select *All*.

4. Correcting geometries

4.1 Separating or combining elements

Elements are separated when, for example, a part is to be deleted. It is also possible to separate two elements at their intersecting points and then to delete the single elements.

Note

If a circle is separated, a (full) arc is automatically generated from the circle. In this manner, circles can be shortened (indirectly).(see "Shortening or extending elements", pg. 4-49).

Separate an element

Application: one part of an element is to be deleted.

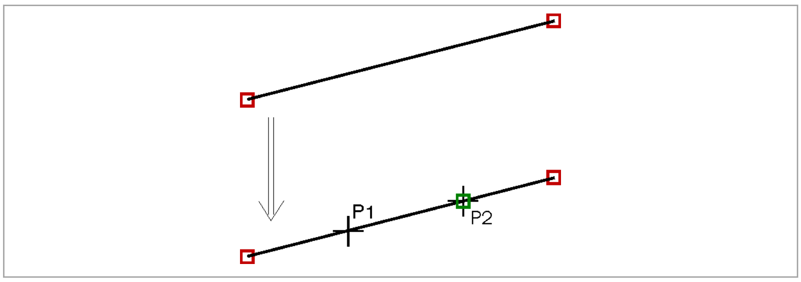


Fig. 4719

1. If need be, select *All points*.

   The start and end points of elements are marked with rot squares (which is helpful when separating elements).

   If the end points are simultaneously element connection points, the squares are shown in green.
2. Select >*Modify* >*Clear* >*Separate elements*.
3. Click on the element (P1).

   The element is displayed in yellow and with dashed lines.
4. Click on point (P2) at which the separation should be done.

   or

   ➢ Enter the coordinates of (P2).

   The element is separated.
5. To refresh the image: press the <F5> key or select *Refresh view*.

Separating two elements at their point of intersection

Application: parts of elements are to be deleted.

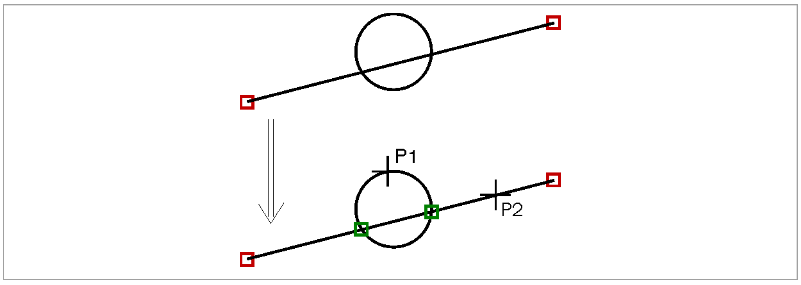


Fig. 4718

1. If need be, select *All points*.

   The start and end points of elements are marked with rot squares (which is helpful when separating elements).

   If the end points are simultaneously element connection points, the squares are shown in green.
2. Select >*Modify* >*Clear* >*Separate at cutting point*.
3. Follow the instructions in the command line.
4. To refresh the image: press <F5> or select *Refresh view*.

Combining elements together

Conditions

- Lines that are to be connected together have to have the same direction or the same angular position. The end points of the lines have to touch each other.
- Two arcs that are to be combined with each other have to have the same radius and center.

Note

It is not possible to combine two lines with different angular positions or two arcs with different radiuses and/or different centers.

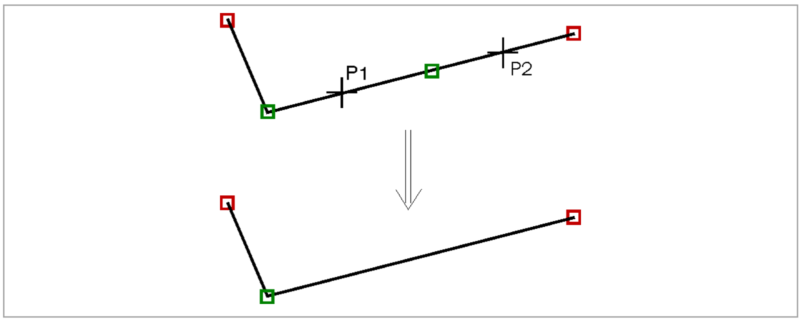


Fig. 4720

1. Select >*Modify* >*Clear* >*Combine elements*.
2. Click on the first element (P1).

   The element (P1) is displayed in yellow and with dashed lines.
3. Click on the second element (P2).

   Both lines are shown in their original color.

4.2 Shortening or extending elements

Shortening or extending an element

Behavior of TruTops:

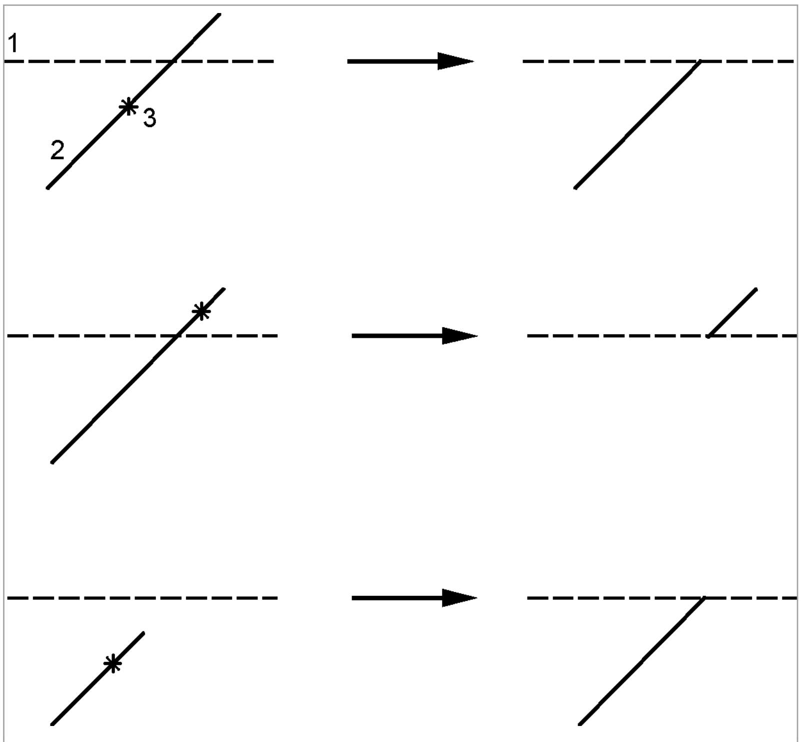
- The element to be modified is shortened or, as the case may be, extended up to the intersecting point.
- If there are **two** shared intersecting points (e.g., arc), the intersecting point is selected which is closest to the first pick point.
- The element to be modified is shortened on the side where the pick point is **not** located. The side that is clicked on is therefore kept.

Conditions

- The element to be modified has to be a line, an arc, a bevel or a rounding. (Bevels and roundings are automatically converted to lines and arcs when they are shortened or extended.)
- The reference element can be a line, arc, circle, rounding, bevel, construction line or construction circle.

Note

If a circle is separated, a (full) arc is automatically generated from the circle. In this manner, circles can be shortened (indirectly).(see "Separating or combining elements", pg. 4-47)



1 Reference element
2 Element to be modified
3 Clicked pick point

Defining modifications by clicking Fig. 26960

1. Select *>Modify >Clear >Shorten/extend 1 element*.
2. Click on the side of the element that is to be kept.
3. Click on the reference element.

   The element to be modified is shortened (or extended) up to the intersecting point.

Shortening or extending two elements

Behavior of TruTops:

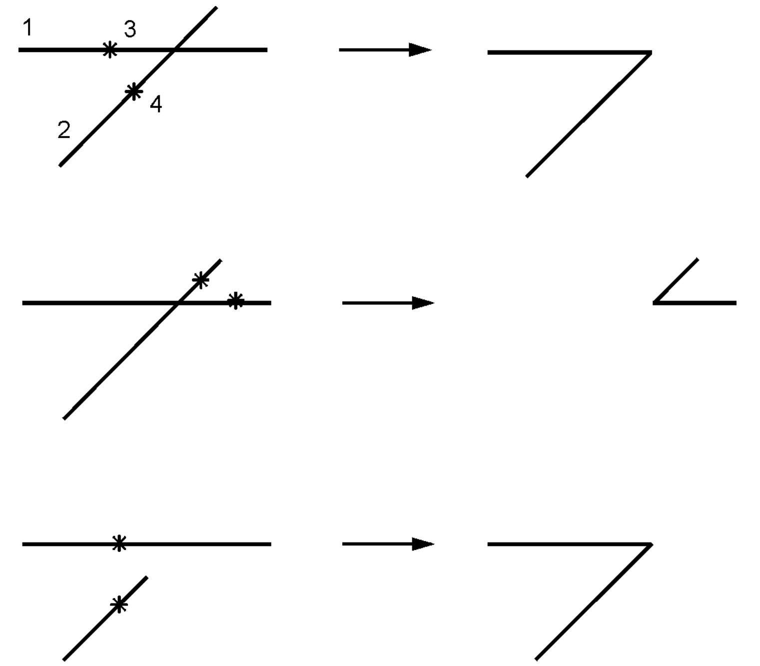
- Both elements will be shortened or, as the case may be, extended up to the intersecting point.
- If there are **two** shared intersecting points (e.g., arc), the intersecting point is selected which is closest to the first pick point.
- Both of the elements to be modified will be shortened on the side where the pick point is **not** located. The sides that are clicked on are therefore kept.

Conditions

- Two elements have to have a common intersecting point (real or virtual).
- The element to be modified has to be a line, an arc, a bebel or a rounding. (Bevels and roundings are automatically converted to lines and arcs when they are shortened or lengthened.)

Note

If a circle is separated, a (full) arc is automatically generated from the circle. In this manner, circles can be shortened (indirectly).(see "Separating or combining elements", pg. 4-47)



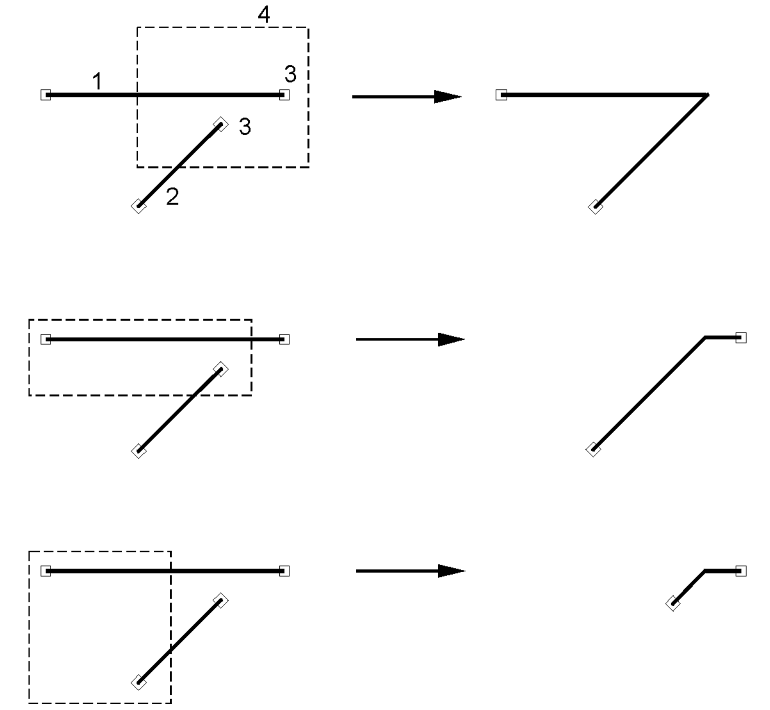
1	First element to be modified	3	Pick point on the first element
2	Second element to be modified	4	Pick point on the second element

Defining a modification by clicking — Fig. 26961

1. Select >*Modify* >*Clear* >*Shorten/extend both elements*.
2. Either
   - ➢ Click on the first element that is to be kept.
   - ➢ Click on the second element that is to be kept.

   or

   - ➢ Use the mouse to box in one each open contour point of the first and second elements to be modified (display in TruTops CAD: red).



1	First element to be modified	3	Open contour points (red)
2	Second element to be modified	4	Frame

Defining modifications by drawing frames Fig. 26962

The elements will be shortened or extended up to their intersecting point.

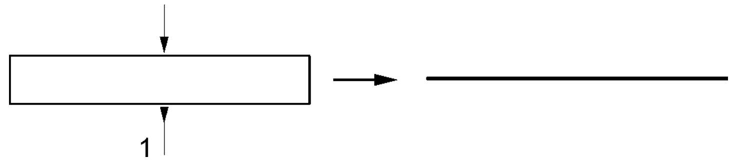
4.3 Converting narrow contours into lines

Converting rectangles or oblong holes into lines

Rectangles or oblong holes that are narrower than the laser kerf can lead to problems with the laser machining. To remedy this, the closed contours can be replaced by single lines and cut as open contours.

1. Load a drawing or create a new one.
2. Select *>Modify >Clear >Replace slitting contours*.

   The "Replace slitting contours" mask is displayed.
3. Select the type of contour that is to be replaced with one single line ("Rectangles" and/or "Oblong holes").
4. Enter the maximum contour width (only rectangles and oblong holes up to this width are then replaced).



1 Maximum contour width

Converting narrow contours into lines Fig. 46138

5. Press *OK*.
6. Using the mouse, click on box in contours that are to be replaced. To replace all contours: Select *All*.

   The original contour is colored in red.
7. To finish, click on *End*.

4.4 Converting lines

Converting lines into bevels

1. Select *>Modify >Elements >Bevel*.
2. Follow the instructions in the command line.

5. Editing multiple outer contours

The assistant for outer contours can be used for drawings that include only one part, but nevertheless possess several outer contours (e.g. for double circles that have a core hole with thread and/or an extrusion, or for open outer contours). TruTops CAD displays the found outer contours and the outer contours can be marked, highlighted or deleted if required.

5.1 Displaying outer contours

If a part has several outer contours, this is reported during saving.

Causes for several outer contours:

- The drawing consists of several parts.
- The drawing contains open outer contours.
- There is a "contour in the contour".

Checking for several outer contours

1. Select >*View* >*Display outer contours*.

   All outer contours are presented in cyan.

Removing several outer contours

2. If the drawing consists of several parts: generate and save several '*.GEO'.
3. For open outer contours: select *Show open points* (). Close open contour points.
4. For a contour in the contour: delete the innermost contour.

5.2 Displaying the first outer contour

Condition

- A drawing (*.GEO or foreign format) with several outer contours has been loaded or created.

➢ Select >*Assistants* >*Outer contours* >*First*.

The outer contour is displayed in cyan and the contour points are outlined.

Contour points are displayed in green if the outer contour is closed and in red if it is open.

5.3 Going to the next outer contour

- Select >*Assistants* >*Outer contours* >*Next*.

The next outer contour is active. The current outer contour remains unchanged.

5.4 Highlighting the outer contour

1. Select >*Assistants* >*Outer contours* >*Highlight*.
2. Click on the desired type of selection line or selection color.

The outer contour is displayed with the new line type and in the new line color and the next contour is active.

5.5 Marking outer contour

Note

If *Highlight characteristics* is activated, TruTops CAD shows the elements or contours in orange. The line color yellow is "covered over".

- Select >*Assistants* >*Outer contours* >*Marking*.

The element is allocated the "Marking" property and is presented in yellow. The next contour is active.

5.6 Deleting contour

1. Select >*Assistants* >*Outer contours* >*Delete*.

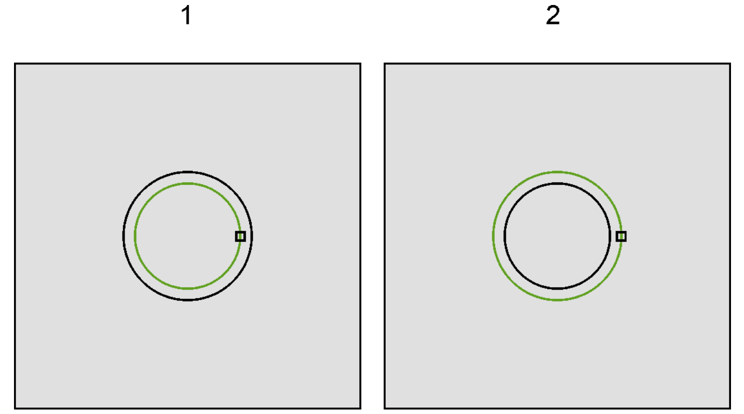
   The highlighted contour is deleted.
2. Delete the superfluous contours one after the other.

5.7 Selecting outer contour

When there are several contours present (contour in the contour), TruTops CAD always initially recognizes the out of line inside contour as an outer contour. To select the enveloped inside contour as an outer contour:

1. Select >*Assistants* >*Outer contours* >*Enveloping*.

   The enveloping contour is defined as an outer contour.



1 Enveloping contour deactivated: The smaller circle is shown as an outer contour (contour in contour).

2 Enveloping contour activated: The larger circle is shown as an outer contour.

Example for an enveloping contour Fig. 46275

2. Highlight, mark or delete the other contours.
3. Create and save several geometries if the drawing consists of several parts (see section Saving parts of a sheet in a foreign format as *.GEO).

5.8 Zooming outer contour

➢ Select >*Assistants* >*Outer contours* >*Zoom*.

The outer contour selected is displayed as large as possible.

6. Deleting geometries

6.1 Deleteing single elements or contours

1. Select >*Edit* >*Delete* >*Element* or >*Contour*.
2. One after the other, click on or box in all elements or contours using the mouse.

6.2 Deleting elements with particular line properties

In TruTops CAD, the properties of elements to be deleted can be combined in the "Delete selection" mask. (Example: all red and blue dashed circles.)

1. Select >*Edit* >*Delete* >*Selection*.

   The "Delete selection" mask is displayed.
2. Check the desired box.

   Notes

   - If "lines" or "arcs" are selected, "bevels" and "roundings" are also activated automatically. If **only** "lines" or **only** "arcs" are to be deleted: deselect "bevels" or "roundings".
   - The "others" line type comprises all lines that do not belong to the four categories ("dashed", "dotted", "point/dash", "solid"). The "Others" line type cannot be created in TruTops CAD. It can however occur when loading foreign formats (e.g. '.DXF').
3. Select *Delete*.

6.3 Delete small contours

1. Select >*Edit* >*Delete* >*Small contours*.

   The "Delete contours" mask is displayed.
2. Enter the surface area of the contour in [mm^2].
3. Press *OK*.

All contours having an area that is equal to or smaller than the entered value are deleted.

6.4 Deleting short elements

Displaying short elements

1. Select >*View* >*Display short elements*.

   The "Highlight elements" mask is displayed.
2. Enter length.
3. Press *OK*.

   All elements up to this length are displayed in cyan.

Deleting short elements

4. Select >*Edit* >*Delete* >*Short elements*.

   The "Delete elements" mask is displayed.
5. Enter the length up to which the elements should be deleted.
6. Press *OK*.

All elements having length that is equal to or less than the entered value are deleted.

6.5 Deleting attributes

1. To display elements with the "Characteristic" property in orange: select *Highlight characteristics*.
2. To delete the "Characteristic" property at selected contours: select >*Edit* >*Delete* >*Characteristics*.
3. Using the mouse, click or box in contours with a "Characteristic" property that you want to delete. To remove all characteristics: select *All* or press <Ctrl>+<Enter> (↲) or <Ctrl>+<Return> (↲).

6.6 Delete dimensioning

1. Select >*Edit* >*Delete* >*Delete dimensioning*.
2. Using the mouse, click or box in dimensioning to be deleted. To remove all dimensioning: select *All* or press <Ctrl>+<Enter> (↲) or <Ctrl>+<Return> (↲).

6.7 Delete duplicate single holes

"Duplicate single holes" are single holes which, for example, have two rows of holes in common. "Duplicate single holes" are equally sized, are positioned exactly over each other and are created by >*Create* >*Macros*. (Duplicate single holes which are **not** created by macros are automatically deleted during saving.)

Display duplicate single holes

1. Select *>Create >Macros >Display duplicate single holes*.

   or

   ➢ Select *>View >Display duplicate single holes*.

Deleting duplicate single holes

2. Select *>Create >Macros >Delete duplicate single holes*.

   or

   ➢ Select *>Edit >Delete >Delete duplicate single holes*.

In case needed, the rows of holes and circles of holes will be deleted. The duplicate single holes are removed.

Then the remaining holes are brought together into smaller rows of holes.

Example: for two rows of holes with one common hole, one row of holes is kept. The other is separated into two small rows of holes (respectively "left" and "right" of the other rows of holes).

6.8 Deleting construction geometries

1. Select *>Edit >Delete >Delete auxiliary geometry*.

   or

   ➢ Select *>Auxiliary tools >Delete auxiliary geometry*.

2. Using the mouse, click or box in auxiliary geometries that are to be deleted. To remove all auxiliary geometries: select *All* or press <Ctrl>+<Enter> (↲) or <Ctrl>+<Return> (↲).

Tip

To rebuild the interface after deleting the auxiliary geometries: press the <F5>- button or select *Update drawing*.

7. Dimensioning geometries (option)

Note

This option will be grayed out if the corresponding password is not available. Should you be interested in a quotation for this option, please contact "software@de.trumpf.com".

Dimensionings show sizes, lengths or other dimensions of objects in a drawing.

The dimensioning in TruTops CAD offers the following functions:

- Dimensionings are dynamic objects that automatically recalculate their dimensions when they are scaled, rotated, moved or angled.
- Dimensionings are displayed in yellow by default so that they can be distinguished from the measured points.
- A dimensioning text can be added before and after the dimensioning.
- Various parameters e.g. font height, tolerances... can be defined for the dimensioning value.
- The dimensioning is saved.
- Dimensioning is adopted when loading foreign formats from CAD systems. The dimensioning can be processed.
- For point dimensionings (2 points, short, axis), the special points of elements such as center, 0 degree position etc., and the original points of a corner (virtual corners) can also be dimensioned.

7.1 Showing and hiding dimensionings

1. Select icon *Show dimensioning* to display the dimensioning (press the button).
2. Select the pressed icon *Show dimensioning* again to hide the dimensioning.

7.2 Lines

Condition

- A drawing has been loaded and the option enabled (the necessary password has been entered).

1. Select >*Dimensioning*.
2. Select the >*Elements* area of application, for example.

- Elements: to dimension an element on the part.
- Contours: to dimension a contour.
- 2 points: to dimension the distance between two points.
- Short: to dimension the distance between two points. Only the end point is dimensioned.
- Axis: to dimension the distance of a point from the X or Y axis.

3. Select the alignment of dimensioning.
   - Horizontal: The dimensioning can be created only horizontally.
   - Vertical: The dimensioning can be created only vertically.
   - Arbitrary: Dimensioning can be created (aligned) in any direction on the dimensioned element or the dimensioned points. To do this, define the specified angle in the input line.
   - Aligned: The dimensioning can be created by aligning on the dimensioned element or the dimensioned points (not for >*Contours* area of application).
4. If >*Elements/Contours* is selected, click or box in part(s).
5. If >*Points* is selected, click on the start and end points.

   The distance between two points is dimensioned.
6. If >*Short* is selected, click on the start and end points.

   The distance between two points is also dimensioned. Dimensioning is however displayed differently and only the end point is dimensioned.

   Note

   When several dimensionings are created, then the start point always remains the same and the end point is redefined each time. The >*Short* area of application is suitable for dimensioning several elements in a compact manner. The start point must be easy to recognize (e.g. bottom left corner).
7. For >*Axis*: click on the line (rectangle) or point on the circle.

   In case of rectangles, the distance of a line (element) to the X or Y axis is dimensioned. In case of circles, the distance from the center of the circle to the respective axis is dimensioned.
8. Position the dimensioning as desired.

The dimensioning is displayed.

7.3 Circles, radiuses, arc lengths

Condition

- A drawing is loaded.

Dimensioning diameter

Note

When dimensioning the diameter, the diameter of a circle, an arc, or a rounding is dimensioned.

1. Select >*Dimensioning*, >*Circles / radiuses* >*Diameter*.
2. Mark the element to be dimensioned.

   The dimensioning is linked to the mouse.
3. Place the dimensioning in the desired position.

   The dimensioning is displayed at the circle.

Dimensioning radii

Note

When dimensioning the radius, the radius of a circle, an arc or a rounding is dimensioned.

4. Select >*Dimensioning*, >*Circles / radiuses* >*Radius*.
5. Mark the element to be dimensioned.

   The dimensioning is linked to the mouse.
6. Place the dimensioning in the desired position.

   The dimensioning is displayed in the circle.

Arc length dimensioning

Note

When dimensioning the arc lengths, the length of an arc or a rounding is dimensioned.

7. Select >*Dimensioning* >*Circles/radiuses* >*Arc length*.
8. Mark the element to be dimensioned.

   The dimensioning is linked to the mouse.
9. Place the dimensioning in the desired position.

   The dimensioning is displayed at the arc.

7.4 Angle

When dimensioning the angle, the angle between two lines is dimensioned. Both lines do not have to intersect.

Condition

- A drawing is loaded.

1. Select *>Dimensioning >Angle >Angle*.
2. Mark the first line for the dimensioning.
3. Mark the second line for the dimensioning.

   The dimensioning is linked to the mouse.
4. Select the angle to be dimensioned.
5. Place the dimensioning in the desired position.

The angle dimensioning is displayed.

7.5 Moving dimensioning

An existing dimensioning can be repositioned at any time. When dimensioning angles, the angle to be dimensioned is defined by the position.

Condition

- A dimensioned drawing is loaded.

1. Select *>Dimension >Modify >Move dimensioning*.
2. Click on the existing dimensioning.

   The dimensioning is linked to the mouse.
3. Place the dimensioning in the desired position.

7.6 Adapting dimensioning text

The following parameters for modifying the dimensioning text can be defined in the "Dimensioning text" mask:

- Calculated dimensioning value:
  - Adopt.
  - Ignore and enter a new value using the displayed calculator.
  - Ignore and enter a new value in the input line
- Tolerances: the tolerances to be displayed are defined here.
- Text: define text before and after the dimensioning value.
- Tolerance values: define upper and lower tolerance.
- Select the font direction.
- Define the font height.

Condition

- A dimensioned drawing is loaded.

Modifying a dimensioning text

1. Select *>Dimensioning >Modify >Modify text*.
2. Mark the dimensioning text to be modified.

   The "Dimensioning text" mask is displayed.

Moving the dimensioning text

3. Select *>Dimensioning >Modify >Move text*.
4. Select dimensioning text.

   The dimensioning is linked to the mouse.

   Note

   As long as the dimensioning text is touching the dimensioning line, text will appear on the line. Otherwise, a comment line is drawn between the text and the dimensioning line to allow the dimensioning text to be allocated to the dimensioning.

5. Place the dimensioning text in the desired position.

Size, changing

6. Select *>Dimensioning >Modify >Text size*.
7. Enter the new text size in the input line.
8. Press ↵.
9. Mark the texts to be modified.

   or

   ➢ Select *All*.

   The text size is modified.

Changing text alignment

10. Select *>Dimension >Modify >Text alignment*.

    The "Text alignment" mask is displayed.

11. Select the font direction.
    - Right: horizontal alignment.
    - High: vertical, upwards.
    - Low: vertical, downwards.
12. Mark the texts to be modified.

    or

    ➢ Select *All*.

    The text alignment is modified.

7.7 Setting the dimensioning format

Condition

- A dimensioned drawing is loaded.

1. Select >*Dimensioning* >*Modify* >*Parameters* >*Format*.

   The "Dimensioning format" mask is displayed.
2. Enter the number of digits after the decimal point :
   - Dimensioning value
   - Tolerances
3. Enter the length of the arrow tip.

The formats for all dimensionings are modified.

7.8 Deleting dimensioning

You can delete individual dimensionings or all dimensionings at once.

Condition

- A dimensioned drawing is loaded.

1. Select >*Dimensioning* >*Delete dimensioning*.
2. Click on the dimensionings that are to be deleted one after the other.

   or

   ➢ Select *All*.

The dimensionings are deleted.

7.9 Setting standard text size

1. Select >*Dimensioning* >*Modify* >*Parameters* >*Text size*.

   The "Standard values for dimensionings" mask is displayed.
2. Enter the value for the text height.
3. Press *OK*.

All texts are displayed with this value.

8. Element groups

All elements can be combined to an element group. Element groups have the "Characteristic" property(see "Characteristics: a property of elements and contours", pg. 1-49).

In the "fixed" characteristic setting, certain actions are therefore either applied to the entire unit, (i.e. on **all** elements) or the actions are declined.

Advantage: If several actions are to be executed on the elements of an element group one after the other, the area of application can be selected by clicking **once** (instead of clicking on and boxing in the single elements several times).

An element group can be resolved by removing the "element group" characteristic. Other characteristics (e.g. "single hole", "notch", "marking"...) are retained.

8.1 Forming element groups

1. Select >*Create* >*Element group* >*Generate*.
2. Using the mouse, click on or box in all elements one after the other which are to be combined into one element group using the mouse.
3. Press *OK*.

If Highlight characteristics is active (), the element group is displayed in orange.

8.2 Dissolving element groups

1. Select >*Create* >*Element group* >*Dissolve*.
2. One after the other, click on or box in all elements using the mouse whose element group is to be dissolved.

9. Marking and hatching

9.1 Setting marking

- Either
  - To set a "marking" attribute: select >*Create* >*Marking*. (see "Characteristics: a property of elements and contours", pg. 1-49).
  - Select the >*Element* or >*Contour* area of application.
  - Click on all elements or contours which are to be marked or box them in with the mouse.

  or

  - Manually assign the contours that are to be marked with the line color "yellow" (to do this, no "marking" attribute is set).

If Highlight characteristics is active ( ), the elements or contours are shown "orange". The line color "yellow" is covered. (The line color "yellow" must not be changed!)

9.2 Deleteing marking

- Either
  - To set a "marking" attribute: select >*Create* >*Marking* >*Remove*.
  - Click on all elements or contours which are no longer to be marked or box them in with the mouse.

  or

  - Contours that have the "yellow" line color but not the "marking attribute": manually reset them to the normal line color.

  The "marking" attribute is deleted. The elements or contours are shown in normal line color again.

9.3 Hatching closed contours

Closed contours (for example, texts) can be marked by hatching. To get a hatched pattern (for example, horizontal and vertical hatched lines), contours can be hatched several times.

Hatched lines receive their own "marking" attribute (see "Characteristics: a property of elements and contours", pg. 1-49).

1. Select *>Create >Marking >Hatch...*

   The "Definition of hatch" mask is displayed.
2. Select one of the predefined definitions for hatched lines.

   or

   ➢ Enter (inclination) angle of the hatching.
3. Enter the distance between the hatched lines.
4. Select the "Elements" or "Contours" area of application.

   Note

   If the elements or contours which are to be hatched are **boxed in the with the mouse**, the "Elements" or "Contours" application area is not relevant.

   When clicking with the mouse, all elements in the "fixed" attribute setting and the "Element" application area will be hatched for which there is a shared attribute. For the "Contour" application area, only the contour which has been clicked on will be hatched.
5. Press *OK*.

   Notes

   - Only the outer contours will be hatched for nested contours. The inner contours are excluded. To hatch nested contours: repeat steps 1 to 6 at the inner contour.
   - Be sure for the "free" attribute setting and the "elements" selection that the entire (closed) contour is boxed in with the mouse. Contours that are not completely covered will only be marked and not hatched.
6. Click on all elements or contours which are to be hatched or box them in with the mouse.

If Highlight characteristics is active ( ), the hatching is shown in orange. The "yellow" line color (for markings) is covered over.

9.4 Hatching contours with closed inner contours

It is possible to exclude the inner contours during hatching.

1. Select *>Create >Marking >Hatch...*

   The "Definition of hatch" mask is displayed.
2. Select one of the predefined definitions for hatched lines.

   or

   ➢ Enter (inclination) angle of the hatching.

3. Enter the distance between the hatched lines.
4. Select the "Elements" or "Contours" area of application.
5. Press *OK*.
6. Use the mouse to box in the complete contour with the inner contour.

   or

   ➢ Press <Ctrl> and click on the individual contours one after the other.

The inner contour will be shown without hatching.

9.5 Changing the hatching

1. Delete old hatching.
2. Create new hatching.

9.6 Deleting the hatching

Note

Hatched lines have their own "marking" attribute (see "Characteristics: a property of elements and contours", pg. 1-49).

In the "fixed" attribute setting (via >*Edit* >*Fixed characteristics*) the hatching of a contour can be removed by a click.

1. Select >*Edit* >*Delete* >*Element*.
2. Click on the hatching(s) that are to be deleted one after the other. (The hatched lines are completely removed.)

10. Inserting or modifying text in geometries

In TruTops CAD, texts can be pasted or edited in geometries.

Note

Neither approach flags nor withdrawal flags are created when cutting text in '*.FNT' format. The beam directly pierces the contour. Processing cannot be rearranged. This can result in uncontrolled scrap (depending on the font).

TruTops CAD supports FNT fonts (TRUMPF custom fonts) and TTF fonts. The TTF fonts are converted into lines and arcs and given a special characteristic so that TruTops can treat them as text. To achieve a greater coordination between both font types, the following modifications were implemented.

10.1 Entering text and defining parameters

Showing set text parameters mask

1. Select >*Create*, >*Special elements*, >*Text...* .

   The "Text input" mask is displayed.

Entering text in a single line

2. Enter the text in the input line.

   Example: `Single line of text`.

Entering several lines of text

3. Mark the end of a line with `\n`.

   Example: `Multiple-line text\nMultiple-line text \n`.

Defining text parameters

4. Enter the text height, the width/height ratio and the line spacing.
5. Enter "Angle of inclination".

   0° corresponds to vertical characters; positive values incline the text to the right.

6. Enter "angle of rotation".

   0° corresponds to horizontal lines; positive values rotate the font counter-clockwise.

   "Entry via 2 points": The angle must be defined by means of a start point and end point before the text is drawn.

7. Select the "font direction".

   In case of "right", the characters are arranged next to each other, and in case of "Bottom", they are arranged one below the other.

8. Select the "reference point position".

   Determines the position of the cursors in the text.

Selecting fonts (*.FNT) or true type font file (*.TTF)

Notes

- Two true type fonts are available in TruTops CAD, which have been developed especially for machining using lasers: TC_LaserScript.TTF and TC_LaserSand.TTF.
- Both true-type fonts are stored in the '...\TRUMPF.NET \Data' directory during the installation. Furthermore links to the true type fonts are stored in the font directory of the Windows version used.
- The font group (e.g. true type fonts) can be changed only as long as the text is being created. You can change to a font within the same group even if the text has already been created.

9. Select *Select font*.

   The "Select font" mask is displayed.

10. Select file type '*.FNT' or '*.TTF'.
11. Select the desired font.
12. Press *OK*.

    The new font is shown in the "Set text parameters" mask.

Marking text

Note

The text is assigned the >*Characteristic* property and automatically displayed in orange.

13. Select "Mark text".
14. Select "From below", if the text "from below" should be marked.
15. Either
    - ➢ Select *OK* once all text parameters have been defined.

      The text and the reference point are shown if a value has been entered under "Angle of rotation".

    or

    - ➢ Select *OK* once all text parameters have been defined and "Entry via 2 points" has been selected under "Angle of rotation".
    - ➢ Click on the start and end point.

      The angle is now defined. The text and the reference point are shown.

    or

    - ➢ Rotate the text using wheel mouse, position it and click.

    The text is drawn at the selected position.

Standard texts

Note

If the stored standard texts are selected, they are added in the text row and the appropriate text is created after closing the mask. Here, the standard texts must be previously selected in the file information (>*File* >*Properties*).

16. Select >*Create*, >*Special elements*, >*Text...* .
17. Open the selection field behind the text entry line with ▼.

    The "Text selection" mask is displayed.
18. Select text.
    - Note: Drawing designation from the file information.
    - User: Name from the file information.
    - Part number: A text "###" is set using this function, which is replaced with an unequivocal part number in the nest, laser or punch. Only one such text can be created for each part.
    - Last text: The last selected text is set. A text can also be searched for from a list of last texts.
19. Press *OK*.

    The text is linked to the cursor.
20. Place the text at the desired position.

10.2 Modifying text parameters

Condition

- The drawing contains texts.

1. Select >*Modify*, >*Elements*, >*Text*.
2. Click on the text.
3. Modify text parameters.
4. Select *OK*.

10.3 Mirroring FNT texts

FNT texts can be mirrored horizontally, vertically, centrally or arbitrarily. Using >*Mirror*, >*Horizontal* as an example, the procedure is described here.

1. Select >*Modify*, >*Mirror*, >*Horizontal*.
2. Click on the point on the horizontal mirror line, through which the text is to be mirrored.
3. Click on or frame text.

The text is mirrored.

10.4 Changing FNT text to TTF text (and vice versa)

1. Select >*Modify*, >*Elements*, >*Text*.
2. Click on the text.
3. Select *Select font*.

   The "Select font" mask is opened.
4. Select the desired font.
5. Press *OK*.
6. Press *OK*.

The text is modified.

10.5 Loading DXF, DWG and MI files with TTF texts

1. Select >*File*, >*Open...*.

   The PDM browser opens.
2. Select the DXF/DWG file.
3. Press *OK*.

   The "DXF/DWG loading options" mask is opened.
4. Select "Transfer texts".
5. Select the font selection arrow.

   The "Select font" mask is opened.
6. Under "File type" select *TTF files*.

   All TTF fonts are displayed.
7. Select the desired fonts.
8. Press *OK*.
9. Press *OK*.

The DXF/DWG file is loaded with the selected TTF font.

10.6 Creating texts with curvature

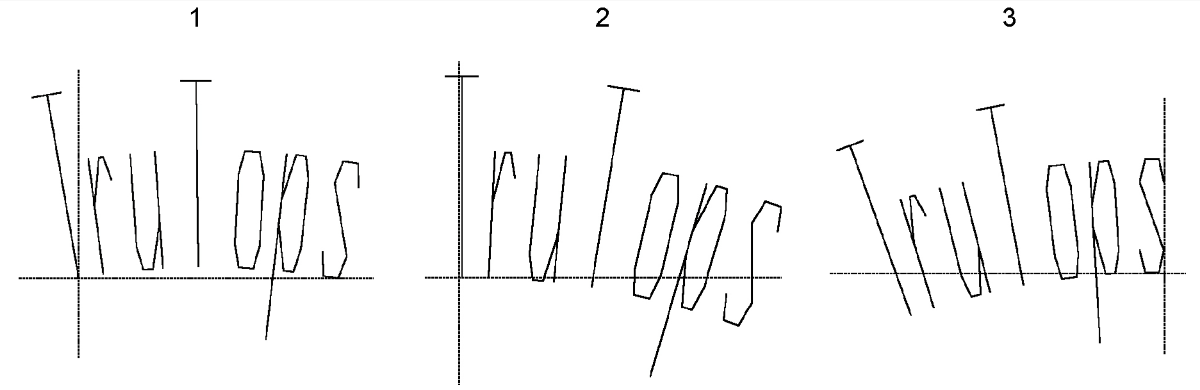
1. Select >*Create*, >*Special elements*, >*Text...*.

   The "Set text parameters" mask is opened.
2. Enter text.
3. Select the "Radius (clockwise)" check box.

4. To modify the rotational direction of the text, select the drop down arrow.

   The "Define type of round text" mask is opened.

5. Select the rotational direction.
6. Press *OK*.
7. Enter the radius.



1 Center 2 Left (center) 3 Right (center)

Reference point position Fig. 57788

8. Select the reference point position.
9. Press *OK*.

The text is displayed on a curve.

10.7 Modifying TTF texts in Variants

1. Select >*Variants*, >*Start*.

   The variant module is opened.

2. If variants have already been generated, select >*Variants*, >*Manage variants*.

   or

   ➢ If no variants have been generated as yet, select >*Variants*, >*Generate variants...* .

   Note

   The following describes the procedural steps for a newly generated variant.

3. Enter a designation for the variant.
4. Open the "Variant type" selection using the arrow.
5. Select `Modify text`.
6. Enter the new text in quotation marks, e. g "TruTops".

or

- ➢ Enter a variable. These must be written in quotation marks. It is possible to link variables to form long chains out of several short ones.

7. Press *OK*.
8. Click on the TTF text to be modified.

The new text is adopted.

10.8 Creating your own text font

The bold.fnt, iso.fnt and isoprop.fnt fonts belong to the standard scope of TruTops.

To create your own fonts, a reference file and a text font file must be created.

The names of all of your font files are entered into the reference file. The text font file itself contains the coordinates of all characters.

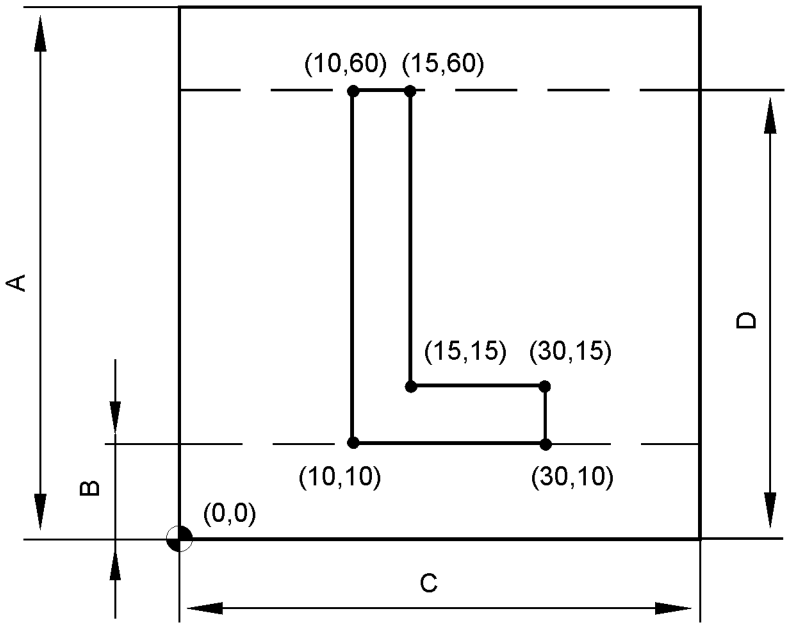
Example:

Condition

- A program must be available for editing txt files such as Notepad, Wordpad or Editor.

Character	Name of reference file (prescribed)	Name of font file (arbitrary)	txt editor program (arbitrary)
L	font.ref	selfdef_font.fnt	Notepad

Tab. 4-1



A	Cell height	C	Cell width
B	Character length	d	Character height

Example sketch: Dimensions and coordinates of a character Fig. 29052

Creating a reference file

1. Open Notepad.

   An empty page is displayed.

2. Enter `1000` in the first line.

   The numbers up to 1000 are reserved for the standard fonts of TruTops CAD.

3. Enter `selfdef_font.fnt` in the second line.
4. Select >*File* >*Save as*.
5. Under "Save in", change the 'Trumpf.net\Data' directory (located on the Tops installation drive).
6. For the "File type", enter `font.ref`.

   This file name cannot be changed. All additional fonts of your own are entered in this file. Each font is given a consecutive number 1001, 1002... A file name must appear in the row after each number.

7. Select *Save*.

   The reference file is stored in the 'Trumpf.net\Data' directory.

Creating a font file

8. In Notepad, select >*File* >*New*.
9. Make entries line by line as in the following example.

Example: Entries in Notepad (line by line)	Meaning	File format description
#~HEADER	Beginning of first block	Beginning of file header
selfdef_font.fnt	Name of font file	
40	Cell width	
80	Cell height	
70	Distance from zero point to the end of character height	
10	Start line of the character without descender	
1	Number of subsequent characters	
0.5	Ratio of cell width to cell height	
##~~	End of block	End of file header
#~2	Beginning of character description	Beginning of character
L	Character to be displayed	
1	Number of groups	
#~2.1	Beginning of group	
7	Number of subsequent points	
M 10 10	**M**ove to coordinate 10, 10	
D10 60	**D**raw a line from the coordinate 10,10 to the coordinate 10,60	
D15 60	etc.	
D15 15	etc.	
D30 15	etc.	
D30 10	etc.	
D10 10	etc.	
C	**C**lose the contour	
I~	End of group description	
##~~	End of character description	End of character
###~~~EOF		End of file

Font file consisting of one character Tab. 4-2

10. Select >*File* >*Save as*.
11. Under "Save in", change the 'Trumpf.net\Data' directory.
12. Select "*.TXT" as 'File type'.
13. Enter `eigene_schrift.fnt`"File name".
14. Select >*Save*.

    The font file with the letter L is created and can be used after restarting TruTops CAD.

Tip

Creating a complete character set is a time-consuming procedure. Therefore, simply copy and rename a TruTops CAD text font file. Then overwrite the entries in the file with your own coordinates and data.

11. Embossed texts

11.1 Creating embossed text

Embossed texts such as ID numbers can be created in TruTops and machined on machines which support 10-station or 12-station Multitools.

Embossed texts are created with a separate mask, are assigned the "characteristic" property and are saved as templates (*.VLG). The function must be switched on in the data.

Two tools are available for embossed texts:

- Digital fonts.
- Embossed text.

Condition

- The fonts have been activated in the Data Management.

Note

TruTops has the numbers 0 to 9 and the alphabets A and E as characters (only for 12-station multitool). They are saved as tools. For example: 'PR_A.wzg'.

1. Select *>Extras >Change data >User >TruTops functionsDigital fonts* or *Embossed text*.
2. Select *Exit*.
3. Select *>File >Execute...*

   The "Execute user-defined TruTops function" mask is displayed.
4. Select *Digital fonts* or *Embossed text*.

   The "Digital fonts input" mask is displayed.
5. Enter the distance between the characters, the segment length and the digital text.
6. Press *OK*.
7. Enter a point or angle in the input line.

   or

   ➢ Rotating the text using wheel mouse.
8. Position the embossed text and left click.

The embossed text is drawn.

11.2 Using special characters

Example: A slash as a special character.

When entering the embossed text, a place holder must be used instead of a slash.

The letter A can be used as a place holder, because in addition to the numbers 0 to 9, the letters A and E are also available (only for 12-station MultiTool). Example: 123A456 and not 123/456. In MultiTool the embossing punch "A" must be replaced by the embossing punch "/".

If the embossed text display in TruTops matches the embossed text, a slash must be drawn and saved as a tool.

1. Draw a slash.

   The slash should correspond to the height of a digit.
2. Select >*Auxiliary tools* >*Displace zero point*.
3. Click on the center of the slash.
4. Select >*File* >*Save as*.

   The TruTops file browser is displayed.
5. Select '*.WZG' as the file type.

   The TruTops file browser automatically goes to the directory '...\TRUMPF.NET\Data\PunchTools'.
6. Save the drawing with the name `PR_s.wzg`.
7. Press *OK*.

   "Save tool" is displayed.
8. Select "Punching tool".
9. Press *OK*.
10. Select *Yes* to overwrite the file.

    The slash is saved under the file name 'PR_s.wzg'. `S` is the place holder for the embossing text input.
11. Select >*File* >*Executeembossed text*.
12. Enter the distance between characters.
13. Enter the embossing text with place holder `s` and select *OK*.
14. Enter a point or angle in the input line.

    or

    ➢ Rotate the text with the wheel mouse.
15. Position the embossed text and left click.

The embossed text is drawn and the s is displayed as a slash.

12. Comparing geometries with one another (geometry comparator)

The elements of two drawings can be compared with one another in TruTops CAD. The differences can also be displayed.

The drawings can be in the TruTops formats *.GEO and *.VLG or in the foreign formats *.DXF, *.MI, *.IGS or *.GRA.

Notes

- This function is not available in TruTops Tube.
- Actions cannot be undone in the geometry comparator (no Undo).
- Reference drawing and comparison drawing are automatically prepared prior to the inspection.

12.1 Opening and closing geometry comparator

Opening the geometry comparator

1. Select *>Assistants >Start comparison*.

   The geometry comparator is opened and the function browser is adapted.

Closing geometry comparator

2. Select *>Assistants >Compare >Exit comparison*.

   The current drawing is displayed again.

12.2 Comparing a reference drawing and a comparison drawing with one another

Note

If a drawing is created or loaded in the "normal" mode and the geometry comparator is then opened, TruTops CAD automatically selects the current drawing as reference drawing.

Loading reference drawing

1. Either
   - ➢ Select *>File >Open reference*.
   - ➢ Select the file format and change the directory if required.
   - ➢ Mark the desired file and select *OK*.

or

- ➢ Open the list of last loaded files in the mask head with ▼.

The path and the name of the reference drawing is displayed in the mask head.

Loading comparison drawing

2. Select *>File >Open comparison*.
3. Select the file format and change the directory if required.
4. Mark the desired file and select *OK*.

   The comparison drawing is linked to the mouse.

Notes

- The reference drawing and the comparison drawing must be congruent, on top of each other. The position of the comparison drawing can be corrected using Move comparison part and Rotate comparison part (moving and rotating are self-holding.) The reference drawing cannot be moved.
- Comparison drawings can be loaded as often as desired. Each comparison drawing replaces the previous one. When a new reference drawing is loaded, TruTops CAD removes the comparison drawing.

5. Enter the target point or angle of the comparison drawing or position the comparison drawing with the mouse.

Setting comparison colors

6. Select *>Assistants >Compare >Reference, comparison color* or *>color of shared elements*.
7. Select color.
   - Reference color: highlights deviating elements in the reference drawing.
   - Comparison color: highlights deviating elements in the comparison drawing.
   - Color of shared elements: Displays all shared elements of the drawings in the selected color.

Extending comparison to line color and line type

8. Select *>Assistants >Compare >Calculate >Parameters*.

   The "Compare drawings" mask is displayed.
9. Select "Consider color" and/or "Consider line type".
10. Press *OK*.
11. Under "Execute", select *Start* to start the geometry comparator.

    or

    - ➢ Select *>Assistants >Compare >Calculate >Start*.

The elements of both the drawings are compared. If the comparison drawing is moved using *>Move* or *>Rotate*, the comparison is automatically updated.

12.3 Merging a reference drawing with a comparison drawing

➢ Select >*Assistants* >*Compare* >*Merge*.

The two parts are merged into one.

12.4 Deleting a reference drawing or a comparison drawing

➢ Select >*Assistants* >*Compare* >*Delete reference part* or >*comparison part*.

13. Prepare geometries for editing

When geometries are drawn, the machining and sequence order can already be determined with the preparation.

For loading and editing of the geometry in TruTops Laser, the preparation is taken into account.

13.1 Loading and preparing geometries

1. Select >*File* >*Prepare*.

   The file browser opens.
2. If need be, change over to the directory in which the drawings are to be managed.
3. Select the desired file.

   or

   ➢ Enter the file names.
4. Press *OK*.

   The drawing with the preparation is automatically displayed.

   Explanation of the colors and symbols for the preparation:

Color	Symbol/line type	Explanation
Green	Star	Start point of the preparation on the contour
Green	Line type: solid	Direction for the processing
Yellow	Line type: solid	Positioning paths between the contours
Blue	Line type: solid	Positioning path for outer contour or for the end point
Red	Star	Start point of the machining

Explanation of the colors and symbols for the preparation Tab. 4-3

13.2 Modifying the processing sequence of prepared geometries

The processing sequence within the parts can be modified via >*Positioning sequence* or >*Rearrange*:

With >*Rearrange*, single contours can be rearranged and positions for insertions can be preset. With >*Rearrange*, the existing (possibly optimized) processing sequence is retained. Only the selected contour is set at another place in the processing.

>*Positioning sequence* is used to globally rearrange whole groups or all contours.

In both cases, linear positioning paths are created.

Freely modifying the processing sequence

Notes

- If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.
- Always process outer contours last.

1. Load and prepare geometry.
2. Select >*Assistants* >*Prepare* >*Modify position sequence*.
3. Click on the contour from which the positioning sequence should be modified.
4. Click on further contours or box them in with the mouse in the sequence in which they should be machined.
5. In order to automatically arrange the rest of the positioning sequence: select >*Assistants* >*Prepare* >*Positioning sequence path optimized*.
6. Select *End*.

Setting a particular contour at the start of the processing

Notes

- If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.
- Always process outer contours last.

7. Load and prepare geometry.
8. Select >*Assistants* >*Prepare* >*Rearranging at the beginning*.
9. Click on the contour.

   Set the selected object at the beginning of the machining. The remaining processing sequence remains unchanged.

Setting a particular contour at the end of the processing

Notes

- If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.
- Always process outer contours last.

10. Load and prepare geometry.
11. Select >*Assistants* >*Prepare* >*Rearranging to the end*.
12. Click on the contour.

    The selected object is set at the end of the machining. The remaining processing sequence remains unchanged.

Machining particular contours before or after others

Notes

- This function can be used to machine a particular contour **before** or **after** another one. The rest of the processing will not be recalculated afterwards.
- If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.

- Always process outer contours last.

13. Load and prepare geometry.
14. Select >*Assistants* >*Prepare* >*Rearranging before* or >*Rearranging after*.
15. Click on the contour before/after which the other contour is to be arranged.
16. Click on the contour after/before which the other contour is to be arranged.

    The current, possibly optimized processing sequence remains and only the selected contour is set at the desired position in the processing sequence.

13.3 Modifying the start point for the machining

This function can use to do the following:

- Manually shift the start point of the machining preparation on the contour.
- Invert the direction of machining.
- Recalculate the start point.

Note

If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.

Shifting the start point as desired

1. Load and prepare geometry.
2. Select>*Assistants* >*Prepare* >*Displace start point*.
3. Click on the desired start point on the contour or box it in using the mouse.
4. Select *End*.

   The existing processing sequence remains and only the selected start point is set at the desired position.

Inverting the start point

5. Select >*Invert start point*.
6. Click on the contour whose machining is to be inverted.

Recalculating all new start points

7. Select >*Calculate start point*.

   All start points will be recalculated.

13.4 Selecting preferred direction

For the preparation of the geometry, the preferred direction can be determined (Execution) in X or Y direction.

Note

If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.

1. Load and prepare geometry.
2. Select >*Assistants* >*Prepare* >*Calculation* >*Parameter*.

   The "Prepare parameters" mask is displayed.
3. Move the controller in the X direction: the contours will be machined by row.

   or

   ➢ Move the controller in the Y direction: the contours will be machined by column.

13.5 Determining the contour start

This function can use to do the following:

- Approach to a corner = tangential approach.
- Approach to an element = vertical approach.

Note

If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.

1. Load and prepare geometry.
2. Select >*Assistants* >*Prepare* >*Calculation* >*Parameter*.

   The "Prepare parameters" mask is displayed.
3. Select"Contour start""Corner" or "Element".
4. Select >*Assistants* >*Prepare* >*Calculation* >*Start*.

The contour start is shifted to the selected position.

13.6 Prefer contours

It is advisable to machine those contours first that are in danger of tilting.

Note

If a new suggestion for the preparation is calculated after the modification of the processing sequence, all modifications will be lost.

1. Load and prepare geometry.

2. Select >*Assistants* >*Prepare* >*Calculation* >*Parameter*.

   The "Prepare parameters" mask is displayed.

3. To determine the maximum contents of surfaces for contours to be processed first:
   - ➢ Select *Determine via picking*.
   - ➢ Click on the contour.
   - ➢ Select >*Start*.

   or

   - ➢ At "Area until", enter the maximum surface that the contours to be machined first can go up to.
   - ➢ Press *OK*.

13.7 Recalculating the preparation

If manual modifications are made on the preparation and these are reset or settings are to be modified (for example, preferred direction, contour start...), then a new suggestion for the preparation can be calculated.

1. Load and prepare geometry.
2. Select >*Assistants* >*Prepare* >*Calculation* >*Start*.

The preparation of all contours will be recalculated.

13.8 Saving prepared geometries

1. Load and prepare geometry.
2. Either
   - ➢ Select >*File* >*Save*.

   or

   - ➢ Select >*File* >*Save as...*-

     The file browser opens. This displays the '*.geo' file format; the file format can not be modified.

   - ➢ If need be, change the file name.
   - ➢ Press *OK*.

The geometry will be saved as '*.geo' with preparation.

For loading and editing of the single part in TruTops Laser, the preparation is taken into account.

14. Processing films

14.1 Film separation

With the film separation function, individual elements or contours can be defined as film separating. They are then processed with a special tool for separating films in TruTops Punch.

1. Select >*Create*, >*Film separation*, >*Element* or >*Contour*.
2. Click on individual element or contour.

   The element is displayed in red and has the characteristic "film separating".
3. Select >*Remove*, if the characteristic has to be removed again.

14.2 Film peeling assistant

In order to make the peeling off of films easier, a special "peeling contour" in the shape of a row of arches which form sharp tips between each other can be made with a film slitting tool. In doing so the film detaches from the sheet at the end of the tip. The peeling contour is automatically integrated in the contour and receives the characteristic "film separating". The contour is then processed with the film slitting tool in TruTops Punch.

The set standard parameters are suitable for a variety of film types. They can be adapted for special cases (>*Create*, >*Film separation*, >*Parameter...*):

- **Height**: peeling the film off can be made easier with a larger tip height, which makes it possible to peel it off with protective gloves.
- **Angle**: sharper angles lead to the film tips being withdrawn more strongly. Tips which are too narrow however may lead to premature tearing!
- **Lower radius**: prevents the film from being torn in the direction of peeling.
- **Upper radius**: can prevent scratching on very sensitive material surfaces. In such a case, values between 0.01 and max. 0.10 mm should be selected. The larger the radius, the less effective the peeling assistant is.
- **Quantity**: several tips increase process reliability.

Notes

- The peeling contour should not be placed too far from the margins of the remaining contours. The contour can be turned using the mouse wheel when it is hanging on the mouse pointer.

- Parameter modifications remain in force until they are reset via the *Default* button.

1. Load part.
2. Select >*Create*, >*Film separation*, >*Peeling assistant*.

   The peeling contour is now hanging on the mouse pointer.

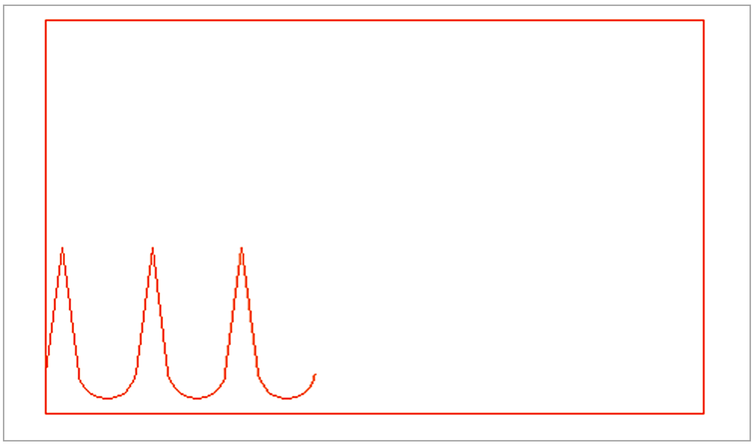


Fig. 62046

3. Place the peeling contour on the desired spot and click on the left mouse button.
4. Press *End*.

Chapter 5

Variants

1. Variants

1.1 Generating geometries in different variants (optional)

Generated geometries can be flexibly modified with the help of variants in TruTops CAD. The modifications are controlled by means of pre-defined variables and expressions.

The following key words are integrated in the variant module. They cannot be defined as variables.

abs(x)	Provides the absolute value of a number
acos(x)	Inverse cosine of x
acosh(x)	Hyperbolic inverse cosine of x
and	Logical operation
asin(x)	Arc sine of x
asinh(x)	Hyperbolic arc sine x
atan(x)	Arc tangent of x
atan2(y,x)	atan(y/x)
cos(x)	Cosine of x
cosh(x)	Hyperbolic cosine of x
degrees(x)	Conversion of arc dimensions into degrees
e	The mathematical constant e == 2.7182...
exp(x)	ex
fabs(x)	Absolute value of the floating point number x
false	Incorrect
False	Incorrect
floor(x)	The value of x rounded down as floating point number
fmod(x,y)	The value of x module y as floating point number
frexp(x)	Provides Mantissa (numbers before the comma of an logarithm) and exponent of x as pair (m, e). Here, m is a floating point number and e is a whole number, so that x = m*2e.
hypot(x,y)	Euclidean distance _x² + y²
int(x)	Converts a floating point number into a whole number
lexp(x,i)	x*2i
log(x)	Natural logarithm of x (base e)
log10(x)	Logarithm of x base 10
max(x)	max(1,2,3) = 3 provides the largest element
min(x)	min(1,2,3) = 1 provides the smallest element
not	Logical operation
or	Logical operation
Pi	The mathematical constant pi = 3.1415
pow(x,y)	x hoch y

radians(x)	Conversion of degrees into arc dimensions
round(x)	Rounding a floating point number
sin(x)	Sine of x
sinh(x)	Hyperbolic sine of x
sqrt(x)	Square root of x
tan(x)	Tangent of x
tanh(x)	Hyperbolic tangent of x
true	true
True	true
+	Addition
-	Subtraction
*	Multiplication
/	Division

Key words Tab. 5-1

Starting/ending the variant module

Note

ive, the relevant icon bar can be opened (under >*View*).

Starting the variant module

1. Select >*Variants*, >*Start*.

   The variant module is displayed.

Ending the variant module

2. Select >*Variants* >*Exit variants*.

   The variant module is closed.

Generating variants

Variants are individual modifications dependent on variables and expressions. Variants can be single parts or entire assemblies.

1. Create or load a drawing.
2. Select >*Variants* >*Generate variant*.

   The "Variant data" mask is opened.
3. Enter a "designation" for the variant.

   If no designation is entered, the designation of the variant type is automatically adopted.
4. Either
   - Enter "Condition". The variant will only be executed when the condition is met.

or

- Select *Variables*.
- Mark the desired variable.
- Select *OK* (see section "Working with variables").

  The selected variable is adopted.

Generating conditions:	true	true
	True	true
	false	Incorrect
	False	Incorrect
Comparison:	<	smaller than
	<=	less than or equal to
	>	greater then
	>=	greater than or equal to
	==	(exactly) equal
	!=	Not the same
Logical operators:	and	and
	or	or

Conditions Tab. 5-2

5. Open the selection field with ▼.
6. Select "Variant type".
   - Move: moving elements and bendings in X and Y direction
   - Rotate: rotating elements and bendings around an angle.
   - Mirror: mirroring elements and bendings horizontally or vertically on a center or two points
   - Move copy: creating and moving the desired number of copies of the elements and bendings.
   - Rotate copy: creating and rotating the desired number of copies of the elements and bendings.
   - Mirror copy: creating and mirroring the desired number of copies of the elements and bendings.
   - Stretch: stretching elements and bendings horizontally, vertically or in both directions
   - Delete: deleting selected elements or bendings
   - Delete selection: selecting and deleting elements and bendings according to the color, line type and element type
   - Modify rounding: entering a new radius for roundings
   - Modify circle: entering a new diameter for circles
   - Modify text: Enter the new text content in quotation marks, e.g. "TruTops".
   - Modify bending angle: entering a new bending angle

- Set variable: makes it possible to set a variable through a variant. The variant requires the name and the expression for the variable (similar to setting in variable management).
- Set property: sets a property of the drawing, e.g. material ID. This is useful when values are read in from a variable file. The text **must** be in quotation marks since it will otherwise be treated as a text variable.
- Cancel variants: Enables canceling of variants. Any text can be defined, which is output during cancellation. This variant can be used for checking.
- Set template: Adding a *.vlg file at the specified position.
- Close contour: Closes the selected contour elements (by specifying a gap to be closed maximally).

7. Either
   - ➢ Further parameters can be entered depending on the selected variant type: Left click in the input field and enter the desired value or the mathematical formula.

     Example: randx - 20 (variable name - old value) or randy - loch.cy (border Y - center Y of reference hole). The old value can be entered either as value or reference here.

   or

   - ➢ Select *Variables*.
   - ➢ Mark the desired variable.
   - ➢ Select *OK* (see section "Working with variables").

   The selected variable is adopted.
8. Press *OK*.
9. One after the other, click on or box in all elements or bendings whose variants are to be allocated.

   Selected elements are displayed in magenta.
10. Press *OK*.

Undoing variants

➢ Select >*Variants* >*Undo step* or >*Undo all*.

The last executed variants or all the variants so far are undone. With >*Undo all*, the program returns to its original state.

Executing variants

➢ Select >*Variants* >*Execute step* or >*Execute all*.

The last undone or all undone variants are executed again.

Selecting an overview of the variants

1. Open the selection field with ▼ in the toolbar.

   All variants of the drawing are displayed.
2. Mark the variant.

   The variant is displayed in the selection and all the variants before it are executed.

   or

   ➢ Mark *Final condition*.

   All variants are executed.

Allocating a variant to an element or a bending

Condition

- One or more variants have been created and one variant is active.

1. Select >*Variants* >*Select elements*.
2. One after the other, click on or box in the elements in the drawing.

   Selected elements are displayed in magenta.
3. Press *OK*.

Adopting element selection of an existing variant

Condition

- One or more variants have been created.

Note

TruTops makes a note of the adoption, i.e. if all elements of the previous variant are adopted and the selection of this variant is modified later, the selection of the current variant also changes.

1. To execute a series of actions on the same elements: select >*Variants* >*Adopting elements*.
2. If the previous variant contains copied elements, the "Transfer elements from predecessor" mask is displayed.

3. Select "Element selection", "Originals".

   Only original elements of the previous variant are adopted.

   or

   ➢ Select "Element selection", "Copies".

   Only copied elements of the previous variant are adopted.

Example

Create variables (manage variables)

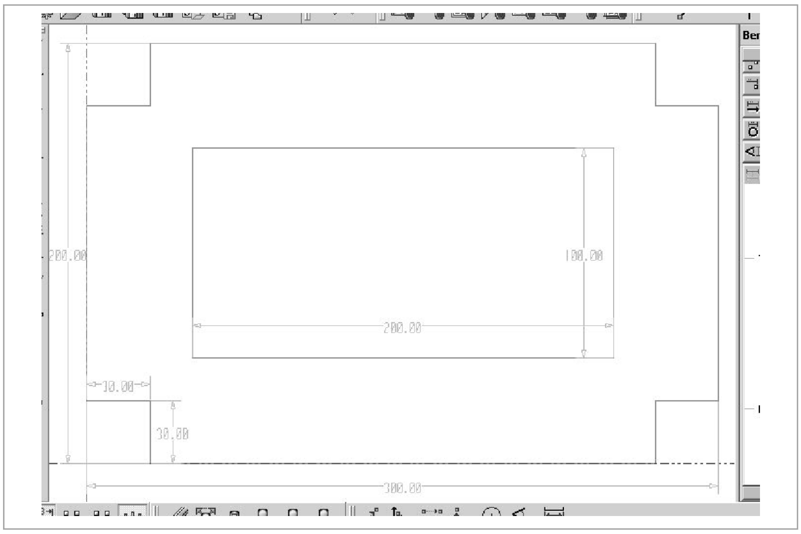


Fig. 51516

1. As "Name", enter `teilx`.
2. For "Expression", enter `500`.
3. Select *Set*.
4. Enter the further variables.

Variable name	Variable expression
teily	200
klx	30
kly	30

Tab. 5-3

Generating variants

5. As "designation", enter `long`.
6. As "Condition", enter `true`.
7. As "Variant type", select *Stretch*.
8. Select "Horizontal".
9. Select *variable*.
10. Mark the desired variable *teilx*.

11. Press *OK*.

    Teilx is applied
12. After *teilx*, enter `-300`.
13. Box in elements that are to be stretched in the X direction.

    The elements that are boxed in are displayed in magenta.
14. Press *OK*.

    The geometry is stretched in the X direction.
15. Follow the same steps for the Y direction.
16. As "Designation", enter `klinkx1`.
17. As "Condition", enter `true`.
18. As "Variant type", select *Stretch*.
19. Select "Horizontal".
20. Select *variable*.
21. Mark the desired variable *klx*.
22. Press *OK*.

    Klx is applied
23. After *klx*, enter `-30`.
24. Box in the notch element.

    The element that is boxed in is displayed in magenta.
25. Press *OK*.

    The notch is stretched in the X direction.
26. As "Designation", enter `klinkx2`.
27. As "Condition", enter `true`.
28. As "Variant type", select *Stretch*.
29. Select "Horizontal".
30. Enter `- (klx-30)`.
31. Box in the notch element.

    The element that is boxed in is displayed in magenta.
32. Press *OK*.

    The notch is stretched in the X direction.

1.2 Working with variables and reference elements

Creating and editing variables

Creating variables

Note

Variables that have no value can not be executed in variants.

1. Select >*Variables* >*Manage variables*.

The "Variables" mask appears.

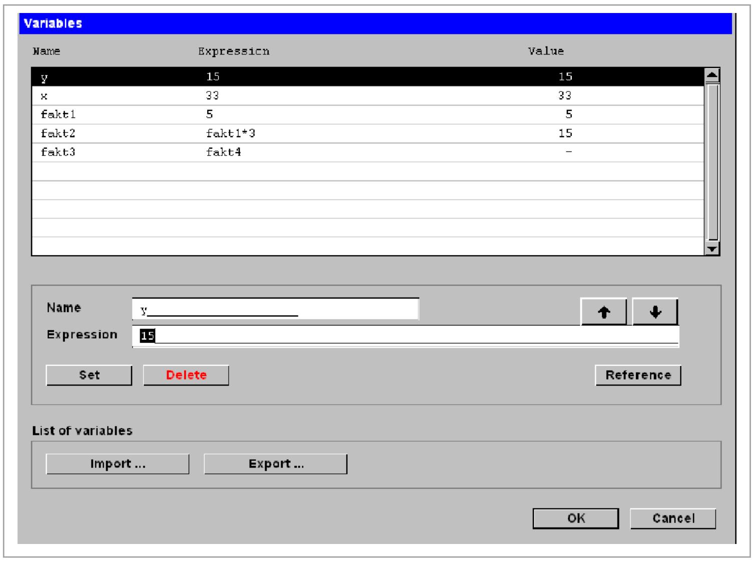


Fig. 41125en

2. Enter the designation for the variable at "Name".
3. Enter "Expression".

   The expression can consist of simple digits and include arithmetic operations or mathematical functions. Other variables can also be used in the expression, but these have to be in the list above.

   Example:

Name:	fakt2
Expression:	fakt1*3 (fakt1=5)
Value:	15

Tab. 5-4

Name:	fakt3
Expression:	fakt4
Value:	- (no value)

Example for variant without value Tab. 5-5

4. To use a reference element in an expression: select *Reference* (see "Working with reference elements").
5. The "Define reference value" mask is displayed.
6. Open the selection field with ▼.
7. Select the name of the reference value from the list.
8. Select the "desired value".

   Values that are not permitted for the selected reference element are grayed out.

9. Press *OK*.

   The reference is attached to the expression.
10. Select *Set*.

    The inputs are applied in the above list.

Modifying variable

11. Mark the variable in the selection.

    Only the expression of one variable can be modified.
12. Modify the expression for the variable.
13. Select *Set*.

    The modification will be applied; the position of the variable in the list will be kept.

Modifying the order of the variables

14. Mark variables.
15. Use the arrow keys to shift the variables upwards or downwards ((see "Fig. 41125en", pg. 5-10)).

Importing variables

If external variables are to be imported (e.g., from SAP), the structure of these variables must correspond to that of TruTops CAD variables.

Conditions

- The files are ASCII files.
- The entry per variable is: variable = expression (e.g., x=a+b).

1. Select "List of variables"*Import* if an available list with variables is to be imported.

   The file browser opens.
2. If need be, change over to the directory in which the variables are managed.
3. Mark the list with variables.
4. Press *OK*.

The new variables are applied and are placed at the start of the selection.

Exporting variables

1. Select "List of variables"*Export* if variables are to be exported.

   The file browser opens.

2. If need be, change over to the directory in which the variables are managed.
3. Select the desired file.

   or

   ➢ Enter the file names.
4. Press *OK*.

Delete variables

1. Mark the variables to be deleted.
2. Select *Delete*.

The variable will be deleted.

Set reference element

A name is assigned to an element or a bending with a reference element. This reference can then be used for variants for conditions and parameters or for expressions of variables.

Reference elements can be used to use values of an element (e.g., radius, length) or a bending in an expression. To do this, you only have to enter the name of the reference element followed by a point, followed an identifier.

- sx: X coordinate of the starting point.
- sy: Y coordinate of the starting point.
- ex: X coordinate of the end point.
- ey: Y coordinate of the end point.
- cx: X coordinate of the center.
- cy: Y coordinate of the center.
- r: radius.
- d: diameter.
- l: length.

Example: to receive the length of the reference element with the name "Line", enter the expression `Line.I`.

Condition

- A drawing is loaded.

1. Select >*Variants* >*Set reference element*.

   The "Define reference element" mask is displayed.
2. Enter name for reference element.
3. Press *OK*.

4. Click on the reference element in the drawing.

The reference element receives the "attribute" property and is shown in orange if *Highlight characteristics* has been selected.

Remove reference element

1. Select >*Variants* >*Remove reference element* if the reference element is to be removed again.
2. Click on the reference element.

The "attribute" property will be deleted at the element.

Show reference element

Condition

- A drawing with set reference elements is loaded.

1. Select >*Variants* >*Show reference element*.
2. Click on the reference elements one after the other.

   or

   - Select *All* to display all reference elements.

The names of the reference elements are displayed.

1.3 Manage variants

Positioning variants

1. Select >*Variants*, >*Manage variants*.

   The "Variants" mask is displayed.
2. Mark the desired variant in the list.

   The ID of the current variant is displayed under the selection field.
3. The position of the variants in the list can be modified using the arrows.

The position of the variant corresponds to the sequence in which the variant is performed.

Creating new variants

➢ Select *Generate*.

The "Variant data" mask is displayed. The new variant is set at the highest position. Elements can be allocated for this later (see section "Allocating variants to an element".

Copying variants

1. Mark any variant.
2. Select *Copy*.

The copied variant has the same parameters or conditions and the same elements selection as the original variant.

Modifying variants

1. Mark the variant to be modified.

   The "Variant data" mask is displayed.
2. Modify any parameter.
3. Press *OK*.

Deleting variants

1. Mark the variant to be deleted.
2. Select *Delete*.
3. Press *OK*.

The last selected variant is adopted.

Renumbering variants

1. Select >*Variants*, >*Start*.

   The variant control is opened.
2. Select >*Variants*, >*Manage variants*.

   The "Variants" mask showing the available variants is opened.

3. Select *New IDs* to renumber the variant IDs consecutively.

The IDs are numbered consecutively from the top down.

Executing variants several times

1. Select *Loop*.

   The "Variant data" mask is displayed.
2. Enter the required parameter.

   Example: Loop variable counter runs from 1 to 25.
3. Press *OK*.

The loop is introduced in the "Variants" mask in place of the variant that was last marked. The previously marked variant is placed in the loop.

All variants belonging to the loop are moved in closer together. Further variants can be allocated to the loop or can be removed from it using the arrow. The loop is dissolved through *Dissolve*.

Loops can also be placed in loops. Loops can thus be nested in any desired manner.

Saving geometry with variant as *.GMV

A GMV file contains the geometry and the variants. It is independent of a Geo file that has been used as the basis. No geometry preparation is made when saving the file. GMV files cannot be nested or further processed.

Condition

- A variant has been created or modified.

1. Select >*File* >*Save as*.

   The file browser opens.
2. Select '*.GMV' as the file type.
3. Enter the file name (without file extension).
4. Press *OK*.

The drawing is saved as a geometry with variant in the '*.GMV' format. There are no other saving options, i.e. no other formats.

Exporting variant as *.GEO

Condition

- A variant (*.GMV) has been loaded.

Note

Intermediate statuses can also be saved. To export the end status, it is important that all variants be executed up to the end status.

1. Select *>File >Export Geo...*.

   The TruTops CAD file browser is displayed.
2. Go to the directory in which the '*.GEO' file is to be exported.
3. Change the file name if required.
4. Press *OK*.

The file is saved in the '*.GEO' format.

Exporting variants to Excel

The list of variables can be exported from a GMV file into Excel, modified there and re-imported into TruTops CAD as a new drawing variant.

Conditions

- A '*.GMV' has been loaded.
- The Excel program is installed.

Saving a file in XLS format

1. Select *>File >Execute...*.

   The "Execute user-defined TruTops functions" mask is displayed.
2. Select *GMV → XLS*.

   The file browser opens.
3. Save the file under the current '*.GMV' name.

   or

   ➢ Enter the new name under "File name".
4. Press *OK*.

   The file is saved in '*.XLS' format.

Opening an XLS file

5. Select *>File >File-Browser*.

   or

   ➢ Select *>Start >Programs >TRUMPF.NET >PDM Browser*.

6. Go to the directory containing the '*.XLS' file.
7. Double-click on the '*.XLS' file.

   The XLS file opens. The designations of the variables are adopted directly from the GMV file.

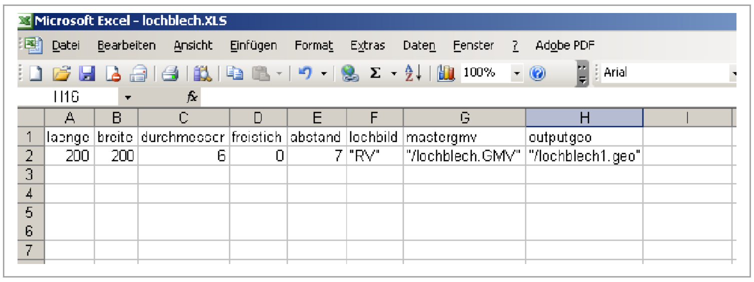
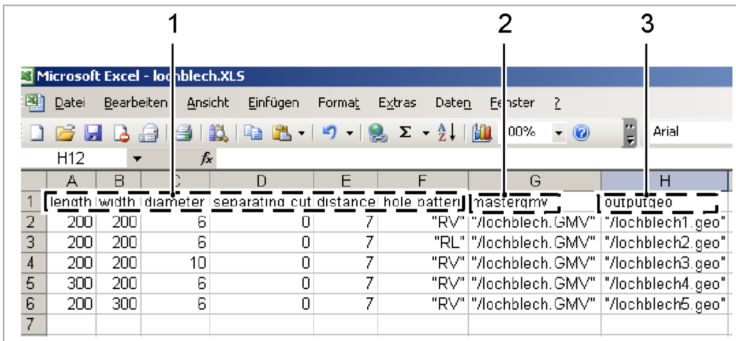


Fig. 44141de

Creating further variants

8. To create further variants, copy line 2.
9. Paste the copied line repeatedly into the following lines until you have the desired number of variants.



1 Designation of the variables
2 mastergmv: storage location of the output file (*.GMV)
3 outputgeo: storage location of the new geometries (*.GEO, by default the same directory which contains the GMV file as well).

Fig. 44139EN

10. Go to the copied line to be modified.
11. Enter new value.
12. Change the "file name" and/or path of the storage location for each new variant in the "outputgeo" column (3), e. g. add an ascending number to the file name.
13. Save the file.
14. Close Excel.

Importing variants from Excel

Creating a *.GEO from *.XLS

1. Select >*File* >*Execute...*

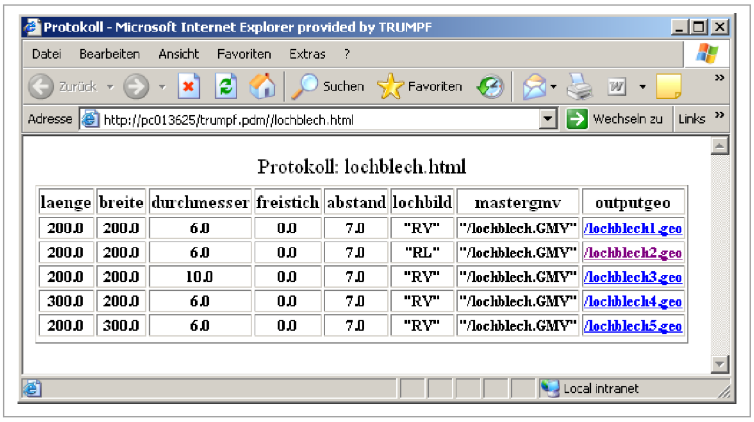
The "Execute user-defined TruTops functions" mask is displayed.

2. Select *XLS* → *GEO*.

   The file browser opens.

3. Double-click on the desired file ('*.XLS').

   From the '*.XLS' file, a new '*.GEO' file is created for each variant and a log with all variants is displayed.



laenge	breite	durchmesser	freistich	abstand	lochbild	mastergmv	outputgeo
200.0	200.0	6.0	0.0	7.0	"RV"	"/lochblech.GMY"	/lochblech1.geo
200.0	200.0	6.0	0.0	7.0	"RL"	"/lochblech.GMY"	/lochblech2.geo
200.0	200.0	10.0	0.0	7.0	"RV"	"/lochblech.GMY"	/lochblech3.geo
300.0	200.0	6.0	0.0	7.0	"RV"	"/lochblech.GMY"	/lochblech4.geo
200.0	300.0	6.0	0.0	7.0	"RV"	"/lochblech.GMY"	/lochblech5.geo

Variant protocol Fig. 44140

Opening a variant with Geo Viewer

4. Double-click the desired drawing in the "outputgeo" column in order to show the generated drawings.

   TruTops CAD opens the Geo Viewer with the selected GEO.

5. Close Geo Viewer with *OK*.
6. Select *>File >Open* to open the drawing in TruTops CAD.
7. Go to the directory containing the '*.GEO' file.
8. Mark the drawing and select *OK*

Chapter 6

Bending

1. Bending lines

1.1 Generating bending lines

Notes

- Bending lines are automatically adapted to the contour.
- If the *Pre-/final bending* bending technique has been selected, then the pre-angle applies for the pre-bend. The opening angle corresponds to the final angle.
- The bending radius is important for the display. If the notches are not large enough, then a small bending radius has to be defined since otherwise there could be problems when creating the bending part (insofar as this edge is not struck).
- Geometries may not include drawing components except for unfolding and for bending lines. If external formats such as IGS or DXF are imported, drawing frames, views, fonts, etc., have to be removed from the drawing.
- In addition to material and thickness, the selection when choosing bending tools can be additionally restricted by a TruTops Bend tool list.
- **Rule of thumb for bending radius**: The notch and the sheet thickness should be larger than the bending radius.

1. Select >*Bend*, >*Bending*, >*Create*.

   The "Define bending line" mask is displayed.
2. Enter the corresponding values.
   - "Bending radius": The bending radius is limited to the tool radius.
   - "Bend factor": Is not converted for acute angles.

   Tip

   If TruTops CAD is connected to a TruTops Bend database, the *Auto* button can be used to automatically determine the bending radius and bend factor for the bending line.
3. Select upper and lower tool.
   - ➢ Click on the arrow button.
   - ➢ Select the desired tool from the list.

   or

   - ➢ Press *Catalogue*.

     The tool catalog is opened.
   - ➢ Select the desired tool.
4. Select "Bending type", "Bending technique", and "Bending method".

- *Folding* (bending type): enter 180° for opening angle and pre-angle.

5. Press *OK*.
6. Click on, for example, the line drawn in cyan.

The line is defined as bending line and shown by the program in green.

1.2 Assigning bending information

Bending information is not adopted. It must be allocated. Exception: files in '*.MI' format from SheetAdvisor.

Note

In TruTops CAD, files can also be loaded in the formats '*.DXF', '*.IGS', '*.MI', or '*.GRA'. The bend factors used when creating the model must be entered here and later converted in TruTops Bend. The upper and lower tools must likewise be entered here.

1. Select >*File* >*Properties*.
2. Enter part ID, material ID, material thickness etc.

   The latter two **must** be entered.
3. Press *OK*.
4. Select >*File* >*Save*.

   or

   ➢ Select >*File* >*Save as* if a name is to given to the program.

1.3 Modifying bending lines

1. Select >*Bending*,>*Bend* >*Modify*.
2. Click on the bending line to be modified.
3. Enter the modifications in the "Define bending lines" mask.
4. Press *OK*.

The modifications are accepted.

1.4 Inverting bending lines

1. Load bending part.
2. Select >*Bend*, >*Bending*, >*Invert*.

3. Click the bending line to be inverted.

The bending line is inverted.

1.5 Marking bending lines

This function can be used to mark the ends or the entire bending lines. This is necessary to be able to correct bending lines that are unfavorably located in the part.

1. Select *>Bend >Bending >Mark*.

   The "Mark bending lines" mask is displayed.
2. Select the length of the marking or the entire bending line.
3. Press *OK*.
4. Click on the bending line.

   or

   ➢ Select *All* if all the bending lines are to be marked.

The selected bending lines are displayed in orange.

1.6 Removing the bending line marking

This function is used to reset the marking of bending lines.

1. Select *>Bend >Bending >Remove marking*.
2. Click on the bending line whose marking is to be removed.

The bending line marking is removed and the line is shown in green.

1.7 Modify unfolding

For each bending line a new bend deduction can be entered. The drawing will be automatically adjusted.

Entering a new bend deduction

1. Select *Modify unfolding*.

   The screen called "Modify unfolding" will be displayed.
2. Mark the required bending line.
3. Enter a new bend deduction under "New".

4. Select *Set*.

   The new bend deduction will be shown in the list of bending lines.

5. Press *OK*.

   The unfolding (developed view) will be recalculated and the drawing will be modified.

Join bendings

6. In the "Modify unfolding" screen, select "Combine bendings".

   All bends with the same bending angle and lower tool will be combined.

Set automatic bending notches

7. Select "Set automatic bending notches".

   If this checkbox is set, the developed view will be changed and, at the same time, a bending notch will be set.

1.8 Generating bending notches

Bends on sharp corners and bends that do not end exactly at a contour can be notched with this function.

1. Select *Bending notches*.
2. Click on the bending line.

Depending on whether a notch is necessary or not (decided by TruTops), corners or contours where the bending line ends are notched.

2. Bending tools

2.1 Transferring bending tools

Tool data can be transferred from one bending to other bendings.

1. Select *>Bend >Bending >Adopt tools*.
2. Click on the bending line from which tools are to be transferred (reference bending line).
3. Click on the bending lines which are to obtain the machining of the reference bending line.

   or

   ➢ Press *All*.

   The data is modified for all bending lines.

2.2 Restricting the selection of bending tools

The facility for filtering by material can be switched off.

1. Select *>Bend*, *>Bending*, *>Create...*.
2. To select a tool, select the drop down arrow.

   The "Upper tools/lower tools" mask is opened.
3. Select the "Restrict selection" check box.

   The "Filter selection" mask is opened.
4. Deselect the "Material" check box, if the selection is only to be sorted by list.
5. Press *OK*.
6. Select a tool.

The tool is adopted.

2.3 Drawing bending tool

Note

TruTops Bend needs a drawing of the tool to display special tools saved in the database in TruTops Bend properly and make collision calculations. A drawing of the side view is usually sufficient.

Modifying the bending tool

Condition

- The drawing of a tool drawn with the assistant is open.

1. Select *>Bend >Bending tool >Modify*.

   The "Bending tool assistant" mask is displayed.
2. Delete the corners or lines that are to be modified.
3. Redraw the corners or lines.
4. Press *OK*.

   The modified tool is displayed.
5. Save tool.

Drawing an upper tool

Defining height and type of tool

1. Select *>Bend >Bending tool >Create*.

   The "Bending tool assistant" mask is displayed.
2. Enter the height of the tool.
3. Select "Upper tool" under "Type of tool".

   A preview of the selected type of tool is displayed in the preview window.
4. Select *Continue*.

Selecting a view

Note

Here, it is defined whether a side view or a front view of the tool is to be drawn.

5. Select "Side view".
6. Select *Continue*.

Selecting tool type

7. Select "Normal" as tool type.
8. Select *Continue*.

Selecting a tool adapter

9. Select "Head position" tool adapter.
10. Select *Continue*.

Entering tool tip parameters

11. Enter "Opening angle" and "Radius".
12. Select *Continue*.

Drawing the right side of the tool

Note

The arrow keys can be used to enter angles in steps of 45°.

13. Enter the "length" of the line.
14. Enter "Angle".
15. Select "Absolute" angle or "Relative" to the marked point.

16. Select *Preview*.

    The drawn line is shown in the window.

17. Use *Continue* to go to the end of the drawn line.

    The line is drawn in black and the end is marked with a red X.

    or

    ➢ Use *Back* to delete the drawn line.

18. Select *Auto*.

    The right side of the tool is automatically drawn to the specified height.

Finishing drawing the upper tool

19. Select *Mirror*.

    The tool is drawn completely.

    If the sides of the tool are different, the same procedure as for the right side must be followed for the left.

Modifying a tool

20. Mark a corner or line with *Last* or *Next*.
21. Select "Corners" and enter "Rounding" or "Bevel".

    or

    ➢ Select "Lines" and enter "Length" or "Angle".

22. Select *Modify*.
23. Press *OK*.

    The upper tool is displayed.

24. Save the tool as '*.WZG'.

Drawing a lower tool

Defining height and type of tool

1. Select *>Bend >Bending tool >Create*.

   The "Bending tool assistant" mask is displayed.

2. Enter the height of the tool.
3. Select the "lower tool".

   A preview of the selected type of tool is displayed in the preview window.

4. Select *Continue*.

Selecting a view

Note

Here, it is defined whether a side view or a front view of the tool is to be drawn.

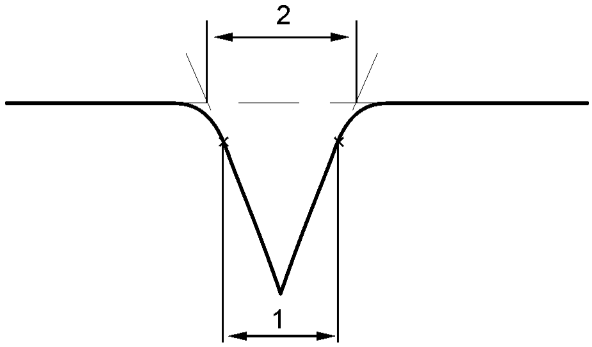
5. Select "Side view".
6. Select *Continue*.

Selecting tool type

7. Select "Normal" as tool type.
8. Select *Continue*.

Entering die lower tool parameters

9. Enter the "tool width", "opening angle", "die width", "groove ground radius" and "feed radius".
10. Activate "Die width without feed radius" if the feed radius is to be **ignored** for the die width.



1 Die width **without** feed radius
2 Die width **with** feed radius

Fig. 41731

11. Select *Continue*.

Drawing the right side of the tool

Note

The arrow keys can be used to enter angles in steps of 45°.

12. Enter the "length" of the line.
13. Enter "Angle".
14. Select "Absolute" angle or "Relative" to the marked point.
15. Select *Preview*.

    The drawn line is shown in the window.

16. Use *Continue* to go to the end of the drawn line.

    The line is drawn in black and the end is marked with a red X.

    or

    ➢ Use *Back* to delete the drawn line.

17. Select *Auto*.

    The right side of the tool is automatically drawn to the specified height.

Drawing the left side of the lower tool

Modifying a tool

18. Draw the left side as described above.
19. Mark a corner or line with *Last* or *Next*.
20. Select "Corners" and enter "Rounding" or "Bevel".

    or

    ➢ Select "Lines" and enter "Length" or "Angle".

21. Select *Modify*.
22. Press *OK*.

    The lower tool is displayed.

23. Save the tool as '*.WZG'.

Drawing holder

1. Enter "Height" and select "Upper tool" or "Lower tool".
2. Select *Continue*.
3. Select "Side view".
4. Select *Continue*.
5. Select "holder".

   A preview of the selected type of tool is displayed in the preview window.
6. Select *Continue*.
7. With upper tools, select the "type of tool adapter".
8. Select *Continue*.
9. Draw the right side.
10. Draw the left side.
11. Modify the corners or lines.
12. Press *OK*.

    The holder is displayed.
13. Save the tool as '*.WZG'.

Drawing Z tool

1. Enter "Height" and select "Upper tool" or "Lower tool".
2. Select *Continue*.
3. Select "Side view".
4. Select *Continue*.
5. Select "Z tool".

   A preview of the selected type of tool is displayed in the preview window.
6. Select *Continue*.
7. With upper tools, select the "type of tool adapter".
8. Select *Continue*.
9. Enter the "angle" and the "flange length" of the Z tool.
10. Select *Continue*.
11. Draw the right side.
12. Draw the left side.
13. Modify the corners or lines.
14. Press *OK*.

    The Z tool is displayed.

15. Save the tool as '*.WZG'.

Drawing an adapter

1. Enter "Height" and select "Upper tool" or "Lower tool".
2. Select *Continue*.
3. Select "Side view".
4. Select *Continue*.
5. Select "Adapter".

   A preview of the selected type of tool is displayed in the preview window.
6. Select *Continue*.
7. Press *OK*.

   The adapter is displayed in TruTops CAD.
8. Save the tool as '*.WZG'.

Drawing front view of a horn tool

Note

A front view of upper or lower tools is required only if they vary from the standard front view. This applies, for example, to horn tools.

1. Enter "Height" and select "Upper tool" or "Lower tool".
2. Select *Continue*.
3. Select "Front view".

   The front view is displayed in the preview window.
4. Select *Continue*.
5. Enter the parameters of the tool.
6. Select *Continue*.
7. Draw the right side.
8. Draw the left side.
9. Modify the corners or lines.
10. Press *OK*.

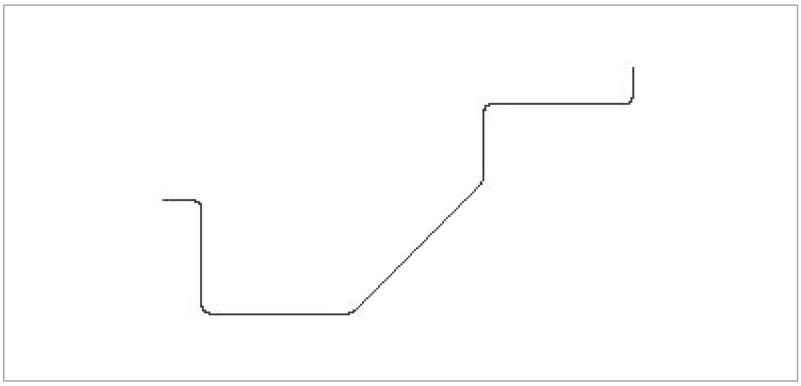
    The horn tool is displayed.
11. Save the tool as '*.WZG'.

3. Bending profiles

3.1 Unfolding a bending profile

Bending parts with a rectangular unfolding and parallel bendings can also be drawn simply in the form of the desired profile after bending, with and without bending radii.

The profile can then be converted in the unfolding drawing with the >*Unfold* function.



Profile drawing Fig. 24586

1. Select >*Bending*, >*Profile* >*Unfold*.
2. Enter the "material name", "material thickness" and "profile depth" (width of the unfolding part).
3. Enter the "Upper tool group" and "Lower tool group".

   The tool groups are required in order to determine appropriate tools and the respective bend factors.
4. Press *OK*.
5. To define the placement of profile materials:
   - Click on a line element of the profile.
   - Click next to the element in the direction in which the material can be found.

The profile is replaced by the respective unfolding.

3.2 Creating a bending profile

Bending profiles can be created to check and modify an unfolding drawing. If the bending part contains inside contours, they are deleted.

The outer contour is still rectangular during reunfolding.

1. Select >*Bend*, >*Profile* >*Fold*.

2. Enter the "material name" and *material thickness*.
3. Select whether the profile is to be for the "top side" or "bottom side" of the bending part.
4. If the profile is to contain bending radiuses: select "With radiuses".
5. Press *OK*.
6. Click on a suitable bending line.

   The bending direction and cross-section are defined for the desired profile.

The profile is created at a right angle to the selected bending line. All bendings which are parallel to the selected bending line are taken into account.

3.3 Creating a profile as auxiliary geometry

The created profile can be displayed as auxiliary geometry in the current drawing.

Note

Bending lines are separated automatically. They do not need to be separated manually. The only exception is if the outer contour is not closed.

1. Define bending line.
2. Select *>Bend*, *>Profile >Fold*.

   The "Create bending profile" mask is displayed.

3. Select "Create profile as construction geom.".
4. Press *OK*.
5. Follow the instructions in the command line.

The profile is added as auxiliary geometry.

3.4 Generating bumping (round bends)

Round bends can be created at the at the corner points in the *>Bending* menu. Either the corner points are clicked directly or the two following segments are clicked.

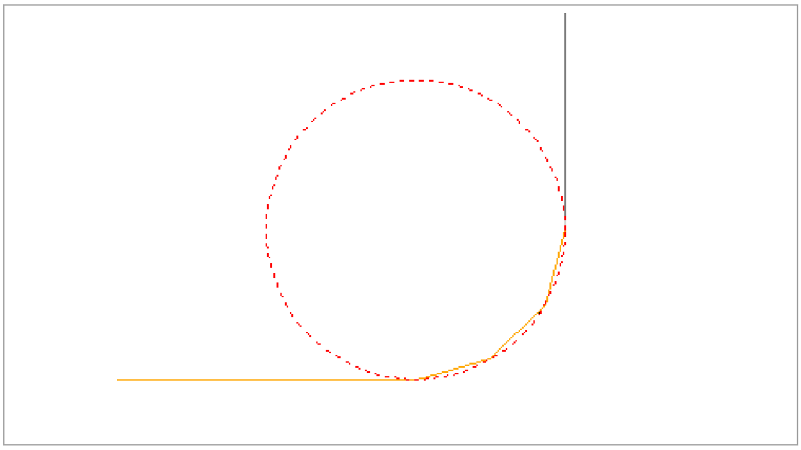
Notes

- Bumping bendings can be created only in a profile.

- In order to modify round bendings, they must first be deleted and then recreated.

1. Open a profile.
2. Select *>Bend >Profile >Create bumping*.
3. Enter the number of segments.
4. Enter the radius.
5. Select *OK* or press ↲.
6. Click on the end points.

The round bending is created.



Round bending with auxiliary geometry Fig. 31039

3.5 Showing material

Note

The material thickness is indicated by the second profile. The points of this profile cannot be caught or modified. The second profile is automatically adapted each time the first profile is changed.

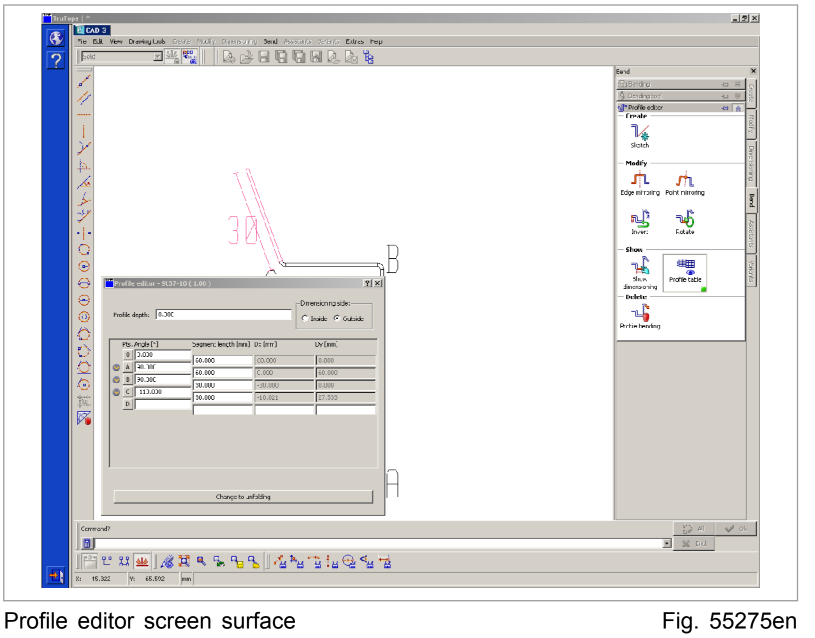
1. Select *>Bend >Profile >Show material*.
2. Use the cursor to click on the side of the profile, which is to be thickened.

The profile is thickened.

4. Creating profiles in the profile editor

4.1 Working in the profile editor

Profiles can be generated easily and quickly in the profile editor. Saved profiles can be opened and modified without difficulty. If the material thickness is changed, for example, the profile is adapted automatically.



Profile editor screen surface Fig. 55275en

Opening the profile editor

➢ Open profile editor

➢ Select >*Bending*, >*Start profile editor*.

The "Select material" mask is opened.

➢ Select "unprocessed material ID" and "material thick-ness".

➢ Press *OK*.

The profile editor is opened. The "Profile editor" screen appears simultaneously, where the material is displayed on screen. Profile segments can be entered via this screen.

or

➢ Select the "profile editor" icon.

The profile editor is opened and the profile currently being processed appears on the drawing sheet.

4.2 Opening and saving profiles

Opening a profile

1. Select >*File*, >*Open*.

   "Open file" is opened.
2. Select the corresponding GEO file.
3. Select *OK*.

   The profile unfolding appears.
4. Open profile editor.

The profile appears on the drawing sheet.

Saving a profile

After the profile is drawn, it cannot be saved until the unfolding is generated.

1. Select >*Bending*, >*Profile editor*,>*Change to unfolding*.

   The development is created.
2. Select >*File* , >*Save as*.

   "Save file" is opened.

   Note

   The profile will not be automatically saved if the unfolding is saved as a *.geo file.
3. Select *.GEO as "File type".
4. Enter the file name.
5. Select *OK*.

4.3 Creating a profile

Profiles can be created in two ways:

Icon	Function
	Sketch
	Profile table (two options): ▪ Input of angle and segment length. ▪ Input of the horizontal and vertical changes in distance between two points.

Tab. 6-1

Sketch

A new profile can be drawn or further segments attached to an existing profile.

1. Select >*Bending*, >*Profile editor*, >*Sketch*.

   The first mouse click (left mouse button) defines the origin. The cursor is positioned at the desired point for further additional segments.

   When there is already an existing profile, the cursor is automatically placed at the previous segment and the new segment can be drawn. The relevant data is stored in the profile table.
2. Position the cursor at the desired point and press the left mouse button.

   The segment is drawn and the length of the segment is displayed.
3. Select *End*.

The profile is drawn.

Profile table - entering angle and segment length

Each segment of the profile is defined by its length and position (angle).

Programming the first segment

1. Select *Profile table*.
2. Enter the first angle in relation to the coordinate system under "Angle" and confirm with <Enter>.

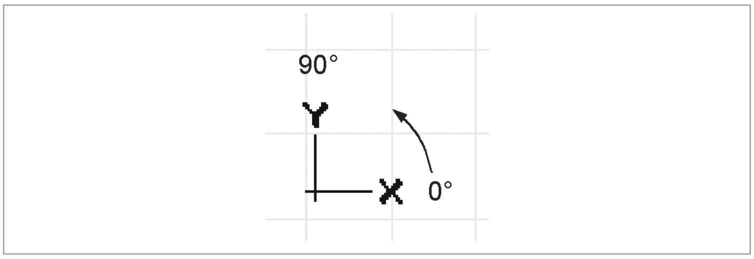


Fig. 46120

3. Enter the "Segment length" and confirm with <Enter>.

   The segment is drawn.

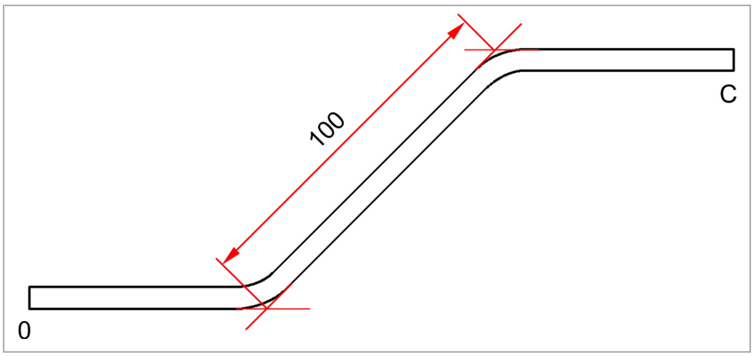
Programming further segments

4. Enter the other angles in relation to the previous segment and confirm them with <Enter>.

   Note

   Dimensioning is always performed from outside radius to outside radius.

5. Enter the "Segment length" of the other segments and confirm with <Enter>.



Dimensioning from outside radius to outside radius Fig. 46121

Profile table - entering angle and delta X or delta Y

Programming the first segment

1. Select *Profile table*.
2. Enter the first angle in relation to the coordinate system under "Angle" and confirm with <Enter>.

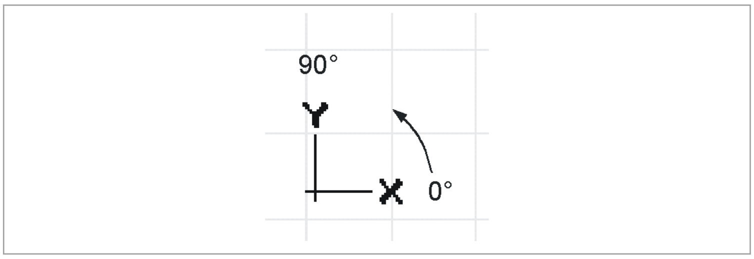


Fig. 46120

3. Enter the distance in X or Y direction under "dx" or "dy" and confirm it with <Enter>.

   The segment is drawn.

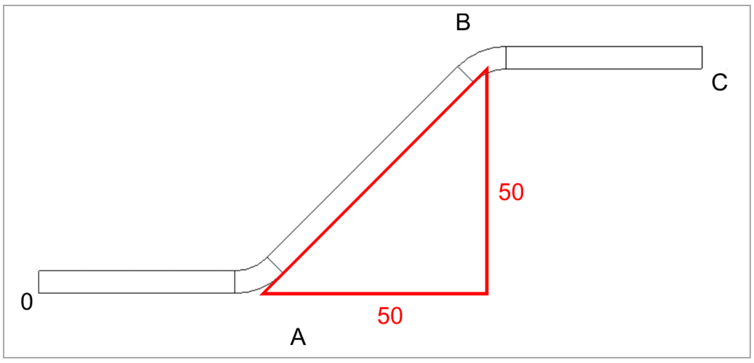
Programming further segments

4. Enter the other angles in relation to the previous segment and confirm them with <Enter>.

Note

The dimensioning always remains on one sheet side. The dimensioning changes from outside to inside dimensioning if the bending direction is changed.

5. Enter the distance in X or Y direction under "dx" or "dy" and confirm it with <Enter>.



Dimensioning on one sheet side

Fig. 45206

Profile table - entering delta X and delta Y

In this case the angle and the segment length are calculated on the basis of the triangle between the distance in X direction and the distance in Y direction.

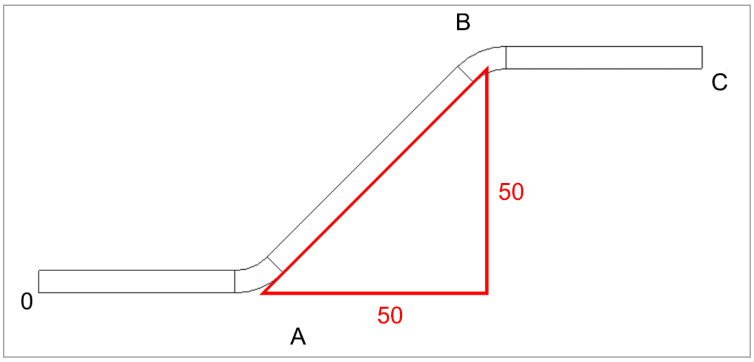
Dx and dy always refer to the coordinate system displayed in the profile editor.

1. Select *Profile table*.
2. Enter the distance in X direction under "dx" and confirm it with <Enter>.

Note

The dimensioning always remains on one sheet side. The dimensioning changes from outside to inside dimensioning if the bending direction is changed.

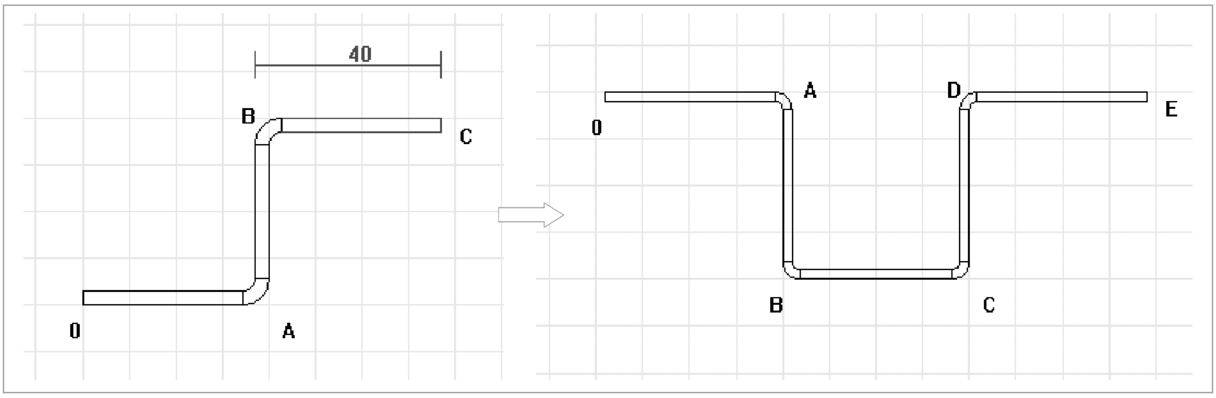
3. Enter the distance in Y direction under "dy" and confirm it with <Enter>.



Dimensioning on one sheet side Fig. 45206

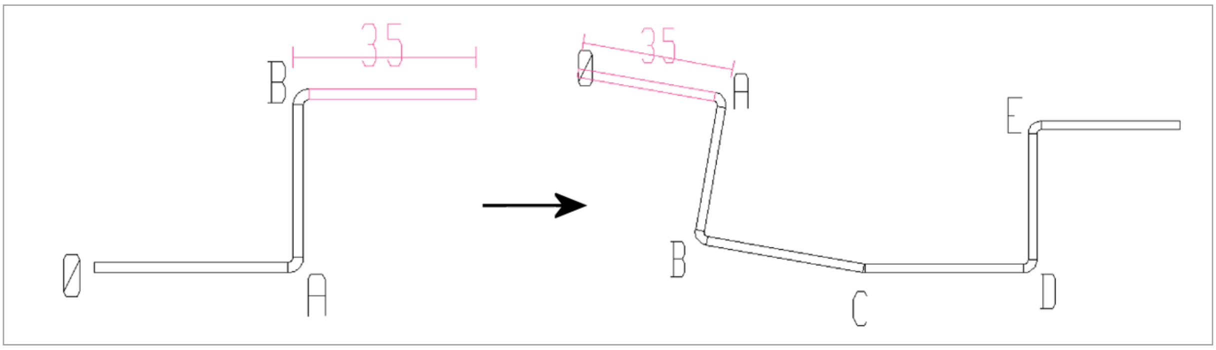
4.4 Modifying profiles

Edge mirroring For edge mirroring the first segment is divided at the center and the mirrored profile attached there. The origin is moved to the end of the new part.



Edge mirroring Fig. 55389

Point mirroring The profile is mirrored at its point of origin (at an angle of 180°). The origin is moved to the end of the added part and the "old" origin, like all segments, is renamed (In the below example the "old" origin is now named "C"). The angle at C can be changed by means of the profile table.



Point mirroring

Fig. 55390

Invert The origin is shifted to the end of the profile and the order of the segments is reversed. This is useful if segments are to be added at the origin as well.

Note

A new segment will always be attached to the last segment.

Turning The selected segment is rotated against the origin.

4.5 Deleting profiles

1. Select >*Bending*, >*Profile editor*, >*Deleting profile bending*.
2. Press *All*.

4.6 Deleting an element

1. Select >*Bending*, >*Profile editor*, >*Deleting profile bending*.
2. Select element.

All segments from the end of the profile up to the marked segment are deleted (the marked segment is not deleted). If a bend is marked, that bend is deleted too.

4.7 Displaying the dimensioning

The outside or inside dimension for every marked segment is displayed, depending on which dimensioning side was selected in the profile table.

4.8 Profile table

A profile table is created automatically when the profile editor is opened. To return from the settings to the profile table, select the button *Change to unfolding*.

Whenever a profile is drawn, the corresponding data (angle, segment length, dx, dy) is automatically entered in the profile table. These values can be edited at any time and the changes are automatically applied to the drawn profile (absolute coordinate system). The individual values can be selected with the cursor keys.

When a profile is drawn in profile editor using the profile table, the current effective length of the part is displayed in the mask header of the profile table after each new element is added.

Mask header of the profile table Fig. 55317

Field	Description
Profile depth	The depth of the profile corresponds to the length of the bending line.
Dimensioning side	Choice between inner and outer dimensioning.
Point	The origin and the bendings of the profile are automatically named with consecutive letters and are assigned to the respective angle in the table (infinitely, AA is assigned after Z etc.).
Angle	The angle can be changed in the table. The value and the profile are adapted. ▪ Select the input field. ▪ Enter the angle. ▪ Select <Enter>.
Segment length	The segment length indicates the length of a segment, measured from outer edge to outer edge or inner to inner edge. The length of the individual segments can be changed.
Dx/Dy	The horizontal and vertical changes in distance between two points are indicated by Dx/Dy. These two values can be used to directly enter and draw an angle.
Grayed out fields	The grayed out values from Dx and Dy can be changed. ▪ Select the respective gray field. ▪ With the context menu open, select the lines whose table values are to be changed (check marks indicate the values to be changed). If the first line is selected, the angle and segment length values are applied to the profile. If the fourth line is selected, the values for dx and dy are applied. Irrelevant values are grayed out.

Field	Description
	This button is used to open a new screen in which the bending parameters are stored. They can be edited here. The symbol on the button changes depending on the type of bending. Data relevant to the particular segment will be displayed, such as type of bending, tool, angle etc. If changes are made, they are reflected in the profile.
	These buttons are used to switch forwards and backwards between the individual bending operations.

Tab. 6-2

Normal bending

Tool

The upper or lower tool is selected via the buttons or the selection table. The tools are applied for all bendings with *Apply to all*.

Dimensioning method

Method	Description
Tangent	The program applies two tangents to the segments if the bending is calculated with the "Tangent" dimensioning method. The radius is created at their point of intersection.
Radius	If the bend is calculated according to the "Radius "dimensioning method, a tangent to the radius is created.

Tab. 6-3

Notes

- Depending on the setting, TruTops CAD uses the Trumpf bend allowances saved in the database for bend value (radius dimensioning method) or theoretical intersecting point (tangent dimensioning method) when creating an unfolding.
- The bend allowances for the radius dimensioning method are always selected for angles <30°.
- The bending radius depends on the tool and the material thickness.

Angle

You can change the angle here.

Side length

Here you can change the lengths of the bordering segments (on the selected angle).

Bending radius

You can enter the bending radius here. The default setting is 1.0 mm.

Round bending

If you select the "Round bending" bending type, two additional selection fields appear.

Number of segments

Here you can enter the number of segments with which the round bending is to be generated.

Notes

- If you change the number of segments for the bending operation, the required length of the segments is calculated automatically.
- TruTops Bend always selects a segment bending with whichever tool is selected for the bending. A round tool is not selected.

Segment length

This field displays the length of the individual segments.

Note

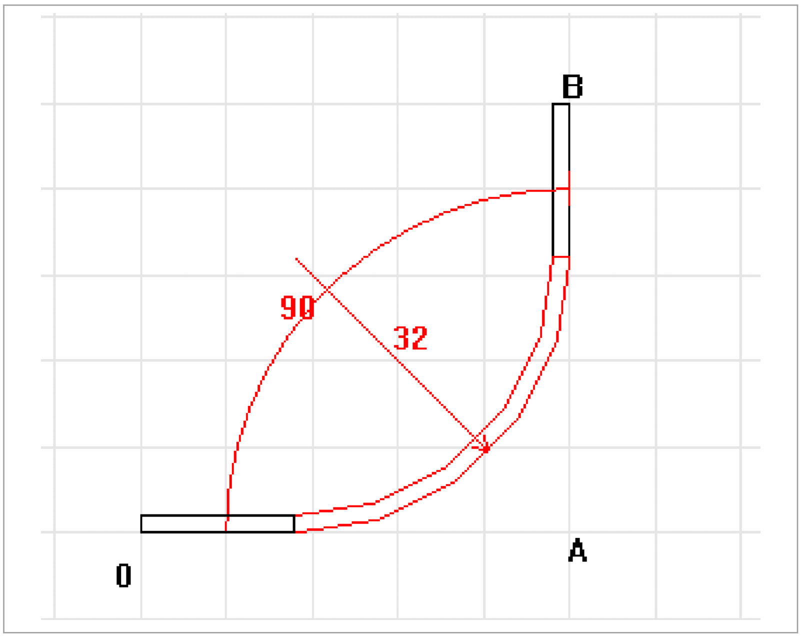
If you change the length of the segments, the number of segments is automatically adapted to generate the desired round bending.

Bending radius

The bend radius is particularly important for the round bending and must not be confused with the angle.

Note

The material thickness is always added onto the entered bending radius.



Round bending with 5 segments (90° angle and 30° bend radius + 2 mm sheet thickness) Fig. 41599

Folding

If you select the "Folding" bending type, the same options as for a normal bend will appear. In addition, here you can specify the folding direction.

Note

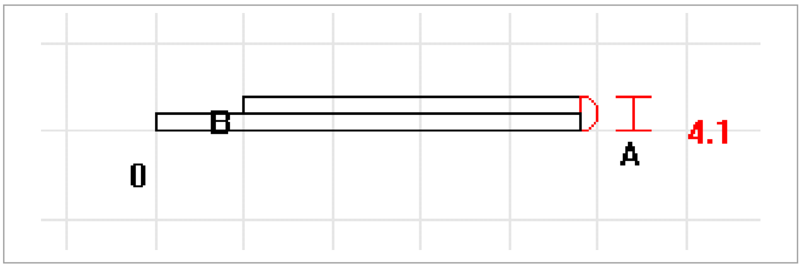
If angle 0° is entered in the profile table, the folding bend type will be selected automatically.

Folding direction

By entering the folding direction, "Up" or "Down", you can specify the direction of the folding.

Notes

- For folding, the bend radius is added twice to the double material thickness (height of the folding = 2x bending radius + 2x material thickness).
- A prebending at 30° is generated automatically for the folding when developing the unfolding.



Top folding (0° angle, 0.05 mm bend radius, 2 mm sheet thickness) Fig. 41600

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Software manual

TruTops

Simple drawing and programming

Software manual

TruTops

Simple drawing and programming

Edition: **10/2006**

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Before you continue reading...

The aim of this software manual is to introduce, after a short overview, the scope of functions of the "Simple drawing" and "Simple programming" modules using two examples.

This software manual complements the software manuals in the TruTops product family.

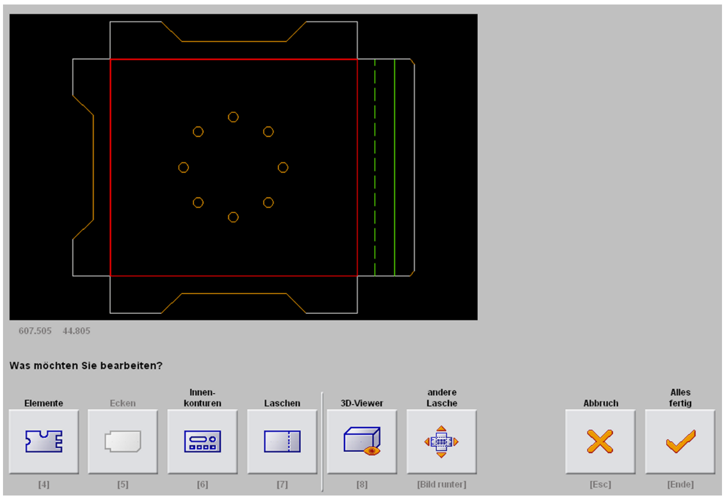
Table of contents

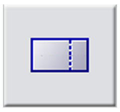
1. Basics

The vast majority of all workpieces are single parts. The "Simple drawing" and "Simple programming" modules in the TruTops production system offer exactly the scope of functions that are needed for single workpieces.

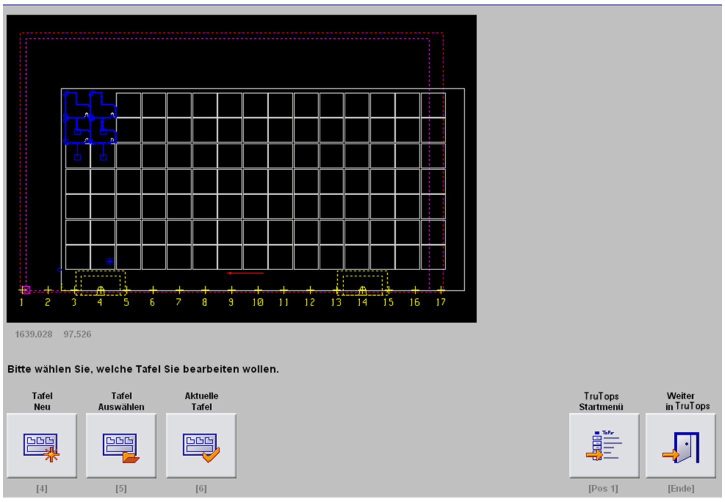
From drawing the workpiece to the reliable NC program - the user-guided, graphical user interface guides you safely and quickly through the program. The design of the operating environment enables operation per touch or completely via the keyboard.

Simple drawing



	Elements
	Corners
	Inner contours
	Flanges

Simple programming



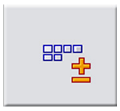
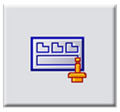
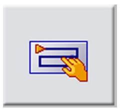
	Layout, changing numbers of parts
	Editing
	Repositioning
	Optimizing
	NC program, generating

Table 1

Note
The simple modules are integrated in TruTops basic, for programming directly at the machine, and integrated in the TruTops full version. In addition, TruTops basic is also available as a PC version.

TruTops basic does not have any nesting modules.

1.1 Opening simple modules

There are two variants for opening and closing the "Simple drawing and programming" modules:

- Opening via TruTops basic directly at the machine.
- Opening via TruTops full version at the PC workstation.

Note
In TruTops basic and in the TruTops full version, an extensive software documentation is available above the question mark in the form of online help.

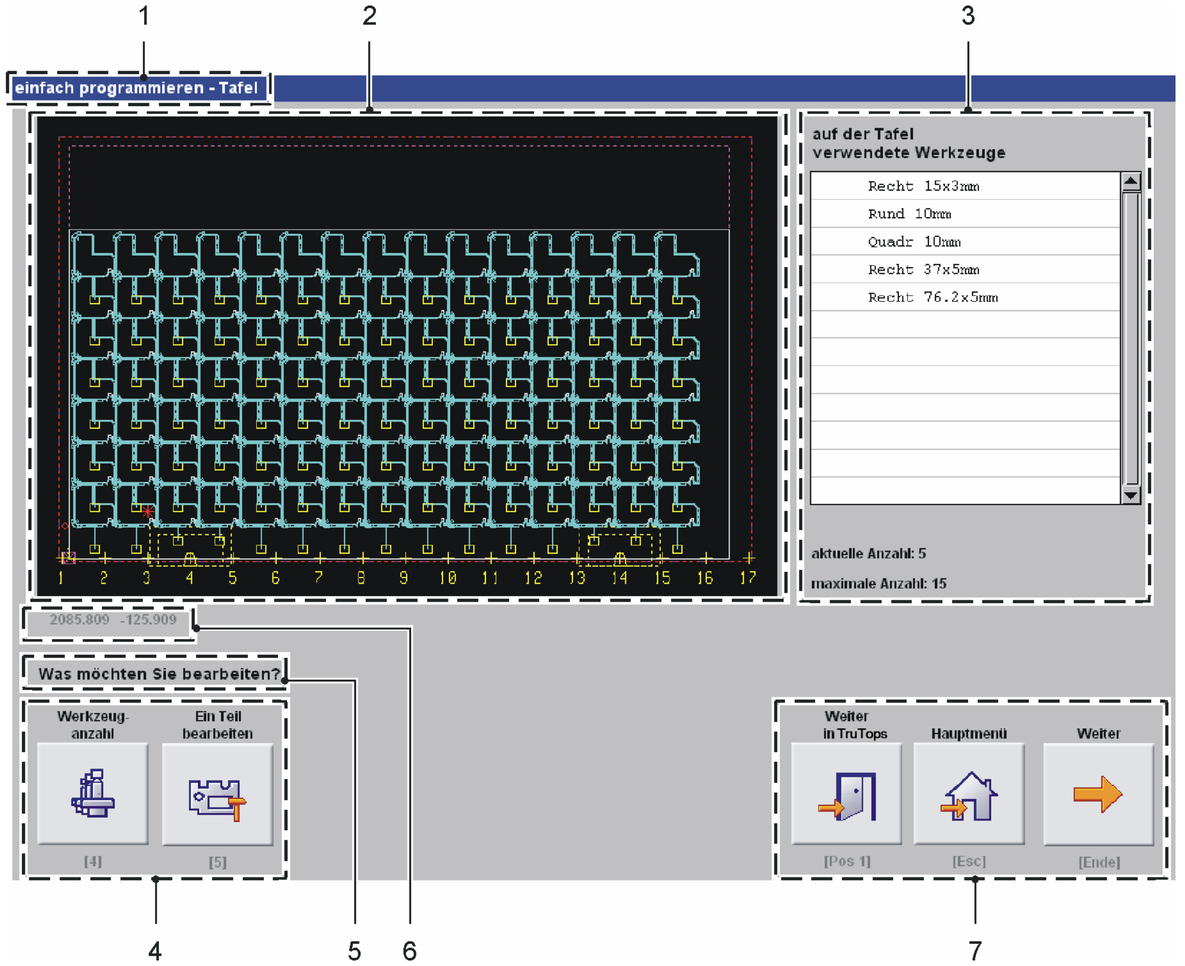
Opening TruTops basic

1. Open TruTops basic at the machine.
   The initial screen is displayed.
2. Press on *Drawing* or *Technology*.
   "Simple drawing" or "Simple programming" is immediately opened.

Opening TruTops full version

1. Open the TruTops full version at the PC workstation.
   The initial screen is displayed.
2. Press on *Drawing* or *Technology*.
   The "Drawing" or "Technology" module is opened.
3. Select *File> Simple drawing* or *>Simple programming*.
   "Simple drawing" or "Simple programming" is opened.

1.2 Subdivision of the user interface



1	Guidance	4	Main actions	6	Coordinates
2	Graphics window	5	Prompt	7	Navigation (continue, undo...)
3	Information or entry				

Subdivision of the user interface Fig. 47752EN

Zooming in the graphics window

The mouse pointer becomes a crosshair cursor when it is above the graphics window. A random, framed in cut out can be enlarged by using the left mouse button. The overall view is displayed again via the right mouse button.

1. Left click on the desired area.
2. Pull up a frame using the mouse.
3. Click once again with the left mouse button.
   The framed cut out is displayed in enlarged form.
4. Right click using the right mouse button.
   The overall view is displayed.

1.3 Operating simple modules using the keyboard

The "Simple drawing and programming" modules can be operated using the mouse, per touch screen, and via the keyboard.

There is a subheading in square brackets beneath every touch button. The subheading matches the designation of a key in the number block on the keyboard.

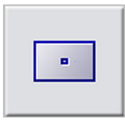
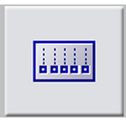
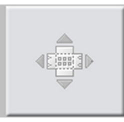
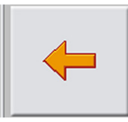
Single hole	Row of holes/ Hole grid	Circle of holes	Select hole geometry	Coordi-nates system	Other flange	Delete/ Restore	Undo	Inner contours finished
								
[4]	[5]	[6]	[Picture up]	[Pos 1]	[Picture down]	[2]	[Esc]	[End]

Table 2

1.4 Configuration

Optimizing keyboard operation

To enable you to operate the "Simple drawing and programming" modules optimally using the keyboard, TruTops key functions must be set.

➢ Select *Data module >User >Configuration,* "Set TruTops key functions".

1.5 Entering values using the formula calculator

Using the formula calculator, you can:

- Carry out basic calculator operations (point before dash).
- Transfer results into the entry fields.

Opening formula calculator

1. **Either**

   ➢ Double click on the entry field using the mouse.

   or

   ➢ Press the <+> key in the entry field.

Executing basic calculator operations

2. Enter for example `(10+20)*5-(40/2)+10`.

   Note
   A special feature of the formula calculator is that the formula stays displayed in the entry field and can be modified.

3. Press *Calculate*.
   The result `140` is displayed.
4. Press *Accept*.
   The result will be adopted in the entry field.

1.6 Setting and operating the 3D-Viewer

The 3D-Viewer shows three-dimensional models of the drawn geometries.

Setting the 3D-Viewer

➢ If the 3D model of the geometry is not displayed: select *View >Filled* in the menu line for the 3D-Viewer.

Rotating 3D model, changing size

1. To rotate the 3D model: keep the left mouse button pressed down.
   A double arrow is displayed.
2. Move the mouse with the left mouse button pressed down.
   The 3D model moves in the corresponding direction.
3. To enlarge or make the 3D model smaller: keep the right mouse button pressed down.
   A double arrow is displayed.
4. Move the mouse upwards with the right mouse button pressed down.
   The 3D model is made smaller.
5. Move the mouse downwards with the right mouse button pressed down.
   The 3D model is enlarged.

TRUMPF

1.7 Meaning of colors and symbols

In the standard settings, colors and symbols in the *Technology* module and in the "Simple drawing and programming" modules have the following meaning:

Color	**Symbol, line type**	**Meaning**
Geometries		
Cyan	Asterisk	Point for dot point marking
Yellow	Line type: solid	Marking and/or engraving (adjustable)
Yellow	Line type: dotted	Vaporizing (adjustable)
Red	All line types	Geometry that is not to be processed
Red	Frame	Only "Simple drawing and programming": • The subsequent action is related to the object framed in red
Red	Line	Only "Simple drawing and programming": • Automatic suggestion that can be confirmed or rejected
White or black	Line type: solid	• Sheet, geometry • Circumscribing rectangle with symbolic parts display
White or black	Line type: solid	Geometry that is to be processed normally (approach and cut)
White or black	Line type: dashed	Geometry that is to be cut without being approached
Operations		
Blue	Line type: solid	• Traversing paths between processed parts • Machining operations in non-active clamping range
Blue	Line type: dashed	Travel limits of the machine; non-active station
Blue	Arrow	Repositioning of non-active clamping
Blue	Square	Ejection via chute
Cyan	Cross	Tool center for punching
Cyan	Line type: solid	Punching operation
Cyan	Square	Ejection via chute
Yellow	Line type: solid	• Non-machined geometries/contours • Current processing in the simulation
Yellow	Coordinates X,Y	Zero point of the machine
Yellow	Circle	Collision point for processing definition
Yellow	Cross	• Positions on the transverse rail for tools and clamps • Tool center when processing with presser foot
Yellow	Rectangle, solid line type	When standard tool is active: Clamping dead range for the standard tool When MultiTool is active: Maximum clamping dead range for the MultiTool type affected

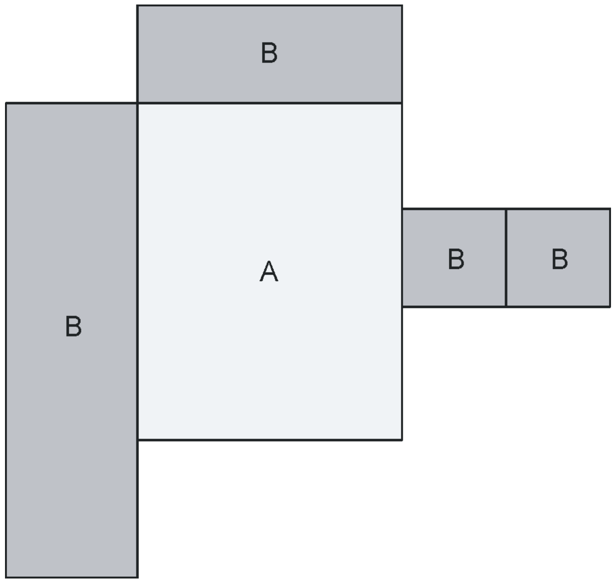
Color	Symbol, line type	Meaning
Yellow	Rectangle, dashed line type	When standard tool is active: Maximum clamping dead range for MultiTools. When MultiTool is active: Clamping dead range for standard tools
Yellow	Trapezoid	Clamping surface for clamps
Yellow	Circle in dashes	Repositioning device
Yellow	Triangle (arrow)	In connection with positioning motion: • Direction of the positioning motion • Start point of the next processing
Yellow	Square	Ejection via chute
Green	Line type: solid	• Laser processing • All other machining operations of a compulsory sequence
Green	Line type: dashed	• Approach flag for laser processing • First processing in a compulsory sequence • Traversing paths between the parts of a processing sequence
Green	Square	Ejection via punching flap
Green	Cross	Tool center when processing with delayed single stroke
Magenta	Line type: solid	Selected objects
Magenta	Line type: dashed	Working range of the machine
Magenta	Square	Ejection via laser flap
Red	Line type: solid	Geometry and processing of the first part in the positioning sequence with interactive optimization
Red	Line type: dashed	Travel limits of the machine, active station
Red	Asterisk	Punching head position during loading
Red	Square	• Ejection with chute during punching • Initial processing with interactive optimization
Red	Square with diagonal	Ejection with push-out subroutine
Red	Triangle	Manual removal after machine stop
Red	Circle/Rectangle	Stop pin without/with sensor
Red	Arrow	Repositioning of active clamping
Red	Frame	Only "Simple drawing and programming": • The subsequent action is related to the object framed in red
Red	Line	Only "Simple drawing and programming": • Automatic suggestion that can be confirmed or rejected

Colors and symbols

1.8 What is important for bending parts

Base plus flanges

A bending part comprises of a base and a random amount of flanges. The base is the surface in the drawing from where the most important or most bends start out.
If the base is determined in the construction drawing of a bending part, then several flanges are added to this.



A Base B Flanges

Configuration of bending parts Fig. 46376

Bend factors and relief grooves

What do bend factors produce?

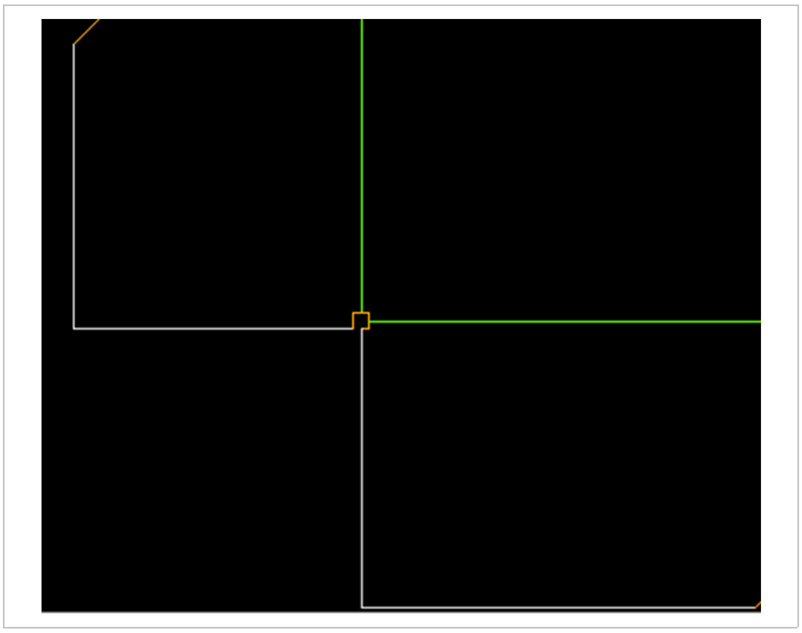
During bending, the material is stretched or flattened. The change in the workpiece length during the bending process is compensated with the use of so-called bend factors.

In "Simple drawing" the drawing is made with the original dimensions from the two dimensional construction drawing.

To enable the finished bent workpiece to correspond with the required dimensions, the dimensions of the flat blank are automatically shortened or extended by the bend factors.

Why are relief grooves necessary?

A relief groove is a "small notch". It reduces deformation of the edge or corner during bending.



Example of a corner relief groove Fig. 46548

Data for bending

The database

In accordance with the material ID and with the sheet thickness, the "Simple drawing" module selects the upper and lower tool combination from a database, and displays the bend factor associated with it.

Where do the data for bending come from?

1st way:

- An individual database has been created. Creating the bending data requires expert knowledge. The data from the individual database are always queried first.

2nd way:

- There is a connection to TruTops Bend. "Simple drawing" reads off the data from this database.

3rd way:

- "Simple drawing" calculates the bend factors in accordance with DIN if the individual database is empty or a connection to TruTops Bend does not exist.

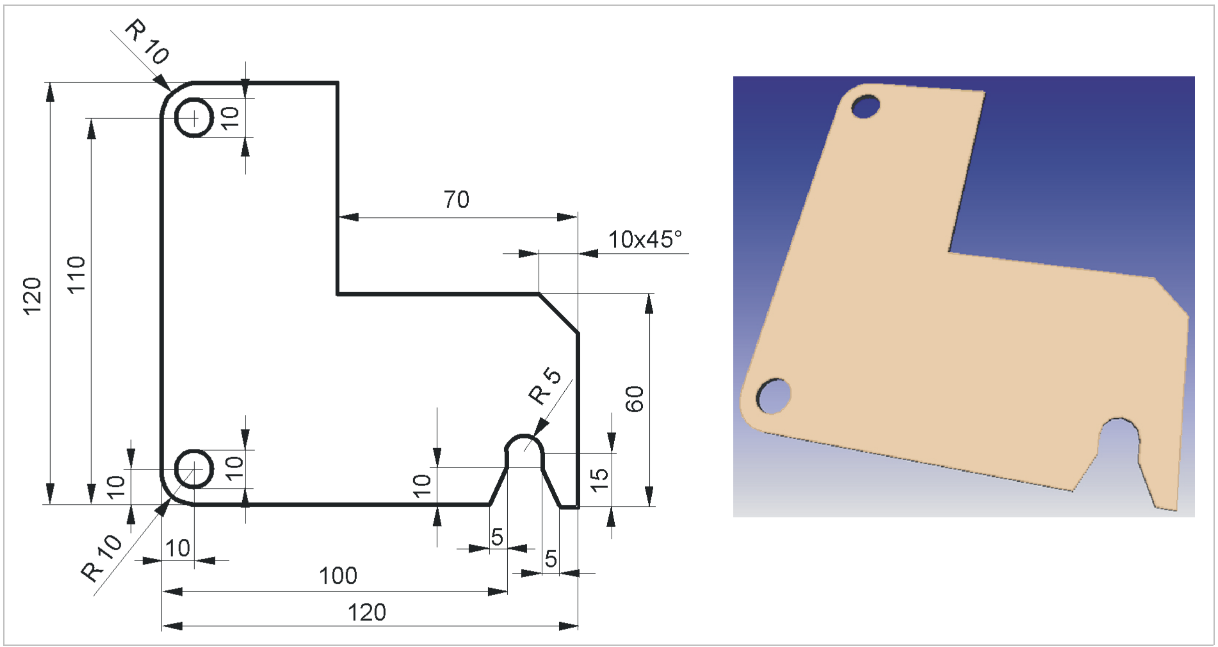
Creating individual bend factors and relief grooves

1. Press *Manage*.
2. Press *Manage bend factors* or *Manage relief grooves*.
3. Press *New*.
4. Enter data for bend factors or relief grooves.
5. Press *Enter*.

The new data are entered into the selection field and into the individual database.

2. Simple drawing

2.1 Example: drawing a flat workpiece

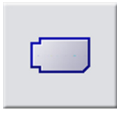


Flat workpiece: "L-plate" Fig. 46531

Creating new drawing

1. Press *New drawing*.
2. Enter or select material:
   - "Designation": `St37-20`.
   - "Thickness" `2` mm.
3. Press *Continue*.
4. Enter "Base measurements":
   - "Horizontally": `120` mm.
   - "Vertically": `120` mm.
5. Press *Continue*.

Drawing rectangular notch



1. Press *Corners*.
2. Press *Rectangular notch*.
3. Enter notch values:
   - "Horizontally": `70` mm.
   - "Vertically": `60` mm.
   - "Relief groove": `0` mm.
4. Press *Continue*.
   A suggestion is displayed in red.
5. If the notch is not positioned at the correct place: press *No* repeatedly.
6. If the correct position has been reached: press *Yes*.
   The notch is drawn.

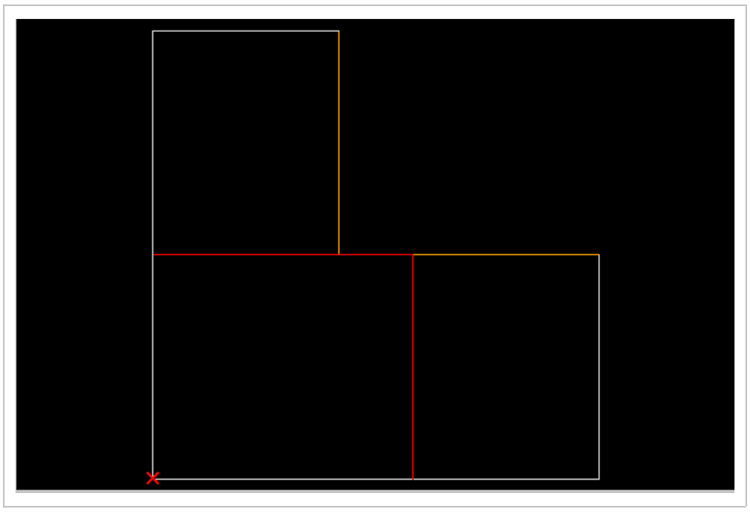


Fig. 46471

7. Press *Finished*.

Drawing square bevels

1. Press *Square Bevel*.
2. Enter bevel values:
   – "Distance": `10` mm.
3. Press *Continue*.

   A suggestion is displayed in red.
4. If the bevel is not positioned at the correct place: press *No* repeatedly.
5. If the correct position has been reached: press *Yes*.

   The bevel is drawn.

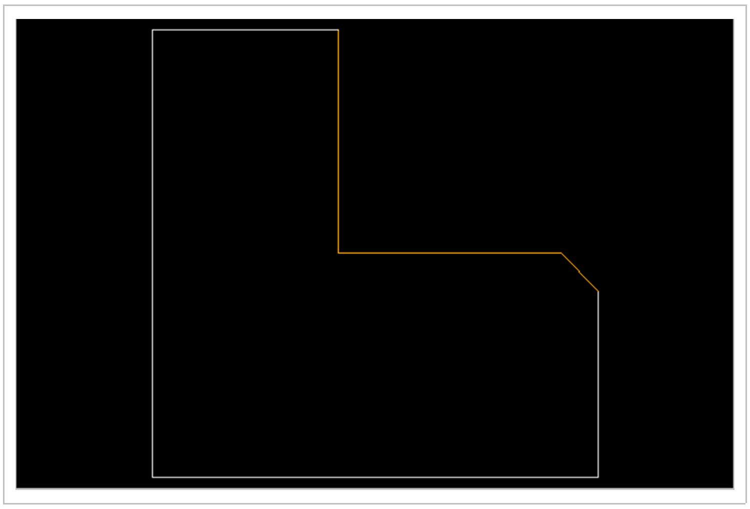


Fig. 46472

6. Press *Finished*.

Drawing roundings

1. Press *Rounding*.
2. Enter rounding data:
   – "Radius": `10` mm.
3. Press *Continue*.
   A suggestion is displayed in red.
4. If the first rounding is not positioned at the correct place: press *No* repeatedly.
5. If the correct position has been reached: press *Yes*.
6. To position the second rounding at the correct place: press *No* repeatedly.
7. If the correct position has been reached: press *Yes*.
   The roundings are drawn.

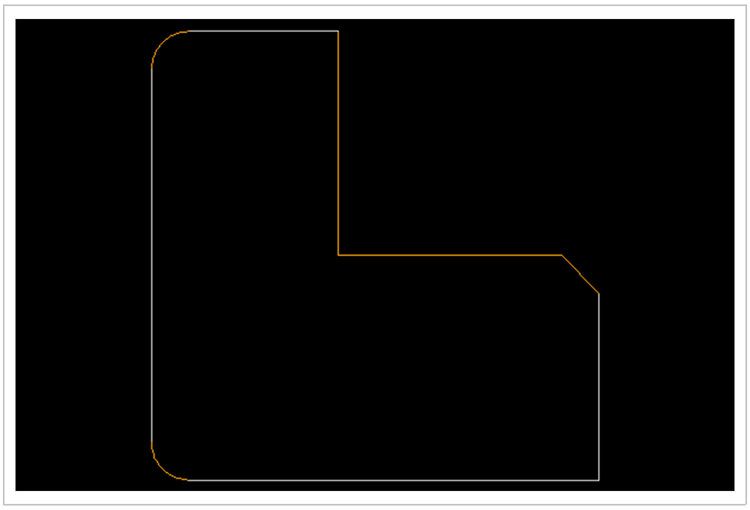


Fig. 46473

8. Press *Finished*.

Drawing a circle

1. Press *Inner contours*.
2. **Either**
   ➢ If an inner contour has not yet been drawn:
   – Press *Single hole*, *Circle*.

   or

   ➢ If an inner contour has already been drawn:
   – Press *Single hole, Select hole geometry*, *Circle*.
3. Enter circle data:
   – "Diameter": `10` mm.
4. Press *Continue*.
5. Enter the single hole position:
   – "to the right": `10` mm.
   – "to the top": `110` mm.
6. Press *Finished*.
7. Press *Single hole, Select hole geometry*, *Circle*.
8. Enter *Circle data*:
   – "Diameter": `10` mm.
9. Press *Continue*.
10. Enter the single hole position:
    – "to the right": `10` mm.
    – "to the top": `10` mm.

    The single holes are drawn.

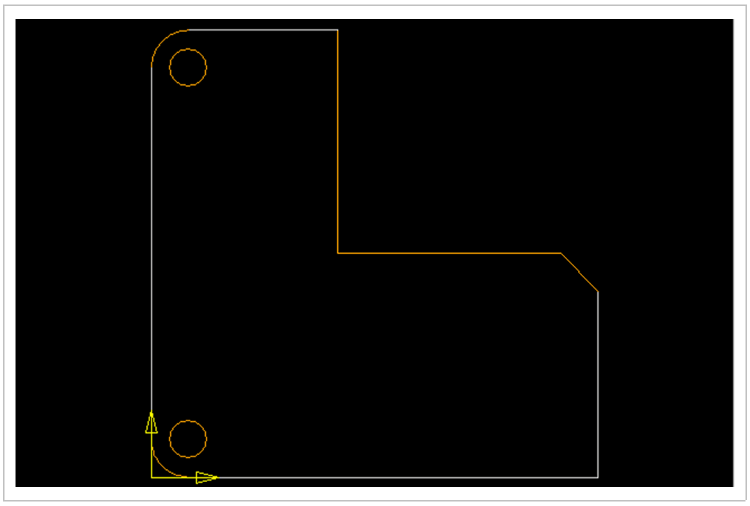
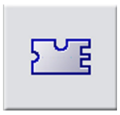


Fig. 46474

11. Press *Finished*.

Drawing oblong hole



1. Press *Elements*.
2. Press *Oblong hole*.
3. Enter oblong hole data:
   – "Depth": 20 mm.
   – "Width": 10 mm.
4. Press *Continue*.
   A suggestion is displayed in red.
5. If the edge on which the oblong hole is to be positioned is not correct: press *Next* or *Previous* repeatedly.
6. If the correct position has been reached: press *Continue*.

Changing reference to edge

7. Press *Change reference*.
8. Press *Right edge end*.
9. Enter position of notch center:
   – "Distance": 15 mm.
10. Press *Continue*.
    The oblong hole is drawn.

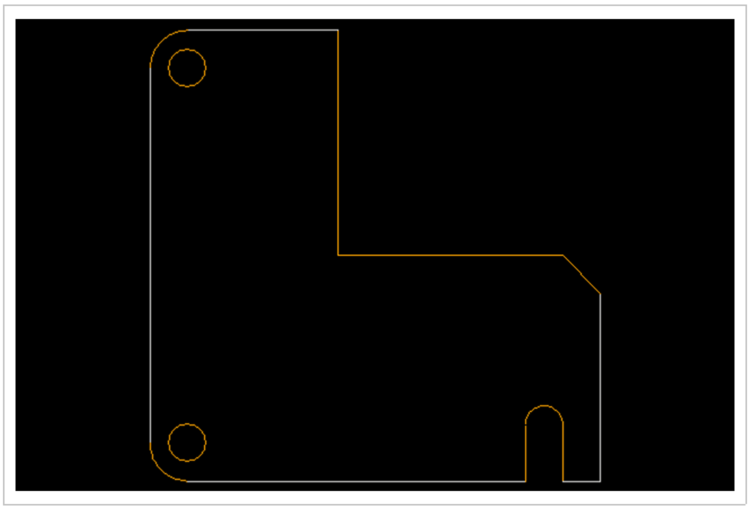
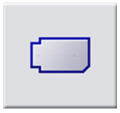


Fig. 46475

11. Press *Element finished*.

Drawing rectangular bevels



1. Press *Corners*.
2. Press *Rectangle Bevel*.
3. Enter bevel values:
   – "Horizontally": `5` mm.
   – "Vertically": `10` mm.
4. Press *Continue*.
   A suggestion is displayed in red.
5. If the first bevel is not positioned at the correct place: press *No* repeatedly.
6. If the correct position has been reached: press *Yes*.
7. If the second bevel is not positioned at the correct place: press *No* repeatedly.
8. If the correct position has been reached: press *Yes*.
   The bevels are drawn.

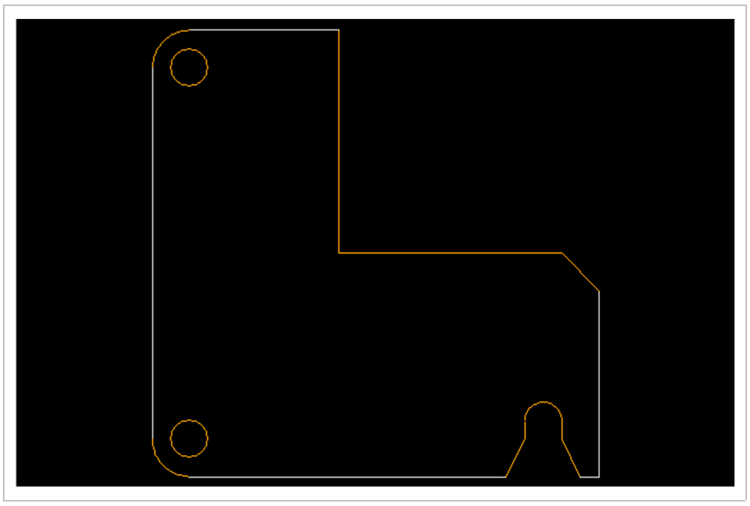


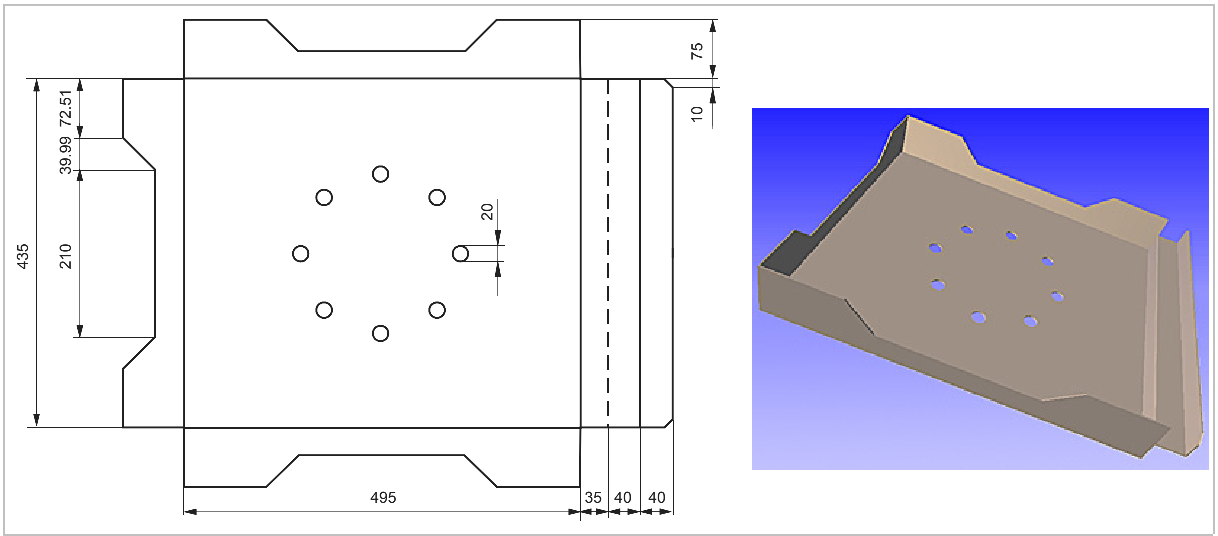
Fig. 46476

9. Press *Finished*.

Closing and saving drawings

1. Press *Corners finished*.
2. Press *All finished*.
3. Save part: press *Yes*.
   "File information" is opened.
4. Enter file information.
5. Select *OK*.
   "Save file" is opened.
6. Enter `L-plate` next to "File name".
7. Select *OK*.

2.2 Example: drawing a bending part



Bending part: "Box"

Fig. 46547

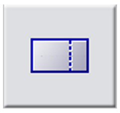
Creating new drawing

1. Press *New drawing*.
2. Enter or select material:
   – "Designation": `St37-20`.
   – "Thickness": `2` mm.
3. Press *Continue*.

   Note
   The base is the surface in the drawing from where the most important or most bends start out.

4. Enter "Base measurements":
   – "Horizontally": `495` mm.
   – "Vertically": `435` mm.
5. Press *Continue*.

Drawing flanges



1. Press *Flanges*.
2. Press *Generate*.
   The lower edge is marked in red.
3. If the correct edge upon which the flange is to be drawn is not selected:
   – Press *Previous* or *Next*.
4. Press *Continue*.

Drawing lower flange

5. Enter flange values:
   – "Height": `75` mm.
6. Enter the distances:
   – "from the left": `0` mm.
   – "from the right": `0` mm.
7. Press *Continue*.
8. Specify flange direction:
   – Press *To the top*.
   The flange is drawn. The bending line is displayed in green.
9. To display the 3D model of the bending part: press *3D-Viewer*.

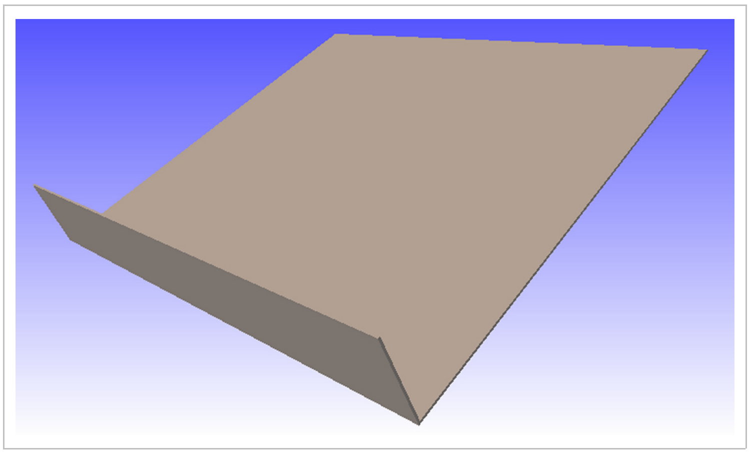


Fig. 46477

10. Press *Generate*.
11. To select the upper edge:
    – Press *Previous* or *Next*.
12. Press *Continue*.

Drawing upper flange

13. Enter flange values:
    – "Height": `75` mm.
14. Enter the distances:
    – "from the left": `0` mm.
    – "from the right": `0` mm.
15. Press *Continue*.
16. Specify flange direction:
    – Press *To the top*.

    The flange is drawn. The bending line is displayed in green.
17. Press *Generate*.
18. To select the left edge:
    – Press *Previous* or *Next*.
19. Press *Continue*.

Drawing left flange

20. Enter flange values:
    – "Height": `75` mm.
21. Enter the distances:
    – "from the left": `0` mm.
    – "from the right": `0` mm.
22. Press *Continue*.
23. Specify flange direction:
    – Press *To the top*.

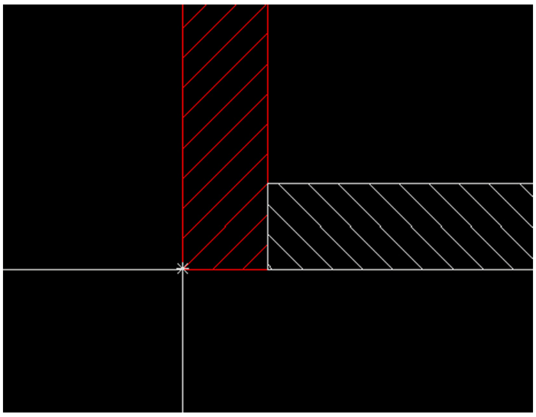
    The flange is drawn. The bending line is displayed in green.

Defining flange overlapping

24. To define flange overlapping: press *Left*.

Notes

- The touch buttons for the overlapping are not pressed in.
- The overlapping of the flanges is represented with shaded bar displays. The overlapping can be controlled via the positioning of the bars to one another.



Overlapping of flanges Fig. 46532

25. Press *Continue*.

    The "*" symbol is displayed at the next corner.

26. To define flange overlapping: press *Left*.
27. Press *Continue*.
28. Press *Generate*.

Drawing first right flange

29. To select the right edge:
    – Press *Previous* or *Next*.
30. Press *Continue*.
31. Enter flange values:
    – "Height": `35` mm.
32. Enter the distances:
    – "from the left": `0` mm.
    – "from the right": `0` mm.
33. Press *Continue*.
34. Specify flange direction:
    – Press *To the top*.

    The flange is drawn. The bending line is displayed in green.
35. To define flange overlapping: press *Bottom*.
36. Press *Continue*.

    The "*" symbol is displayed at the next corner.
37. To define flange overlapping: press *Top*.
38. Press *Continue*.
39. To display the 3D model of the bending part: press *3D-Viewer*.

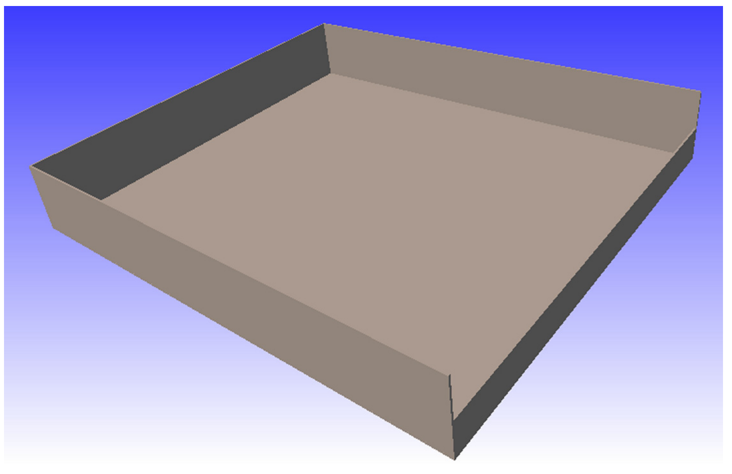


Fig. 46478

Drawing second right flange

40. Press *Generate*.
41. To select the right edge:
    – Press *Previous* or *Next*.
42. Press *Continue*.
43. Enter flange values:
    – "Height": `40` mm.
44. Enter the distances:
    – "from the left": `0` mm.
    – "from the right": `0` mm.
45. Press *Continue*.
46. Specify flange direction:
    – Press *to the bottom*.

    The flange is drawn.
47. To define flange overlapping: press *None*.
48. Press *Continue*.

Drawing third right flange

49. Press *Generate*.
50. To select the right edge:
    – Press *Previous* or *Next*.
51. Press *Continue*.
52. Enter flange values:
    – "Height": `40` mm.
53. Enter the distances:
    – "from the left": `0` mm.
    – "from the right": `0` mm.
54. Press *Continue*.
55. Specify flange direction:
    – Press *To the top*.

    The flange is drawn.

56. To display the 3D model of the bending part: press *3D-Viewer*.

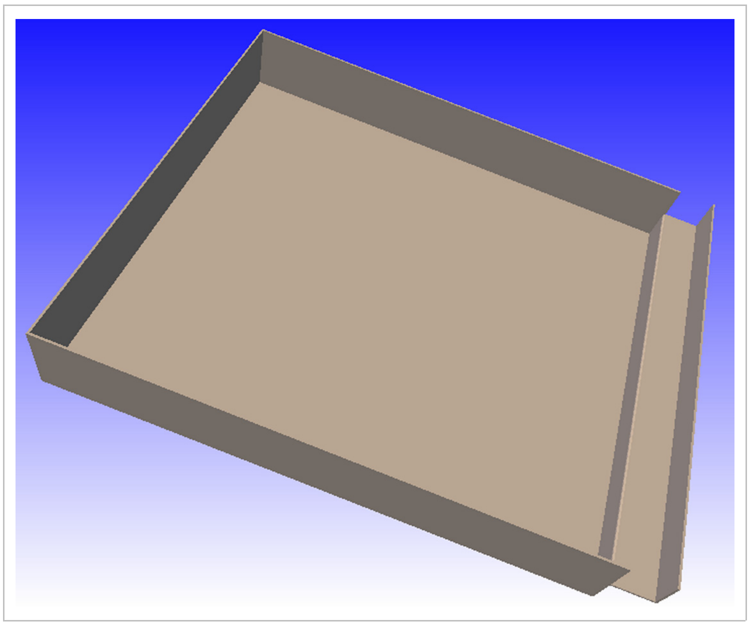
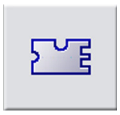


Fig. 46479

57. Press *Flanges finished*.

Drawing notches at elements



1. Press *Elements*.
2. If the correct lower flange is not selected:
   - Press *Other flange*.
   - Navigate to the lower flange using the navigation touch button.
   - Press *Continue*.
3. Press *Rectangle*.

Drawing rectangular notch at lower flange

4. Enter rectangular values:
   - "Depth": `40` mm.
   - "Width": `210` mm.

   Note
   The entered values are self-sustaining. They don't have to be reentered for more notches.

5. Press *Continue, Continue*.

Changing reference for the notch

6. Press *Change reference*.
   - Press *Central*.
   - Press *Continue*.

   The "*" symbol marks the notch center at the element.
7. Press *Continue*.

   The notch is drawn.

Drawing rectangular notch at upper flange

8. To select the upper flange:
   - Press *Other flange*.
   - Navigate to the upper flange using the navigation touch button.
   - Press *Continue*.
9. Press *Rectangle*.
10. Press *Continue, Continue, Continue*.

    The notch is drawn.

Drawing rectangular notch at left flange

11. To select the left flange:
    - Press *Other flange*.
    - Navigate to the left flange using the navigation touch button.
    - Press *Continue*.
12. Press *Rectangle*.
13. Press *Continue, Continue, Continue*.

    The notch is drawn.

14. Press *Element finished*.
15. To display the 3D model of the bending part: press *3D-Viewer*.

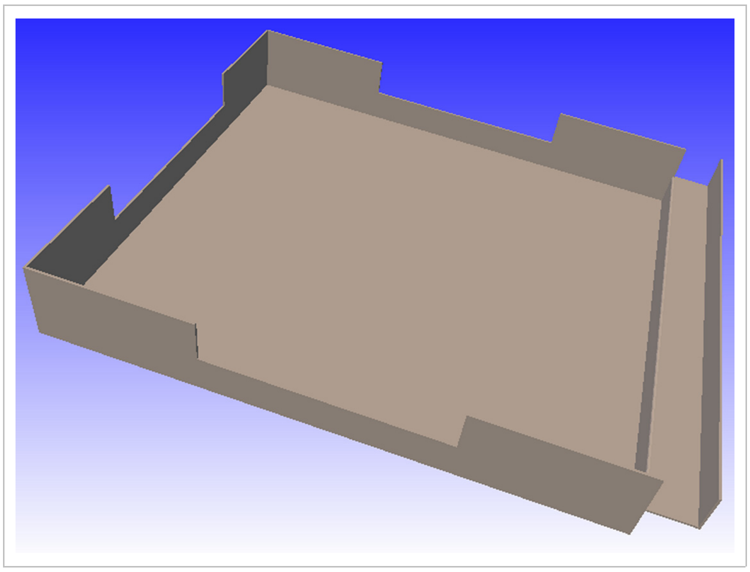
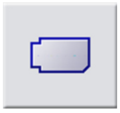


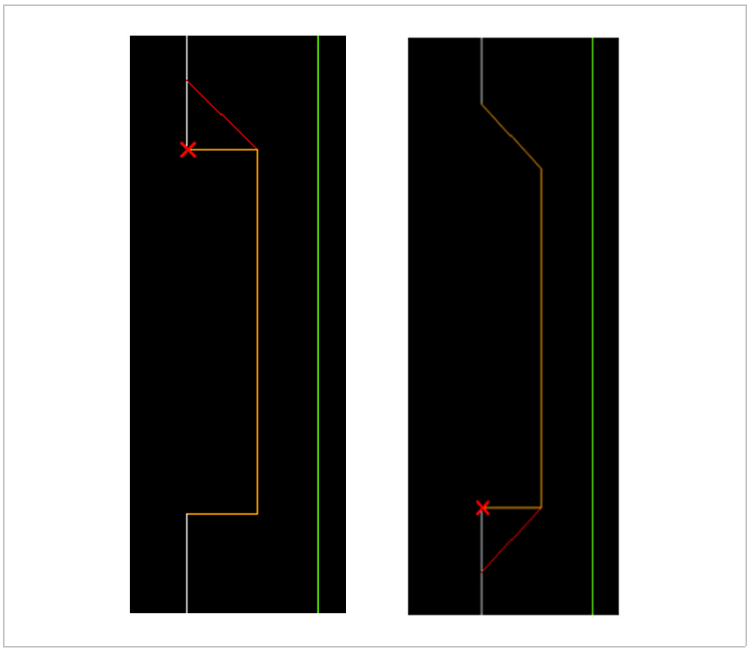
Fig. 46480

Drawing rectangular bevels



Drawing bevels for the upper, lower, and left flange

1. To select the left flange:
   – Press *Other flange*.
   – Navigate to the left flange using the navigation touch button.
2. Press *Corners*.
3. Press *Square Bevel*.
4. Enter bevel values:
   – "Distance": `39.99` mm.
5. Press *Continue*.
   A red cross marks the corner for the next action.
6. If the first bevel is not positioned at the correct place: press *No* repeatedly.



Corners to be processed (red cross) Fig. 46533

Note
Make sure that the red cross is positioned at the correct corner.

7. If the correct position has been reached: press *Yes*.
8. If the second bevel is not positioned at the correct place: press *No* repeatedly.
9. If the correct position has been reached: press *Yes*.
   The bevels are drawn.
10. Press *Finished*.

Drawing bevels for the right flange

11. Repeat steps for the upper and lower flange.
12. To select the right flange:
    - Press *Other flange*.
    - Navigate to the right flange using the navigation touch button.
13. Press *Corners*.
14. Press *Square Bevel*.
15. Enter bevel values:
    - "Distance": `10` mm.
16. Press *Continue*.
17. If the first bevel is not positioned at the correct place: press *No* repeatedly.
18. If the correct position has been reached: press *Yes*.
19. If the second bevel is not positioned at the correct place: press *No* repeatedly.
20. If the correct position has been reached: press *Yes*.
    The bevels are drawn.
21. Press *Finished*.
22. To display the 3D model of the bending part:
    - Press *Corners finished*.
    - Press *3D-Viewer*.

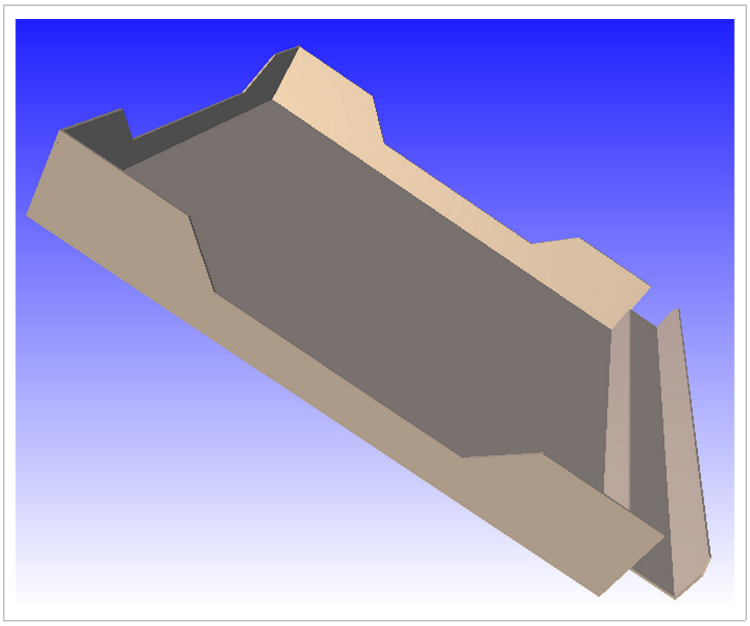


Fig. 46534

23. Press *Inner contours finished*.

Displacing coordinates system

1. Press *Inner contours*.
2. Press *Other flange*.
3. Press *Base*.
4. Press *Continue*.
5. Press *Coordinates system*.
6. Press *Bottom left*.

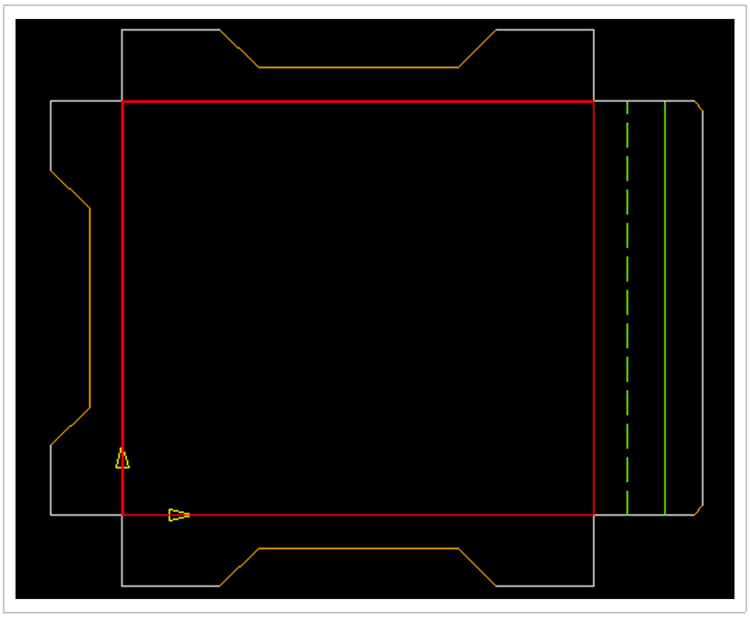


Fig. 46535

Drawing a circle of holes

1. Press *Inner contours*.
2. **Either**
   - ➢ If an inner contour has not yet been drawn:
   - – Press *Circle of holes*, *Circle*.

   or

   - ➢ If an inner contour has already been drawn:
   - – Press *Circle of holes, Select hole geometry*, *Circle*.
4. Enter circle data:
   - "Diameter": `20` mm.
5. Press *Continue*.
6. Enter center of hole circle:
   - "to the right": `247` mm.
   - "to the top": `217.5` mm.
7. Press *Continue*.
   The center is marked with a red cross.
8. Press the lower symbol under "Point/Angle definition".
9. Enter hole circle dimensions:
   - "Start angle": `0`°.
   - "End angle": `360`°.
   - "Radius": `100` mm.
   - "Number": `8`.
10. Press *Circle of holes finished*.
    The circle of holes is drawn.
11. Press *Inner contours finished*.

12. To display the 3D model of the bending part:
    – Press *3D-Viewer*.

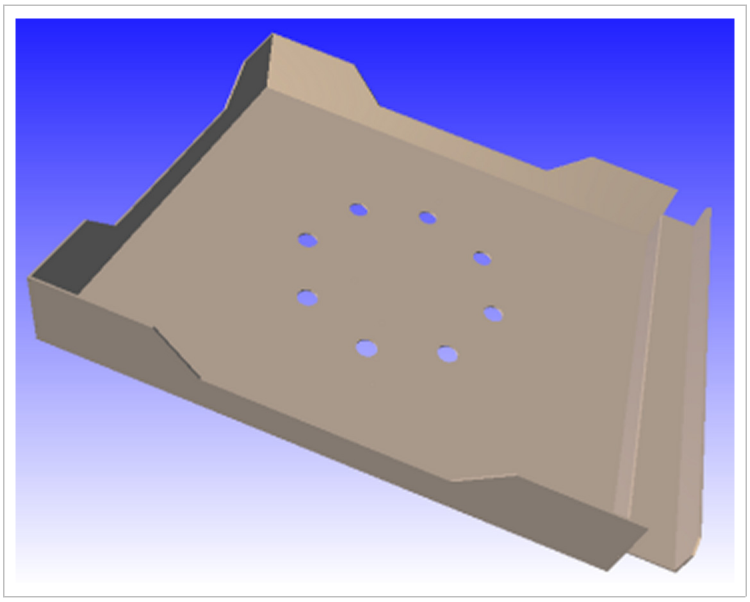


Fig. 46536

13. Press *All finished*.

Managing bend factors and relief grooves

(see Section 1.8, p. 19)

Drawing relief grooves - saving bending part

1. Press the lower symbol under "Corner relief groove", and upper symbol under "Edge relief groove".
2. Enter corner relief groove:
   - "Diameter": `4` mm.
3. Enter edge relief groove:
   - "Length": `5` mm.
   - "Width": `2` mm.
4. Press *Continue*.

   Note
   For the selected material ID and thickness, the entered values for the corner and edge relief grooves can be entered into the database.

5. Transfer values to database? Press *Yes*.

   Note
   The relief grooves are saved after saving the drawing.

6. Save part: press *Yes*.
   "File information" is opened.
7. Enter file information.
8. Select *OK*.
   "Save file" is opened.
9. Enter `Box` next to "File name".
10. Select *OK*.

The file is saved. The relief grooves are displayed. The bending part has been shrunk in accordance with the bend factor.

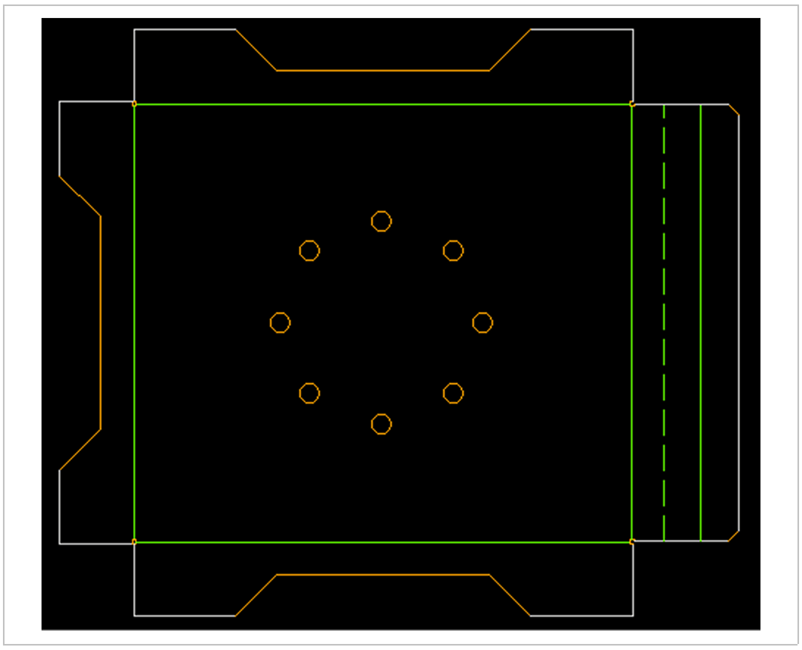


Fig. 46537

3. Simple programming

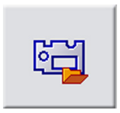
3.1 Example: programming L plate

Creating new sheet

1. Press *New sheet*.
2. Enter machine and sheet data:
   – "Machine": `TruPunch 1000`.
   – "Material ID": `ST37-20`.
   – "XxYxZ": `2000x1000x2` mm.
3. Enter margins:
   – "Left": `20` mm.
   – "Right": `20` mm.
   – "Top": `20` mm.
4. To set the lower margin: press on the lower icon under "Block clamping range".
   – Enter `100` mm.
5. Press Continue.

The empty sheet is displayed.

Load part



1. Press *Load part*.
2. Select file 'L-Platte.geo'.
3. Select *OK*.
   The part is displayed under "Part information".
4. Enter part information:
   – "Number of pieces": `30`.
   – "Angle": `0`°.
5. Enter web width for parts:
   – "X": `15` mm.
   – "Y": `15` mm.

Laying out the sheet

1. To change the start corner: press *Start corner*.
2. Press *Top left*.
3. Press *Continue*.
   The sheet layout is displayed:

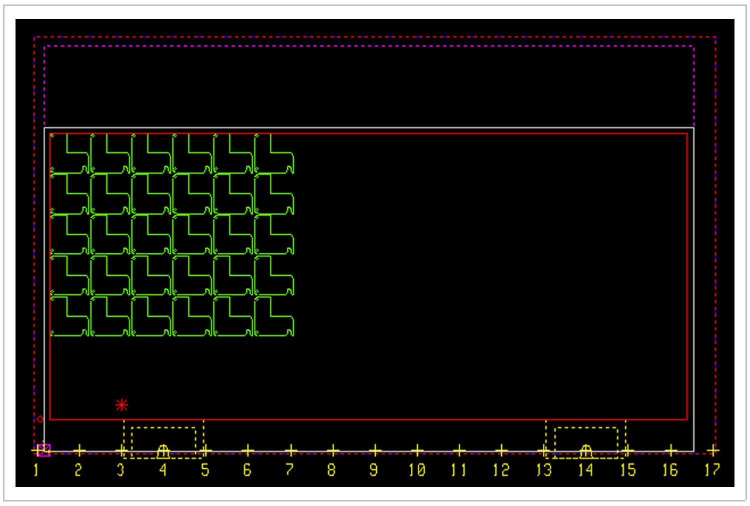


Fig. 46538

Changing block layout

4. Press *Format block*.
5. Press *+X* repeatedly.
   The modified sheet layout is displayed:

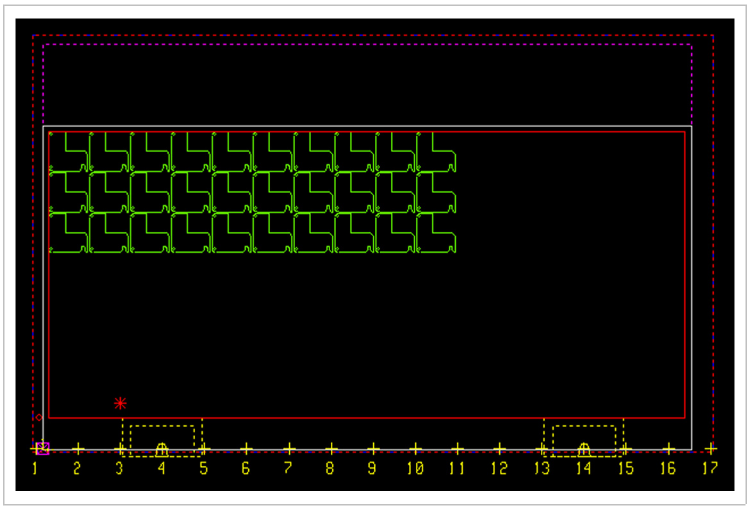


Fig. 46539

6. Press *Continue*.

Adding parts

7. Press *Change numbers of parts*.
   – Press +*X* repeatedly.
   – Press +*Y* repeatedly.

   The modified sheet layout is displayed:

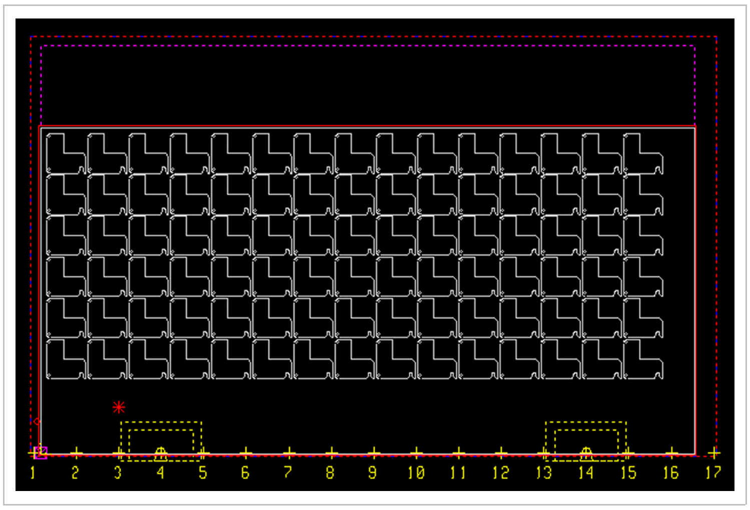


Fig. 46540

8. Press *Continue*, *Continue*.
   The final sheet layout is displayed.
9. Press *Continue to technology*.
   "Separating strategy" is opened.

Defining slitting strategy

➢ Press *Grid*.
"Automatic processing" is displayed.

Processing parts automatically

1. To create your own tooling list: press *Tooling lists*.
   "Tooling list management" is opened.
2. Select *New*.
   "Create new tooling list" is opened.
3. Enter name: e.g. `User1`.
4. In the tooling list: select *Circular 10 mm*.
5. Select *Undo*.
6. Define the tooling lists for automatic processing:
   – Select "Single hole": *User1*.
   – Select "Notch": *Trumpf*.
   – Select "Contour": *Trumpf*.
7. Press *Edit*.

   Automatic processing begins. The tools used are displayed in the information area on the right-hand side. Processings are marked in red.

   Note
   Tooling lists can still be changed at the present time for processings shown in red. *Undo* cancels the complete processing. All processings are shown in cyan after the next step. The named actions are then no longer an option.

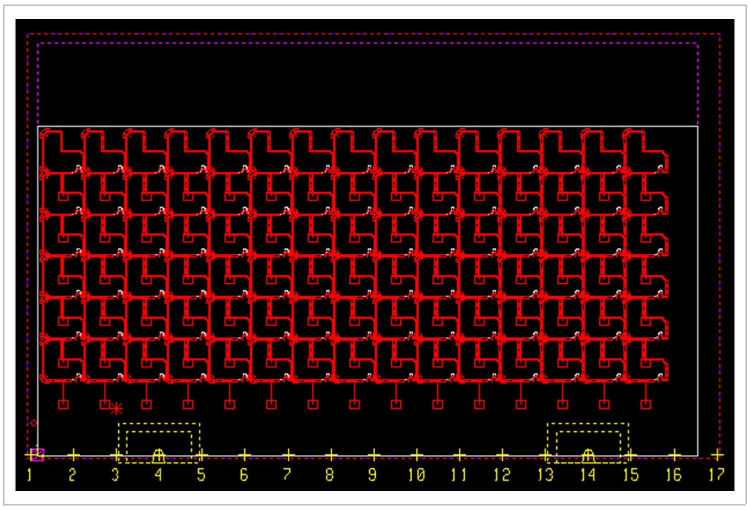


Fig. 46543

8. To check whether the processing is complete: make a frame around a part using the left mouse button.

   The part is shown in enlarged form.

Processing oblong hole manually

9. If the oblong hole is not processed: press *Process one part*.
   - Press *Elements, At element*.

   The element not processed is framed in red (in this example: straight element at oblong hole).
   - Press *Tool*.
   - In the selection list: select *Rectangle 10x5 mm*.
   - Select *OK*.
   - Press "Confirm processing for this element": press *Yes*.

   The next element not processed is framed in red (in this example: circular element at oblong hole).
   - Press *Tool*.
   - In the selection list: select *Circular 10 mm*.
   - Select *OK*.
   - Press "Confirm processing for this element": press *Yes*.

   The next element not processed is framed in red (in this example: straight element at oblong hole).
   - Proceed as described above for this element.
   - Press "Confirm processing for this element": press *Yes*.
   - Press *Finished*, *Elements finished*.

   The added processing is transferred to all equal elements. All processings are shown in cyan.

Changing number of tools, processing rounding

10. If the rounding has been processed with "Square 10": press *Number of tools*.
    - To display the processing of the rounding: press *Previous* or *Next* repeatedly.

    In the *Tools already used in part* list, the tool for the rounding is marked with a star.
    - Press *Delete*.

    Processing with this tool is removed.
    - Press *Add*.

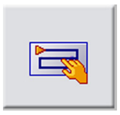
    "Tools" opens.
    - In the selection list: select *Square 5 mm*.
    - Select *OK*.
    - Press *Recalculate*.
    - Press *Finished*.

    The processing is recalculated. The processing of the rounding is executed using "Square 5".

Concluding processing

11. Press *Part finished*.
    The entire sheet is displayed.
12. Press *Continue*.
    "Trimming" is displayed.
13. Press *Continue*.
14. Confirm the message for repositioning with *Yes*.
    "Optimization" is displayed.

Optimizing the processing sequence



1. Press *Slitting sequence*.
   - Press *Meander top left*.
   - Press *Continue*.
2. Enter "Optimization parameters":
   - "Grouping of rows": `2`.

   "Tool sequence" opens.
3. Press *Previous tool*.

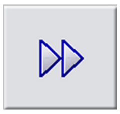
   In the list of tools already used, the "*" symbol wanders downwards by one position. The tool is marked.
4. Press *Tool to the bottom*.

   The marked tool is sorted in the list by one position downwards.
5. Press *Continue*.

   The "Simulation of tool change" message list is displayed.
6. Select *OK*.

   "Simulation" is displayed.

Simulating processing



1. **Either**
   - Press *Next processing*.

   The simulation is displayed step-for-step.

   or

   - Keep key <6> on the keyboard pressed down.

   The longer the key is pressed down, the quicker the simulation proceeds.
2. To the start the simulation from the beginning: press *Simulation to start*.
3. Press *Continue*.
   "Generate NC Program" is displayed (see Section 4, p. 56).

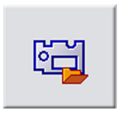
3.2 Example: programming bending part

Creating new sheet

1. Press *New sheet*.
2. Enter machine and sheet data:
   – "Machine": `TruPunch 1000`.
   – *Material ID*: `ST37-20`.
   – "XxYxZ": `2000x1000x2` mm.
3. Enter margins:
   – "Left": `20` mm.
   – "Right": `20` mm.
   – "Top": `20` mm.
4. To set the lower margin: press on the lower icon under "Block clamping range".
   – Enter `100` mm.
5. Press Continue.

The empty sheet is displayed.

Loading part



1. Press *Load part*.
2. Select the 'Kiste.geo' file.
3. Select *OK*.
   The part is displayed under Part information.
4. Enter part information:
   – "Number of pieces": `2`.
   – "Angle": `0°`.
5. Enter web width for parts:
   – "X": `15` mm.
   – "Y": `15` mm.

Laying out the sheet

1. To change the start corner: press *Start corner*.
2. Press *Top left*.
3. Press *Continue*.
   The sheet layout is displayed:

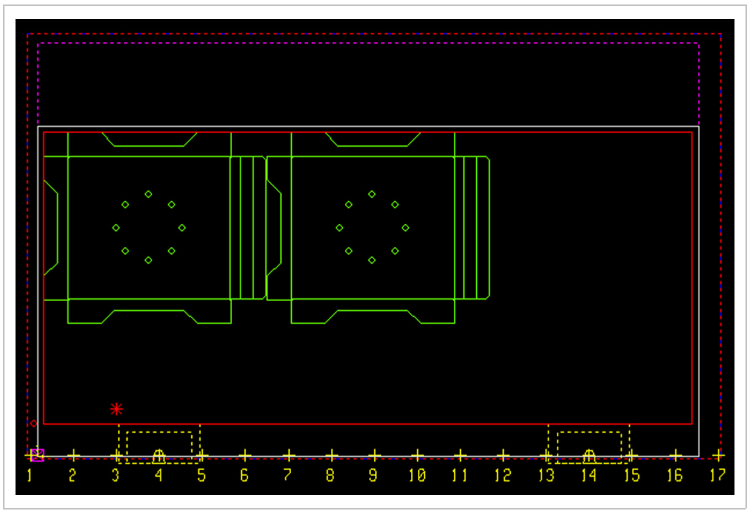


Fig. 46545

4. Press *Continue*, *Continue*.
   The final sheet layout is displayed.
5. Press *Continue to technology*.
   "Separating strategy" is opened.

Defining slitting strategy

➢ Press *Grid*.
"Automatic processing" is displayed.

Processing parts automatically

1. To create your own tooling list: press *Tooling lists*.
   "Tooling list management" is opened.
2. Select *New*.
   "Create new tooling list" is opened.
3. Enter name: e.g. `User1`.
4. In the tooling list: select *Circular 20 mm*.
5. Select *Undo*.
6. Define the tooling lists for automatic processing:
   – Single hole: select *User1*.
   – Notch: select *Trumpf*.
   – Contour: select *Trumpf*.
7. Press *Edit*.

   Automatic processing begins. The tools already used are displayed. The processed parts are shown in red.

   Note
   Tooling lists can still be changed at the present time for processings shown in red. *Undo* cancels the complete processing. All processings are shown in cyan after the next step. The named actions are then no longer an option.

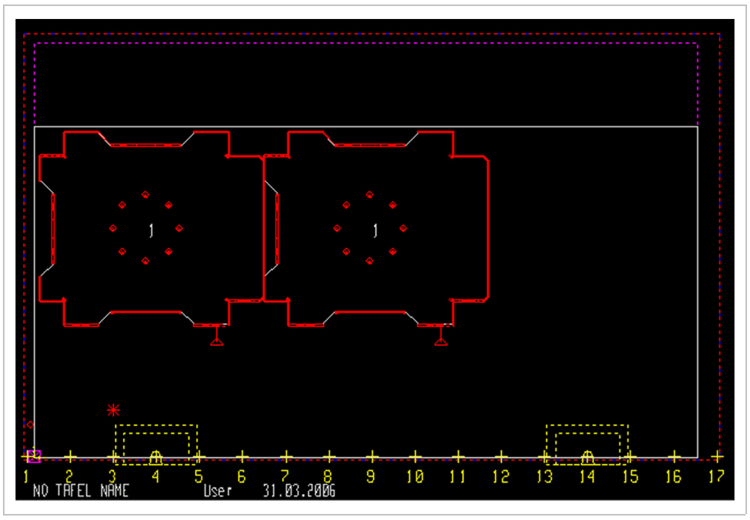


Fig. 46546

8. Press *Continue*.
   "Trimming" is displayed.

Trimming sheets

1. Press *Trimming, Strategy, U stripes*.
2. Press *Tool*.
3. In the selection list: select *Rectangle 70x5 mm*.
4. Select *OK*.
5. Press *Continue*.

   The start part (magenta color) at which the trimming cut begins is suggested.
6. Press *Continue*.

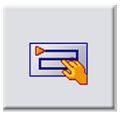
   The start part is confirmed.
7. To define the end part at the bottom right on the sheet: press *Previous* repeatedly.

   The end part (magenta color) at which the trimming cut ends is suggested.
8. Press *Continue*.
9. Confirm the messages for the trimming edge with *Yes* and *OK*.

   The trimming edge is displayed.
10. Press *Continue*.
11. Confirm the message for repositioning with *Yes*.

    "Optimization" is displayed.

Optimizing the processing sequence



1. Press *Slitting sequence*.
   - Press *Meander top left*.
   - Press *Continue*.
2. Enter "Optimization parameters":
   - "Grouping of rows": 1.

   "Tool sequence" opens.
3. Press *Previous tool*.

   In the list of tools already used, the "*" symbol wanders downwards by one position. The tool is marked.

4. Press *Tool to the bottom*.

   The marked tool is sorted in the list by one position downwards.

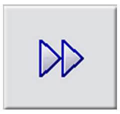
5. Press *Continue*.

   The "Simulation of tool change" message list is displayed.

6. Select *OK*.

   "Simulation" is displayed.

Simulating processing



1. **Either**

   ➢ Press *Next processing*.

   The simulation is displayed step-for-step.

   or

   ➢ Keep key <6> on the keyboard pressed down.

   The longer the key is pressed down, the quicker the simulation proceeds.

2. To the start the simulation from the beginning: press *Simulation to start*.

3. Press *Continue*.

   "Generate NC Program" is displayed (see Section 4, p. 56).

4. Generating an NC Program

4.1 Defining settings

➢ Select "Settings".

Connection between name of "NC file" and "Program name"

Machines with Bosch control systems

TruTops uses the digits from the name of the NC file. If the name of the NC file contains no digits, then TruTops will use the date and the time in sequence to generate a name that is as unequivocal as possible.

The program name and the name of the NC file can be modified at any time (see following pages).
For machines with Bosch control systems, only numerical program names (digits) can be used.

Machines with Siemens control systems

For machines with Siemens control systems and alphanumeric program names, TruTops offers the name of the NC file as the default program name.

Note
With alphanumeric program names no distinction is made between lower case and upper case letters.

Modifying the name of the NC file

TruTops uses the following NC file names:

- The file name of the first loaded part when generating multi-copy areas.
- The sheet name of the respective sheet when machining a nesting job.

1. Delete the names to be found under "NC file".
2. Enter New name.

Modifying NC program path

TruTops offers in the default settings the NC program path '/TRUMPF.NET/Workfiles/user1/'.

1. Open "NC program path" with "▾".
   The file browser opens.
2. Select New directory.
3. Select *OK*.

TruTops shows the new path under "NC program path."

Defining path for saving setup plan

TruTops offers in the default settings the path for setup plan '/TRUMPF.NET/Workfiles/user1/'.

1. Put a check mark at "Path for setup plan".
2. Open "Path for set-up plan" with ▾.
   The file browser opens.
3. Select New directory.
4. Select *OK*.

TruTops shows the new path under "Path for setup plan".

Modifying program name

1. Delete old program name.
2. Enter new program name.
   Note
   For machines with Bosch control systems, only numerical program names (digits) can be used.

NC title, customer, programmer, remark ...

TruTops writes this information in the setup plan. The remarks for the setup plan are self-holding until they are overwritten by new text.

➢ Enter the desired information.

Defining additional settings

1. Go to one of the tabs "Parameters", "Subroutine techn.", "Laser" or "Setup plan" (see following descriptions).
2. Select *OK*.

TruTops generates the NC program.

4.2 Defining parameters

➢ Select "Parameters".

Parameter	Effect
Number of program runs	Indicates how often the NC program at the machine is to be executed. The information is outputted in the setup plan. If a production plan has been created with TruTops or NcLink, then the NC program will be executed accordingly often, one time after another, as specified in the NC program.
Monitoring clamp positions	The machine executes the clamp checking program with position monitoring.
Monitor space between clamps	The machine executes the clamp checking program with space monitoring.
Prepare loading at end of program	Useful when the NC program is to be executed several times in succession. The loading of the next identical sheet starts immediately at the end of the NC program even while the sheet is still being processed.
Reload first tool at end of program	Useful when the NC program is to be executed several times in succession. The first tool in the NC program will be automatically newly mounted at the end of the NC program.
Use M24	Using high number of strokes.
Block numbering	To reduce the file size of the NC program: deselect option.
Transferring technology data	The laser technology data and the basic tooling data are transferred to the machine. Depending on the machine setting, the data is then adopted only if it is not yet present at the machine.
Transferring push-out subroutine	The standard push-out subroutine Q9998 is outputted.
Transferring sheet layout subroutine	The sheet layout subroutine "P9994" is applied to the control system of the machine with this option. It must be applied after each initial installation or after each update. The sheet layout subroutine is the user interface for the layout with single parts.
Overwriting NC files without warning	The message "Do you wish to overwrite" will not be displayed.
NC post-processing program	The post-processing program is started immediately after the generation of the NC program (see Section below).
Starting transfer immediately	The mask for data transfer is automatically opened after the generation of the NC program. The NC program can be copied or moved into the transfer folder.

Table 3

Using NC post-processing programs

Entries for post-processing programs or complete post-processing programs are supplied with TruTops. Post-processing programs are started directly after the NC program is generated.

TruTops Punch contains as default settings the following NC post-processing programs:

	Name of the NC post-processing program
Open NC program using an editor.	• LST_EDITOR
Print the NC program with the default printer.	• STD_PRINTER • STD_PRINTER_MO (MO = modal, i.e. TruTops waits while the NC program is being printed)
	INTERNETEDITOR
Collect data for printing labels.	PREPARE_LABEL

NC post-processing programs Table 4

Example: the NC text is to be opened and printed with the "Notepad" editor after the NC program is generated.

1. Select "NC post-processing program".
2. Open the input box with ▾.
   "Program selection" opens.
3. Make a check mark at NC post-processing program (here: STD_PRINTER).

TruTops imports the STD_PRINTER in the input box.

Using one's own post-processing programs

You can create your own NC post-processing programs and add them to the selection list (see TruTops software manual, Data module).
TruTops then offers these post-processing programs here.

4.3 Defining options for subroutine technique

Subroutine technique means: certain processing sequences which are repeatedly executed on a single sheet are written in a separate program (subroutine). This subroutine can be called up repeatedly within the main program.

1. Select "Subroutine techn.".
2. **Either**
   - ➢ Select "Part-oriented".

   The NC program is generated in a part-oriented manner. This means that the system is searched within a multi-copy area for identical processing sequences, which are written in the subroutine according to the defined rules.
   - ➢ Enter "Minimum number of machining operations for a subroutine".
   - ➢ Enter "Minimum number of activations for a subroutine".

   or
   - ➢ Select "Compulsory sequence".

   The machining operations in compulsory sequences are written into subroutines.

4.4 Defining vaporizing with the laser

Prerequisite

- The "Vaporizing" option is activated (Rule).

1. Select "Laser".

   Note
   The options under "Vaporizing" control the automatic vaporizing of all laser contours. No manually programmed vaporizing is affected by this, neither "Sheetwise" nor "Partwise".

2. Select "Vaporizing" "No" or "Sheetwise" or "Partwise".

4.5 Defining options for outputting the setup plan

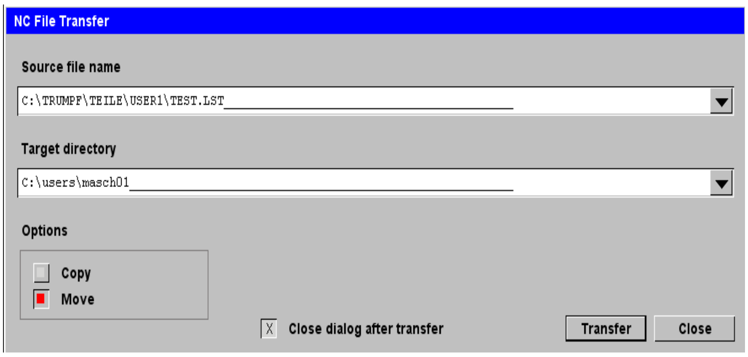
➢ Select "Setup plan".
(refer to TruTops software manual)

4.6 Sending the NC program to the machine

1. Select the "Parameters" tab.
   – Select "Start transfer immediately".
   – Select *OK*.

   TruTops displays the "NC file transfer" mask.

   Note
   'C:\DH\ToPsmanu.dir' is the default directory for NC programs on the machine.



"NC file transfer" Fig. 27067EN

Selecting the NC program

2. Open "Source file name" with ▾.
   The TruTops file browser is displayed.
3. Search the file if necessary.
4. Mark the NC program ('.lst') you are searching for.
5. Select *OK*.
   TruTops displays the "NC file transfer" mask again. The file name and the path of the NC program are entered under "Source file".

Changing the target directory on the machine

When transferring, TruTops always offers the directory that is entered for the machine (see TruTops software manual, Data module).

Note
'C:\DH\ToPsmanu.dir' is the default directory for NC programs on the machine.

1. To define the target directory permanently: select *Data module*, *Machines >Data*, *Basic machine*, *Data 7*, "Name of transfer path".
2. To modify the destination directory on a short-term basis (until next time the machine is changed): open "Target directory" via ▾.

   TruTops displays the "Select directory" mask.
3. Select directory.
4. Select *OK*.

   The "NC file transfer" mask will be displayed again. The new target directory is entered. It remains in effect until the next time the machine is exchanged.

Starting transfer

1. To send a copy of the NC program to the machine: select "Copy".

   or

   - To send the original of the NC program to the machine: select "Pan".
2. To transfer additional NC programs after transferring the first NC program, deactivate "Close dialog after transfer".
3. Select *Transfer*.

The data is transferred to the machine.

5. Glossary

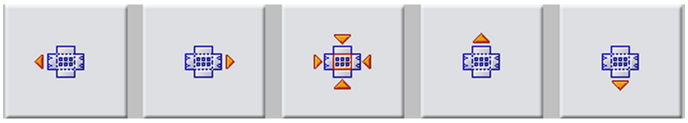
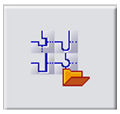
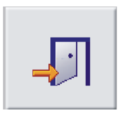
Term	Icon	Action/Explanation
Bend factors	-	During bending, the material is stretched or flattened. The change in the work-piece length during the bending process is compensated with the use of so-called bend factors. To enable the finished bent workpiece to correspond with the required dimensions, the dimensions of the flat blank are automatically shortened or extended by the bend factors.
Manage bend factor		Opens the database for bend factors.
Other flange		Opens a navigation window for single selection of the flange to be changed: 
Relief grooves	-	A relief groove is a "small notch". It reduces deformation of the edge or corner during bending.
Manage relief grooves		Opens the database for relief grooves.
Base	-	A bending part comprises of a base and a random amount of flanges. The base is the surface in the drawing from where the most important or most bends start out. If the base is determined in the construction drawing of a bending part, then several flanges are added to this.
TruTops start menu		Goes to TruTops start mask.
Manage		Opens the database for bend factors or relief grooves.
Continue in TruTops		Goes to TruTops basic or into the TruTops full version.

Table 5